GHOST AVIATOR · DGCA CPL & ATPL

Human Performance & Limitations

Aviation Physiology · Aviation Psychology · Human Factors — a complete guide for the DGCA CPL & ATPL examinations.

Capt. Pankaj Pahil

Pilot · Flight Instructor · Ground Instructor

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Foreword

DRAFT — IN THE CAPTAIN'S VOICE, FOR HIS APPROVAL

The foreword below is drafted for Capt. Pahil to edit and make his own before publication.

Of all the subjects a pilot must master, this is the only one about *you*. Navigation teaches the earth; meteorology teaches the sky; technical general teaches the machine. Human Performance teaches the one component that flies them all — and the one that most often fails.

In seventeen years of flying and instructing I have watched good pilots undone not by weather or metal, but by an ache in a sinus, a drink the night before, a light stared at too long in the dark, a slow bank that felt like level. None of it was mysterious. All of it was in a textbook. The gap between knowing a limit and respecting it at altitude is where this subject lives — and where accidents are quietly prevented, one informed decision at a time.

I wrote these notes for my own students, and I have gathered them here for every student pilot in India who cannot afford an expensive course but deserves a correct one. Learn the numbers until they are reflex. Understand the mechanisms until they are obvious. And then carry them into the cockpit, where the only exam that matters is the one you pass every time you land safely.

— Capt. Pankaj Pahil

How to Use This Book

The book is built in **eleven modules** and **twenty-six chapters**, running from the air outside the aircraft to the mind inside the pilot. Every chapter follows the same rhythm, and the furniture is always the same:

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The furniture of every chapter	
You will see…	**It means…**
Blue box	A core definition or principle to anchor on.
Amber box	An exam tip or mnemonic — the thing the examiner actually asks.
Red box	A danger, a killer item, or a number to memorise exactly.
Green box	A standard operating procedure — what a professional does.
Cinematic plate	A full diagram — anatomy, cockpit or process — to make the mechanism unforgettable.
Mnemonic pill	A memory aid; collected together in Chapter 25.

AN EIGHT-WEEK PLAN

- **Weeks 1–2:** the air & the body (Modules A–B, Ch 2–5).
- **Weeks 3–4:** the assaults of altitude (Modules C–D, Ch 6–10).
- **Weeks 5–6:** the senses & their deceptions (Modules E–F, Ch 11–15).
- **Week 7:** fitness & the mind aloft (Modules G–I, Ch 16–24).
- **Week 8:** the Exam Arsenal — numbers, mnemonics, mock papers (Module J, Ch 25–26).

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HUMAN PERFORMANCE & LIMITATIONS · MODULE 0 — THE THRESHOLD

The Human Factor

Chapter 1 — Every other subject teaches you the machine. This one teaches you the pilot flying it: the one component that cannot be re-certified, swapped out, or rebooted at altitude.

BOOK
Human Performance & Limitations

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EXAM
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CHAPTER
1 of 26 · Module 0

WHY THIS SUBJECT KEEPS YOU ALIVE

In modern aviation the aircraft rarely fails — **the human does**. Roughly **three out of every four accidents** trace back not to metal, but to a human limitation that was understood in a textbook and forgotten in a cockpit. Human Performance & Limitations is the study of that gap between what a pilot's body and mind can do and what the flight demands of them. Close it, and you become the safest system on the aircraft.

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Telegram channel
Telegram community

Plate 1.0 — The last line of defence. When the reports are read, it is almost never the wing or the radio that is named — it is the pilot.

1 What Human Performance & Limitations Is

Human Performance & Limitations (HPL) is the applied study of human behaviour, physiology and psychology as they relate to the operation of an aircraft, and of how those human factors influence the **safety and efficiency of flight**. It takes the tools of medicine and psychology into the cockpit for one reason only: to understand the pilot well enough to keep the pilot alive.

The **aim** of the subject is to **understand and predict** the behaviour of a person in the aviation environment, so that we can improve **safety, efficiency and comfort**. A practitioner works through four actions — fix them in memory now, because the whole subject is built on them.

EXAM TIP — THE FOUR ACTION VERBS

The four things HPL sets out to do with human behaviour: **D – P – U – I → Describe, Predict, Understand, Influence.** If a question asks what aviation psychology aims to do, the answer lives in those four verbs.

2 Why the Human Fails

Start from an uncomfortable truth the examiner expects you to state plainly: **human error is inevitable.** It is not the mark of a bad pilot; it is a property of being human. The professional's task is therefore not to *never* err — that is impossible — but to **reduce the frequency of error, and to trap and mitigate the errors that do occur** before they reach the ground.

Most pilot-related errors are not failures of stick-and-rudder skill. They are failures in four areas: **interpersonal skills, communication, decision-making, and leadership.**

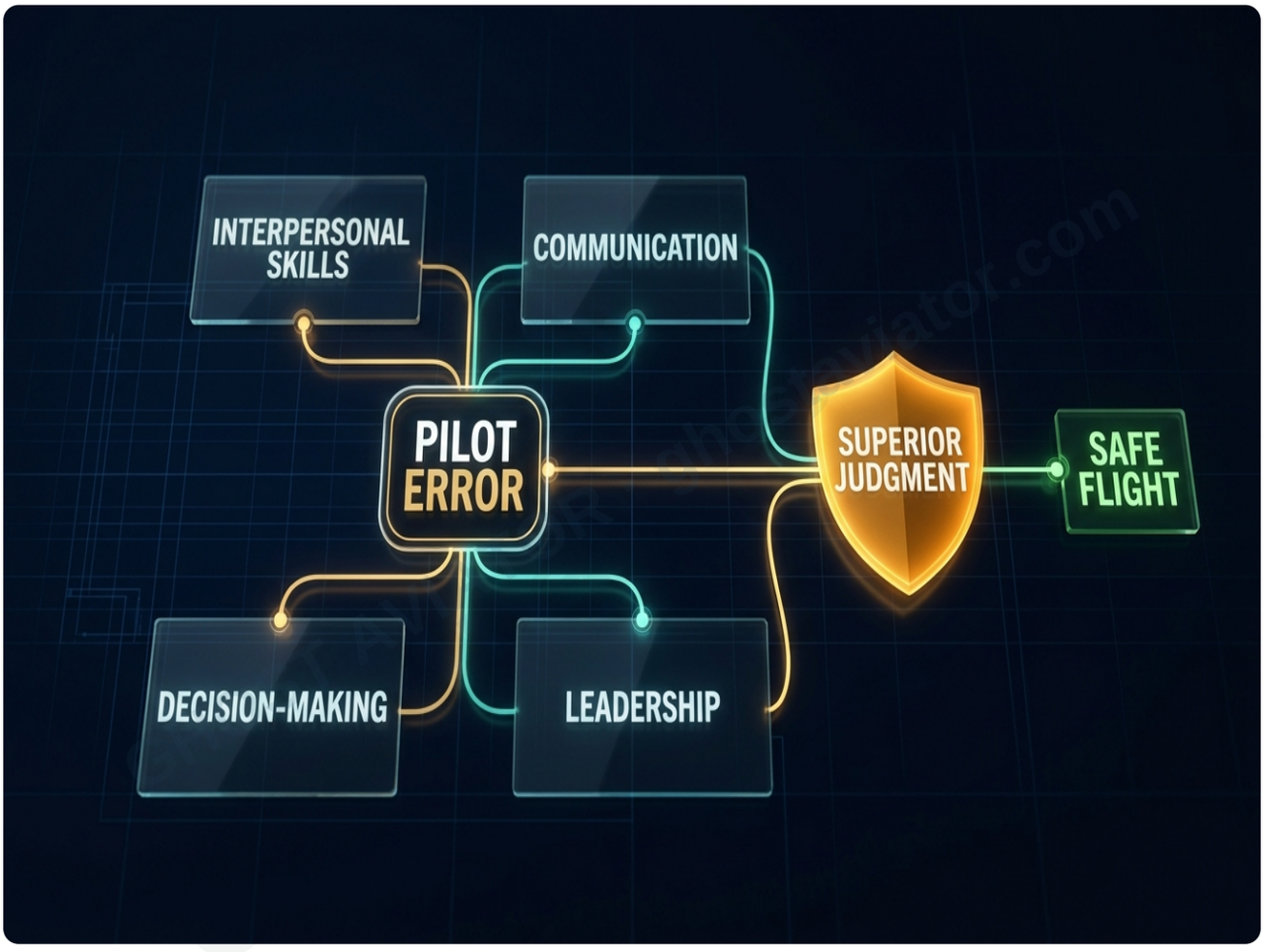

Figure 1.1 — The four origins of pilot error, all trapped by one thing: superior judgment.

GOLDEN MAXIM

"A superior pilot uses his superior judgment to avoid situations that would require his superior skill."

3 The SHELL Model — Where the Pilot Meets the System

No pilot flies in isolation. You sit at the centre of a system, connected to the machine, the paperwork, the world outside, and the other people in the loop. The **SHELL model** is the standard way of picturing those connections. Its central insight: accidents rarely happen inside a component — they happen at the **joints between them**, where the human interface is imperfect.

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Figure 1.2 — The SHELL model. Liveware (you) sits at the centre; the ragged edges show the imperfect fit of every human interface.

The four SHELL interfaces — what sits at each joint with the pilot		
Interface	**What it is**	**Where it bites in the cockpit**
Liveware – Software	The pilot and the non-physical system: procedures, checklists, SOPs, symbology, charts.	A confusing checklist, an ambiguous chart symbol, a badly worded procedure.
Liveware – Hardware	The pilot and the machine: controls, displays, seats, switches.	A switch shaped like its neighbour; a display read wrong in a hurry.
Liveware – Environment	The pilot and the world, inside and out: weather, cabin altitude, noise, heat, plus the organisation's rules and pressures.	Glare, hypoxia, time pressure from the schedule.
Liveware – Liveware	The pilot and other people: the other crew member, ATC, cabin crew, engineers.	A missed read-back; an authority gradient too steep to question the Captain.

EXAM TIP — THE CENTRE BLOCK

The component at the **centre** of the SHELL model is **Liveware — the human being.** It is drawn with ragged edges because humans are the least standardised part of the system. Every other block must be shaped to fit the human, not the other way round.

4 The Dirty Dozen — Twelve Ways the Trap Is Set

If the SHELL model shows you *where* error lives, the **Dirty Dozen** shows you *how* it is invited in — the twelve most common human preconditions that precede an error. For each there is a **safety net**. Learn them as pairs.

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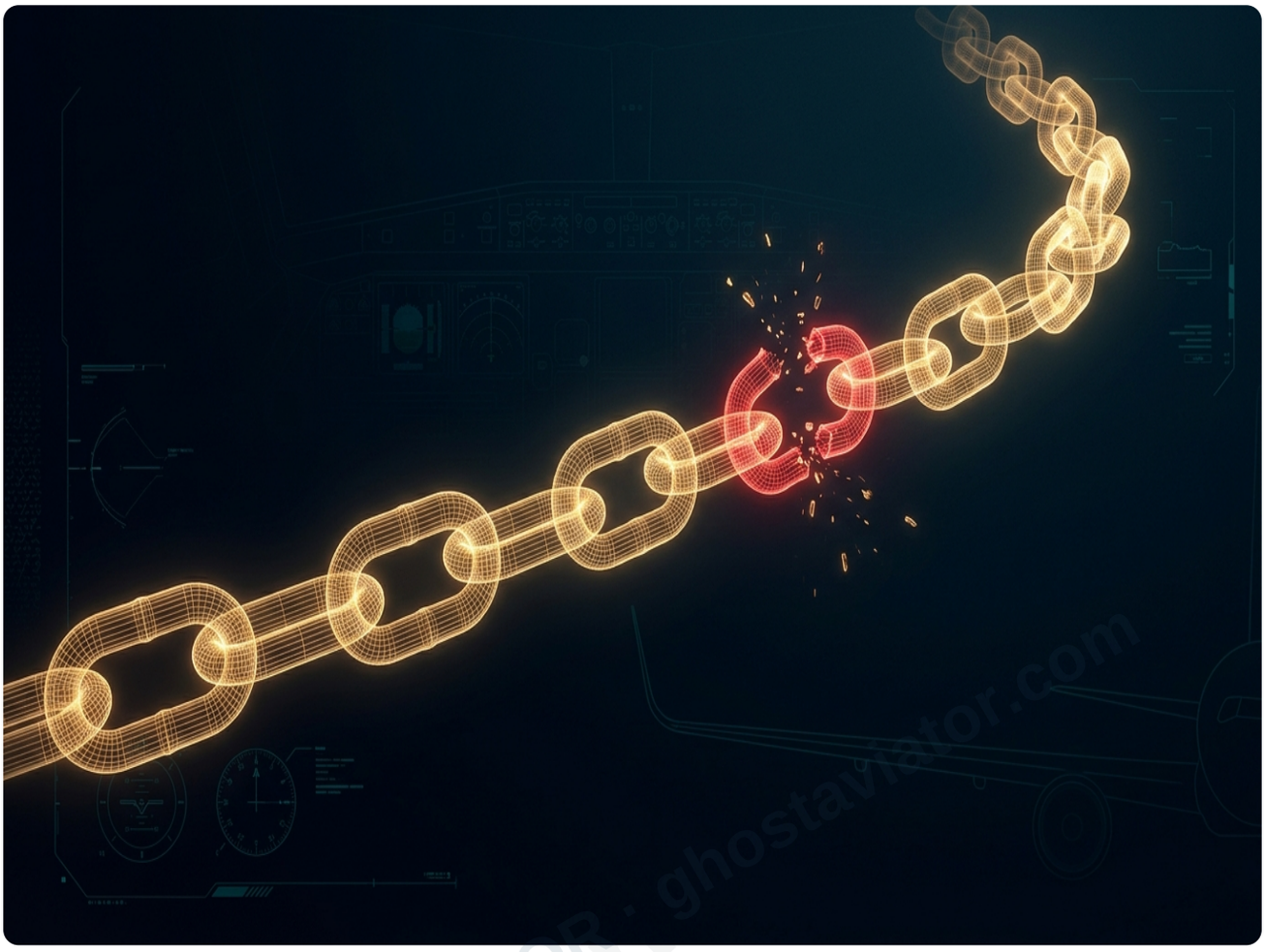

Figure 1.3 — The accident chain. Catch a single precondition and the chain breaks before it reaches the ground.

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Capt. Pat

The Dirty Dozen and their safety nets	
The precondition	**The safety net**
1. Lack of communication	Use read-backs; say the whole message and confirm it was received.
2. Complacency	Expect to find a fault; never sign off on "it's always been fine".
3. Lack of knowledge	Don't guess — ask, or open the book. This is why the book exists.
4. Distraction	When interrupted, go back three steps before carrying on.
5. Lack of teamwork	Brief, share the plan, agree who does what.
6. Fatigue	Know the symptoms in yourself; rest is a duty, not a luxury.
7. Lack of resources	Have the right parts, charts, fuel and people before you commit.
8. Pressure	Refuse self-imposed haste; a delay is cheaper than an accident.
9. Lack of assertiveness	Speak up. "I am not happy with this" is a complete sentence.
10. Stress	Recognise it, manage it, ask for help — don't fly through it silently.
11. Lack of awareness	Think one step ahead; ask "what if?" before it happens.
12. Norms	Just because "everyone does it" does not make it safe or legal.

WHY THIS MATTERS

Every one of the Dirty Dozen has a body count. They are insidious precisely because each feels harmless in the moment — a small skipped step, a small silence, a small assumption. HPL is the discipline of catching the small thing before it becomes the last thing.

5 How to Use This Book

This book is built to be studied, not just read. Every chapter follows the same rhythm — learn the rhythm and you move fast.

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The furniture of every chapter	
You will see…	**It means…**
Blue box	A core definition or principle to anchor on.
Amber box	An exam tip or a mnemonic — the thing the examiner actually asks.
Red box	A danger, a killer item, or a number that must be memorised exactly.
Green box	A standard operating procedure — what a professional actually does.
Cinematic plate	A full diagram — anatomy, cockpit or process — built to make the mechanism unforgettable.
Self-check	End-of-chapter questions and mnemonics. Answer before you reveal.

AN EIGHT-WEEK PLAN

HPL rewards steady memorisation over cramming. A workable rhythm:

- **Weeks 1–2:** the air & the body — atmosphere, gas laws, the body's systems (Modules A–B).
- **Weeks 3–4:** the assaults of altitude — hypoxia, pressure, poisons (Modules C–D).
- **Weeks 5–6:** the senses & their deceptions — eye, ear, illusions, G (Modules E–F).
- **Week 7:** fitness to fly & the mind aloft — health, psychology, stress (Modules G–I).
- **Week 8:** the Exam Arsenal — master tables, mnemonics, mock papers (Module J).

6 Self-Check & Memory Aids

ANSWER THESE WITHOUT LOOKING BACK

1. State the four action verbs that describe the aim of aviation psychology.
2. Human error is inevitable — name the four areas from which most pilot error originates.
3. Name the four interfaces of the SHELL model, and state which component sits at the centre and why.
4. What are the "Dirty Dozen"? Give any six, with the safety net for each.
5. Quote the Golden Maxim on superior judgment.

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ANSWERS

1. Describe, Predict, Understand, Influence (D-P-U-I). **2.** Interpersonal skills, communication, decision-making, leadership. **3.** Software, Hardware, Environment, Liveware; *Liveware (the human)* sits at the centre because it is the least predictable, least standardised component — everything else must be shaped to fit it. **4.** The twelve common preconditions for error (lack of communication, complacency, lack of knowledge, distraction, lack of teamwork, fatigue, lack of resources, pressure, lack of assertiveness, stress, lack of awareness, norms). **5.** "A superior pilot uses his superior judgment to avoid situations that would require his superior skill."

MNEMONIC — SHELL **S**oftware · **H**ardware · **E**nvironment · **L**iveware · **L**iveware — the human **L** sits in the middle.

MNEMONIC — AIM OF HPL **"Do Pilots Understand Instruments?" → D**escribe · **P**redict · **U**nderstand · **I**nfluence.

✦ END OF CHAPTER 1 ✦

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HUMAN PERFORMANCE & LIMITATIONS · MODULE A — THE OCEAN OF AIR

The Atmosphere

Chapter 2 — The atmosphere and its constituent gases, the five Gas Laws, and the International Standard Atmosphere. The foundation on which every physiological effect of flight is built.

BOOK
Human Performance & Limitations

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CHAPTER
2 of 26 · Module A

In this chapter

1. The Atmosphere & its Constituent Gases
2. The Gas Laws (Boyle · Henry · Charles · Graham · Dalton)
3. Variation of Pressure & Temperature with Altitude (ISA)

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Plate 2.1 — The ocean of air: from the dense, weather-filled troposphere to the thin stratosphere the pilot must respect.

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WHY THIS CHAPTER MATTERS TO A PILOT

Aviation Physiology deals with the **physical and mental effects of flight on aircrew personnel and passengers**. In aviation, the demands upon the compensatory mechanisms of the body are *numerous and of considerable magnitude*. The environmental changes of greatest physiological significance involved in flight are:

- Marked changes in **barometric pressure**
- Considerable variation in **temperature**
- **Movement at high speed in three dimensions**

This part of the syllabus familiarises you with the physiological problems of flight and assists in human compensation for the numerous environmental changes encountered in flight. **Remember: every human is physiologically different and can react differently in any given situation.**

§ 1

The Atmosphere

Human beings live their lives in the **lower reaches of the atmosphere** where temperatures, pressures and oxygen supply are able to support life. The moment we climb, every one of those three variables shifts against us. To fly safely, you must first understand the layer you're flying in and what it is made of.

1.1 The Layers — Troposphere & Stratosphere

TROPOSPHERE – AVERAGE TOP

13 km · 8.1 mi · 43,000 ft

STRATOSPHERE – REACHES ABOVE

100,000 ft

WHERE ALMOST ALL WEATHER LIVES

Troposphere

WHERE AIR IS DENSEST

Lowest Layer (Surface)

The Troposphere height **varies**. On average it stretches from the Earth's surface to about `13 km (8.1 mi; 43,000 ft)`. The Stratosphere reaches up to over `100,000 ft`. The troposphere contains **almost all the weather**, the air is densest in this lowest layer, and in fact the troposphere contains **three-quarters of the mass of the entire atmosphere**.

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Figure 2.2 — Vertical structure of the atmosphere: the troposphere holds ¾ of all atmospheric mass and nearly all weather.

1.2 Constituent Gases of the Atmosphere

The earth's atmosphere near the surface is composed primarily of **Nitrogen** and **Oxygen**. Together, the two comprise **about 99%** of the gas in the atmosphere. The remaining 1% is made up of Argon plus traces of other gases.

Composition of the lower atmosphere — exact percentages

Gas	Percentage	Gas	Percentage
Nitrogen (N_2)	`78.084 %`	**Oxygen (O_2)**	`20.95 %`
Argon (Ar)	`0.934 %`	Carbon Dioxide (CO_2)	`0.036 %`
Neon (Ne)	`0.0018 %`	Helium (He)	`0.0005 %`
Methane (CH_4)	`0.00017 %`	Hydrogen (H_2)	`0.00005 %`
Nitrous Oxide (N_2O)	`0.00003 %`	Ozone (O_3)	`0.000004 %`

In addition, **water vapour** is variable but typically makes up about `1 – 4 %` of the atmosphere.

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CRITICAL PRINCIPLE TO REMEMBER

The relative proportions of these gases remain CONSTANT in the Troposphere and Stratosphere. What changes with altitude is not the *percentage* of oxygen — it is the *total pressure*, and therefore the **partial pressure** of each gas. Oxygen continues to make up 21 % of the air by volume **at every altitude**; only the partial pressure of oxygen, pO_2, falls.

MNEMONIC **"Never Ought An Cool Naval Helmsman Make His Nasty Owl"** — Nitrogen, Oxygen, Argon, Carbon dioxide, Neon, Helium, Methane, Hydrogen, Nitrous oxide, Ozone — in descending order of abundance. (Optional aid — feel free to invent your own.)

§ 2 The Gas Laws

The body responds to barometric pressure changes in temperature, pressure, and volume. These changes are **rapid and continuous** in the aviation environment. It is therefore essential to know the implication of these changes on our body and take preventive measures to counter them. The gas laws explain to us the **science behind what goes on within our body** when exposed to changes in pressure and temperature.

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Figure 2.3 — The five gas laws of flight and what each one explains in the cockpit.

2.1 Boyle's Law

DEFINITION

At a **constant temperature**, a given volume of gas is **inversely proportional to the pressure** surrounding the gas. The volume of gas **expands as the pressure surrounding the gas is reduced**.

- **As altitude increases → gas expands.**
- **As altitude decreases → gas compresses.**
- The amount of volume expansion is **limited by the pliability** of the structure or membrane which encloses the gas.

EXAMPLES IN AVIATION & MEDICINE

PASG / Air Splints · Respiratory Rate & Depth changes · Flow rates of **IV sets** · **ETT** or **Tracheal cuff** pressures · **Trapped gas effects** within the body (middle ear, sinuses, GI tract — every pilot has felt these).

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Figure 2.4 — Boyle's Law: as pressure halves, a fixed mass of gas doubles in volume. Trapped gas in sinuses, ears, gut and fillings expands as you climb.

2.2 Henry's Law

DEFINITION

The amount of **gas in solution** is proportional to the **partial pressure of that gas over the solution**. As the pressure of the gas above a solution *increases*, the amount of that gas dissolved in the solution *increases*; the reverse is also true — as the pressure of the gas above a solution *decreases*, the amount of gas dissolved *decreases* and forms a **"bubble" of gas within the solution**.

In normal physiologic function, this law can be seen in the **transfer of gas between the alveoli and the blood**. This is **significant physiologically** for the occurrence of evolved gas disorders, e.g. **decompression sickness**. It explains the hypoxia experienced with increasing altitude — as the pressure of gases is reduced with ascent, the amount of gases dissolved in solution decreases, and this leads to hypoxia and may lead to **nitrogen bubble formation**.

EVERYDAY EXAMPLE

Bottle of soda. With the cap on, the gas within the solution is at equilibrium. With the cap removed, the gas pressure decreases and bubbles are released into the solution. Your blood does exactly the same thing on a sudden cabin depressurisation.

CLINICAL LINK — DCS (DECOMPRESSION SICKNESS)

A pilot who has been SCUBA diving cannot climb shortly after surfacing because Henry's Law is still working against them — dissolved nitrogen in their blood will come out of solution as bubbles when ambient pressure falls. *(Strict wait-times before flying after diving are covered in a later section.)*

2.3 Charles' Law

DEFINITION

The **pressure of a gas is directly proportional to its temperature** with the **volume remaining constant**. Temperature increases make gas molecules move faster, greater force is exerted, and volume expands.

The law explains:

- The **temperature changes associated with rapid decompression**.
- The **pressure changes that induce temperature changes with an oxygen cylinder** (a freshly-charged O_2 bottle feels warm; a discharging one feels cold).

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CLASSIC ILLUSTRATION

Shaving cream can be placed into a fire — heat raises the internal pressure of the can until the structure fails. Same physics governs the rapid heating of an oxygen cylinder during fast filling.

2.4 Graham's Law — Law of Gaseous Diffusion

DEFINITION

Gases **diffuse or migrate from a region of higher concentration (or pressure) to a region of lower concentration (or pressure)** until equilibrium is reached. The physiological significance is in the explanation of **gas exchange**.

SOP — HOW YOUR LUNGS USE THIS LAW

Oxygen moves from the alveoli into the blood, and from the blood into the tissues due to this phenomenon. The whole architecture of the respiratory system relies on Graham's Law for survival.

2.5 Dalton's Law — Law of Partial Pressures

DEFINITION

Partial pressures describe the **distribution of certain gases in a mixture** and follows Dalton's law which states that the **total pressure of a mixture of gases is equal to the sum of the partial pressures of the gases which compose the mixture**. As altitude increases — gases exert **less pressure**. This explains the **hypoxia that occurs with flight to higher altitudes**.

WORKED EXAMPLE — OXYGEN AT DIFFERENT ALTITUDES (DALTON'S LAW IN ACTION)

At Sea Level: O_2 = 21 % & **pO_2 = 21 % × 760 mm =** `≈ 160 mmHg`

At 8,000 ft: O_2 = 21 % & **pO_2 = 21 % × 565 mm =** `≈ 119 mmHg`

The barometric pressure at `36,000 ft` is **one-fourth** of that at sea level. Hence the quantity of oxygen available is **proportionately low**. *Whatever the air pressure, oxygen continues to make up 21 % of the air by volume.* In other words, the **proportion of oxygen in the air always stays the same whatever the altitude**. **The partial pressure of Oxygen decreases with altitude as does the total pressure of air.**

EXAM-CRITICAL TAKEAWAY

The percentage of O_2 in air does **NOT** change with altitude — it is always `21 %`. What kills the pilot is the falling **partial pressure** of that oxygen, because diffusion across the alveolar membrane depends on *pressure gradient*, not percentage.

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Quick-Reference — The Five Gas Laws of Aviation Physiology		
Law	**One-line Statement**	**What it Explains in Flight**
Boyle's	P × V = constant (at fixed T) — pressure ↑ → volume ↓	Trapped-gas pain in ears, sinuses, teeth, GI tract; tracheal-cuff & IV behaviour
Henry's	Gas dissolved ∝ partial pressure over the solution	Decompression sickness, hypoxia mechanism, alveolar gas transfer
Charles'	P ∝ T (at fixed V) — warming raises pressure	Rapid-decompression temperature drop; O_2 cylinder heating
Graham's	Gases diffuse from high → low concentration	O_2 from alveoli → blood → tissues; CO_2 in opposite direction
Dalton's	Total P = Σ partial pressures	Why high-altitude hypoxia happens despite air being "21% oxygen"

§ 3

Variation of Pressure & Temperature with Altitude

3.1 The International Standard Atmosphere (ISA)

WHAT ISA ACTUALLY IS

The International Standard Atmosphere is used to show **standardised values for temperature, pressure, density and lapse rate**. Pressure decreases with altitude *throughout* the atmosphere. Temperature decreases with altitude *in the Troposphere*.

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ISA Sea-Level Values (memorise these cold)

SEA-LEVEL PRESSURE
1013.25 hPa

EQUIVALENT (MB)
1013.25 mb

EQUIVALENT (PSI)
14.7 lb/in²

EQUIVALENT (HG)
29.92 in Hg

EQUIVALENT (MMHG)
760 mm Hg

ISA LAPSE RATE
1.98 °C / 1000 ft

STRICT ISA LIMITS — EXAMINATION VALUES

Temperature reduces at 1.98 °C per 1000 ft up to `36,090 ft` **.** Thereafter it remains **constant** at `−56.5 °C`. *(The 1.98 °C figure is the precise ISA lapse rate; many texts round to 2 °C / 1000 ft for mental maths — never round in your exam paper.)*

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Figure 2.5 — The ISA temperature profile: 1.98 °C / 1000 ft up to 36,090 ft, then constant at −56.5 °C through the isothermal stratosphere.

✦ END OF CHAPTER 2 ✦

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HUMAN PERFORMANCE & LIMITATIONS · MODULE A — THE OCEAN OF AIR

Altitude & Oxygen

Chapter 3 — The partial pressure of oxygen and the altitude penalty, the four physiological zones of the atmosphere, and the numbers every pilot must know cold.

BOOK
Human Performance & Limitations

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CHAPTER
3 of 26 · Module A

In this chapter

1. Partial Pressure of Oxygen — Effects of Increasing Altitude
2. The Physiological Zones of the Atmosphere
3. Self-Check & Memory Aids (Module A)

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Plate 3.0 — Higher is thinner. As the total pressure falls, so does the oxygen your blood can reach — the altitude penalty.

§ 4 Partial Pressure of Oxygen — Effects of Increasing Altitude

As you climb, the **total barometric pressure** drops, and with it the **partial pressure of oxygen** (pO_2). Because the diffusion of oxygen across the alveolar membrane depends on the pressure gradient, what matters to your blood is not the percentage of O_2 — it is the pO_2. Here are the standard values the DGCA expects you to know:

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Standard barometric pressure & oxygen availability at altitude

Altitude	Standard Barometric Pressure	O_2 Available (% of Sea Level)	Pilot Significance
Sea Level (0000 ft)	`101 kPa` (`760 mmHg`)	`100 %`	Normal physiological baseline.
8,000 ft	`77 kPa` (`574 mmHg`)	`76 %`	**Cabin pressure is maintained between 6,000 – 8,000 ft.**
12,000 ft	`66 kPa` (`496 mmHg`)	`65 %`	Lower edge of the Physiological-Deficient Zone.
18,000 ft	`53 kPa` (`395 mmHg`)	`52 %` (≈ half)	**Half the O_2 of sea level.**
24,000 ft	`41 kPa` (`311 mmHg`)	`41 %`	Without supplemental O_2 — incapacitation imminent.
36,000 ft	`25 kPa` (`187 mmHg`)	`25 %` (one-fourth)	**Only ¼ of sea-level oxygen.**

THREE DEDUCTIONS YOU MUST BE able TO STATE IN THE EXAM

1. The **availability of oxygen and barometric pressure DECREASE with altitude**.
2. The oxygen available is **one-fourth** at `36,000 ft` and **half** at `18,000 ft` of the oxygen available at sea level.
3. **Atmospheric pressure drops faster at lower altitudes** in comparison to the same altitude changes at higher altitudes (the pressure–altitude curve is exponential, not linear).

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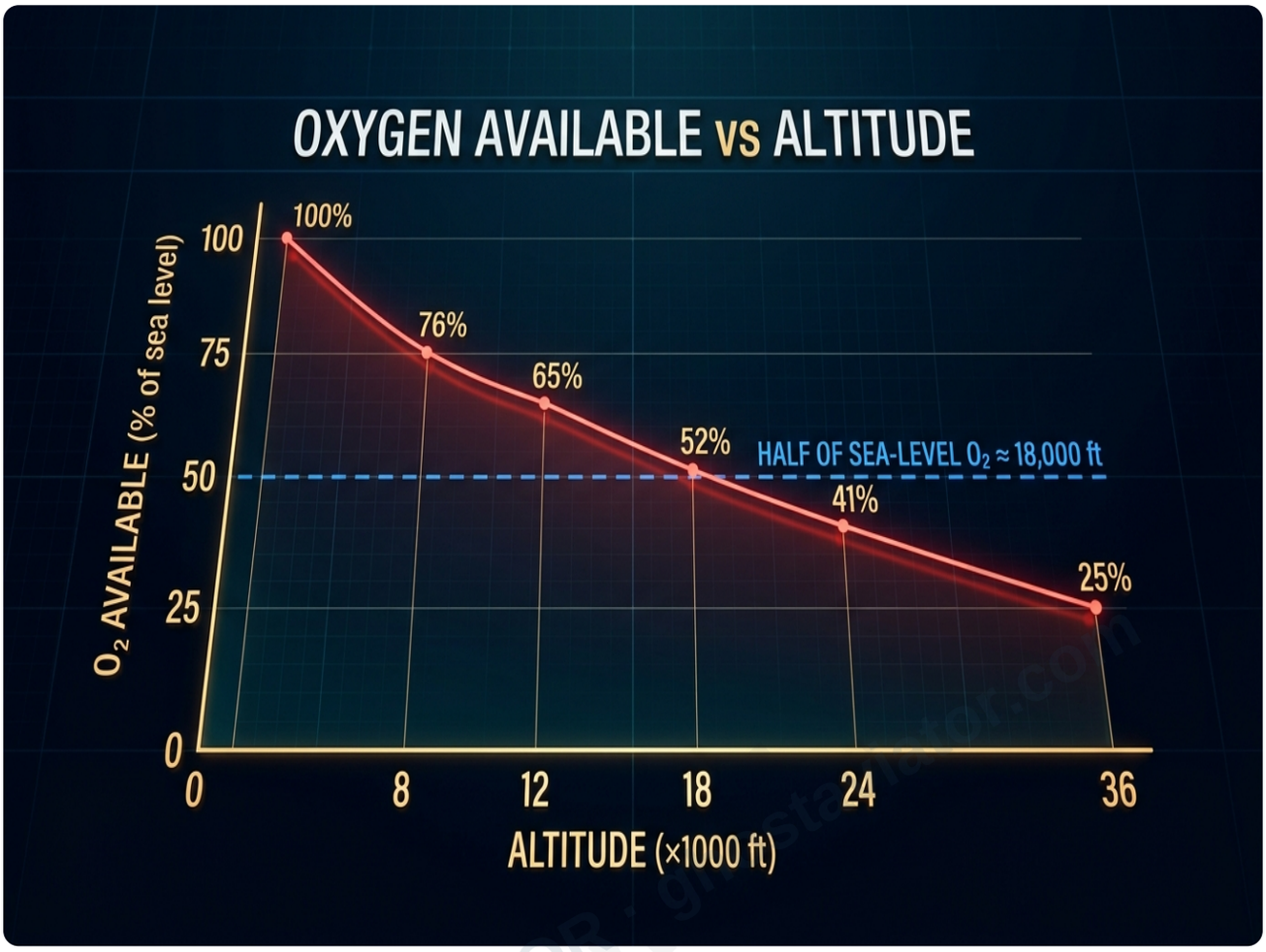

Figure 3.1 — The oxygen available falls non-linearly with altitude: half is gone by 18,000 ft, three-quarters by 36,000 ft.

§ 5 The Physiological Zones of the Atmosphere

The atmosphere is divided — from the pilot's body's point of view — into four functional zones, each demanding a different protective strategy. **You must know the boundaries and the SOP for each.**

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Figure 3.2 — The four physiological zones of the atmosphere and the protection each demands.

5.1 The Physiological Zone

SOP — WITHIN THIS ZONE

Normal healthy human beings used to living near sea level will need **supplementary oxygen to function normally at altitude exceeding** `10,000 to 12,000 ft`. **Pilots will normally begin breathing supplementary oxygen from** `10,000 ft` **above sea level.**

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Figure 3.3 — The oxygen-supply regime by altitude: air only, oxygen+air, 100% oxygen, then oxygen under pressure.

5.2 The Physiological-Deficient Zone

LIMITS & SYMPTOMS — STRICT

Exists from `12,000 ft to 50,000 ft`. **The body is not used to this environment.** The adverse effects include:

- Middle ear and sinus blockage
- Shortness of breath
- Dizziness
- Headache

COMPENSATORY THRESHOLD

At `6,000 – 7,000 ft` altitude the human organism starts with remarkable measures to compensate for the drop in pO_2 when climbing — this is the **threshold for compensatory reactions**. (Hyperventilation, raised heart rate, raised cardiac output, etc. — all later sections.)

CRITICAL-THRESHOLD LIMITS — NON-PRESSURISED FLIGHT

- A pilot climbing in a **non-pressurised aircraft and WITHOUT using supplemental oxygen** will pass the **"critical threshold" at approximately** `22,000 ft`.
- **Breathing 100 % oxygen** will lift the pilot's physiological safe altitude to approximately `38,000 ft`.

5.3 The Partial Space-Equivalent Zone

DEFINITION & EXTENT

This zone extends from `50,000 ft to 120 nm` (nautical miles). Even 100 % oxygen ceases to be enough at these heights — total pressure becomes the limiting factor.

5.4 The Total Space-Equivalent Zone

DEFINITION

The Total Space-Equivalent Zone extends **outwards from** `120 nm`.

MANDATORY LIMITS IN THE UPPER TWO ZONES

- **100 % oxygen does NOT protect from hypoxia** — pressure-breathing required.
- **Sealed cabins and pressure suits are a MUST.**
- Blood and body fluids **BOIL above** `63,000 ft` (Armstrong's Line — vapour pressure of body fluids equals ambient pressure).
- **Gravitational changes affect the body** (micro-G adaptation, fluid shift, motion-sickness, etc.).

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Figure 3.4 — Altitude-versus-protection timeline, from air-only at low level to pressure suits and Armstrong's Line.

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§ 6 Self-Check & Memory Aids

6.1 Numbers you MUST know by heart

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Master cheat-sheet — Part 1 numbers

Parameter	Exact Value	Where it comes from
Top of troposphere (average)	`13 km / 8.1 mi / 43,000 ft`	§1 Atmospheric layers
Top of stratosphere (over)	`100,000 ft`	§1
Troposphere holds	`75 %` of atmospheric mass	§1
Nitrogen / Oxygen / Other	`78.084 / 20.95 / ~1 %`	§1.2
Water vapour content	`1 – 4 %` (variable)	§1.2
ISA sea-level pressure	`1013.25 hPa = 760 mmHg = 29.92 inHg = 14.7 psi`	§3.1
ISA lapse rate	`1.98 °C / 1000 ft`	§3.1
Tropopause altitude (ISA)	`36,090 ft`	§3.1
Isothermal temperature	`−56.5 °C`	§3.1
Cabin altitude maintained	`6,000 – 8,000 ft`	§4
O_2 available at 18,000 ft	`52 %` (half of SL)	§4
O_2 available at 36,000 ft	`25 %` (one-fourth of SL)	§4
Compensatory reactions begin	`6,000 – 7,000 ft`	§5.2
Pilot starts supplementary O_2	`10,000 ft`	§5.1
Physiological-Deficient Zone	`12,000 – 50,000 ft`	§5.2
Critical threshold (no O_2, unpressurised)	`~22,000 ft`	§5.2
Safe altitude on 100% O_2	`~38,000 ft`	§5.2
100% O_2 Under Pressure (PBA) from	`Above 40,000 ft`	§5.1 fig.
Partial Space-Equivalent Zone	`50,000 ft – 120 nm`	§5.3
Armstrong's Line (body fluids boil)	`63,000 ft`	§5.4
Total Space-Equivalent Zone	`Outwards from 120 nm`	§5.4

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6.2 Common DGCA-style probe questions

TRY ANSWERING THESE WITHOUT LOOKING BACK

1. State the percentage composition of nitrogen and oxygen in the atmosphere. *Does it change with altitude?*
2. Give the ISA sea-level values in **five** different units.
3. What is the ISA lapse rate, and up to what altitude does it apply?
4. Why does hypoxia occur with altitude, when the % of O_2 is unchanged? *(Hint: Dalton's Law.)*
5. At what altitude does a pilot normally start using supplementary oxygen?
6. State the four physiological zones with their altitude boundaries.
7. At what altitude does an unpressurised pilot, NOT using supplemental O_2, hit the critical threshold?
8. Why is a pressure suit mandatory above 50,000 ft? Quote Armstrong's Line value.
9. Explain Boyle's Law and give two flight-medical situations where it applies.
10. How does Henry's Law explain both hypoxia *and* decompression sickness?

MNEMONIC — 5 GAS LAWS **"Bad Hens Cluck, Geese Dance"** → **B**oyle · **H**enry · **C**harles · **G**raham · **D**alton.

MNEMONIC — ZONES (LOW → HIGH) **P – P-D – PS – TS** → Physiological · Physiological-Deficient · Partial Space-equivalent · Total Space-equivalent. *(Or: "Pilots Probably Prefer Tea" — P, PD, PS, TS.)*

MNEMONIC — HALVING RULE **"Half by Eighteen, Quarter by Thirty-Six"** → Half of sea-level O_2 at **18,000 ft**; one-quarter at **36,000 ft**.

✦ END OF CHAPTER 3 ✦

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HUMAN PERFORMANCE & LIMITATIONS · MODULE B — THE LIVING MACHINE

The Brain & Nervous System

Chapter 4 — The pilot's control computer: the brain, and the central, peripheral and autonomic nervous systems that carry every command and reflex.

BOOK
Human Performance & Limitations

AUTHOR
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CHAPTER
4 of 26 · Module B

In this chapter

1. The Brain & the Nervous System (CNS · PNS · ANS)

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Plate 4.0 — The control computer. Every decision, reflex and command in the cockpit begins here.

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§ 7 The Brain & the Nervous System

The brain is the master controller of your body — every input you receive in flight (visual, vestibular, proprioceptive, auditory) is processed here, and every output (control inputs, R/T calls, decisions) originates here. The nervous system is the wiring that connects the brain to every organ and muscle.

7.1 The Brain

DEFINITION

The brain controls all bodily functions. It performs an incredible number of tasks, every one of which is potentially flight-critical.

What the brain does — exam list

- It controls **body temperature, blood pressure, heart rate and breathing**.
- It accepts information about the world around you from your various **senses** — seeing, hearing, smelling, tasting and touching.
- It handles your **physical movement** when walking, talking, standing or sitting.

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- It lets you **think, dream, reason and experience emotions**.
- The brain controls **decision-making and speech**. Through the brain we experience **consciousness, vision, hearing, taste, smell, thought and memory**.

> **WHY THIS MATTERS IN THE COCKPIT**
>
> Every one of these functions is **oxygen-dependent**. The brain consumes about **20 % of the body's oxygen** despite being only ~2 % of body mass. Even mild hypoxia degrades judgement, decision-making and speech long before the pilot is consciously aware of it. *(Detailed in §10 Hypoxia — Part 3.)*

7.2 The Nervous System — Three Main Parts

The Nervous System is divided into **three main parts**:

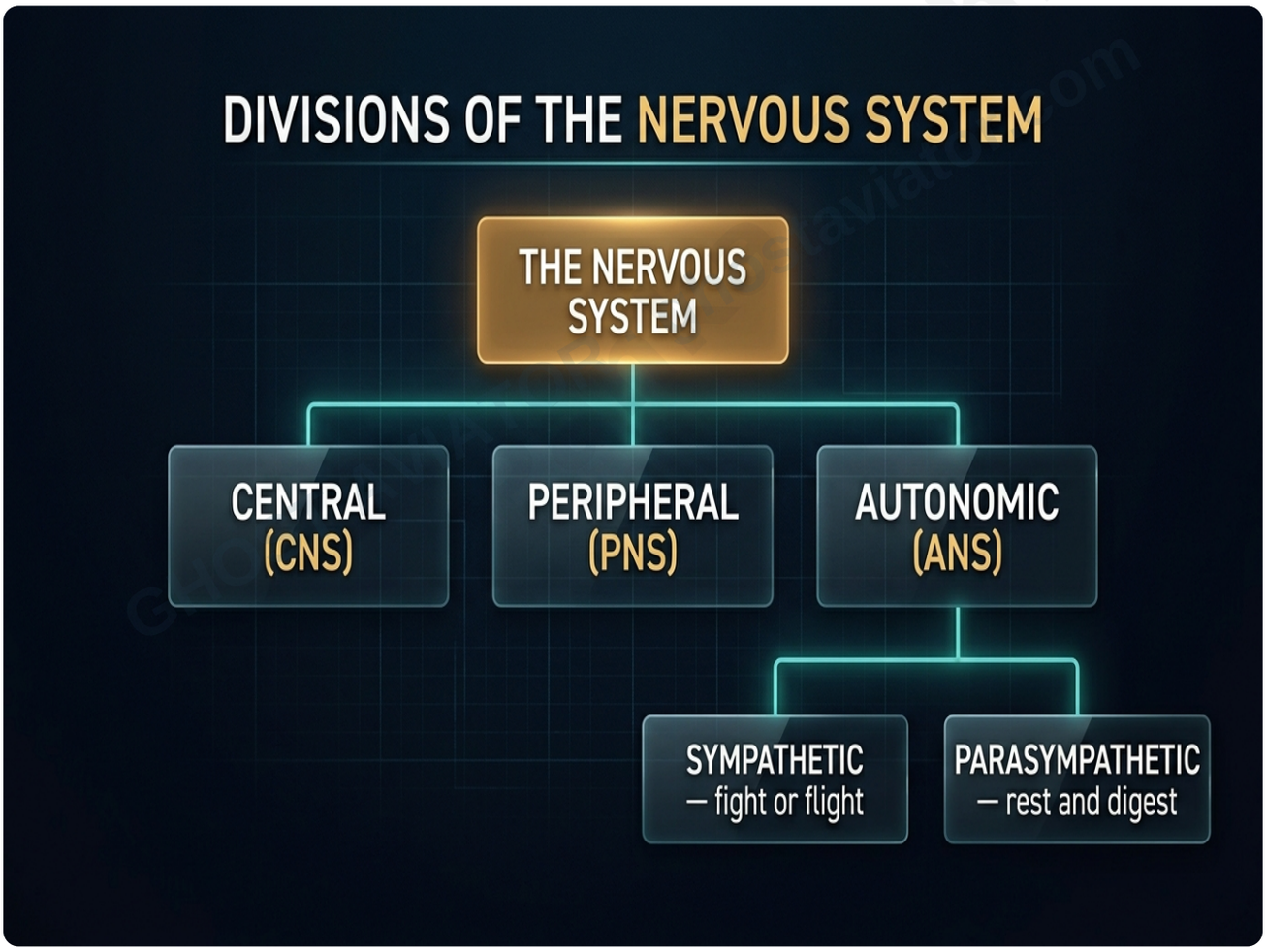

Figure 4.2 — The three divisions of the nervous system, with the autonomic split into sympathetic and parasympathetic.

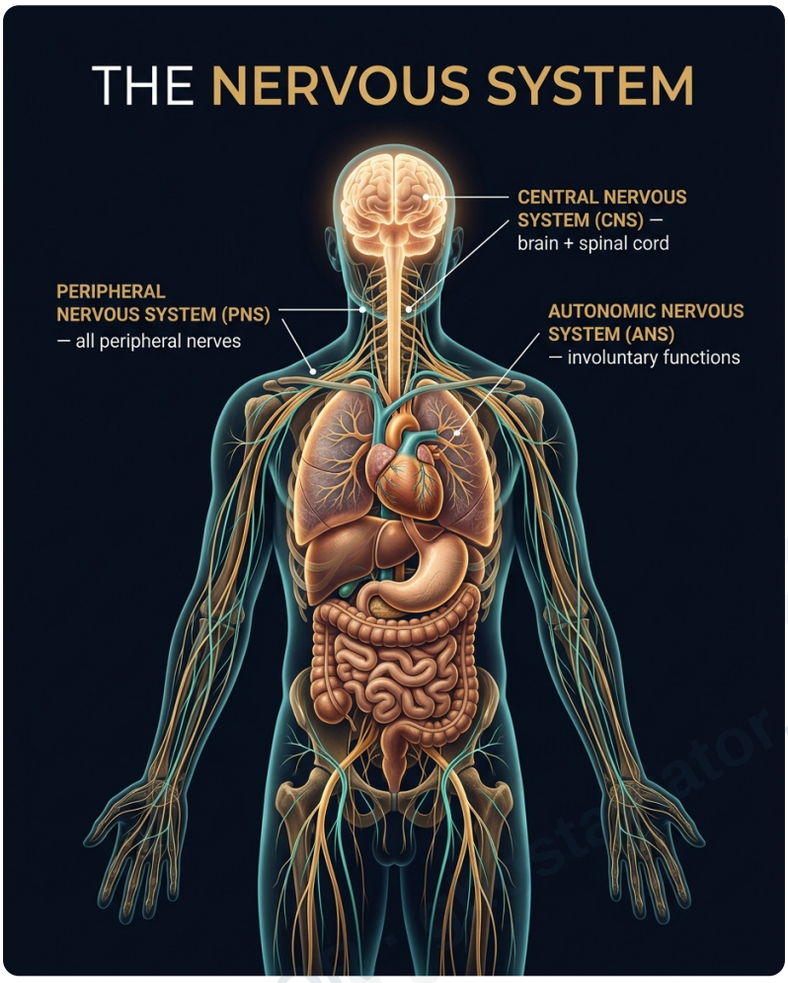

Figure 4.1 — The nervous system: the central (CNS), peripheral (PNS) and autonomic (ANS) divisions.

The Central Nervous System (CNS)

DEFINITION

The central nervous system consists of the **brain and the spinal cord**. The brain, spinal cord and peripheral nerves make up a **complex, integrated information-processing and control system known as Central Nervous System**. In tandem, they regulate **all the conscious and unconscious facets of your life**.

- The scientific study of the brain and nervous system is called **neuroscience** or **neurobiology**.
- The **brain** functions to **receive nerve impulses from the spinal cord and cranial nerves**.
- The **spinal cord** contains the nerves that **carry messages between the brain and the body**.

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The Peripheral Nervous System (PNS)

FUNCTION

The peripheral nervous system includes **all peripheral nerves**. It **connects the central nervous system to the organs and muscles** of the body and regulates **all purposeful and reflex actions**. The peripheral nervous system controls organs and muscles like **skin, eye, blood vessels, heart and stomach**.

The Autonomic Nervous System (ANS)

DEFINITION

The organs (the **"viscera"**) of our body — such as the **heart, stomach and intestines** — are regulated by a part of the nervous system called the **Autonomic Nervous System (ANS)**. The ANS is **part of the peripheral nervous system**, and it controls many organs and muscles within the body.

KEY CHARACTERISTIC — IT WORKS WITHOUT YOU KNOWING

In most situations, we are unaware of the workings of the ANS because it functions in an **involuntary, reflexive manner**. For example, we do not notice when blood vessels change size or when our heart beats faster.

The Autonomic Nervous System exercises its functions **independently of the Central Nervous System** to the extent that it controls parts of the body without having to think about it.

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What the ANS controls — full DGCA list

SENSORY
Eye

CARDIAC
Heart

RESPIRATORY
Breathing

THERMOREGULATION
Temperature

VASCULAR
Blood Pressure

DIGESTIVE
Stomach & Intestines

EXCRETORY
Urinary Output & Bladder

COOLING
Sweating & Glands

STRESS RESPONSE
Fight-or-Flight Reaction

FLIGHT-RELEVANCE — THE "FIGHT OR FLIGHT" REFLEX

The ANS controls the **fight-or-flight response or reaction to stress**. In a sudden emergency — engine failure, severe turbulence, near-miss — the ANS dumps adrenaline into your bloodstream automatically. Heart rate, breathing rate, BP and sweating all spike **without your conscious permission**. *This is why a well-trained pilot's drilled checklist beats raw reaction: the conscious brain must override an autonomic surge.*

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Three divisions of the Nervous System — quick comparison			
Division	**What it Contains**	**What it Controls**	**Conscious / Involuntary**
Central NS	Brain + Spinal Cord	Receives, processes & commands all activity	Both
Peripheral NS	All peripheral nerves	Skin, eye, blood vessels, heart, stomach & muscle action	Both — voluntary & reflex
Autonomic NS	Sub-system of PNS	Eye · Heart · Breathing · Temp · BP · GI · Bladder · Sweat · Stress	**Involuntary only**

✦ END OF CHAPTER 4 ✦

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HUMAN PERFORMANCE & LIMITATIONS · MODULE B — THE LIVING MACHINE

Breath & Blood

Chapter 5 — How the body takes in oxygen and moves it: the respiratory system and gas exchange, the heart and circulation, and blood pressure.

BOOK
Human Performance & Limitations

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CHAPTER
5 of 26 · Module B

In this chapter

1. The Respiratory System & Gas Exchange
2. The Circulatory System & the Heart
3. Blood Pressure
4. Module B — Numbers to Know

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Plate 5.0 — The delivery system. Lungs load the oxygen; the heart and blood carry it to every cell that keeps you flying.

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§ 8 The Respiratory System

OVERALL PURPOSE — ONE SENTENCE

Lungs are complex organs, but what they do is to get rid of carbon dioxide and exchange it for oxygen.

The respiratory process consists mainly of:

- the **diffusion of oxygen through the respiratory membranes into the blood**,
- **transportation to the cells**,
- **diffusion into the cells**, and
- **elimination of carbon dioxide from the body**.

8.1 The Pathway of Air — Nose to Alveoli

As you breathe air in through your nose or mouth, it travels along this path:

Figure 5.2 — The pathway of air, from nose and mouth down to the alveoli where oxygen crosses into the blood.

ANATOMICAL LANDMARKS — MEMORISE THESE LABELS

1. **A — Nasal cavity**
2. **B — Pharynx**
3. **C — Larynx** (houses the vocal cords)
4. **D — Trachea**
5. **E — Alveoli**
6. **F — Bronchial tree**
7. **G — Diaphragm**

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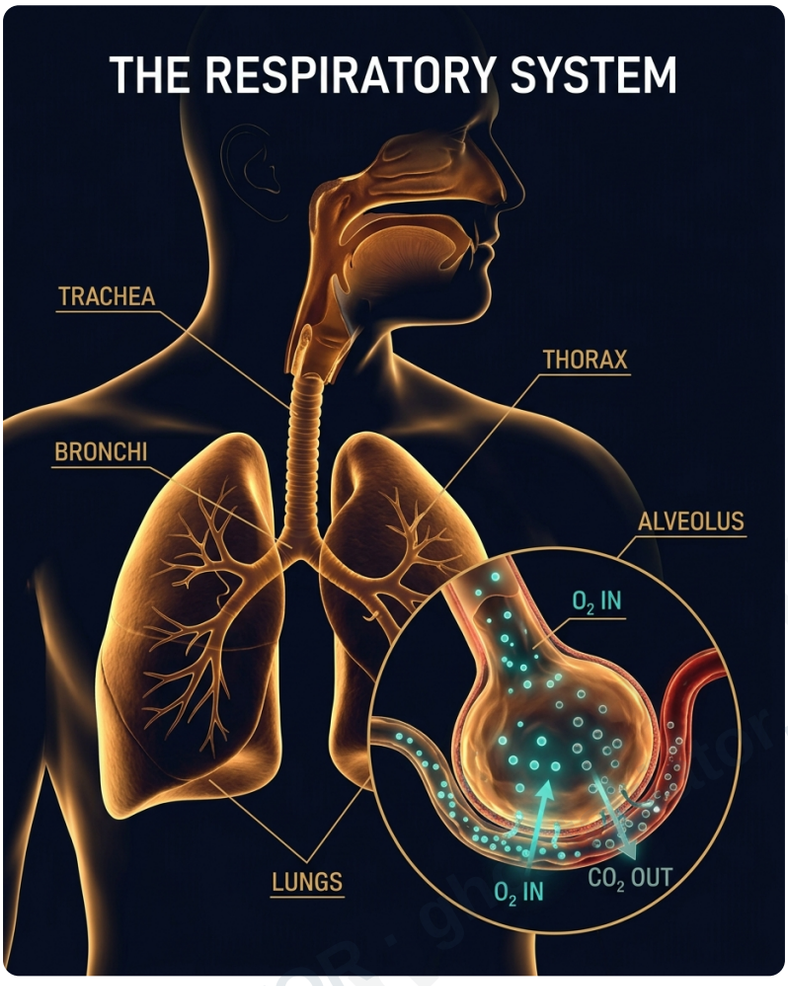

Figure 5.1 — The air pathway and the alveolus, site of all gas exchange; each alveolus is one cell thick.

8.2 What Happens Inside the Alveolus — Gas Exchange

THE DIFFUSION SEQUENCE (GRAHAM'S LAW IN ACTION)

1. Within each air sac, the **oxygen concentration is high**, so oxygen passes or **diffuses across the alveolar membrane into the pulmonary capillary**.
2. At the beginning of the pulmonary capillary, the **haemoglobin** in the red blood cells has **carbon dioxide bound to it and very little oxygen**.
3. The oxygen **binds to haemoglobin**, and the carbon dioxide is **released**.
4. **Carbon dioxide is also released from sodium bicarbonate dissolved in the blood** of the pulmonary capillary.
5. The concentration of carbon dioxide is high in the pulmonary capillary, so CO_2 **leaves the blood and passes across the alveolar membrane into the air sac**.
6. This exchange of gases occurs **rapidly (fractions of a second)**.
7. The carbon dioxide then leaves the alveolus (tiny air sacs of the lungs which allow for rapid gaseous exchange) when you **exhale**, and the **oxygen-enriched blood returns to the heart**.

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THE PURPOSE, IN ONE LINE

The purpose of breathing is to keep the **oxygen concentration HIGH** and the **carbon dioxide concentration LOW** in the alveoli, so this gas exchange can occur.

8.3 External vs Internal (Tissue) Respiration

External Respiration

External Respiration takes place through the **lungs** and refers to:

- The **absorption of Oxygen from the air into the blood**.
- The **excretion of Carbon Dioxide from the blood to the air**.

Internal / Tissue Respiration

Internal or Tissue Respiration refers to the **transfer of Oxygen from the blood to the tissues** of the body. At the same time as this occurs, the tissues **give up Carbon Dioxide to the blood**.

QUICK TEST **External = at the LUNGS** (air ↔ blood). **Internal = at the TISSUES** (blood ↔ cell). Don't mix these up — examiners love this distinction.

8.4 Breathing

DEFINITION

Normal breathing is a **purely automatic process under the unconscious control of the nervous system**. The normal rate of respiration in adults is `14 to 18 breaths per minute`.

WHAT REGULATES YOUR BREATHING RATE — COUNTER-INTUITIVE BUT VITAL

The level of carbon dioxide in the blood effectively regulates the rate and depth of breathing.

It is **NOT oxygen** that triggers your urge to breathe — it is **rising CO_2**. This is the core mechanism behind **Hyperventilation** (over-breathing → CO_2 washes out → urge to breathe drops → tingling, dizziness) which is dealt with separately in Part 3.

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NORMAL ADULT RESPIRATION RATE

14 – 18 / min

CONTROL CENTRE

Medulla (Autonomic)

PRIMARY STIMULUS

Blood CO_2 Level

CONSCIOUS OVERRIDE

Possible (limited)

§ 9

The Circulatory System

WHY IT EXISTS

Blood supplies our organs with life-giving oxygen and carries away waste products. The circulatory system consists of the **heart and the blood vessels**, and maintains the **flow of blood throughout the body**. Without it, gas exchange at the lungs is pointless — there must be a delivery network.

9.1 The Heart

WHAT THE HEART DOES — CONCISE

The arteries **carry blood from the heart at HIGH pressure**, and the veins **return blood to the heart at LOW pressure**. The heart is a **pumping system** which:

- Intakes **de-oxygenated blood through the veins**.
- Delivers it to the **lungs for oxygenation**.
- Then pumps it into the various **arteries to be transmitted** where it is needed throughout the body for energy.

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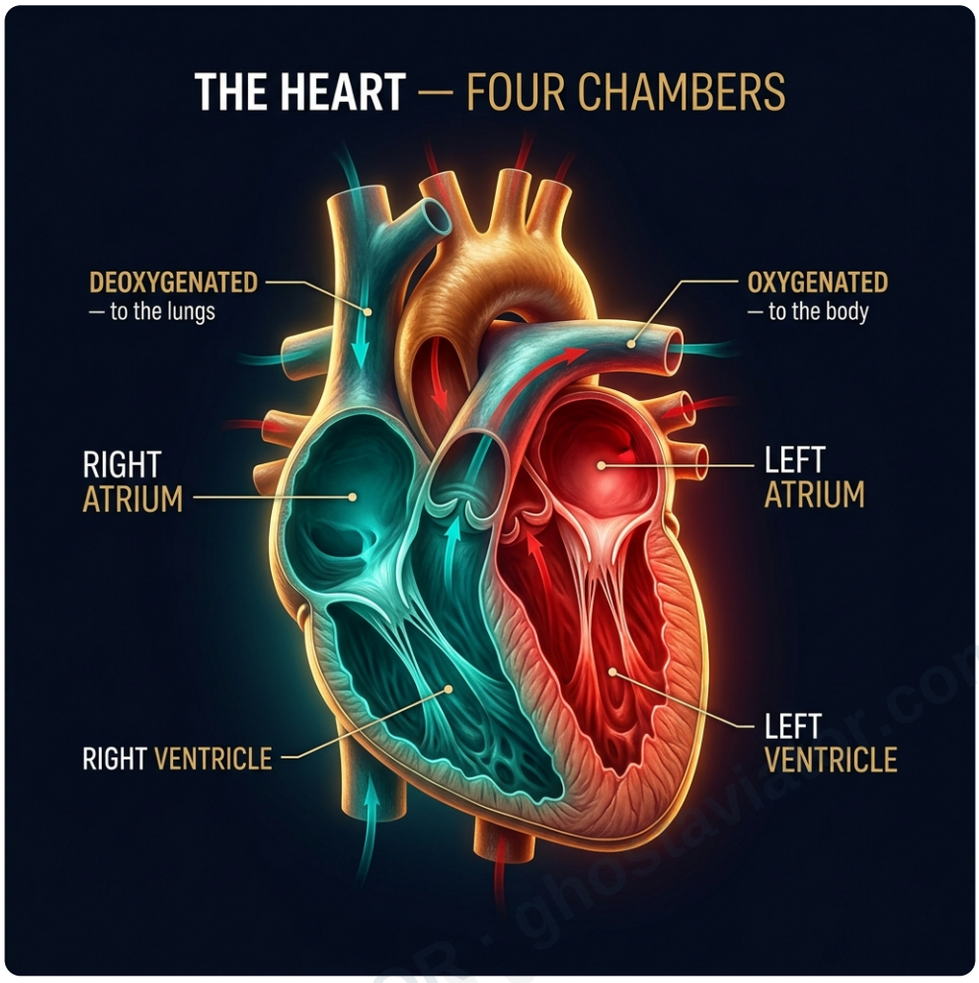

Figure 5.3 — The four chambers of the heart: the right side handles deoxygenated blood to the lungs, the left side oxygenated blood to the body.

MEMORISE — CARDIAC OUTPUT VALUES

At rest the **cardiac output** (the quantity of blood the heart pumps in one minute) of an adult with:

- Heart Rate = `72 beats per minute`
- Stroke Volume = `70 ml`

... is **about** `5 litres per minute`. (Stroke volume × Heart rate = Cardiac output.)

9.2 How the Heart Works — Systole & Diastole

TWO PHASES OF THE CARDIAC CYCLE

When the heart muscle contracts or beats (called **SYSTOLE**), it **pumps blood out of the heart**. Then the heart muscle relaxes (called **DIASTOLE**) before the next heartbeat. This allows blood to **fill up the heart again**.

The two-stage contraction of systole

1. **Stage 1:** The **right and left atria contract at the same time**, pumping blood to the right and left ventricles.
2. **Stage 2:** The **ventricles contract together** to propel blood out of the heart.

The two-sided traffic plan

Right Side of the Heart

Collects **oxygen-poor blood from the body** and pumps it to the **lungs**, where it picks up oxygen and releases carbon dioxide.

Left Side of the Heart

Collects **oxygen-rich blood from the lungs** and pumps it to the **body**, so that the cells throughout your body have the oxygen they need to function properly.

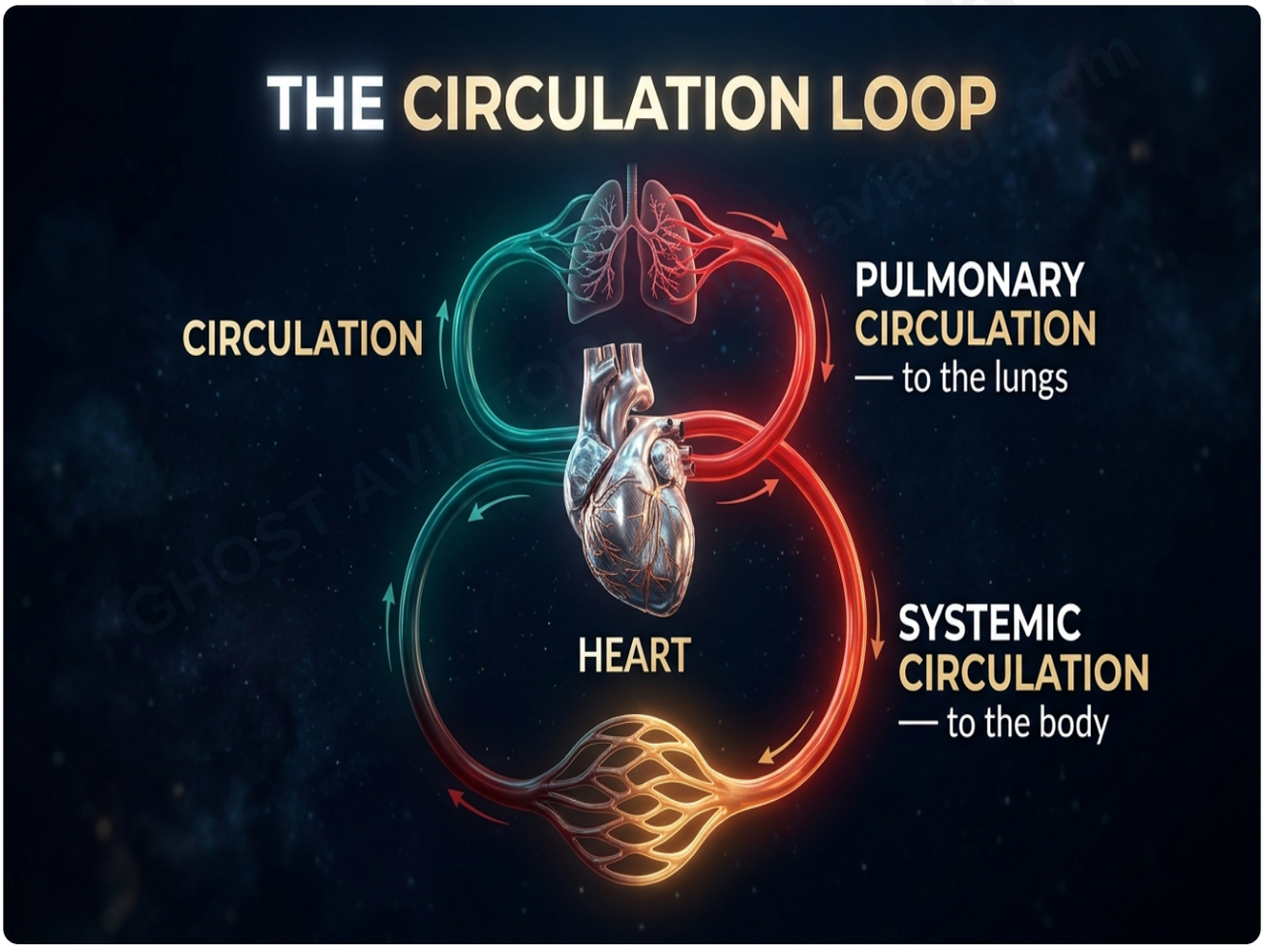

Figure 5.5 — The circulation loop: the pulmonary circuit to the lungs and the systemic circuit to the body.

BLOOD'S JOURNEY FROM LEFT VENTRICLE TO CAPILLARIES

Blood containing oxygen is pumped around the body from the **left ventricle**. The oxygenated blood passes through the **aorta into the arteries** before arriving at the smallest vessels of the system, the **capillaries**. The **Oxygen–Carbon Dioxide exchange takes place through the walls of the capillaries**.

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9.3 Pulse Rate

DEFINITION

The normal rate of the **pulse is the rate of the heartbeat**. A healthy person at rest has a pulse rate of between `60 and 80 beats per minute`. The rate is increased by:

- Exercise
- Emotional inputs
- Disease

STRESS & FEAR — AUTONOMIC ADRENALINE SURGE

When the body experiences **stress or fear**, **adrenaline is released into the bloodstream** causing an **immediate increase in the pulse rate**. (This is the ANS "fight-or-flight" response from §7.4.)

RESTING PULSE (HEALTHY)
60 – 80 bpm

CARDIAC OUTPUT REFERENCE
72 bpm × 70 ml

OUTPUT PER MINUTE
≈ 5 L/min

ADRENALINE EFFECT
Immediate ↑ pulse

9.4 Composition & Function of the Blood

TWO MAIN COMPONENTS OF BLOOD

Blood has **two main components — PLASMA and FORMED ELEMENTS**.

- **Plasma:** Nearly everything that blood carries — including **nutrients, hormones and waste** — is dissolved in plasma, which is **mostly water**.
- **Formed elements:** Cells and parts of cells that also **float in plasma**.

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Formed elements — the three cell types

Cells and cell-fragments in the blood		
Element	**What it is / does**	**Notes**
White Blood Cells (WBCs)	Part of the **immune system**	White corpuscles **produce antibodies** to fight bacteria.
Platelets	Help form **clots**	**Smallest of the blood cells**; assist in the blood-clotting process.
Red Blood Cells (RBCs)	Carry **oxygen & carbon dioxide**	Numerous — make up **more than 90 % of the formed elements in the blood**. Virtually everything about them helps them carry oxygen more efficiently.

Inside a Red Blood Cell — Haemoglobin (Hb)

STRUCTURE OF HAEMOGLOBIN — DGCA-FAVOURITE QUESTION

A red blood cell's lack of nucleus also gives it **more room for haemoglobin (Hb)**, a complex molecule that carries oxygen. It is made of:

- A protein component called **GLOBIN**.
- **FOUR pigments called HEMES**.
- The hemes use **IRON** to bond to oxygen.

Inside each RBC are `≈ 280 million haemoglobin molecules`.

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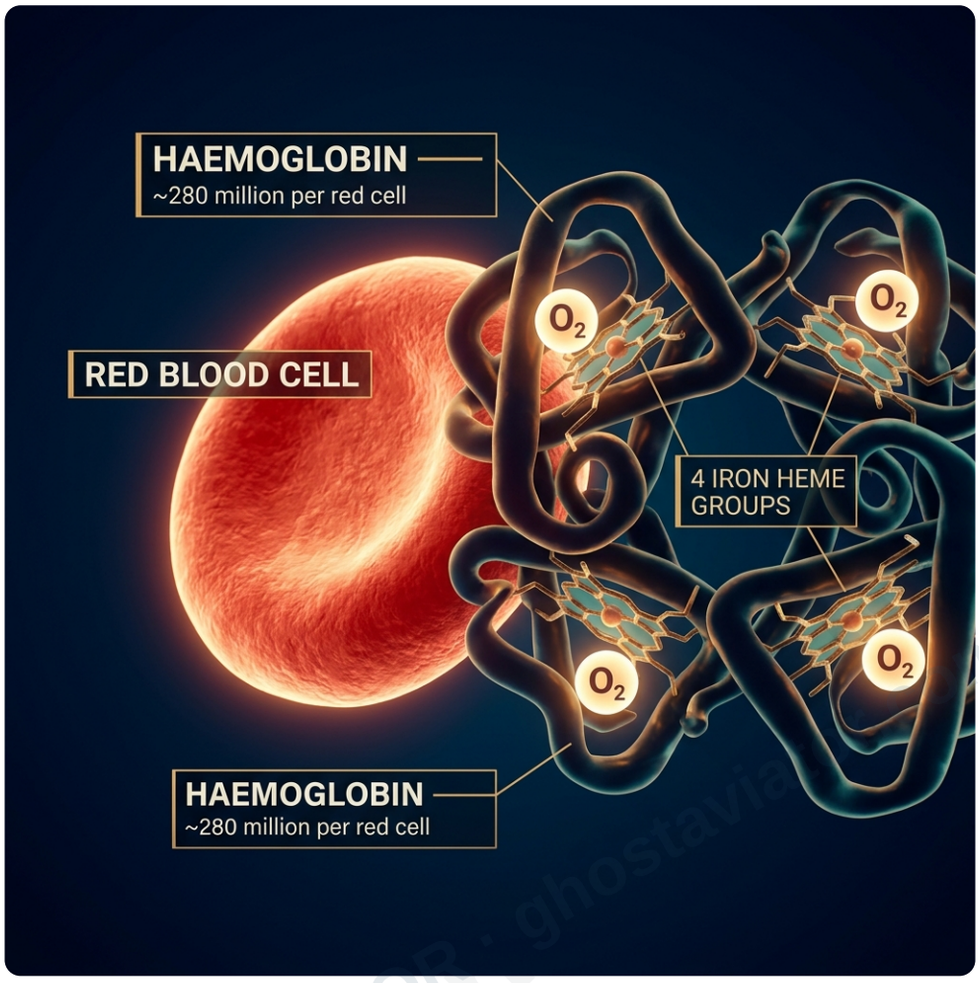

Figure 5.4 — Each red cell packs ~280 million haemoglobin molecules; each carries four iron-bearing hemes that grip oxygen.

If you lose a lot of blood…

WHY BLOOD LOSS IS DANGEROUS

If you lose a lot of blood, you lose a lot of your **oxygen delivery system**. The immune cells, nutrients and proteins that blood carries are important too, but doctors are generally **most concerned with whether your cells are getting enough oxygen**.

EMERGENCY MANAGEMENT

In an emergency situation, doctors will often give patients **volume expanders**, like **saline**, to make up for lost blood volume. This:

- Helps restore **normal blood pressure**.
- Lets the remaining red blood cells **continue to carry oxygen**.
- Sometimes is enough to keep the body going until it can produce new blood cells and other blood elements.

If not, doctors give **blood transfusions** to replace some of the lost blood. Blood transfusions are also fairly common during some **surgical procedures**.

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The principal functions of the blood — full DGCA list

The five principal functions of blood		
#	**Function**	**Mechanism**
1	Carry **oxygen to, and carbon dioxide from**, the various tissues and organs of the body.	Haemoglobin in RBCs
2	Carry **nutrients to tissues** and remove **waste products** from these tissues.	Dissolved in plasma
3	Carry **chemical messengers, such as hormones including ADRENALINE**, to regulate the actions and secretions of various organs.	Plasma transport
4	Transport **cells which can attack and destroy invading micro-organisms**, enabling the body to resist disease.	WBCs / antibodies
5	**Assist in temperature control** of the body.	Vasodilation/ constriction

9.5 Failures or Malfunctions of the Circulatory System

TWO PRINCIPAL FAILURE MODES

The circulatory system can malfunction in **two principal ways**:

1. The main component of the system, the **heart and the blood vessels**, may develop a fault.
2. The **blood may become unable to carry enough Oxygen** for the need of the organs and tissues of the body.

Angina & Heart Attack

DEFINITIONS

A **lack of oxygen supply to the heart** may give rise to symptoms of **Angina**. A **heart attack** is when **low blood flow causes the heart to starve for oxygen**. Heart muscle **dies or becomes permanently damaged**. Your doctor calls this a **myocardial infarction**.

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Causes of angina and heart attack

THE BLOOD-CLOT MECHANISM

- Most heart attacks are caused by a **blood clot that blocks one of the coronary arteries**.
- The **coronary arteries bring blood and oxygen to the heart**. If the blood flow is blocked, the heart starves for oxygen and heart cells die.
- A clot most often forms in a coronary artery that has become **narrow because of the build-up of a substance called PLAQUE** along the artery walls.
- Sometimes, **the plaque cracks and triggers a blood clot to form**.
- Occasionally, **sudden overwhelming stress can trigger a heart attack**.

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Risk factors for angina and heart attack — the full DGCA list

Risk factors — pilots should be screened for these at every medical	
Risk Factor	**Why it matters**
Bad genes (hereditary factors)	Family history is non-modifiable but flagged in flight-medical examinations.
Being male	Higher statistical risk than females (until menopause for women).
Diabetes	Damages blood-vessel linings and accelerates plaque build-up.
Getting older	Arterial elasticity falls; plaque accumulation rises.
High blood pressure	Damages arterial walls — central to plaque mechanism. *(See §14 in next part.)*
Smoking	Reduces O_2-carrying capacity 5–8 %, raises BP, accelerates atherosclerosis. *(§12 — next part.)*
Too much fat in your diet	Raises LDL cholesterol — primary feedstock for plaque.
Unhealthy cholesterol levels — especially **HIGH LDL ("bad") cholesterol** and **LOW HDL ("good") cholesterol**	Direct correlation with coronary artery narrowing.
Lack of exercise	Increases all of the above; lowers cardiac reserve.
Stress	Sustained adrenaline release damages the heart and vessels.
Obesity	Independent risk factor + driver of diabetes, BP, lipids.
Alcohol	Raises BP, damages heart muscle, raises triglycerides. *(§13 — next part.)*

WHY PILOTS CARE — DGCA MEDICAL IMPLICATIONS

Every one of these risk factors is screened at Class-I & Class-II DGCA medicals via ECG, lipid panel, fasting glucose, BMI, blood pressure measurement and history. *Even an unsuspected silent angina is a fitness-defeating condition; routine cardiac screening exists precisely because the in-flight failure mode is sudden incapacitation.*

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Insufficiency of Oxygen — the second failure mode

WHY THIS IS ITS OWN CATEGORY

The second principal way the circulatory system can malfunction is **"Insufficiency of Oxygen"** — the blood becoming **unable to carry enough oxygen for the needs of the organs and tissues**. This is the gateway concept that leads directly into the next major topic — **HYPOXIA** — covered in detail in §10 (Part 3). Hold the term in mind.

9.6 Self-Check & Memory Aids — Part 2

Numbers you must know cold

DGCA-style probe questions

TRY THESE WITHOUT LOOKING BACK

1. State the three main divisions of the Nervous System. Which one controls the heart, lungs and gut *without* conscious effort?
2. Trace the path of air from the nose to the alveoli — name **every** structure.
3. What is the difference between External and Internal Respiration?
4. What is the normal adult respiration rate, and what is the primary regulator of breathing?
5. Define cardiac output. Compute it for HR 72 bpm, SV 70 ml.
6. What is the difference between Systole and Diastole? Describe the two-stage contraction.
7. State the two main components of blood and the three types of formed elements.
8. How many haemoglobin molecules are inside each RBC? How many hemes per Hb?
9. List the five principal functions of blood.
10. What are the two principal failure modes of the circulatory system? Name the medical term for a heart attack.
11. List at least eight risk factors for angina/heart attack relevant to a DGCA medical.
12. What is plaque, and why does it matter to coronary arteries?

MNEMONIC — AIR PATHWAY **"Never Pinch The Lazy Bronchitis Boy's Alveoli"** → **N**ose · **P**harynx · **T**rachea · **L**arynx · **B**ronchi · **B**ronchioles · **A**lveoli. (Note the epiglottis sits between nose/pharynx and trachea acting as the airway gatekeeper.)

MNEMONIC — 3 NS DIVISIONS **"Cops Patrol Automatically"** → **C**NS (Central — Brain + Cord, the HQ) · **P**NS (Peripheral — all peripheral nerves, the patrol) · **A**NS (Autonomic — runs without orders).

MNEMONIC — CARDIAC OUTPUT **CO = HR × SV** → Cardiac Output = Heart Rate × Stroke Volume. **72 × 70 ≈ 5040 ml ≈ 5 L/min.**

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MNEMONIC — RIGHT VS LEFT HEART **"Right is BLUE, Left is RED"** → Right side handles de-oxygenated blood (to lungs); Left side handles oxygenated blood (to body).

§ 15 Blood Pressure

15.1 What Determines Blood Pressure

THE FOUR DETERMINANTS — DGCA-QUOTED

Blood pressure depends on:

- The **cardiac output**,
- The **resistance of the capillaries** (peripheral resistance),
- The **elasticity of the arterial walls**, and
- The **blood volume and viscosity**.

15.2 Definitions — pressure on the arteries

THREE KEY FACTS

- **Blood pressure is the pressure exerted by blood on the walls of the main arteries**.
- The blood-pressure which is measured during flight medical checks is the **pressure in the artery of the upper arm** (representing the pressure at heart level).
- The **permanent pressure against the arterial wall is called DIASTOLIC pressure**.
- The **increased pressure occurring with each beat of the heart is called the SYSTOLIC pressure**.

SYSTOLIC PRESSURE

The **increased pressure occurring with each beat of the heart**.

This is the higher number in the BP reading — generated during ventricular contraction (*systole*, §9.2). It is the pressure peak as blood is ejected from the left ventricle into the aorta.

DIASTOLIC PRESSURE

The **permanent pressure against the arterial wall**.

This is the lower number — the resting pressure between heartbeats, when the heart muscle relaxes (*diastole*) and refills with blood. It represents the baseline strain on the arteries.

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15.3 120/80 — the Normal Benchmark

THE NUMBER EVERY PILOT MUST KNOW

`120/80` **is a normal blood pressure for a healthy young adult.**

The notation reads "120 over 80" — meaning **systolic 120 mmHg, diastolic 80 mmHg**. This is the figure that DGCA aero-medical examiners measure against at every Class-1 and Class-2 medical.

Figure 5.6 — Measuring blood pressure: the cuff inflates around the upper arm, then deflates while the examiner listens for arterial sounds.

15.4 High Blood Pressure (Hypertension) — Causes & Pilot Implications

DGCA-QUOTED — THE UNFITNESS CLAUSE

High blood pressure or hypertension is a MAJOR CAUSE of unfitness in pilots.

This single sentence is the most important takeaway from §15. The DGCA aero-medical examiner will **defer or restrict** a pilot's medical certificate the moment hypertension is detected and not adequately controlled.

Causes of Blood Pressure (Hypertension) — full DGCA list

Seven causes of hypertension a pilot must remember

#	Cause	Pilot-Specific Note
1	**Stress**	Sustained ANS / adrenaline activation. Pilots are exposed to chronic occupational stress — duty rosters, weather, currency requirements.
2	**Smoking**	Direct link to §13. Nicotine constricts vessels; CO damages endothelium.
3	**Poor diet** (excess fat or salt)	Salt → fluid retention → ↑ blood volume. Fat → plaque → ↑ peripheral resistance.
4	**Obesity**	Greater body mass requires more cardiac output; metabolic syndrome adds insulin resistance.
5	**Lack of exercise**	Deconditioned heart, stiffer vessels, higher resting BP.
6	**Age**	Arterial elasticity falls with age (recall §9.5 risk-factor list). Non-modifiable.
7	**Narrowing of the arteries**	Atherosclerosis — plaque-narrowed vessels create higher peripheral resistance. Same plaque that causes angina/MI.

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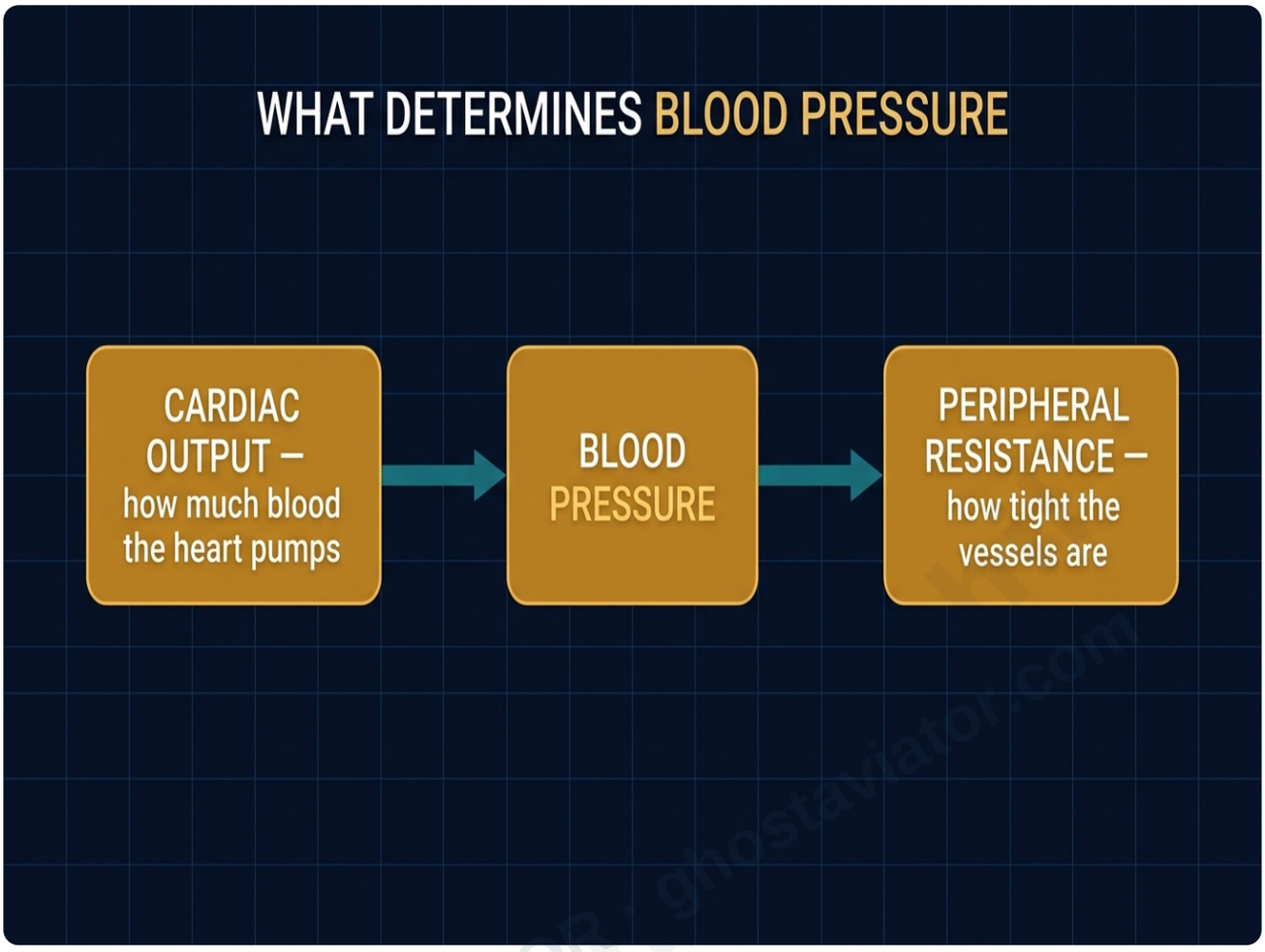

Figure 5.7 — What determines blood pressure: cardiac output and peripheral resistance.

WHAT A PILOT CAN DO — THE MODIFIABLES

Of the seven causes, **five are modifiable** (stress management, no smoking, low-salt/low-fat diet, weight control, regular exercise). Only **age** and (to a large extent) **narrowing of arteries from existing plaque** are non-modifiable. **The pilot who fails his Class-1 medical because of hypertension has lost his career to a condition that was, in most cases, preventable.**

CROSS-LINKS — SEE ALSO

- **§9.1 / 9.2:** cardiac output = HR × stroke volume — directly drives systolic pressure.
- **§9.5:** hypertension is one of the 12 risk factors for angina / heart attack.
- **§13.3:** smoking causes circulatory problems and accelerates plaque.
- **§14.5:** alcohol contributes to liver/heart/brain damage.

The DGCA HPL paper frequently *combines* these topics in one MCQ stem — e.g. "A 48-year-old smoker with BP 145/95 …" — testing whether you can connect the dots.

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B Module B — Numbers to Know

Master cheat-sheet — the body's systems		
Parameter	**Exact Value**	**Where**
Normal adult respiration rate	`14 – 18 breaths/min`	§8.4
Primary regulator of breathing rate	`Blood $CO_2$ level`	§8.4
Resting pulse (healthy)	`60 – 80 bpm`	§9.3
Reference heart rate (for CO calc)	`72 bpm`	§9.1
Stroke volume (reference)	`70 ml`	§9.1
Cardiac output at rest	`≈ 5 L/min`	§9.1
RBCs as % of formed elements	`> 90 %`	§9.4
Haemoglobin molecules per RBC	`≈ 280 million`	§9.4
Hemes per Hb molecule	`4`	§9.4
Metal that bonds O_2 to heme	`Iron`	§9.4
Protein component of Hb	`Globin`	§9.4
Heart attack medical term	`Myocardial Infarction`	§9.5
"Bad" cholesterol	`LDL (high = bad)`	§9.5
"Good" cholesterol	`HDL (high = good)`	§9.5

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✦ END OF CHAPTER 5 ✦

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Hypoxia

Chapter 6 — The single most-tested block in the paper, and a killer that gives no warning: the four types of hypoxia, its insidious stages, and the altitude thresholds every pilot must know.

BOOK
Human Performance & Limitations

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CHAPTER
6 of 26 · Module C

In this chapter

1. Hypoxia — definition, stages, the four types, thresholds

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Plate 6.0 — The silent killer. By the time hypoxia is felt, judgment is already gone — which is why it is recognised, not sensed.

§ 10

HYPOXIA

TECHNICAL DEFINITION — "CEREBRAL HYPOXIA"

The term **cerebral hypoxia** technically refers to **lack of oxygen supply to the cerebral hemispheres** (the outer portion of the brain). However, it is more typically used to refer to a **lack of oxygen supply to the entire brain**.

10.1 Why Hypoxia Matters to a Pilot

TWO WAYS THE BODY CAN BE STARVED OF OXYGEN

1. The body's organs and tissues may be **deprived of the oxygen they need because of illness or disease**.
2. **But the pilot must know that oxygen deprivation or hypoxia can be caused by breathing air at low pressures at high altitude.**

This second category — **altitude-induced hypoxia** — is the aviator's main concern, and the form most commonly tested in DGCA exams.

EFFECT ON THE PILOT — DIRECT QUOTE, MEMORISE

Hypoxia will cause a pilot's intellectual and sensory judgment to become impaired. In **mild cases**, hypoxia causes only:

- **Inattentiveness**,
- **Poor judgment**, and
- **Uncoordinated movement**.

10.2 Stages & Clinical Course — from "tipsy" to lethal

SEVERE CASES — THE CASCADE TO DEATH

Severe cases result in a state of complete unawareness and unresponsiveness (coma) — **brain-stem reflexes**, including **response to light and the breathing reflex**, stop. Only **blood pressure and heart function are maintained**. If this persists, **brain death is inevitable**.

If the lack of oxygen to the brain is limited to a **very brief period of time**, coma **may be reversible** with varying levels of return to function, depending on the extent of injury. Sometimes **seizures** may occur, which may be **continuous with no stop between them** — a condition called **STATUS EPILEPTICUS**.

Serious oxygen deprivation can kill a pilot within minutes.

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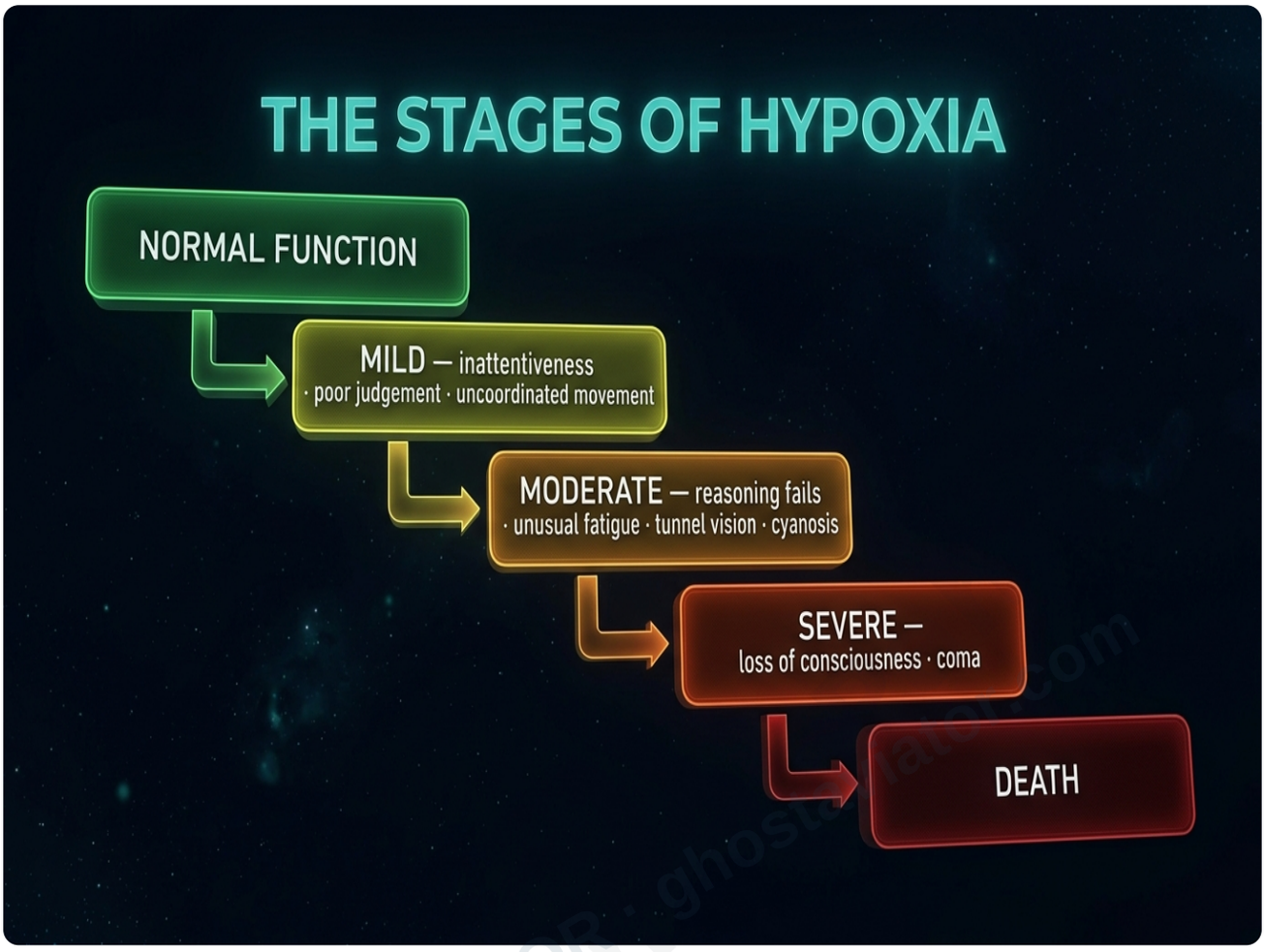

Figure 6.2 — The clinical course of hypoxia, from normal function through mild and moderate to coma and death.

ONSET CHARACTERISTIC — NOTE THE WORD "INSIDIOUS"

The symptoms are **slow but progressive, insidious in onset**, and are most marked at altitudes **starting above** `10,000 ft (3,500 m)`. **Night vision, however, can be impaired starting at altitudes** `5,000 ft`.

"Insidious" means it *sneaks up on you*. You will not realise you are hypoxic — that is the entire problem. Your buddy in the right seat may notice it before you do.

WHY "EUPHORIA" IS THE MOST DANGEROUS EARLY SYMPTOM

The onset of hypoxia **may be accompanied by a feeling of well-being, known as EUPHORIA**. The pilot feels good — even *great* — and is therefore the **last person in the cockpit** who will suspect anything is wrong. This is why CRM and crew cross-checks are critical above 10,000 ft.

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10.3 The FOUR Types of Hypoxia

> **CLASSIFICATION — OPENING LINE OF EVERY DGCA HYPOXIA QUESTION**
>
> **HYPOXIA IS CLASSIFIED INTO FOUR DIFFERENT TYPES:**

TYPE (a)

Hypoxic Hypoxia

Cause: low oxygen *levels* in the bloodstream.

The pilot's type: in pilots, this most often occurs with **exposure to altitude** — **hypobaric hypoxia**. At low altitudes the partial pressure of oxygen is adequate to maintain brain function at peak efficiency. **Atmospheric pressure and the partial pressure of oxygen both decline at higher altitudes.**

TYPE (b)

Anemic Hypoxia

Cause: blood cannot carry enough oxygen even though atmospheric oxygen may be plentiful. Oxygen in blood is carried by **haemoglobin**, which is found in red blood cells. When the **RBC count decreases**, or the **haemoglobin does not function properly**, less oxygen can be carried by the blood.

Occurs in: heavy bleeding · some cancers · sickle-cell anemia · **carbon monoxide poisoning**, to name a few. Symptoms: **breathlessness, fatigue, chest pain** — and they **worsen at higher altitudes**, as the effects of hypoxia and anemia are **additive**.

TYPE (c)

Ischemic / Stagnant Hypoxia

"Ischemia" = inadequate supply of blood. **Ischemic hypoxia** occurs when there is **inadequate blood flow to body tissues**.

Occurs in:

- **Constriction of blood vessels** (e.g. fingers and toes exposed to cold).
- **Low blood pressure / low cardiac output** such as fainting.
- **Exposure to high sustained accelerations** like excessive G-forces (stagnant hypoxia).

Oxygen therapy is not very helpful in this form. The **best remedy is to correct the underlying cause**.

TYPE (d)

Histotoxic Hypoxia

Cause: the **inability of the cells of the body to use the oxygen available**. The oxygen is there in the blood, in the tissues — but the cells can't metabolise it.

Rare in pilots, but can occur with:

- **Cyanide poisoning**,

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- **Chemical poisoning**,
- **Intoxication with certain drugs**.

Can also be caused by HIGH BLOOD ALCOHOL LEVELS. (Direct link to §14 in Part 4.)

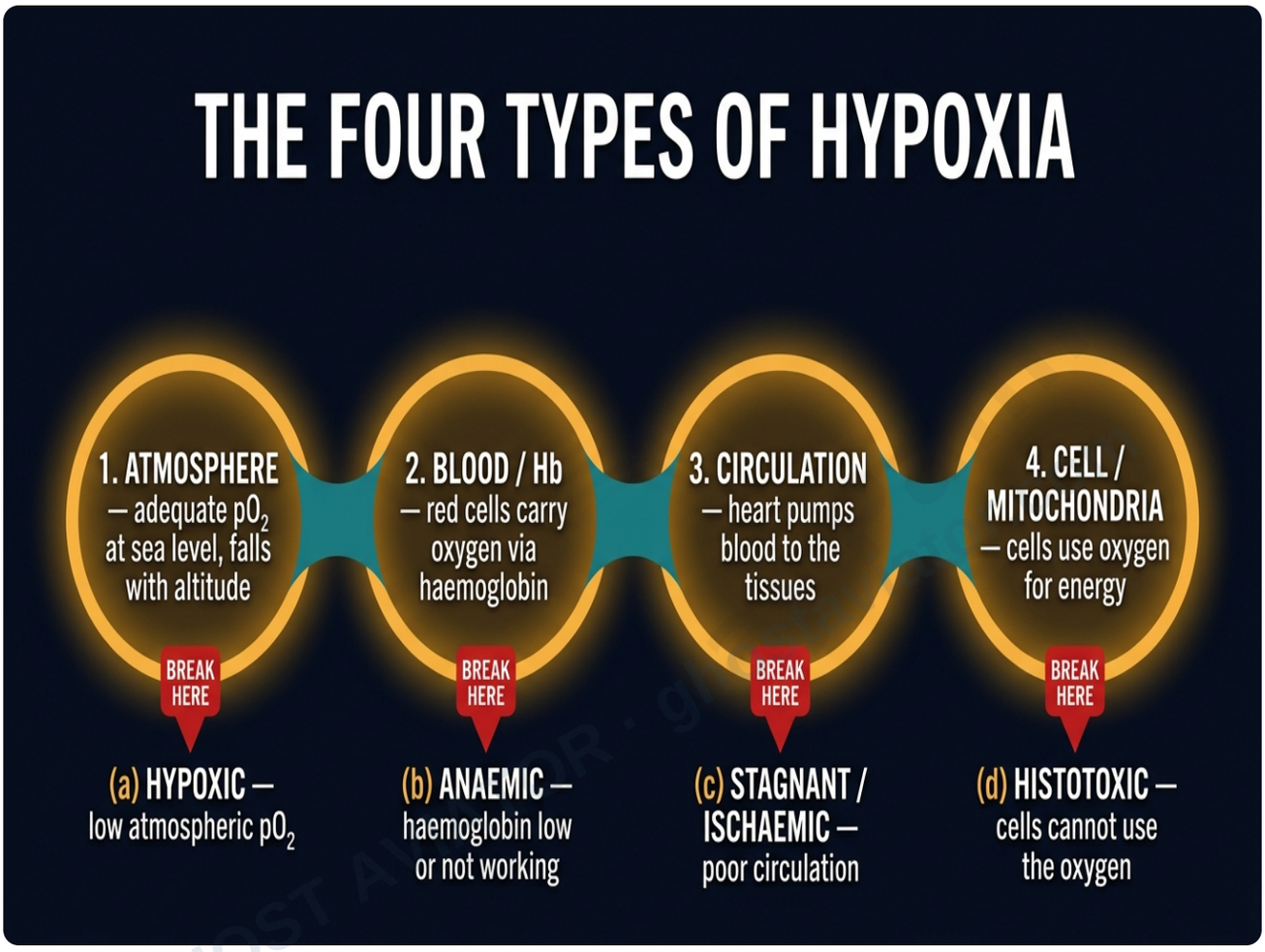

Figure 6.1 — The four types of hypoxia, each breaking the oxygen-delivery chain at a different point.

10.4 Causes · Symptoms · Immediate Actions — Master Table

The DGCA syllabus consolidates Hypoxia management into **one master tabulation** of all causes, all symptoms, and the immediate actions if hypoxia is suspected. This is reproduced verbatim below.

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Hypoxia — Causes, Symptoms & Immediate Actions (verbatim DGCA syllabus table)

Hypoxia Causes	Symptoms	Immediate Actions if Hypoxia is suspected
• Drug overdose / alcohol • Asphyxiation caused by smoke inhalation • Very low blood pressure • Strangulation • Cardiac arrest (when the heart stops pumping) • Carbon monoxide poisoning • High altitudes • Choking • Compression of the trachea • Diseases that paralyze the respiratory muscles • **A tobacco smoker is likely to experience the effects of hypoxia at a lower altitude than a non-smoker**	• The symptoms are **slow but progressive, insidious in onset**, and are most marked at altitudes **starting above 10,000 ft (3,500 m)**. **Night vision can be impaired starting at altitudes 5,000 ft**. • Its onset may be accompanied by a feeling of well-being, known as **EUPHORIA**. • Even minor hypoxia **impairs night vision and slows reaction time**. • More serious hypoxia **interferes with reasoning**, gives rise to **unusual fatigue** and, finally, results in **loss of consciousness / death**. • **Impaired Judgment** • **Headache** • **Tingling in hands & feet** • **Hyperventilation** (yes, hypoxia can *trigger* hyperventilation) • **Muscular Impairment** • **Memory Impairment** • **Sensory Loss** • **Tunnel Vision** • **Cyanosis** (a bluing of the body extremities) • **Formication** (a feeling of ants under the skin)	• **Provide Oxygen**. • **Descend below 10,000 feet** or **Minimum Safe Altitude if it is higher than 10,000 ft**.

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MEMORISE — THE IMMEDIATE-ACTION DRILL (A 2-STEP REFLEX)

1. **PROVIDE OXYGEN** — don the mask, 100 % flow, regulator to "EMERGENCY" if available.
2. **DESCEND BELOW** `10,000 ft` (or below MSA — Minimum Safe Altitude — if MSA is higher than 10,000 ft).

This is the standard DGCA-examined answer. Do not over-elaborate in the exam — write these two steps in this order.

10.5 Altitude Thresholds & Night Vision

NIGHT VISION IMPAIRMENT FROM
5,000 ft

DAY SYMPTOMS MARKED FROM
10,000 ft (3,500 m)

ACTION THRESHOLD — DESCEND BELOW
10,000 ft

SMOKER EFFECTIVE ALTITUDE (RECAP)
≈ 7,000 ft

DGCA REASONING THE EXAMINER IS TESTING

The reason the action threshold is **10,000 ft** and not, say, 14,000 ft is because **symptoms are most marked at altitudes starting above 10,000 ft (3,500 m)**. The reason supplemental O_2 is required **at night above 5,000 ft** in many jurisdictions is because **night vision is impaired starting at 5,000 ft** — the rod cells in the retina are extraordinarily sensitive to even mild hypoxia. *(Vision pathophysiology is dealt with in a later part of this chapter.)*

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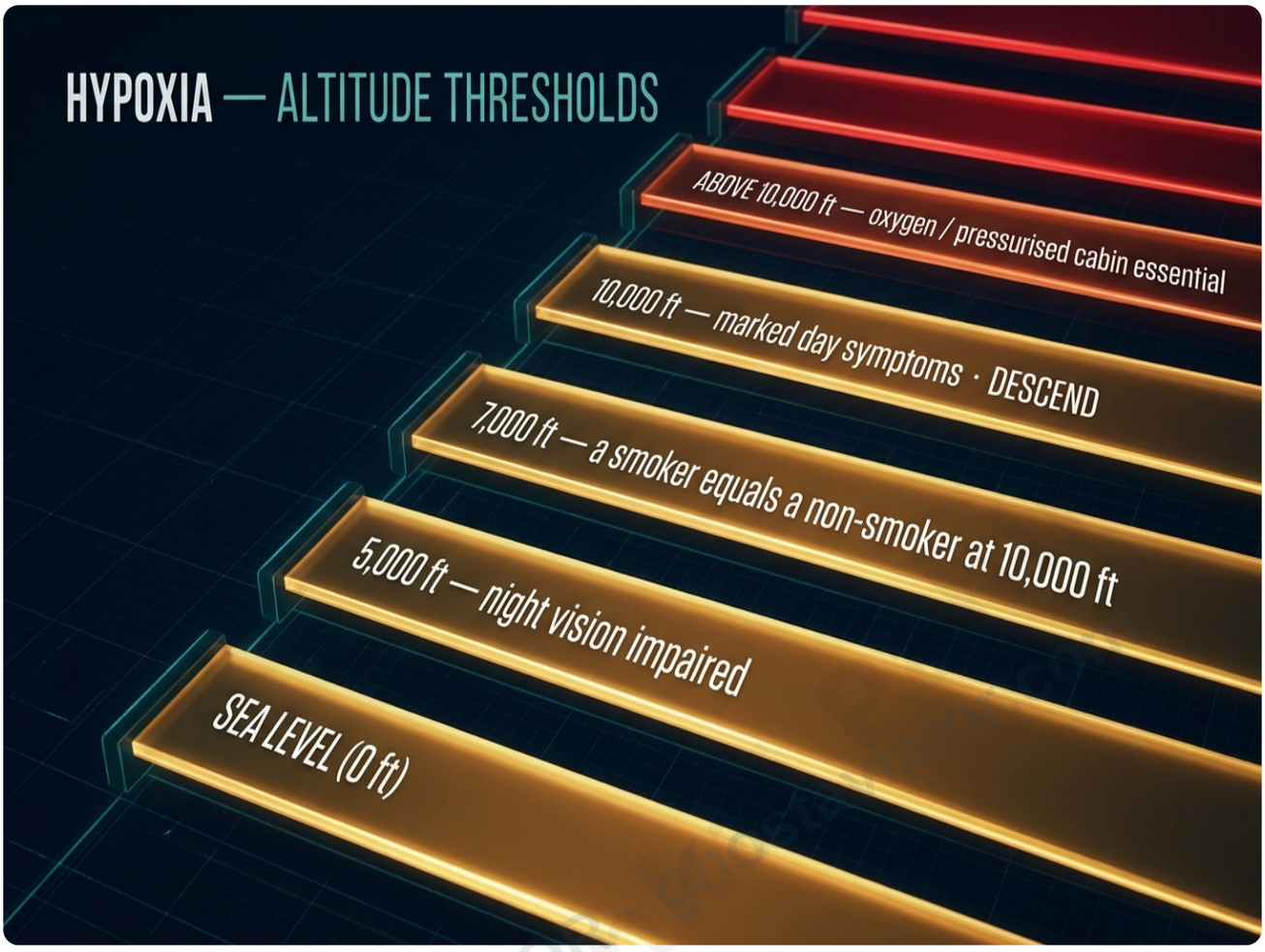

Figure 6.3 — Altitude thresholds: night vision impaired from 5,000 ft, marked day symptoms by 10,000 ft.

✦ END OF CHAPTER 6 ✦

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HUMAN PERFORMANCE & LIMITATIONS · MODULE C — THE THIN AIR

The Silent Killers

Chapter 7 — Two hazards that give no honest warning: carbon monoxide, the colourless killer that steals your haemoglobin, and hyperventilation, the panic that mimics hypoxia.

BOOK
Human Performance & Limitations

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CHAPTER
7 of 26 · Module C

In this chapter

1. Carbon Monoxide (CO) Poisoning
2. Hyperventilation — and how to tell it from hypoxia

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Plate 7.0 — Colourless, odourless, and patient. Carbon monoxide gives no warning until judgment is already gone.

§ 11

Carbon Monoxide (CO) Poisoning

11.1 The Colourless Killer

WHAT CO IS

Carbon monoxide is a **colourless, odourless, tasteless gas** that is a **product of incomplete combustion**.

WHY CO IS SO DANGEROUS — THE 200× RULE

Haemoglobin, the oxygen-carrying chemical in the blood, picks up carbon monoxide `over 200 times more readily` than it picks up oxygen.

Thus, even **minute quantities in the cockpit** (often from improperly vented exhaust fumes) **may result in pilot incapacitation**. Exhaust gases from piston engines can consist of **as much as** `9 % carbon monoxide`. So, gases from **leaking exhausts** can cause carbon monoxide poisoning in pilots.

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Figure 7.1 — Carbon monoxide binds haemoglobin more than 200× more strongly than oxygen, crowding oxygen off the blood.

SOURCES OF COCKPIT CO — THE USUAL SUSPECTS

- **Improperly vented exhaust fumes** entering the cabin via the heating system.
- **Leaking exhaust manifolds / muffler shrouds** (especially common in piston singles where cabin heat is drawn over the exhaust).
- **Engine combustion gases**, which can contain up to 9 % CO.
- **Tobacco smoke** in confined cabins.

11.2 Symptoms · Actions · Prevention

Carbon Monoxide Poisoning – Symptoms

- **Initially, there is an INABILITY TO CONCENTRATE**
- Headache
- Dizziness
- Nausea
- Impaired vision
- Lethargy or weakness
- Impaired judgment

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- Personality change
- Impaired memory
- **Flushed cheeks and cherry-red lips** *(classic CO sign)*
- Convulsions

Actions if CO Poisoning is Suspected

1. **Turn off cabin heating**
2. **Open cabin ventilators**
3. **Consider using oxygen if available**
4. **Land as soon as possible**
5. **Take medical aid**
6. **Do not fly till cleared by doctor**

DGCA-QUOTED PREVENTION RULE — COPY VERBATIM INTO YOUR EXAM ANSWER

AT ALL TIMES WHEN THE CABIN HEATING IS USED, FRESH AIR MUST BE CIRCULATED TO REDUCE PRESENCE OF CO.

WHY THE "CHERRY-RED" COLOUR?

Carboxyhaemoglobin (Hb-CO) is a *bright cherry red* — much brighter than oxygenated Hb. The pilot's cheeks and lips therefore flush bright pink/red. **Don't confuse this with the bluish CYANOSIS seen in hypoxic hypoxia.** They look opposite:

- **Hypoxic Hypoxia →** blue extremities (cyanosis)
- **CO Poisoning →** bright cherry-red lips/cheeks

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Figure 7.2 — Carbon monoxide poisoning: symptoms, immediate actions and prevention.

CO AFFINITY FOR HB (VS O_2)
> 200 ×

CO IN PISTON-ENGINE EXHAUST
up to 9 %

FIRST SYMPTOM
Inability to concentrate

CLASSIC SKIN SIGN
Cherry-red lips

FIRST ACTION
Turn off cabin heat

HYPOXIA TYPE INVOLVED
Anemic (Type b)

§ 12 Hyperventilation

DEFINITION — "OVER-BREATHING"

"Hyperventilation" is another word for "over-breathing" and may be defined as **lung ventilation in excess of the body's needs**. **Good training is the best way to avoid Hyperventilation in pilots.**

MITIGATION FOR PASSENGERS

The chances of Hyperventilation affecting your passengers can be reduced by **giving them a thorough pre-flight briefing on every aspect of the flying sortie**. (Briefed passengers = calm passengers; calm passengers don't hyperventilate.)

12.1 Causes at Low Altitude

At low altitude — where hypoxia is *not* a factor — the most common causes of hyperventilation are psychological or environmental rather than physiological.

The most common causes of Hyperventilation at low altitude

Category	Cause
Cognitive load	**Intense concentration on a difficult task**
Emotional	**Fear**
Emotional	**Anxiety**
Physical	**Motion sickness**
Physical	**Shock**
Environmental	**Vibration**
Environmental	**Heat**
Aerodynamic / Manoeuvring	**High G-forces**

12.2 Symptoms · Treatment · the Paper-Bag Trick

Symptoms of Hyperventilation

- Dizziness
- Tingling
- Visual disturbances
- Hot or cold sensation
- Anxiety

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- Loss of muscular co-ordination
- Increased heart rate
- Spasms
- Loss of consciousness
- **Cramping and spasms of the hands and feet**
- Cold clammy skin
- Paleness

Treatment of Hyperventilation

1. **Breathe oxygen at 100 percent.** If hypoxia is the cause, the symptoms will improve markedly after **three or four breaths**.
2. **If the symptoms persist**, consciously slow the rate of breathing to `10–12 breaths per minute` and **do not breathe deeply**.
3. **If you are flying below 10,000 feet, hypoxia is unlikely and hyperventilation may be assumed.**
4. If you suspect that any occupant of your aircraft is suffering from hyperventilation, try to calm them down. Give them a **simple task to fulfill that might take their mind off their anxiety**.
5. One of the direct causes of hyperventilation is a **reduction in the carbon dioxide level in the blood**. The condition may be alleviated by getting the sufferer to **breathe into a PAPER BAG**. This action will **increase the blood's carbon dioxide level**, causing the brain to **reduce the breathing rate**.

> **THE SCIENCE BEHIND THE PAPER BAG**
>
> Hyperventilation drives off too much CO_2 from the blood (**respiratory alkalosis**). Since the **brain's breathing centre is regulated by CO_2** (see §8.4), low CO_2 causes the urge to keep over-breathing — a vicious circle. A **paper bag** traps exhaled CO_2 which is then re-inhaled, raising blood CO_2 back to normal and breaking the loop.

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Figure 7.3 — Hyperventilation: causes, symptoms and treatment, including the paper-bag technique.

12.3 Hypoxia vs Hyperventilation — Telling Them Apart

A VITAL DISTINCTION — THE SYMPTOMS OVERLAP

Many symptoms of hypoxia and hyperventilation overlap (dizziness, tingling, visual disturbances, loss of consciousness). The DGCA syllabus gives you a clean **altitude rule** to distinguish:

- **If flying BELOW** `10,000 ft` → hypoxia is *unlikely* → **assume HYPERVENTILATION**.
- **If flying AT or ABOVE 10,000 ft** → hypoxia is plausible → **provide O_2 AND descend**. If symptoms clear after 3–4 breaths of 100 % O_2, it was hypoxia.

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Hypoxia vs Hyperventilation — side-by-side		
Parameter	**Hypoxia**	**Hyperventilation**
Typical altitude	**> 10,000 ft**	**Any altitude** (often below 10,000 ft)
Onset	**Insidious, slow**	Rapid — linked to a trigger (stress, fear)
Skin sign	**Cyanosis** (bluish)	**Paleness · cold clammy skin**
Heart rate	Often increased	**Increased**
Feeling of well-being?	**Yes — euphoria**	No — anxiety, fear
Blood CO_2 level	Variable / often normal	**LOW**
First-line treatment	**100 % O_2 + descend below 10,000 ft**	**Slow breathing to 10–12/min, or breathe into paper bag**
How to confirm in flight	Symptoms improve in 3–4 breaths of 100 % O_2	Symptoms persist on O_2; resolve with re-breathing technique

EXAMINER'S FAVOURITE TRAP A pilot at **9,500 ft** reports tingling in the fingers and dizziness. **Hypoxia is unlikely below 10,000 ft → assume hyperventilation**. Slow your breathing, calm down, optionally use a paper bag — do *not* declare an emergency and descend for a hypoxia event that isn't happening. (However — if O_2 is available, the standard advice "**breathe 100 % O_2; if symptoms clear in 3–4 breaths it was hypoxia**" lets you confirm safely.)

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✦ END OF CHAPTER 7 ✦

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HUMAN PERFORMANCE & LIMITATIONS · MODULE C — THE THIN AIR

Pressure, TUC & Decompression

Chapter 8 — What happens when the pressure fails: cabin pressurisation and the shrinking Time of Useful Consciousness, decompression, the bends, and the rules for blood donation and flying after diving.

BOOK
Human Performance & Limitations

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CHAPTER
8 of 26 · Module C

In this chapter

1. The Baroreceptor Reflex
2. Donating Blood — the 48-hour rule
3. Cabin Pressurisation & Time of Useful Consciousness
4. Decompression — Theory
5. Decompression Sickness (DCS)
6. Flying After Diving — the 24-hour rule
7. Delayed DCS After Diving

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Plate 8.0 — When the pressure fails. At altitude the margin between control and unconsciousness is measured in seconds.

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§ 16

Baroreceptor Reflex

WHAT IT IS

The baroreceptor reflex is the body's **automatic blood-pressure regulator**. Any change in your body's **demand for blood** can **trigger your baroreceptor reflex**. Specialised pressure-sensors ("baroreceptors") in the carotid arteries and aortic arch detect changes in BP and signal the brain to adjust heart rate and vessel diameter accordingly.

16.1 When the reflex fires — DGCA examples

For example, your body may need to adjust your blood pressure when you:

- **Change your body position, such as when you stand**.
- **See or experience something that frightens you**.
- **Switch from walking to running**.

COCKPIT RELEVANCE

- **Postural changes:** standing up from a long sit (e.g. after a long flight, in turbulence, or after a low-blood-sugar moment) can cause transient hypotension → light-headedness. The baroreceptor reflex normally corrects this in 1–2 seconds.
- **Sudden fright / startle response:** pulls in adrenaline from the ANS (§7.4), spiking HR and BP. The baroreceptor reflex moderates this so you don't pass out from a hypertensive crisis.
- **Exercise & G-load:** from steady cruise to sudden manoeuvring, blood demand changes rapidly. The reflex keeps cerebral perfusion stable so you don't grey out.

A **well-conditioned, well-rested, well-hydrated** pilot has a sharper, faster baroreceptor reflex. A dehydrated or fatigued pilot's reflex is sluggish — which is one of several reasons hydration matters in long sectors and high-G ops.

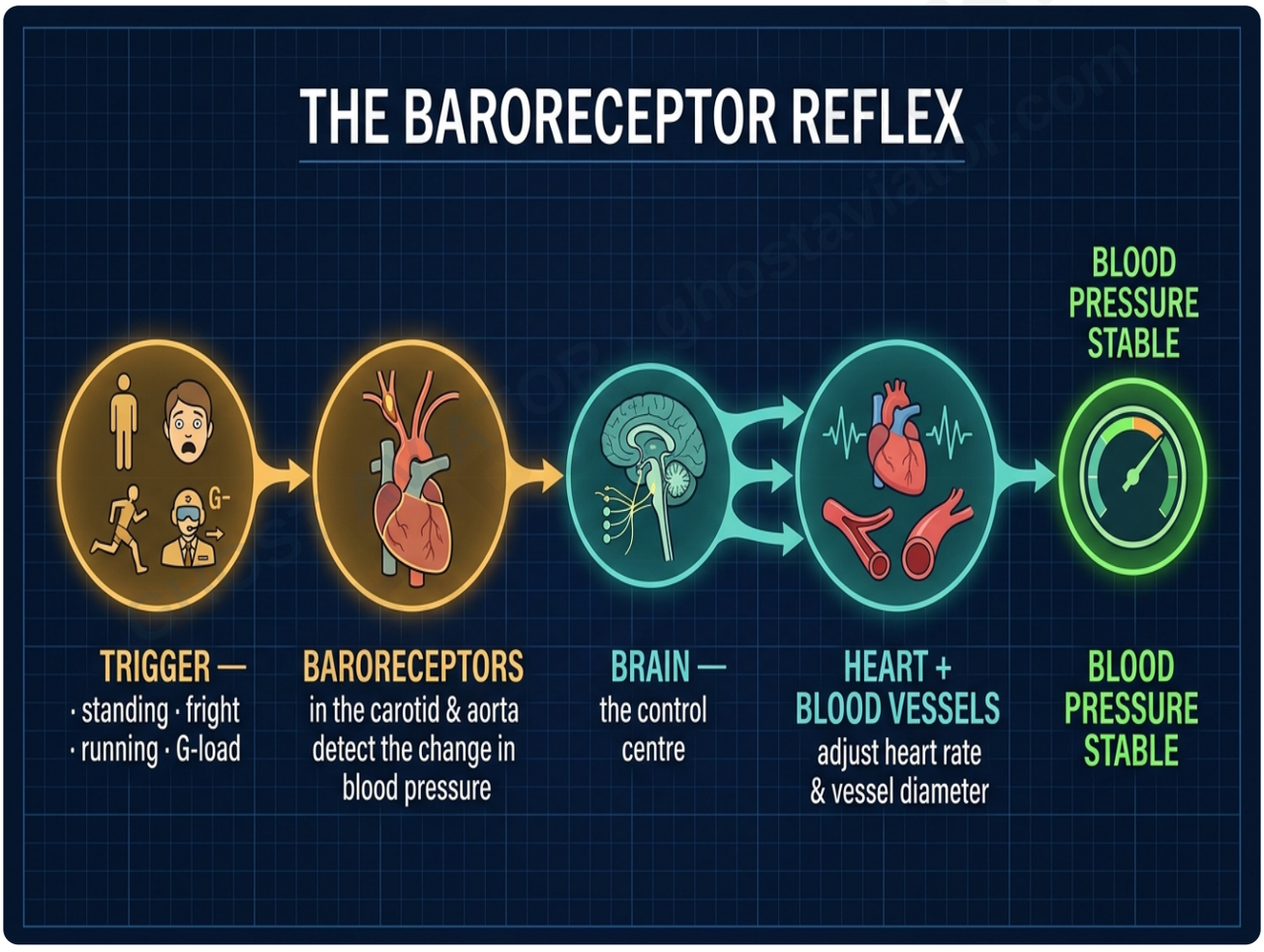

Figure 8.1 — The baroreceptor reflex: how the body senses and corrects a fall in blood pressure.

§ 17 Donating Blood — the 48-hr rule

DGCA-QUOTED

In a **completely healthy individual**, the fluid reduction caused by donating **one unit of blood** is replaced within **several hours**. In some people, however, the loss of blood causes **disturbances to the circulation that may last for several days**.

While the effects at **ground level are minimal**, **flying during this period may entail a risk**.

THE DGCA RULE — EXACT PHRASING

Generally, active pilots should NOT donate blood, but if blood has been donated, they should wait at LEAST `48 hours` before flying.

RECOMMENDED FOR ACTIVE PILOTS

DO NOT donate

IF DONATION HAS occurred

Wait ≥ 48 hr

REASON

Circulatory disturbance

GROUND-LEVEL EFFECT

Minimal

WHY BLOOD DONATION INTERACTS DANGEROUSLY WITH FLIGHT

A single unit of blood (~450 ml) represents about **9 % of total blood volume** in an average adult. While the plasma volume is restored in hours, the **red blood cell mass** takes weeks to fully recover. Reduced RBC mass = reduced haemoglobin = reduced oxygen-carrying capacity = **anemic hypoxia (§10.3b)**. Add altitude (hypoxic hypoxia, §10.3a) and the two stack — exactly the same principle as smoker-at-altitude and alcohol-at-altitude in Part 4.

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§ 18

Cabin Pressurization & Time of Useful Consciousness (TUC)

18.1 What TUC / EPT is

DEFINITION — MEMORISE VERBATIM

Time of useful consciousness (TUC), also **effective performance time (EPT)**, is defined as the **amount of time an individual is able to function effectively (e.g., perform flying duties) in an environment of inadequate oxygen supply**.

It is the **period of time from the interruption of the oxygen supply** or exposure to an oxygen-poor environment **to the time when useful function is lost**, and the individual is **no longer capable of taking proper corrective and protective action**.

CRITICAL CONCEPT — TUC IS NOT TIME TO UNCONSCIOUSNESS

TUC is the time during which a pilot can still **do useful work** — don a mask, push a checklist, initiate descent. After TUC ends, the pilot may still be conscious for several more seconds, but is no longer **functional**. *This distinction is precisely why the DGCA emphasises "corrective and protective action" — the mask must be on the face before TUC expires.*

18.2 Why TUC Drops Sharply at Altitude

DGCA-QUOTED

At the higher altitudes, the **TUC becomes very short; considering this danger, the emphasis is on PREVENTION rather than CURE**.

Cabin pressurization eliminates many problems associated with high altitude flying, but it introduces other potential problems, the most important being the **risk of rapid decompression**.

WHAT DETERMINES TUC AFTER A RAPID DECOMPRESSION

The time of useful consciousness (TUC) following a rapid decompression depends on:

1. **Aircraft altitude**,
2. **The rate at which pressure falls**, and
3. **The level of physical activity of the individual at the time of the event**.

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JET-TRANSPORT TUC REFERENCE VALUES

- At **typical jet transport aircraft altitudes (35,000 feet)** TUC will vary between `33 and 54 seconds`.
- Those average values can be expected to **drop by HALF at 40,000 feet**.
- This emphasizes the importance of **immediate availability of supplemental oxygen to crew members**.
- **RAPID DECOMPRESSION CAN REDUCE TUC BY HALF.**

18.3 TUC Master Table — Verbatim

This is the table you must memorise — both columns. Every DGCA HPL question on TUC is drawn from these figures.

Time of Useful Consciousness (TUC) by Altitude — DGCA Reference

Altitude (measured barometrically)	TUC (normal ascent)	TUC (rapid decompression)
FL180 (18,000 ft; 5,500 m)	`20 to 30 minutes`	`10 to 15 minutes`
FL220 (22,000 ft; 6,700 m)	`10 minutes`	`5 minutes`
FL250 (25,000 ft; 7,600 m)	`3 to 5 minutes`	`1.5 to 3.5 minutes`
FL280 (28,000 ft; 8,550 m)	`2.5 to 3 minutes`	`1.25 to 1.5 minutes`
FL300 (30,000 ft; 9,150 m)	`1 to 2 minutes`	`30 to 60 seconds`
FL350 (35,000 ft; 10,650 m)	`30 seconds to 1 minute`	`15 to 30 seconds`
FL400 (40,000 ft; 12,200 m)	`15 to 20 seconds`	`7 to 10 seconds`
FL430 (43,000 ft; 13,100 m)	`9 to 12 seconds`	`5 to 6 seconds`
FL500 (50,000 ft; 15,250 m)	`8 to 10 seconds`	`5 seconds`

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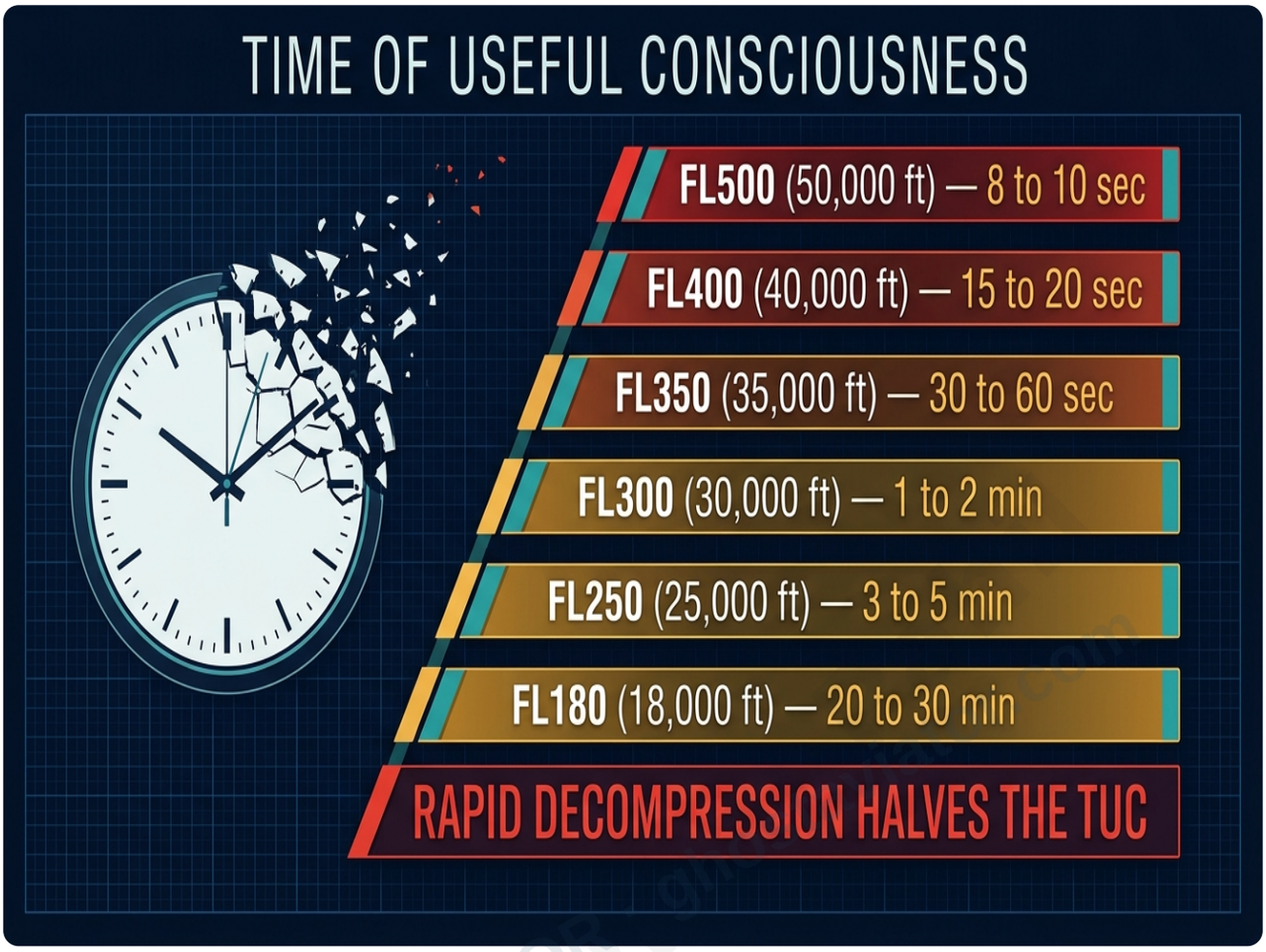

Figure 8.2 — Time of Useful Consciousness collapses with altitude: above FL250 it falls fast; by FL400 it is under 20 seconds.

18.4 Rapid Decompression Halves TUC

THE "HALVING RULE" — DGCA-QUOTED

"Rapid decompression can reduce TUC by half."

Why? Because a rapid decompression doesn't just drop the pilot to ambient pressure — it actually **sucks oxygen out of the lungs into the cabin atmosphere** (Henry's law, in reverse). The pilot is left with **less oxygen in the bloodstream than the new ambient pressure alone would predict**.

33–54ₛ TUC at FL350 (normal ascent)
½ Reduction after rapid decompression
7–10ₛ TUC at FL400 (rapid decompression)

OPERATIONAL PILOT RESPONSE — THE DRILL

After a rapid decompression in a transport jet you have **seconds, not minutes**:

1. **Don oxygen masks** immediately — quick-don, regulator to EMERGENCY/100 %.
2. **Establish crew communication**.
3. **Initiate emergency descent** — target **10,000 ft** or MSA, whichever higher.
4. **Squawk 7700**, declare MAYDAY.

This drill is muscle-memory in transport flight training precisely because there is no time to think it through at FL350+.

§ 19 Decompression — Theory

DEFINITION

"Decompression" means the lowering of pressure.

19.1 The 8,000 ft "Maximum Comfort" Rule

DGCA-QUOTED

The maximum altitude without oxygen at which flying efficiency is not impaired is `8,000 feet`.

DGCA-QUOTED — FLIGHT RISK BELOW 10,000 FT

When flying at altitude below 10,000 feet, the risk of suffering from conditions related to decompression is low.

19.2 Atmospheric Pressure is Halved at 18,000 ft

NITROGEN SATURATION & PRESSURE FACTS

- At ground level, the body tissues are **saturated with nitrogen**, the inert gas which makes up **80 % of our atmosphere**.
- As the aircraft climbs, atmospheric pressure is reduced.
- **By `18,000 ft ASL (5,486 m)` atmospheric pressure is HALVED.**
- **Decompression sickness symptoms may develop at 18,000 ft and above.**
- Pilots flying aircraft with **unpressurised cabins at altitudes greater than 25,000 feet ASL (7,620 m)** may be subject to "the bends".

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19.3 The Bends — Physiology of Nitrogen Bubbles

THE BOTTLE-OF-SODA MECHANISM — DGCA'S OWN ANALOGY

"The bends" condition is caused by **bubbles of nitrogen forming in the tissues because the ambient (atmospheric) pressure is less than the pressure at ground level**. (An example of this phenomenon is the **bubbles formed when a bottle of soda pop is opened**, and the pressure is reduced.)

The bubbles may:

- **Track into joint spaces** causing a **dull, sickening pain**.
- **More dangerously** they may be **released into the lungs or the brain**, giving rise to **chest pain and/or collapse**.

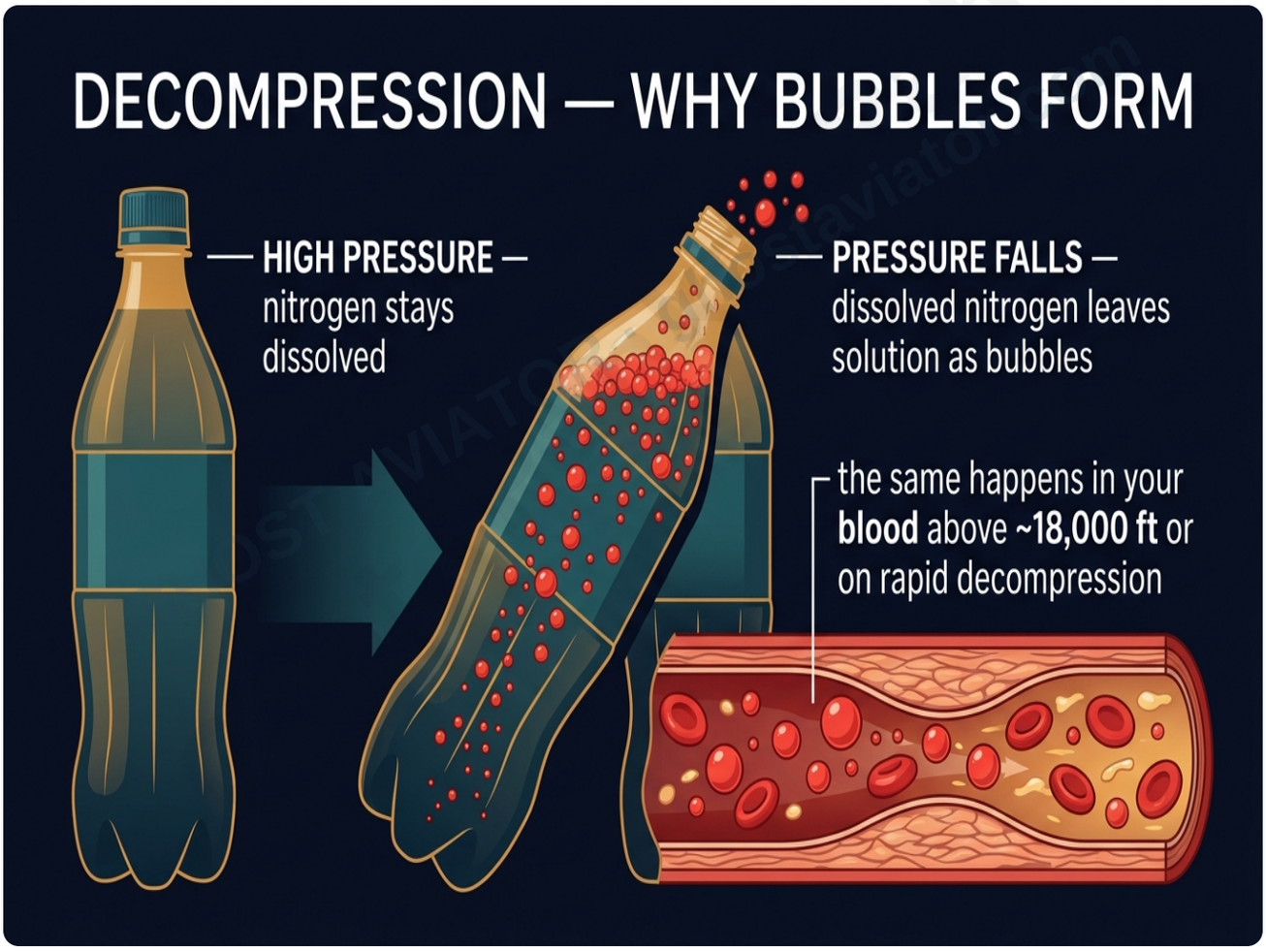

Figure 8.3 — When pressure falls, dissolved nitrogen leaves solution and forms bubbles in the blood and tissues.

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19.4 Cabin Pressurization & Rapid vs Slow Decompression

HOW AIRLINE CABINS SOLVE THE PROBLEM

Airlines and high-performance aircraft have cabin pressurization systems to maintain an artificially "low altitude" within the cabin or cockpit. **The pressurization of a commercial airliner flying at 30,000 ft maintains an internal cabin pressure equivalent to about `6,000 ft`, with a maximum pressure of `8,000 ft`.**

FAST (Rapid) Decompression

A fast decompression is recognizable by:

- **Mist in the cabin**
- **Blast towards the exterior of the aircraft**
- **Expansion of body gases**
- **Blast of air released violently from the lungs**

SLOW Decompression

A slow decompression may be caused by:

- A **slight air-tightness defect**, or
- **Bad functioning of the pressurization**.

Insidious onset — no warning blast, no mist, but cabin altitude creeps up over minutes. Detected via the cabin-altitude warning horn (typically at ~10,000 ft cabin altitude in transport jets).

19.5 Factors that Increase the Tendency to Develop the Bends

FIVE DGCA-QUOTED PREDISPOSING FACTORS

The tendency to develop the bends increases with:

1. **High rates of climb**,
2. **Age**,
3. **Obesity**,
4. **Physical activity**,
5. **Low temperatures**.

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MAX ALTITUDE WITHOUT O_2 — NO IMPAIRMENT
8,000 ft

LOW DCS RISK BELOW
10,000 ft

ATMOSPHERIC PRESSURE HALVED AT
18,000 ft

UNPRESSURISED FLIGHT — BENDS THRESHOLD
> 25,000 ft

TYPICAL AIRLINER CABIN ALTITUDE
≈ 6,000 ft

MAXIMUM CABIN ALTITUDE
8,000 ft

N_2 % IN ATMOSPHERE
≈ 80 %

DCS RISK FACTORS COUNT
5

§ 20

Decompression Sickness (DCS)

DGCA-QUOTED INTRODUCTION

Without a cabin pressurization system, pilots and passengers in high flying aircraft would be exposed to high altitude, like hypoxia, low temperatures and Decompression Sickness/Illness.

20.1 Henry's Law Revisited — the underlying physics

HENRY'S LAW STATEMENT — VERBATIM FROM DGCA

Henry's Law explains the occurrence of decompression sickness. The principle that **at a constant temperature the concentration of a gas dissolved in a fluid with which it does not combine chemically is almost directly proportional to the partial pressure of the gas at the surface of the fluid**.

A rapid reduction in ambient pressure may cause the nitrogen in our blood to come out of solution as small bubbles leading to decompression sickness.

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SYMPTOM LIST & THE "DELAYED ONSET" WARNING

Symptoms of decompression sickness are:

- **Bends**
- **Chokes**
- **Skin manifestations**
- **Neurological symptoms**
- **Circulatory shock**

Symptoms may appear several hours after the exposure.

20.2 Effects of Nitrogen Bubbles — Four Syndromes

The DGCA syllabus names **four specific syndromes** based on the tissue affected. Memorise the names and the body region.

Joints The BENDS
Joints: Bubbles in the joints cause **rheumatic-like pain**, called **the Bends**.

Classic shoulder & knee deep, dull ache — diver's bends, identical mechanism.

Skin The CREEPS
Skin: Nitrogen bubbles released under the skin cause **the Creeps**, a **sensation of movement under the skin**.

Linked to formication (§10.4) — the "ants crawling under the skin" feeling.

Respiratory System The CHOKES
Respiratory System: Shortness of breath and a feeling of burning, gnawing and piercing pain. Known as **the Chokes**.

Bubbles in pulmonary capillaries → ventilation/perfusion mismatch. Burning chest pain.

The Brain The STAGGERS
The Brain: Loss of mental functions and control of movement. Known as **the Staggers**.

Neurological DCS — confusion, ataxia, weakness, paralysis. Most dangerous form.

DCS Syndromes — quick-reference one-liner

Tissue	Syndrome Name	Symptom Type
Joints	**The Bends**	Rheumatic-like joint pain
Skin	**The Creeps**	Sensation of movement under skin
Respiratory System	**The Chokes**	Shortness of breath; burning, gnawing, piercing chest pain
The Brain	**The Staggers**	Loss of mental function & movement control

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20.3 Treatment of Decompression Sickness

DGCA-QUOTED FOUR-STEP TREATMENT

1. **Keep patient warm and on 100 % Oxygen.**
2. **An immediate descent must be initiated.**
3. **Land as soon as possible.**
4. **Seek medical assistance immediately on landing.**

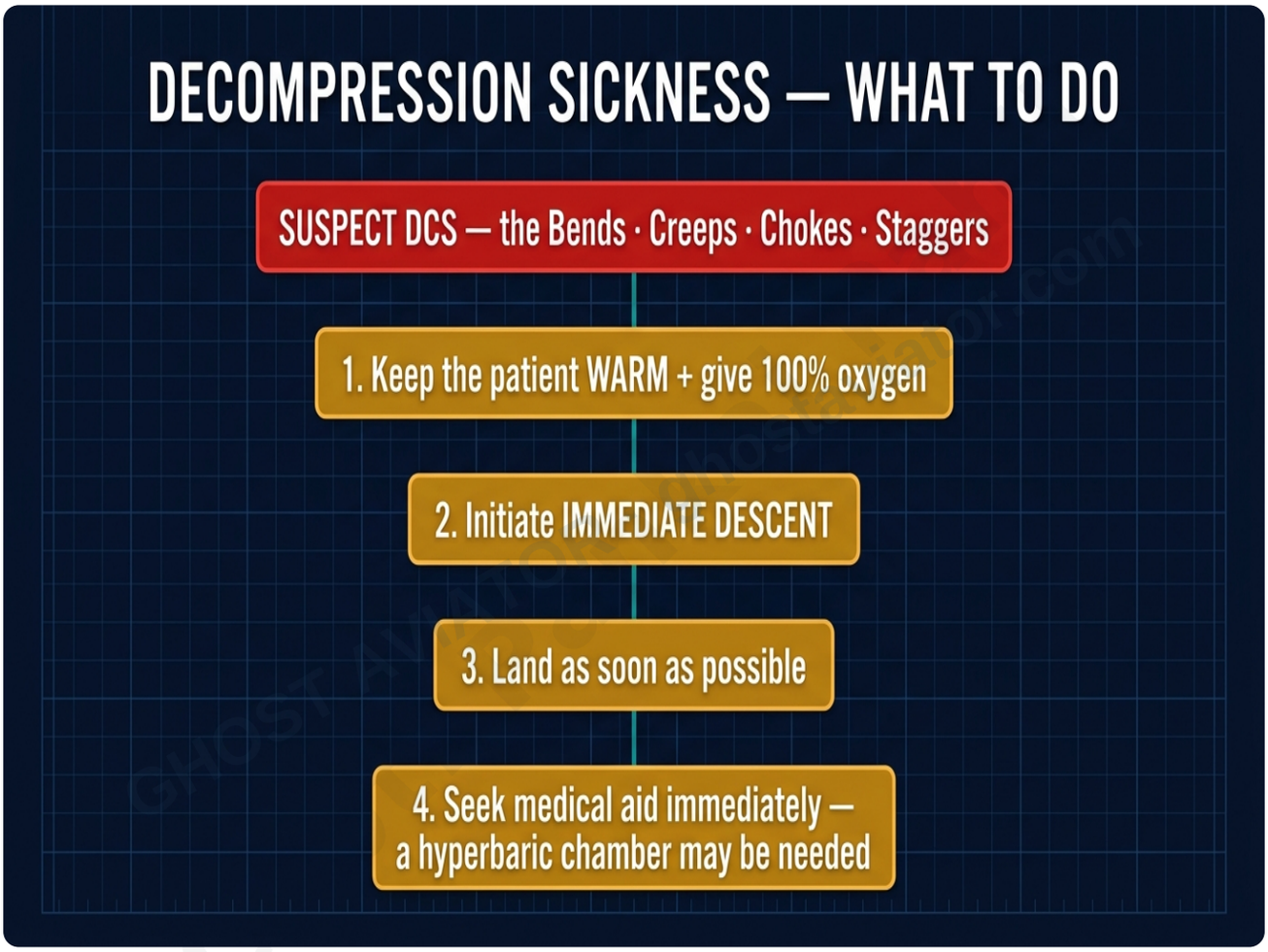

Figure 8.4 — Decompression sickness (the bends): the forms it takes and the response.

20.4 The 12-Hour No-Fly Rule After Rapid Decompression

DGCA-QUOTED

Do not fly for at least `12 hours` after experiencing rapid decompression even though you may be feeling fit.

WHY 12 HOURS?

DCS symptoms can be **delayed for several hours**. Feeling "fit" immediately after a decompression event does **not** mean nitrogen bubbles are not still forming or growing in tissues. The 12-hour pause allows the body to off-gas the residual nitrogen at normal pressure, much like a diver waiting before flying or driving up a mountain.

§ 21 Flying After Diving — the 24-hour rule

DGCA-QUOTED MECHANISM

Decompression sickness can occur when flying at low altitude in individuals who have been diving, using compressed air breathing apparatus shortly before flight at a depth of `30 feet or more`.

THE HARD RULE — MEMORISE VERBATIM

As a general rule, individuals should NOT fly within `24 hours` following diving and certainly not the same day.

30ft Diving depth that creates DCS risk on flying
24hr Minimum wait between diving & flight
15min – 12hr Symptom-onset window after surfacing

WHY DIVING + FLIGHT IS SO DANGEROUS — DGCA-QUOTED

Occasionally a "medical emergency" arises as a result of compressed-air diving, when a diver is in danger of developing **air-embolism (bends) at the surface altitude**, as a result of being unable to decompress before surfacing. In some of these cases **air-evacuation is the only feasible method of getting the patient to a decompression chamber in time to treat this condition**. Flight, however, should be at **the lowest possible altitude to avoid aggravating the condition**.

SYMPTOM ONSET TIMING — DGCA QUOTE

Symptoms and signs **usually appear within `15 minutes to 12 hours` after surfacing**; but in severe cases, symptoms may **appear before surfacing or immediately afterwards**. **Delayed occurrence of symptoms is rare** (the source text terminates mid-sentence at the page break — continued in §21+).

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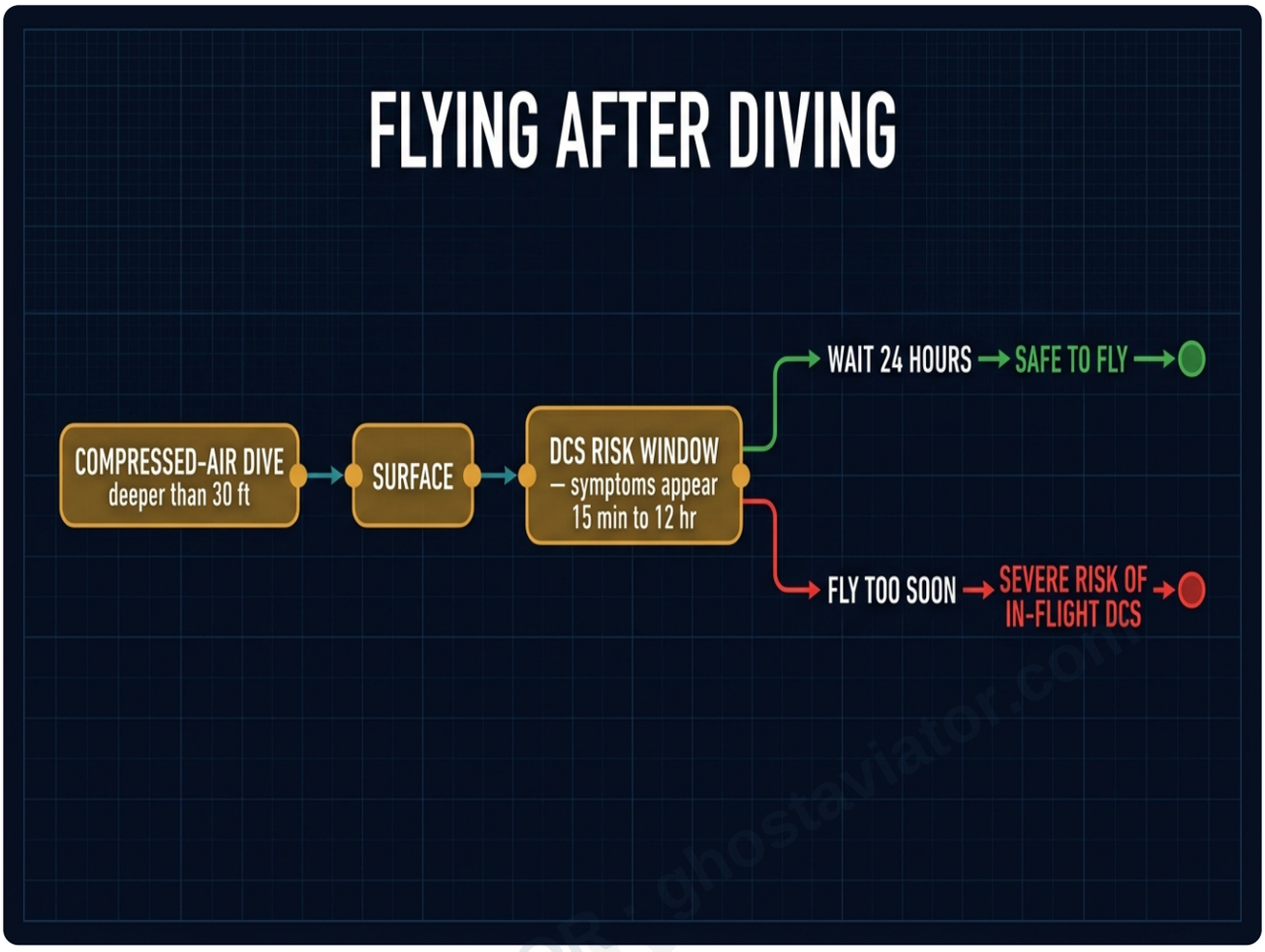

Figure 8.5 — Flying after diving: the wait times that keep dissolved nitrogen from bubbling at altitude.

CROSS-LINK RECAP — THE THREE "WAIT BEFORE FLYING" RULES

You now have three numbers in your head that the DGCA examiner loves to swap around:

- **Alcohol — bottle to throttle:** `24 hr` (§14.1)
- **Blood donation:** `48 hr` (§17)
- **Flying after diving:** `24 hr` (§21)
- **Flying after rapid decompression:** `12 hr` (§20.4)

Do not mix them up. Diving and alcohol share the 24-hour figure; donation gets the longest pause; rapid decompression "only" 12 — but with mandatory medical clearance afterwards.

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§ 22 Closing §21 — Delayed DCS After Diving

PICKING UP THE BROKEN SENTENCE

The Part-5 text ended with "*Delayed occurrence of symptoms is rare...*" — the rest of the sentence (Part 6 source PDF page 22) reads: **"...but it does occur, especially if air travel follows diving."**

In plain English: **even if a diver feels perfectly fine 24+ hours after surfacing, a subsequent flight can re-precipitate DCS symptoms**. This is because residual nitrogen still dissolved in slow-perfused tissues (fat, bone marrow) can come out of solution when the pilot enters the lower cabin pressure of a transport flight. *The 24-hour rule is a minimum, not a guarantee.*

✦ END OF CHAPTER 8 ✦

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HUMAN PERFORMANCE & LIMITATIONS · MODULE D — POISONS & THE CABIN

Smoking & Alcohol

Chapter 9 — Two self-inflicted hazards the exam loves: the smoker's altitude penalty and oxygen-capacity loss, and alcohol — the 24-hour bottle-to-throttle rule, the fixed metabolic rate, and why altitude doubles every drink.

BOOK
Human Performance & Limitations

AUTHOR
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EXAM
DGCA CPL / ATPL — HPL

CHAPTER
9 of 26 · Module D

In this chapter

1. Smoking — the 3,000-ft penalty & oxygen-capacity loss
2. Alcohol — bottle-to-throttle, metabolism & BAC

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Plate 9.0 — Two habits, one cost: both quietly steal the oxygen and judgment a pilot cannot spare.

§ 13

Smoking

13.1 The 3,000-ft Penalty — 10,000 ft becomes 7,000 ft

THE SINGLE MOST-ASKED DGCA FACT ABOUT SMOKING

A pilot who is also a smoker **may experience the symptoms of oxygen deprivation, or hypoxia, at a lower altitude (`7,000 ft`) than a non-smoker (`10,000 ft`)**.

That is a **3,000-ft penalty**. The smoker walks into the cockpit already partway up the hypoxia ladder. Combine that with an unpressurised climb to FL080 and the smoker can be hypoxic *at cruise altitude* when the non-smoker beside him is still asleep.

7,000ft Smoker's effective hypoxia threshold
5–8% O_2-carrying capacity reduction — 1 pack/day
↑ Increased susceptibility to CO poisoning

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13.2 The 5–8 % O_2-Capacity Hit

WHY SMOKING HANDICAPS YOU — DIRECT QUOTE

A person who smokes **one packet of cigarettes per day** will **reduce his capacity to carry oxygen by** `5–8 %`. A smoker also has **increased susceptibility to Carbon Monoxide poisoning**.

THE MECHANISM — ANEMIC HYPOXIA, SELF-INFLICTED

Cigarette smoke contains **carbon monoxide**. Recall from §11.1 that CO binds to haemoglobin **over 200 times more readily** than oxygen. Every cigarette saturates a portion of your haemoglobin with CO — and that Hb is then **unavailable to carry oxygen** until the body slowly clears it (hours). A heavy smoker effectively walks around with **permanent low-grade anemic hypoxia (Type b)**. This is why the smoker's hypoxia threshold drops 3,000 ft.

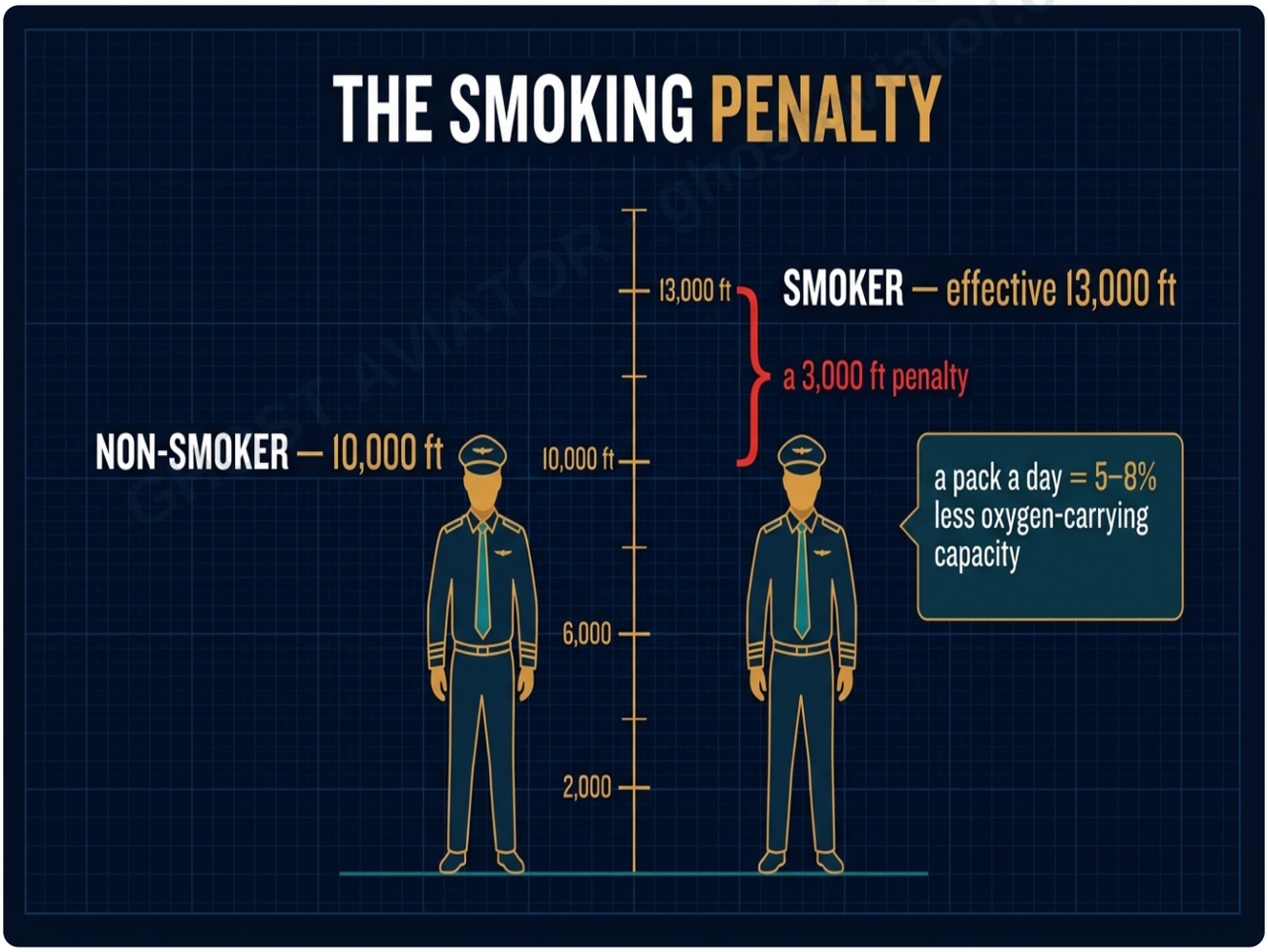

Figure 9.1 — The smoking penalty: a pack a day permanently shaves 5–8% off effective oxygen-carrying capacity, so 10,000 ft feels like 7,000 ft worse.

13.3 The Full Medical-Impact List

DGCA-QUOTED HEALTH EFFECTS ON A SMOKING PILOT

The smoker pilot is exposed to **all six** of the following — every one with direct flight-safety implications:

- **Lung cancer**
- **Breathing problems**
- **Circulatory problems**
- **Reduced tolerance to G forces**
- **Increased risk of heart attack**
- **Degradation of night vision**

Smoking — what it breaks and why it matters in the cockpit

Health Effect	Flight-Safety Implication
Lung cancer	Eventual respiratory failure — medical disqualification.
Breathing problems (COPD, chronic bronchitis)	Reduced ventilation → reduced O_2 uptake → lowered hypoxia threshold (the 7,000-ft figure above).
Circulatory problems	Plaque, atherosclerosis, peripheral vascular disease → narrower arteries → ischemic hypoxia risk (§10.3c).
Reduced tolerance to G forces	Critical in aerobatics, fighter pilots, upset recovery. Less G-tolerance → earlier G-LOC.
Increased risk of heart attack	Direct link to angina/MI risk-factor list in §9.5 (smoking was one of the 12 listed).
Degradation of night vision	Recall night vision is impaired at 5,000 ft for non-smokers (§10.5). Smokers see degradation at *even lower altitudes* — devastating for night IFR and night VFR ops.

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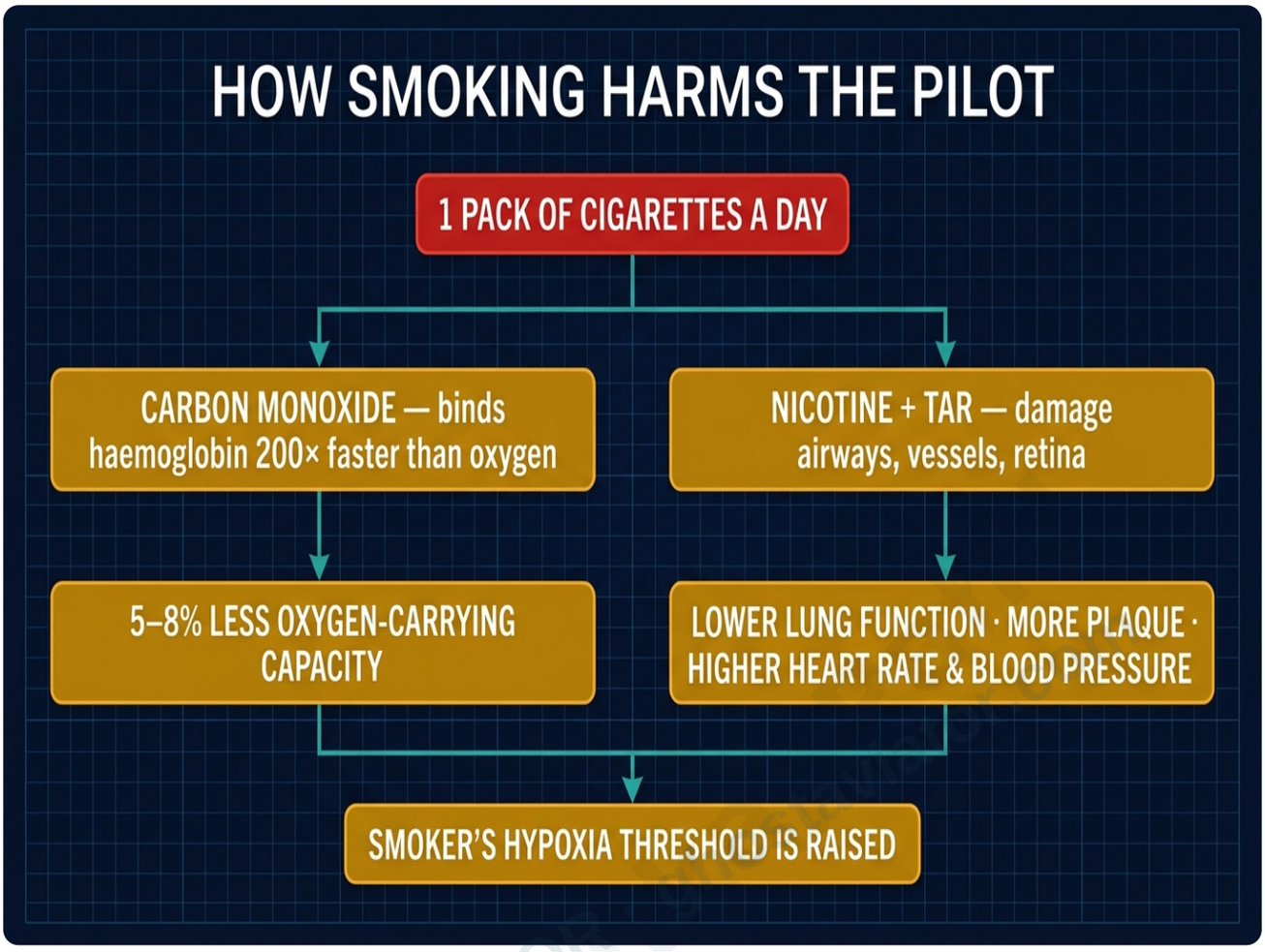

Figure 9.2 — What smoking breaks: carbon monoxide load, reduced oxygen capacity, and the cardiovascular toll.

PRACTICAL CRM NOTE

DGCA Class-1 / Class-2 medicals do *not* automatically disqualify a smoker, but the lifestyle is flagged. **If you smoke and you are reading this, you are accepting a permanent flight-performance handicap.** Even one cigarette before takeoff materially lowers your hypoxia threshold for that sortie. Best practice for examined fitness: **do not smoke at all, especially in the 24 hours before flight**.

§ 14 Alcohol

THE HEADLINE — ALCOHOL IS A DEPRESSANT

Alcohol acts primarily as a depressant. Do not fly while under the influence of alcohol. **Even small amounts of alcohol in the system can adversely affect judgment and decision-making abilities.**

14.1 The 24-Hour Bottle-to-Throttle Rule

THE HARD RULE — MEMORISE VERBATIM

An excellent rule is to allow TWENTY-FOUR HOURS between the last drink and takeoff time.

This is the "*24-hour bottle-to-throttle*" rule. It is more conservative than the statutory minimum in some jurisdictions (often 8 hours), and is the answer DGCA expects in the HPL paper. Some operators enforce 12 hours; some, 24. The / DGCA text recommends **24 hours**.

14.2 Alcohol Metabolism — The Fixed Rate

DGCA-QUOTED METABOLISM RATE

Remember that your body metabolises alcohol at a **fixed rate**, and **no amount of coffee or medication will alter this rate**. Alcohol leaves the body at an average rate of `0.015 g/100 mL/hour`, which is the same as reducing your **Blood Alcohol Content (BAC) level by 0.015 per hour**.

For men, this is usually a rate of about ONE STANDARD DRINK PER HOUR.

24_{hr} Bottle-to-throttle separation
0.015 g/100 mL/hr metabolism rate (≈ 1 drink/hr in men)
$40_{mg\%}$ BAC at which pilot errors spike significantly

14.3 Altitude Amplifies the Effect — "1 = 2" Rule

THE "ONE DRINK AT ALTITUDE EQUALS TWO AT SEA LEVEL" RULE

Relatively small amounts of alcohol **significantly decrease a pilot's tolerance to hypoxia (oxygen lack)**. **At `6,000 ft (1,800 m)`, the effect of ONE drink is that of TWO drinks at sea level.** Even at sea level alcohol impairs judgment and reaction time. **Therefore, alcohol and flying do not mix.**

WHY? — HISTOTOXIC + HYPOXIC STACKING

Recall **Histotoxic Hypoxia (§10.3d)**: high blood alcohol levels prevent cells from *using* the oxygen they receive. Layer that on top of **Hypoxic Hypoxia (§10.3a)** from altitude, and you have two oxygen-starvation mechanisms operating *at the same time*. They are **additive** — exactly the same principle that makes a smoker's hypoxia worse at altitude.

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Figure 9.3 — The 1=2 rule: one drink at 6,000 ft hits the pilot like two at sea level (hypoxic + histotoxic effects combine).

14.4 Recommended Maximum Alcohol Intake — Units & Rules

> **DEFINITION OF "ONE UNIT"**
>
> **ONE UNIT of alcohol = half a pint of beer = a standard glass of wine = one measure of spirits.**

MEN

5 units / day · 21 units / week

Maximum recommended consumption.

WOMEN

3 units / day · 14 units / week

Maximum recommended consumption.

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1 UNIT EQUALS

½ pint beer

1 UNIT EQUALS

1 glass wine

1 UNIT EQUALS

1 measure spirits

MEN – DAILY/WEEKLY

5 / 21

WOMEN – DAILY/WEEKLY

3 / 14

14.5 BAC 40 mg/100 ml — The Pilot Error-Spike Threshold

DGCA-QUOTED: THE BAC LEVEL AT WHICH ERRORS BALLOON

Blood alcohol concentrations of `40 mgs per 100 ml` **results in significant increases in errors committed by pilots** due to the following effects:

- **Impaired judgment and impaired ability to reason**
- **Degraded muscular coordination** and **degraded vision**
- **Lack of inhibitions and self-control**
- **Increased susceptibility to Hypoxia**
- **Damages to the liver, heart, brain and blood cells**
- **Affects short and long-term memory**
- **Slows reaction time**
- **Sufferer may feel that performance is improved** *(the alcohol-euphoria trap)*
- **Balance and sensory illusions**
- **Irregular sleep patterns**

THE EUPHORIA TRAP — ANALOGOUS TO HYPOXIA

"**Sufferer may feel that performance is improved**" is the alcohol equivalent of *hypoxia's euphoria* (§10.2). The pilot is the *least* qualified person in the cockpit to judge whether he's safe to fly. This is precisely why **fixed bottle-to-throttle rules exist** — they take the judgement out of the impaired pilot's hands.

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14.6 Hangover & the "Masked Hangover" Warning

DGCA-QUOTED HANGOVER RULE — VERBATIM

Do not fly with a hangover, or a "masked hangover" (symptoms suppressed by aspirin, caffeine or other medication). High altitude, where oxygen is deficient, aggravates these effects.

WHY A "MASKED" HANGOVER IS WORSE THAN AN OBVIOUS ONE

A normal hangover at least gives you a warning sign — you feel awful, you don't fly. A **masked hangover**, where caffeine, paracetamol, ibuprofen or other OTC medication has suppressed the headache and nausea, leaves you **still impaired** in judgement, coordination, vision and reaction time — but **without the warning**. Several aviation accidents have been traced to a pilot who took an aspirin, "felt fine," and flew with substantial residual BAC.

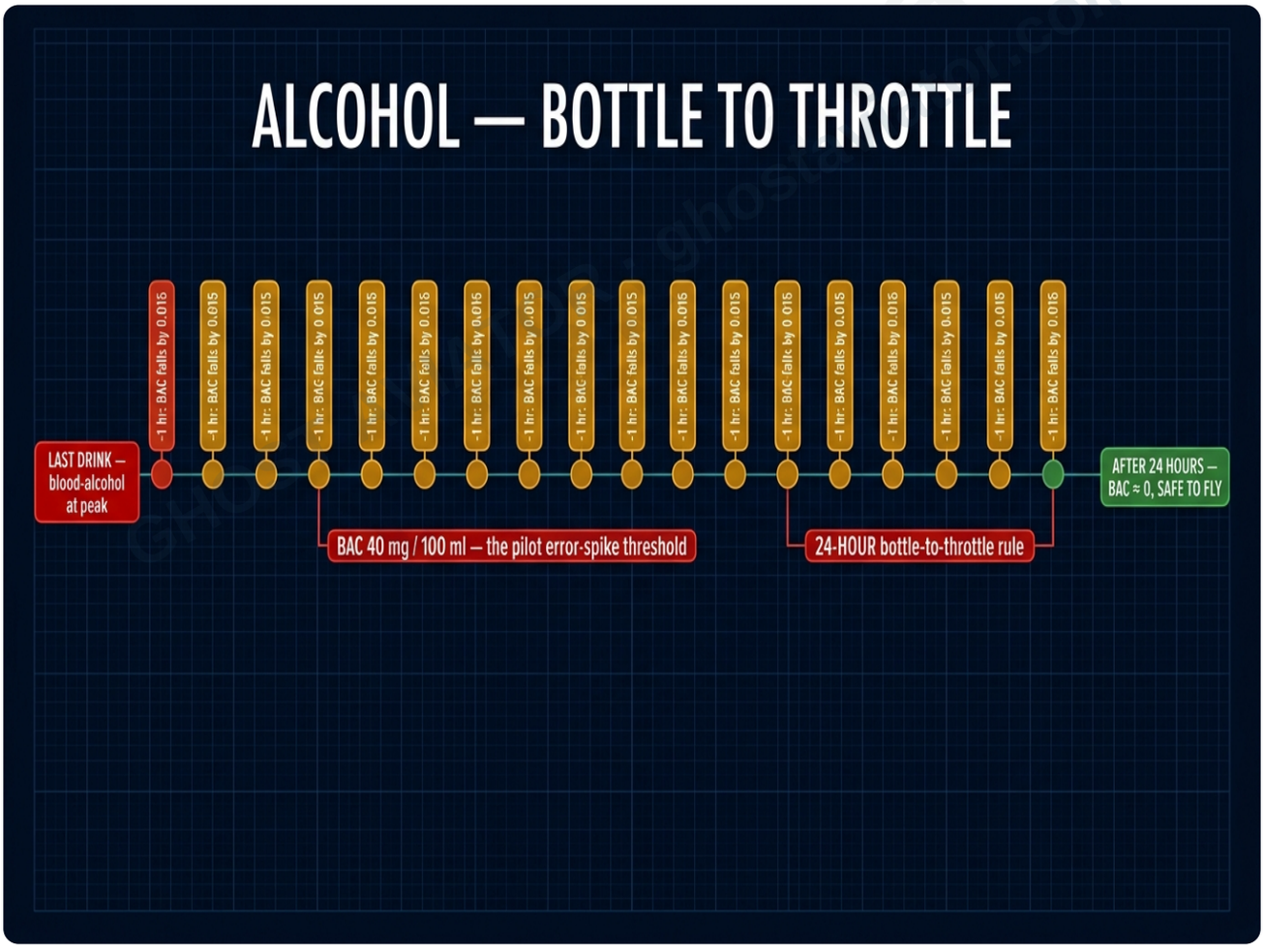

Figure 9.4 — Alcohol and the pilot: the 24-hour bottle-to-throttle rule, the fixed metabolic rate, and the BAC error-spike threshold.

✦ END OF CHAPTER 9 ✦

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The Cabin Environment

Chapter 10 — The physical environment around the pilot: humidity, temperature, vibration, glare and UV, and what happens when a pilot is incapacitated in flight — from a simple faint to a fit.

BOOK
Human Performance & Limitations

AUTHOR
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EXAM
DGCA CPL / ATPL — HPL

CHAPTER
10 of 26 · Module D

In this chapter

1. Humidity
2. Temperature
3. Vibration
4. Glare & UV Radiation
5. Incapacitation in Flight
6. Fits & Faints

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Plate 10.0 — The pilot's physical world: the heat, glare, dryness and vibration that quietly wear performance down.

§ 23 Humidity

23.1 Why Humidity Matters Up High

DGCA-QUOTED

Humidity may become an issue with high-altitude jet transport aircraft because of the low relative humidity at their operational altitudes. The discomfort arising from low relative humidity may not imply physical indisposition.

DGCA-QUOTED HUMIDITY RANGES

- `40 – 60 %` **relative humidity is NORMAL.**
- **<** `20 %` may create **minor discomfort**, such as **skin, eye, nose, throat dryness**.

40–60% Normal relative humidity
< 20% Minor discomfort — skin, eye, nose, throat dryness
3–8% → 22% Modern humidifiers raise cabin RH from 3–8% to ~22%

PREVENTION & MANAGEMENT

- **Overall dehydration can be prevented with adequate fluid intake.**
- **Diuretics like coffee or tea should be avoided.** They drive fluid out of the body via increased urination — the opposite of what is needed at altitude.

23.2 Aircraft Humidifier Systems

DGCA-QUOTED

The installation of humidifiers on aircraft **raises cabin/cockpit humidity**, but there are potential problems such as:

- **Weight penalty**,
- **Condensation**, and
- **Mineral contamination** that the designer must consider.

Aircraft such as the **Airbus A350, Boeing 787, and the future Boeing 777X** are compatible with onboard humidifiers, either in **crew areas or the entire cabin**. These technologies **increase humidity from 3–8 % to 22 %** and **reduce contaminants** in the air we breathe on a flight.

§ 24 Temperature

DGCA-QUOTED

Temperature affects performance as follows:

24.1 The Three Reference Temperatures — memorise verbatim

Temperature effects on pilot performance (DGCA verbatim)	
Temperature	**Effect on Performance**
`20 °C`	**Comfortable temperature for most people in normal clothing.**
`30 °C`	**Increased heart rate, blood pressure, and sweating.**
`> 15 °C`	**Discomfort, loss of feeling in hands, poor control of fine muscle movement.** *(Note: source phrasing uses "> 15 °C" — referring to deviations beyond 15 °C from a comfortable baseline, i.e. cold-stress.)*

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COCKPIT-TEMPERATURE CONSIDERATIONS

- **At 20 °C** the pilot is comfortable — fine motor skills, judgement and reaction time are optimal.
- **At 30 °C** the body's ANS responds to heat stress by raising HR, BP and triggering sweating to cool the skin. Cognitive performance starts to degrade.
- **In cold conditions** — far below the comfort baseline — the pilot loses sensation in the fingers, struggles with fine-motor tasks like switch selection or fast checklist scanning, and may shiver involuntarily.
- For both extremes, **cabin temperature management is a flight-safety task**, not merely a comfort one. Climate control should be set to keep cockpit temperature near 20 °C for sustained operations.

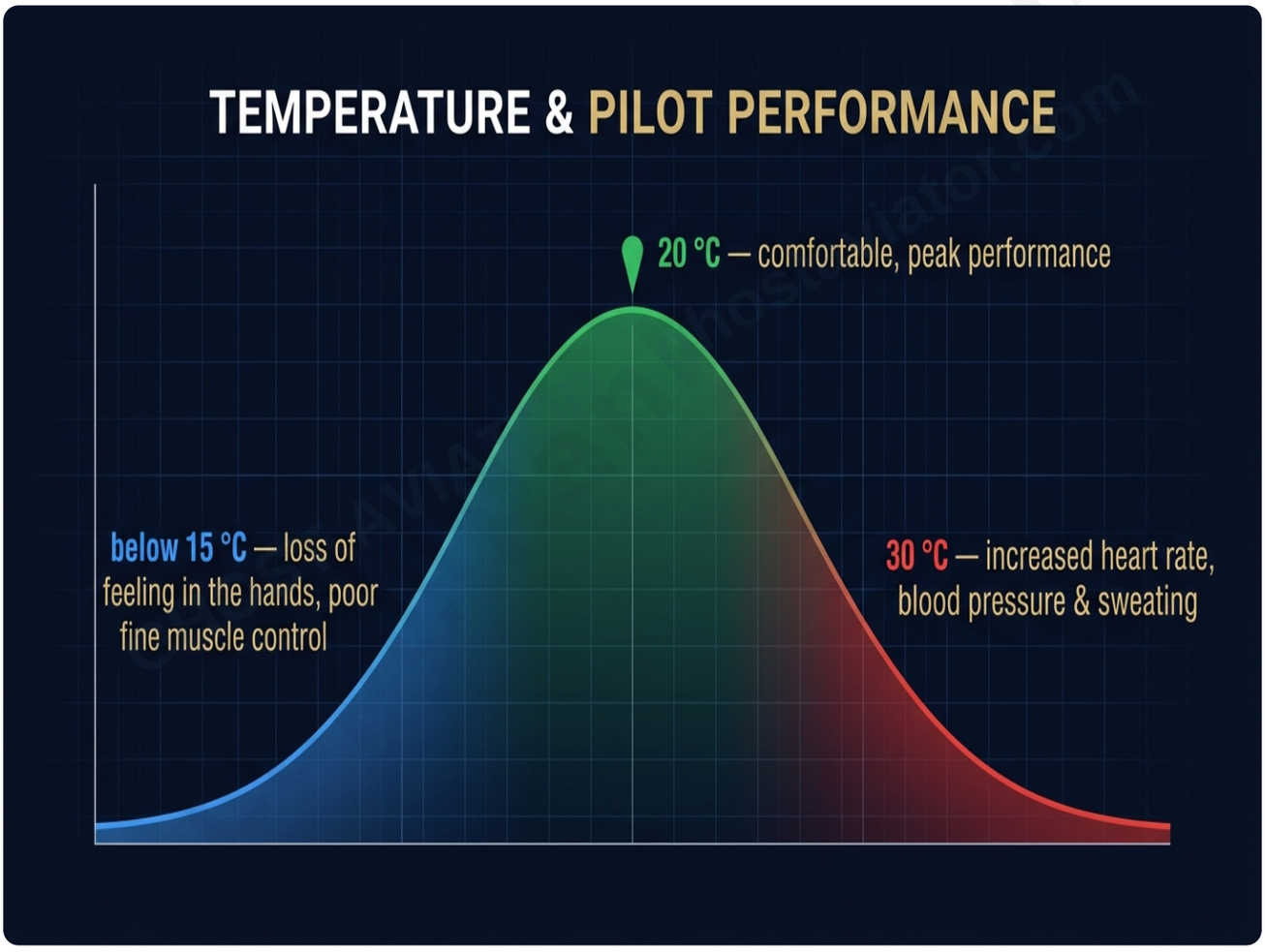

Figure 10.1 — Pilot performance peaks around 20 °C; both cold and heat degrade it in their own way.

§ 25

Vibration

25.1 Natural Resonance — Why Frequency Matters

DGCA-QUOTED OPENING

Different parts of the body show a natural resonance at different periods of vibration. For example:

- The **natural resonance of the eyeball is** `30 – 40 Hz`, and
- The **skull is** `1 – 4 Hz`.

WHAT "NATURAL RESONANCE" MEANS IN PLAIN ENGLISH

Every physical structure has a frequency at which it *wants* to vibrate — push it at that frequency and small inputs build up into huge oscillations (think of pushing a child on a swing in time with the swing's natural period). When external aircraft vibration matches a body part's resonant frequency, **that part suffers maximum disturbance**. The eyeball resonating at 30–40 Hz is precisely why **vision becomes blurred at certain helicopter or propeller-vibration frequencies**.

25.2 Effects of Vibration — Full DGCA Frequency Table

MEMORISE THIS TABLE — DIRECT EXAM TARGET

Effects of vibration include:

Vibration frequency vs body effect (DGCA verbatim)

Frequency Range	Effect on the Body
`1 – 4 Hz`	**Interference with breathing; neck pain.** *(Skull resonance range.)*
`4 – 10 Hz`	**Chest and abdominal pain.**
`8 – 12 Hz`	**Backache.**
`10 – 12 Hz`	**Headache, eyestrain, throat pain, speech difficulty, muscle tension.**
`30 – 40 Hz`	**Interference with vision.** *(Eyeball resonance range.)*

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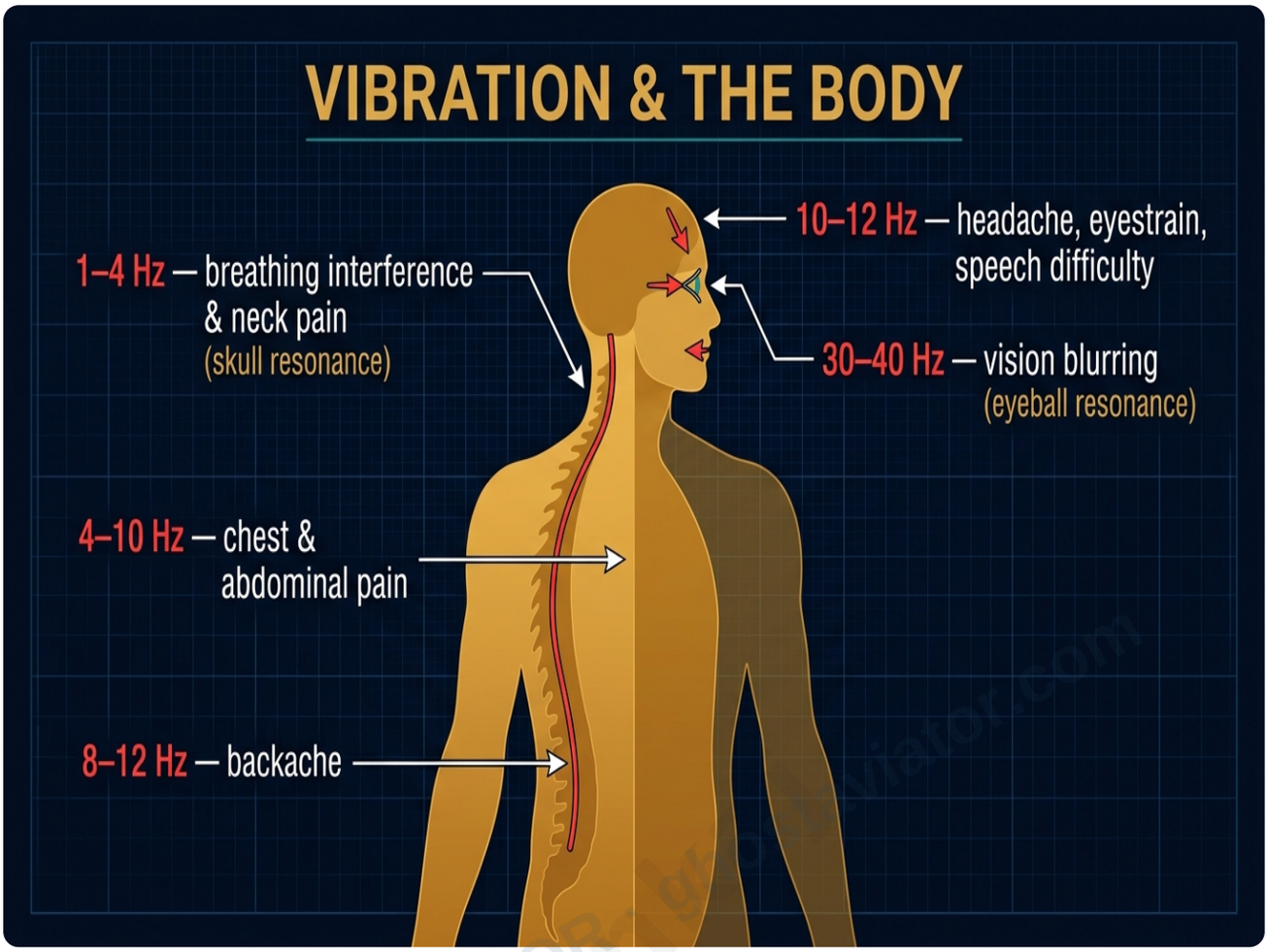

Figure 10.2 — Each vibration frequency band affects a specific body region; the natural-resonance bands are the dangerous ones.

OPERATIONAL VIBRATION SOURCES IN AVIATION

- **Helicopter rotors:** typically 4–7 Hz (main rotor) → chest/abdominal pain & backache range.
- **Piston / turboprop engines:** 10–30 Hz on the airframe → headache, eyestrain, vision interference.
- **Jet engine buffet:** higher-frequency, but resonant peaks in the seat may still strike critical body-part frequencies.
- **Turbulence:** low-frequency buffeting (1–10 Hz) → fits the skull / neck / chest range.

This is why aircraft seats incorporate **damping and isolation systems** — to keep the airframe's vibration spectrum from delivering energy at the worst body-resonance frequencies.

§ 26 Glare & UV Radiation

DGCA-QUOTED

UV radiation from sunlight can cause visual fatigue, as well as affect visual health.

WHY GLARE AND UV DESERVE THEIR OWN SECTION

At cruise altitudes (FL300+), the protective filtering of the atmosphere is reduced. Cockpit windows admit **significantly more UV-A and UV-B** than ground-level glass would. Over a long career, this contributes to:

- **Visual fatigue** — the immediate effect of glare on the working pilot. Squinting, eyestrain, reduced contrast sensitivity, headache.
- **Cumulative visual health damage** — cataract formation, macular degeneration risk, conjunctival/skin damage.

PRACTICAL PILOT PRECAUTIONS

- Use **aviation-grade polarised or non-polarised UV-blocking sunglasses** (note: certain LCD instruments may distort under polarised lenses — check before adoption).
- Use **cockpit sun-visors** appropriately — don't leave them stowed when the sun is over the nose.
- For long-haul crews, schedule **vision rest breaks** on autopilot legs.
- Periodic **aero-medical eye examinations** — already mandated by DGCA — pick up early cataract / retinal changes.

§ 27

INCAPACITATION IN FLIGHT

27.1 Why Pilot Incapacitation is a Flight-Safety Issue

DGCA-QUOTED INTRODUCTION — VERBATIM

The risk of seizure in flight is obvious. Incapacitation is in most cases:

- **Sudden**,
- **Unpredictable**,
- **Unavoidable**,
- **Prolonged**,
- **Complete**, and
- **Potentially more frequent in the stressful flying environment**,

... and constitutes a **direct threat to the health and safety**.

DGCA-QUOTED PREVENTION MEASURE

Periodical medical examinations minimize the risk of incapacitation in flight. The frequency of medical checks increases with advancing age.

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CLASS-1 MEDICAL (CPL / ATPL), UP TO AGE 60

12 months

CLASS-1, ABOVE 60 (ABOVE 40 FOR SINGLE-CREW OPERATIONS)

6 months

CLASS-2 MEDICAL (PPL / SPL), UP TO AGE 50

2 years

CLASS-2 MEDICAL, ABOVE 50

1 year

(DGCA CAR Section 7 Series C Part I, para 5.1, effective 15 Nov 2025.)

27.2 Obvious Incapacitation

DGCA-QUOTED DEFINITION

Obvious incapacitation normally refers to a state in which **all of a crew member's physical or mental functions are lost**, including:

- **Loss of consciousness**, OR
- **Being unable to move while retaining consciousness**,

... **rendering them completely unable to carry out their duties**.

ONSET PATTERNS & PRESENTATION

- Obvious incapacitation can sometimes involve **convulsions**,
- OR the victim may **lapse into unconsciousness several minutes after the incapacitation occurs**.

DGCA-QUOTED CAUSES OF OBVIOUS INCAPACITATION

- **Cardiac arrest**
- **Myocardial infection** *(textual variant — "myocardial infarction" — heart attack)*
- **Interracial hemorrhage** *(intended: "intracranial haemorrhage" — bleed in the brain)*
- **Cerebral apoplexy** *(stroke)*
- **Epilepsy**

The source PDF uses the phrasings shown — students should be aware of the modern medical terminology in parentheses.

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27.3 Subtle / Insidious Incapacitation

DGCA-QUOTED DEFINITION

SUBTLE (develops slowly and gradually), incapacitation refers to a state of a **partial or temporary loss of physical or mental function** which manifests itself in the form of:

- **Partial paralysis**,
- **A dulling of perception, judgment, or responses** or lack thereof,
- **A state of absence of mind, distraction of attention**,
- **Trouble with speech**,
- **Inadequate responses**,
- **Meaningless utterances**, etc.

THE HIDDEN-DANGER WARNING — DGCA VERBATIM

Particular attention must be paid to the fact that a crew member may become incapacitated EVEN THOUGH THEIR APPEARANCE IS NO DIFFERENT FROM NORMAL.

POSSIBLE CAUSES OF SUBTLE INCAPACITATION — DGCA LIST

- **Temporary hypoglycemia** (low blood sugar — pilot skipped meals)
- **Reduction in blood pressure**
- **Cerebropathy or psychopathy**
- **Excessive muscular fatigue**
- **Excessive drinking** (see §14)
- **Insufficient sleep**
- **Emotional instability**
- **Toothache**
- **Stomachache**
- **Headache**

27.4 Obvious vs Subtle — Why Subtle is MORE Dangerous

DGCA-QUOTED — THE COUNTERINTUITIVE TRUTH

Since other crew members are often unable to detect subtle incapacitation quickly, from a flight safety point of view, it is possible that SUBTLE INCAPACITATION MAY LEAD TO A SITUATION OF COMPARATIVELY GREATER DANGER THAN OBVIOUS INCAPACITATION.

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WHY THIS MATTERS IN TWO-CREW OPERATIONS

With **obvious incapacitation**, the other crew member sees the captain slumped or convulsing, declares emergency, takes control, gets the aircraft on the ground. The hand-off is instant.

With **subtle incapacitation**, the captain is still moving switches, talking on the radio, even responding to questions — but his judgement is impaired. He may set the wrong altitude, mis-tune a NAV, mishear an ATC clearance, or fail to initiate a missed approach. The other pilot may not detect this until *after* the wrong outcome unfolds. This is the rationale behind **two-crew CRM**: a healthy independent cross-check that catches subtle drift before it becomes an accident.

OBVIOUS Incapacitation

- **All functions lost** — LoC or paralysed-conscious
- May include convulsions / delayed LoC
- **Detected instantly** by the other crew
- Causes: cardiac arrest, MI, intracranial bleed, stroke, epilepsy
- **Handover happens fast**

SUBTLE Incapacitation

- **Partial or temporary** loss of function
- Appearance may be normal
- **Often undetected for some time** by the other crew
- Causes: hypoglycemia, ↓ BP, fatigue, sleep loss, drinking, emotional/dental/stomach/ headache
- **Greater overall flight-safety danger**

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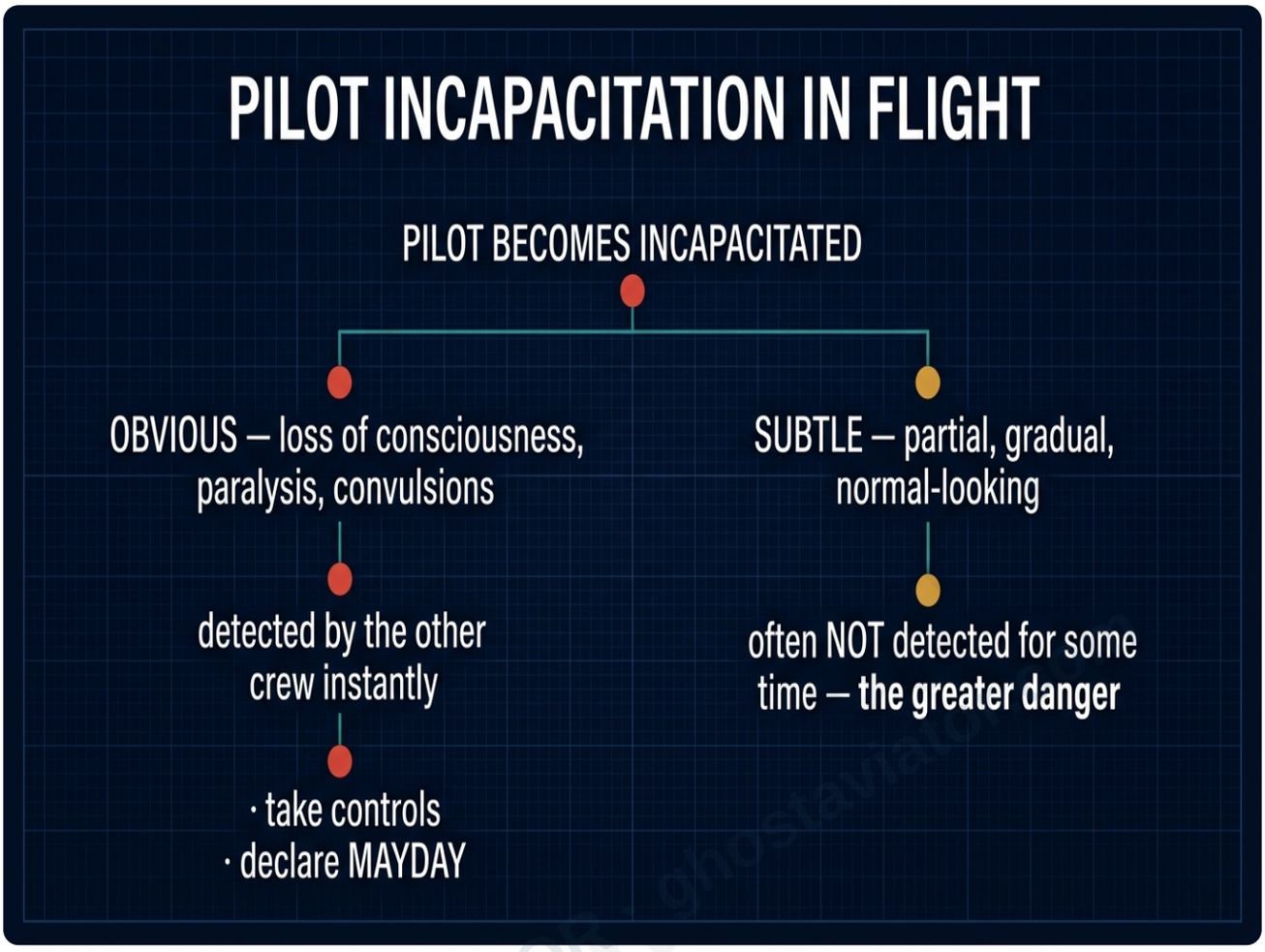

Figure 10.3 — Pilot incapacitation: recognising it and the other crew member's response.

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§ 28

FITS & FAINTS

28.1 Epilepsy — Grand Mal & Petit Mal

DGCA-QUOTED DEFINITION

A **fit or seizure is usually referred to as "epilepsy"**. A fit or a seizure is **not a specific disease** but **a set of signs or symptoms in response to a disturbance of the electrical activity in the brain**.

Grand Mal Epilepsy

- Manifests as a **generalized seizure**
- **Associated with a transient loss of consciousness**
- May be associated with a **prodromal phase** *(warning aura before the seizure)*
- Normally accompanied by **convulsions and uncontrolled physical movement**

Petit Mal Epilepsy

- Also a **generalized seizure**
- **Not associated with a loss of consciousness**

- **Petit Mals are a MINOR attack**
- Often presents as brief "absence spells" — staring, blanking out for a few seconds

DGCA-QUOTED — THE ABSOLUTE BAR

A seizure may or may not be associated with a loss of consciousness... but **ANY FIT, MAJOR OR MINOR, IS ASSOCIATED WITH AN UNPREDICTABLE LOSS OF CONSCIOUSNESS AND IS THEREFORE AN ABSOLUTE BAR TO THE HOLDING OF A FLYING LICENCE.**

PILOT-MEDICAL IMPLICATION

This is one of the very few *absolute* medical disqualifications in aviation. Even controlled epilepsy on medication — even a single documented seizure event in adult life — is grounds for medical refusal. The reason is the word "**unpredictable**" — DGCA cannot certify the probability of an in-flight seizure as acceptably low.

28.2 Faint & Vasovagal Syncope

DGCA-QUOTED DEFINITION

Faint is a common cause of a loss of consciousness in adults. The most common causes of faints are:

Common causes of faint — DGCA verbatim list	
#	**Cause**
1	**Standing up quickly after prolonged sitting especially when hot or dehydrated**
2	**A sudden shock**
3	**Loss of blood after an accident**
4	**Lack of food or fluid**
5	**Other physiological stress**

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DGCA-QUOTED — SYNCOPE & VASOVAGAL SYNCOPE

Syncope is a temporary but sudden loss of consciousness when blood flow to the brain is compromised. In young individuals, **fear, anxiety, sight of blood, etc., can result in a temporary loss of consciousness**. This is referred to as **VASOVAGAL SYNCOPE**.

Frequently, syncope is associated with symptoms like:

- **Light-headedness**,
- **Muscle weakness**, and
- **Dizziness** before the actual fainting occurs.

DGCA-QUOTED — FLIGHT-LICENCE IMPLICATION

A faint has NO SIGNIFICANCE as far as future flying is concerned, so long as the cause is clearly understood.

EPILEPSY VS FAINT — THE CRITICAL COMPARISON

This is the most frequently tested contrast in DGCA HPL papers. Memorise it:

Epilepsy vs Faint (Syncope) — side-by-side

Parameter	Epilepsy (Grand/Petit Mal)	Faint (Syncope)
Underlying cause	**Disturbance of brain electrical activity**	**Compromised blood flow to the brain**
Loss of consciousness	Often (Grand Mal) / Absent (Petit Mal)	**Yes, temporary & sudden**
Convulsions / uncontrolled movement	**Yes (Grand Mal)**	No
Predictability	**Unpredictable**	Cause is usually identifiable (heat, hunger, fear, blood loss)
Typical warning signs	Prodromal aura (Grand Mal)	**Light-headedness, muscle weakness, dizziness**
Effect on flying licence	**ABSOLUTE BAR — no licence**	**No significance — fit to fly if cause is clear**

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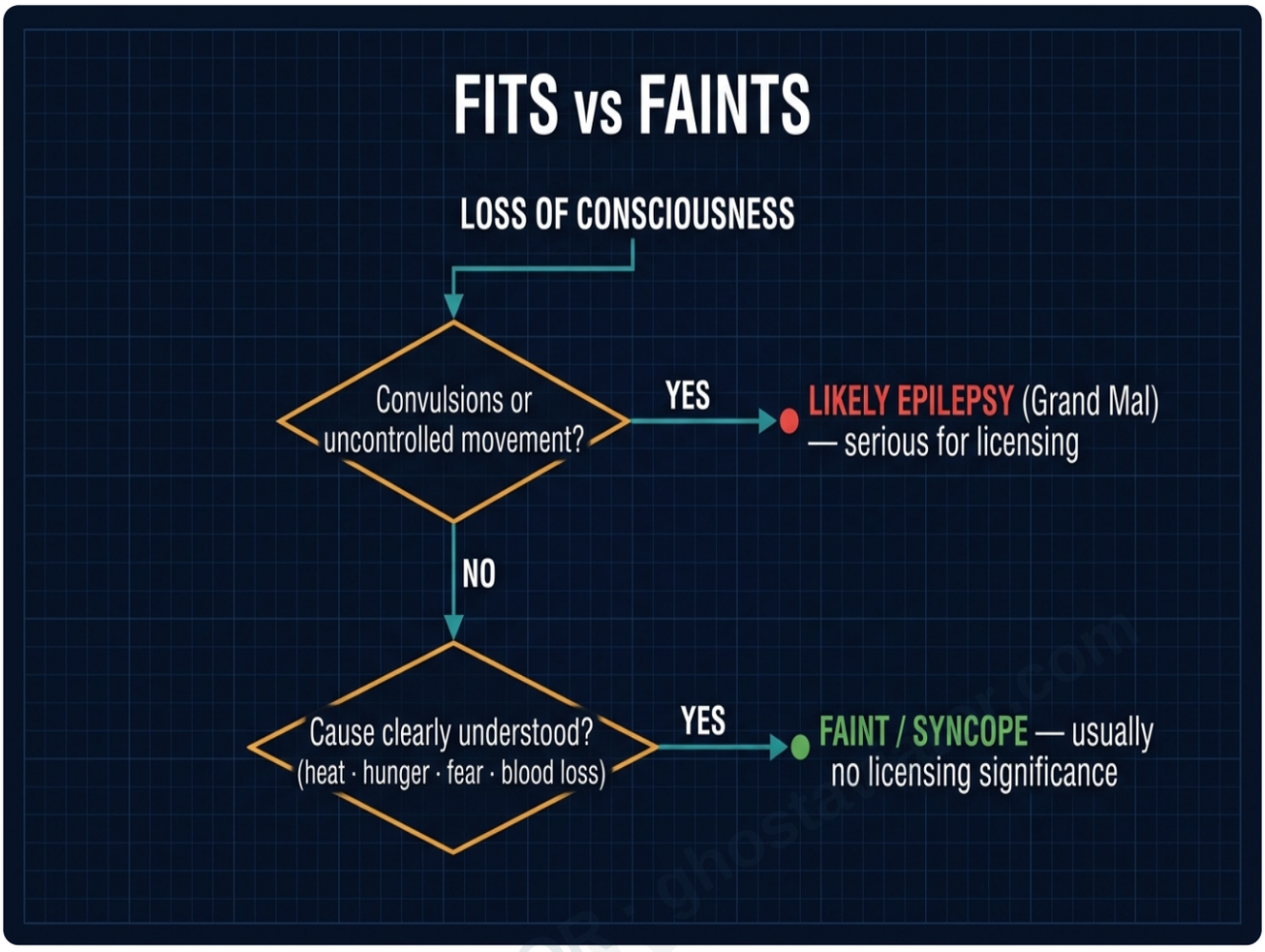

Figure 10.4 — Fits and faints: causes of syncope and how epilepsy differs from a simple faint.

✦ END OF CHAPTER 10 ✦

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HUMAN PERFORMANCE & LIMITATIONS · MODULE E — THE SENSES

The Eye

Chapter 11 — The pilot's primary instrument: how the eye focuses light like a camera, the retina's rods and cones, and the sharpness (visual acuity) the examiner tests.

BOOK
Human Performance & Limitations

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CHAPTER
11 of 26 · Module E

In this chapter

1. The Eye — Anatomy & How it Focuses
2. Limitations of Acuity — Sharpness of Central Vision

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Plate 11.0 — The pilot's primary instrument. Ninety per cent of the information a pilot uses arrives through the eyes.

GHOST AVIATOR Capt. Parv

§ 29 THE EYE

DGCA-QUOTED OPENING — THE EYE IS THE MOST IMPORTANT SENSOR

The eye delivers to the brain information about the outside world at a much faster rate than any other sensory organ. The eye is the organ of sight. The eye is the most sensitive of our sensory organs.

Its basic structure is similar to a **camera with an aperture (Iris), a lens (lens) and a light sensitive screen (Retina)**.

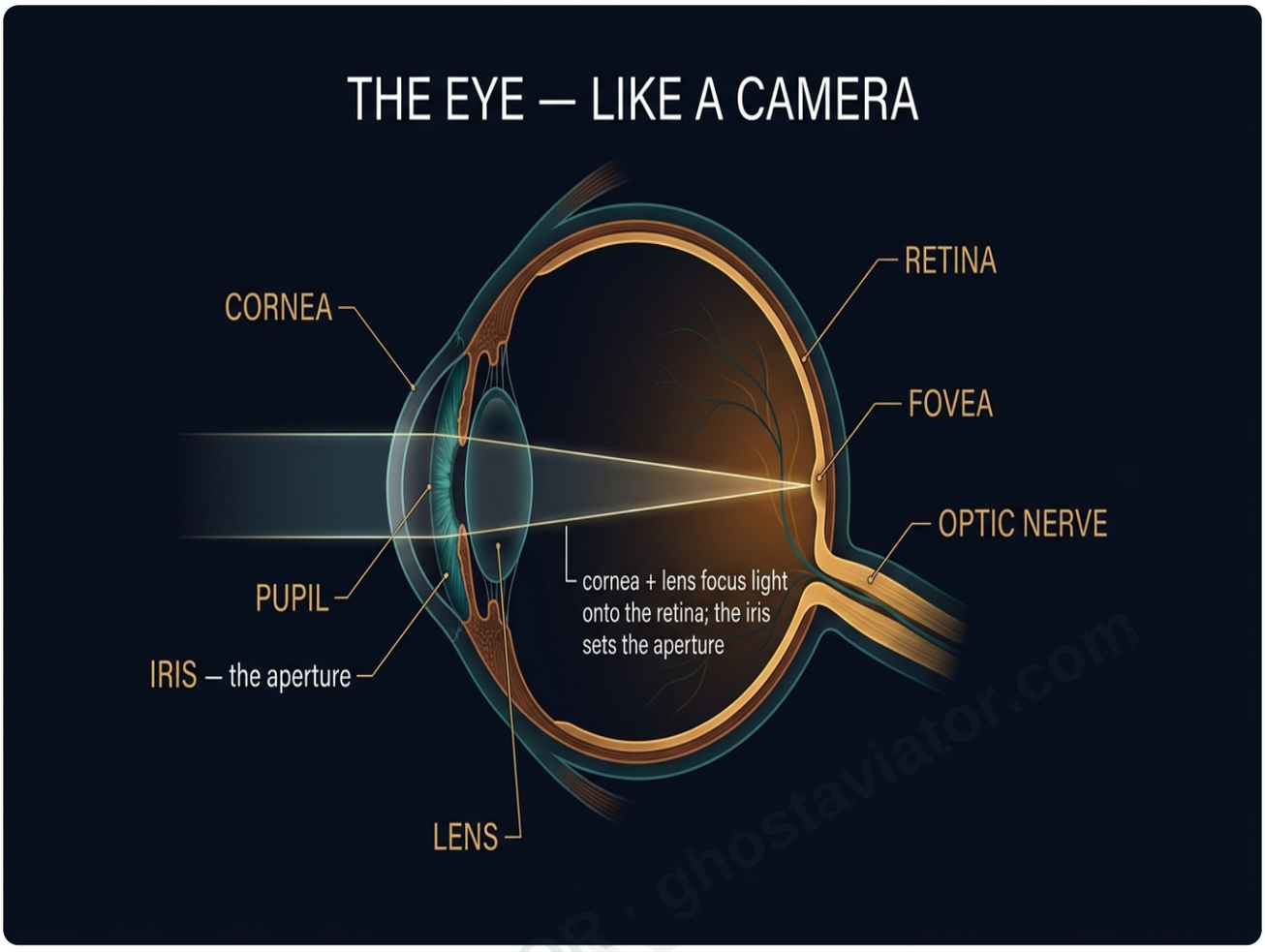

Figure 11.1 — The eye as a camera: the cornea and lens focus light onto the retina; the iris sets the aperture.

29.1 The Cornea, Iris & Pupil, Lens

THE CORNEA — DGCA-QUOTED

Light enters the eye through the **Cornea**, a **clear window at the front of the eyeball**. The Cornea is capable of contributing `70 % to 80 %` **towards the total focusing ability of the eye**.

THE IRIS AND THE PUPIL — DGCA-QUOTED

The **amount of light allowed to enter the eye is controlled by the Iris**. The **Pupil**, the **hole in the center of the Iris**, **adjusts to the flow of light**.

THE LENS — DGCA-QUOTED

The **shape of the lens is changed by muscles**. This controls **final focusing onto the fovea**.

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29.2 The Retina — Rods, Cones, and the Fovea

THE RETINA — DGCA-QUOTED

The **Retina is a light sensitive screen lining the inside of the eyeball**. On this screen are **light-sensitive cells**. When light falls on them, it generates a **small electrical charge which is passed to the brain by nerve fibres (neurons) which combine to form the Optic Nerve**.

CRITICAL NUMBERS — BURN INTO MEMORY

The retina has **rods in its peripheral zone** and **cones in its central zone**. The retina contains the **receptors for vision**:

- **About** `100 million rods`
- **About** `6 million cones`

100M Rods (peripheral · night vision · black & white)
6M Cones (central · day vision · colour & detail)
~17:1 Rod-to-cone ratio

The Rods

The Rods can **only detect black and white** but are **much more sensitive at lower light levels**.

Rods are responsible for our PERIPHERAL VISION.

- Located in the peripheral retina
- ~ 100 million in number
- Sensitive in dim light → night vision
- No colour information — monochrome
- Slow dark-adaptation (~ 30 min)

The Cones & The Fovea

The **central part of the retina is called the FOVEA**. The fovea centralis is a **small, central pit composed of closely packed cones** in the eye.

This is the area with the **best day vision and NO night vision at all**.

Any object that **needs to be examined in detail is automatically brought to focus on the fovea**. This is called **"Central Vision"**. The rest of the retina fulfils the function of **attracting our attention to movement and change**.

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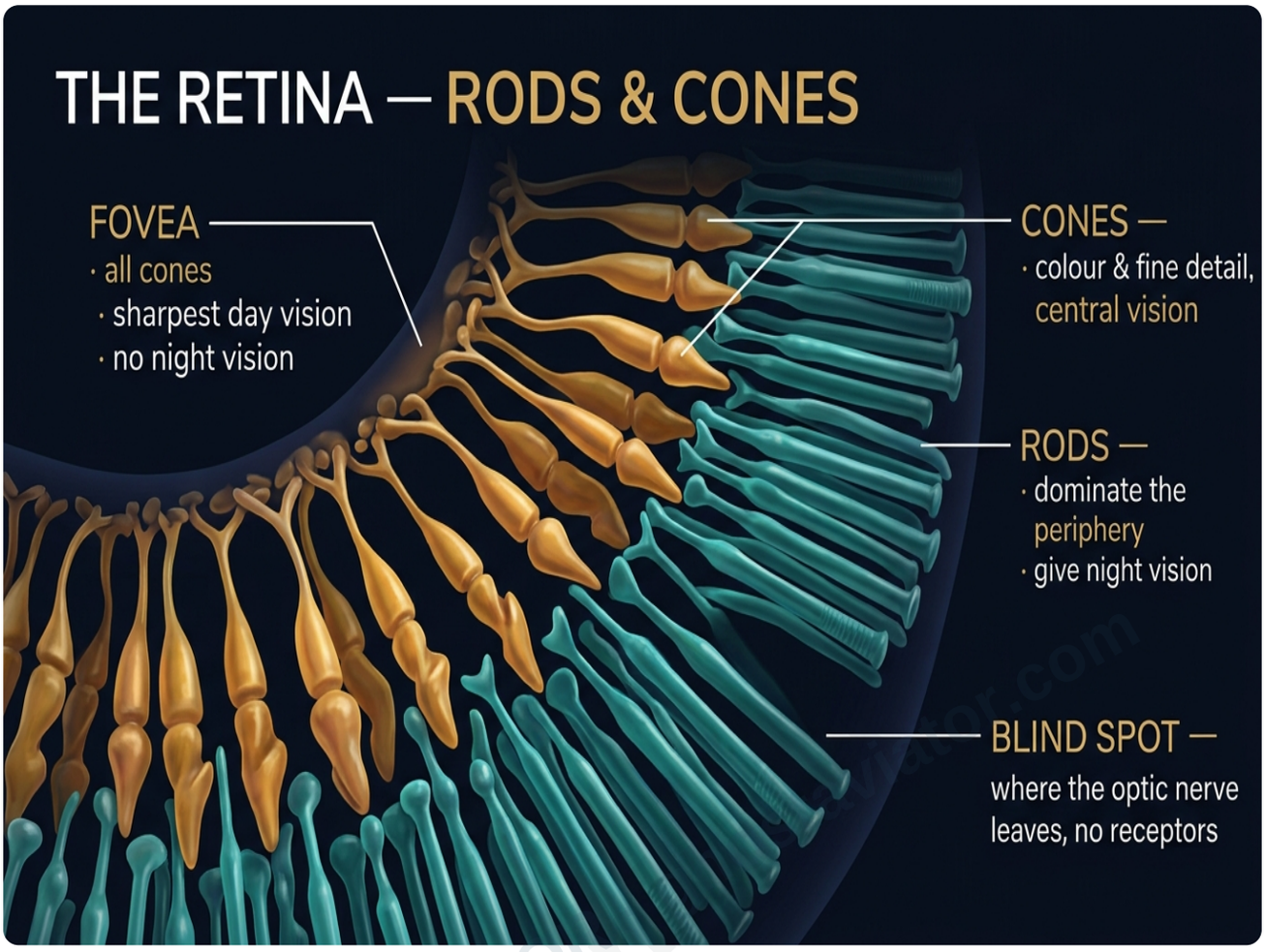

Figure 11.2 — The retina: the fovea is all cones (sharp day vision, no night vision); rods dominate the periphery and give night vision.

29.3 Eye Movement & Binocular Vision

EYE MOVEMENT — DGCA-QUOTED

To **track an object successfully, or to focus on an object, the eyes need to move in harmony with one another**. This means that the brain must **co-ordinate control of the muscles of the two eyes**. **In a fatigued person, double vision can occur.**

BINOCULAR VISION — DGCA-QUOTED

Binocular Vision means seeing with two eyes. With binocular vision **each eye sees an object from two slightly different angles**. The brain **merges the two images into one** and is thus able to **perceive that the image has depth**.

A further advantage of binocular vision is that the **blind spot of one eye is covered by the other eye**. **Depth perception when objects are close is achieved through binocular vision.**

29.4 Visual Acuity

DGCA-QUOTED DEFINITION

Visual acuity is a measure of the capacity of the eye to determine SMALL DETAIL, UNDISTORTED, at a given distance.

WHY ACUITY MATTERS IN FLIGHT

The **sharpest visual acuity occurs when the retinal image is sharply focused on the fovea**, so that the pilot needs to look **exactly in the direction of the on-coming aircraft to detect it**. It is thus **essential for pilots to have normal visual acuity**, either with the **naked eye, or by wearing spectacles**, in order that they may detect **objects clearly at safe distance**.

29.5 Accommodation & Reading Glasses

ACCOMMODATION — DGCA-QUOTED

As well as being able to see objects clearly at a distance, **pilots also need good near vision in order to read instruments and maps**. Being able to **focus on close objects is a function of the eye's ability to accommodate**.

READING GLASSES — DGCA-QUOTED

Pilots and drivers who have reached middle-age normally wear bi-focal spectacles to allow them to **see clearly at a distance and to read their instruments and maps, while wearing the same spectacles**.

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CORNEA — % OF TOTAL FOCUSING

70 – 80 %

RODS TOTAL

~ 100 million

CONES TOTAL

~ 6 million

BEST DAY VISION AREA

Fovea (all cones)

FOVEA NIGHT VISION

NONE

PILOT MIDDLE-AGE GLASSES

Bi-focal

§ 30

Limitations of Acuity — Sharpness of Central Vision

DGCA-QUOTED

The sharpness of central vision **drops as light falls on retina at increasing angles from the fovea**. The following factors affect the sharpness of central vision:

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Twelve factors that affect the sharpness of central vision (DGCA verbatim)		
#	**Factor**	**Pilot Note**
1	**Angular distance from the fovea**	Acuity is maximum on the fovea, drops sharply off-axis.
2	**Physical imperfections within the visual system**	Refractive errors, cataract, retinal disease.
3	**Age**	Presbyopia, lens stiffening, declining contrast sensitivity.
4	**Hypoxia**	Retina has very high O_2 demand — first system to suffer (§10).
5	**Smoking**	CO + nicotine reduce retinal perfusion (§13).
6	**Alcohol**	Histotoxic mechanism degrades cells including retinal photoreceptors (§14).
7	**Amount of light available**	Photopic / mesopic / scotopic — acuity scales with luminance.
8	**Size and contours of an object**	Large object with clear contour = easier; small or amorphous = harder.
9	**Distance of the object from the viewer**	Resolvable detail falls off with distance.
10	**Contrast of an object with its surrounding**	Low contrast = hard to see (e.g. white aircraft in cloud).
11	**Relative motion of a moving object**	A constant-bearing target on collision course is *hardest* to detect.
12	**Drugs or medication**	Many OTC and prescription meds cause blurred vision or accommodation problems.

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✦ END OF CHAPTER 11 ✦

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HUMAN PERFORMANCE & LIMITATIONS · MODULE E — THE SENSES

The Limits of Sight

Chapter 12 — Where vision fails the pilot: night vision and dark adaptation, the blind spot, protecting the eyes, refractive defects, colour vision and colour blindness, and how speed steals reaction time.

BOOK
Human Performance & Limitations

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CHAPTER
12 of 26 · Module E

In this chapter

1. Limitations of the Visual System — Night Vision
2. Protection of the Visual System — Sunglasses
3. Visual Defects — Myopia & Hypermetropia
4. Colour Vision
5. Colour Blindness
6. Vision & Speed — Reaction Times

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Plate 12.0 — The eye is a superb instrument in daylight — and a treacherous one in the dark, at speed, and against glare.

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§ 31

LIMITATIONS OF THE VISUAL SYSTEM

31.1 Night Vision & Dark Adaptation — the critical 7 vs 30 minutes

DGCA-QUOTED DEFINITION

Adaptation is the adjustment of the eyes to high or low levels of illumination.

The time required for:

- **Complete adaptation for HIGH levels of illumination is** `10 seconds`, and
- **For full DARK adaptation,** `30 minutes`.

WHY PILOTS CARE — DGCA-QUOTED

When **passing from bright ambient surroundings into the dark**, **visual capacity is severely reduced** until the eyes have adapted to the dark. It is **especially important for pilots to allow sufficient time for dark adaptation to take place before flying at night**.

CONES VS RODS — THE TWO-STAGE DARK ADAPTATION

Dark adaptation takes time:

- **About `7 minutes` for the CONES**,
- **And `30 minutes` for the RODS**.

10s Bright-light adaptation (dark → bright)
7min Cones — partial dark adaptation
30min Rods — FULL dark adaptation

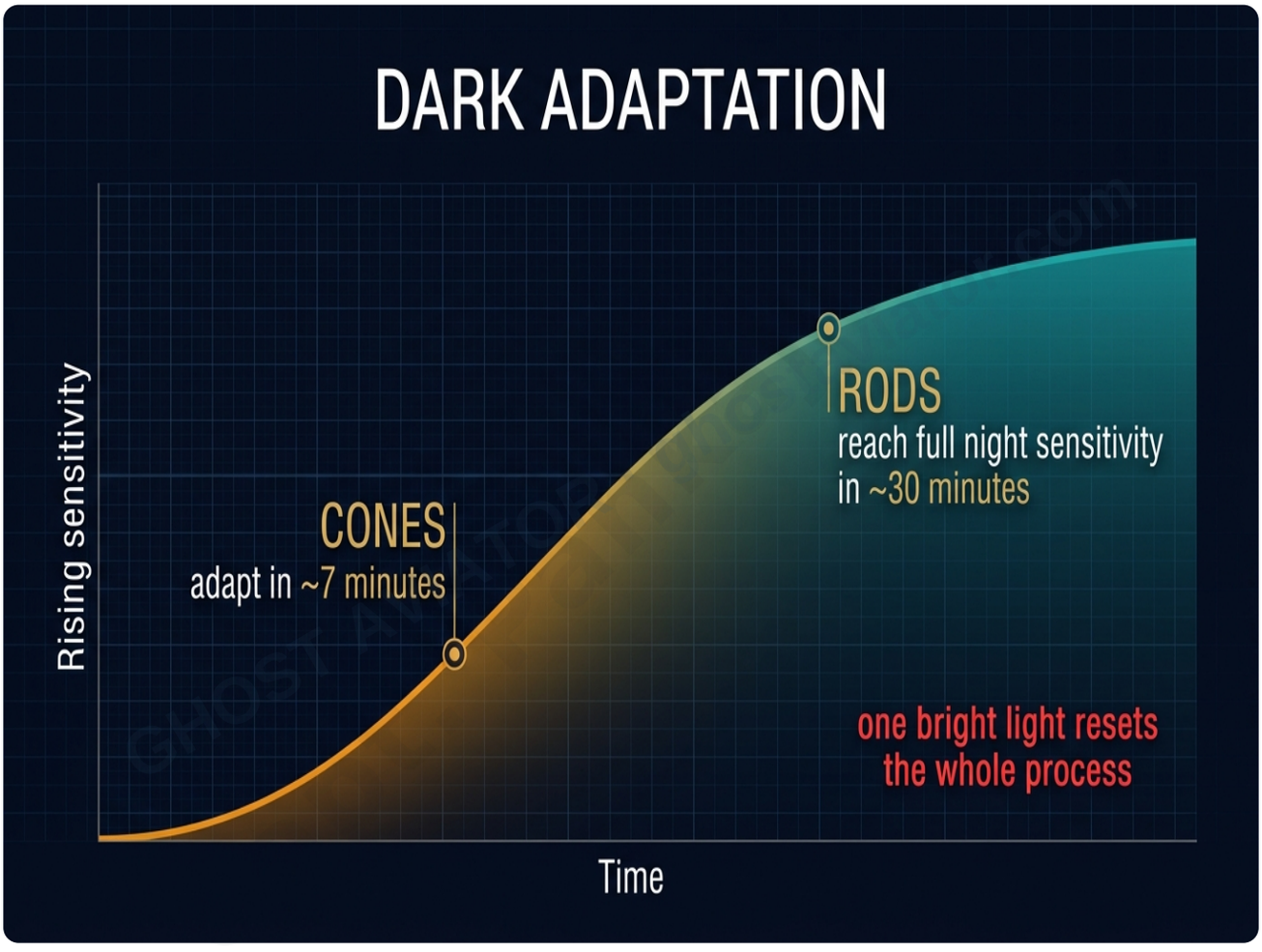

Figure 12.1 — Dark adaptation: cones adapt in ~7 minutes, but full rod (night) sensitivity takes ~30 minutes — and one bright light resets it.

31.2 Vision Under Dim & Bright Illumination

DGCA-QUOTED

Under conditions of **dim illumination**, **small print and colors on aeronautical charts and aircraft instruments become unreadable unless adequate cockpit lighting is available**. Moreover, another aircraft must be much closer to be seen **unless its navigation lights are on**.

In darkness, vision becomes more sensitive to light, a process called **dark adaptation**. Although exposure to **total darkness for at least 30 minutes is required for complete dark adaptation**, a pilot can achieve a **moderate degree of dark adaptation within 20 minutes under dim red cockpit lighting**.

WHY RED LIGHT — AND WHEN NOT TO USE IT (DGCA-QUOTED)

Since red light severely distorts colors, especially on aeronautical charts, and can cause serious difficulty in focusing the eyes on objects inside the aircraft, its use is advisable ONLY where optimum outside night vision capability is necessary.

Even so, white cockpit lighting must be available when needed for map and instrument reading, especially under IFR conditions.

DGCA-QUOTED — DARK-ADAPTATION KILLERS

Dark adaptation is impaired by:

- **Exposure to cabin pressure altitudes above 5,000 feet**,
- **Carbon monoxide inhaled in smoking and from exhaust fumes**,
- **Deficiency of Vitamin A in the diet**, and
- **By prolonged exposure to bright sunlight**.

DGCA-QUOTED — THE "CLOSE ONE EYE" PRESERVATION TRICK

Since any degree of dark adaptation is lost within a few seconds of viewing a bright light, a pilot should close one eye when using a light to preserve some degree of night vision.

31.3 Off-Centre Viewing — "Look 15–20° to the Side"

DGCA-QUOTED — THE NIGHT-VISION TECHNIQUE

Look to the side (`15 – 20 deg`) of the object.

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WHY OFF-CENTRE VIEWING WORKS AT NIGHT

The fovea has **no rods — and therefore no night vision**. When you stare *directly* at a dim object at night, its image falls on the fovea, where there are zero rods → you cannot see it. By looking **15–20° to the side**, you place the image on the **rod-rich peripheral retina** where night vision is sharpest.

Counterintuitive but vital: at night, the way to see something is NOT to look at it. This is taught in night-VFR training as "**off-centre / scan viewing**".

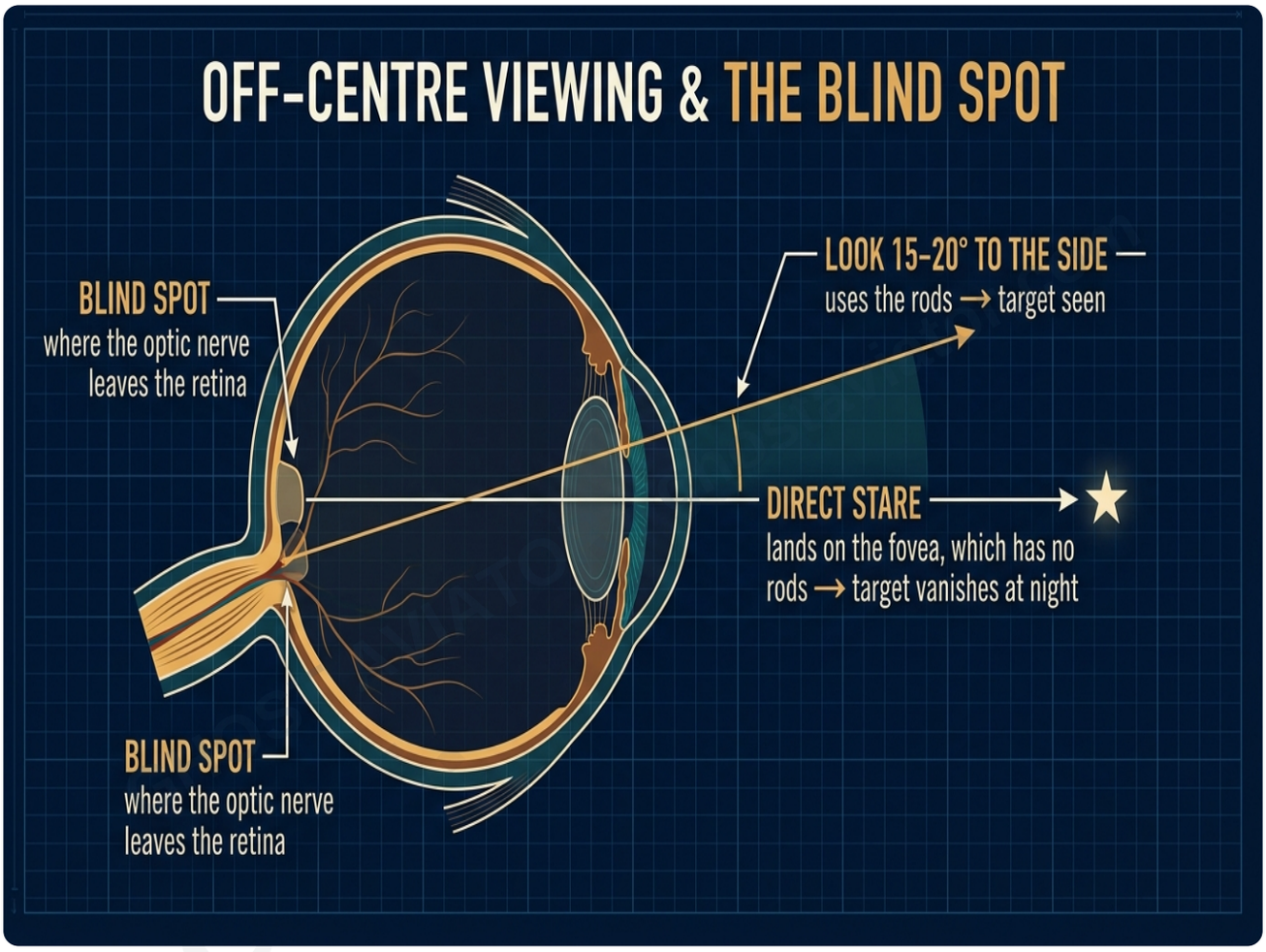

Figure 12.2 — Off-centre viewing: at night look 15–20° to the side, because a direct stare lands the target on the rod-free fovea.

31.4 The Blind Spot

DGCA-QUOTED

The **blind spot is point on the retina where the optic nerve enters the eyeball**. Here the retina has **no covering of light-detecting cells**.

WHY THIS MATTERS FOR SEE-AND-AVOID

If the eye remains **looking straight ahead**, it is possible for a **closing aircraft to remain in the blind spot until a very short time before impact**. To lessen the danger of collision, pilots are taught to **carry out a SYSTEMATIC LOOK OUT at all times**.

With both eyes open, the blind spot of one eye is covered by the other eye. But **be aware of obstructions to your visual field** such as passengers or canopy structures.

31.5 Empty Visual Field (Empty Field Myopia)

DGCA-QUOTED

In the **absence of anything to focus on** (that is when your visual field is empty), the **natural focus point of the eye is, on average, at a distance of between** `1 and 2 meters in front of the eye`.

Pilots should **minimize the risks associated with empty visual field by periodically and deliberately focusing on objects, both close and at a distance**.

THE DANGER — "EMPTY FIELD MYOPIA"

At high altitude with a clear blue sky and no clouds, your eye relaxes to its **resting focal length of 1-2 m** — well inside the cockpit. A distant aircraft becomes **blurry and easy to miss**. Mitigation: deliberately re-focus the eyes by looking at the wingtip, then at the horizon, then at a distant ground feature. This pumps the ciliary muscles and breaks empty-field myopia.

31.6 Damage to the Visual System — UV at Altitude

DGCA-QUOTED

Very high light occurs at altitude. At altitude, light contains **more of the high energy blue and ultraviolet wavelengths than is experienced at sea level**. Over a long period, such light can cause **cumulative damage to the retina and lens of the eye**. However, **most harmful wavelengths are filtered out by the cockpit windows**.

31.7 Vibrations & Blurred Vision

DGCA-QUOTED

Vibrations can cause blurred vision. This is due to TUNED RESONANCE oscillations of the eyeballs.

*(Recall §25 — the eyeball's natural resonance is **30–40 Hz**. Aircraft vibration that hits that band → vision-blur. This is one of the most direct cross-links between Part 6 and Part 7.)*

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§ 32 Protection of the Visual System — Sunglasses

DGCA-QUOTED — FLASH-BLINDNESS PROTECTION IN THUNDERSTORMS

When flying through a thunderstorm with lightning you can protect yourself from FLASH BLINDNESS by:

- **Turning up the intensity of cockpit lights**,
- **Looking inside the cockpit**,
- **Wearing sunglasses**, and
- **Using face blinds or face curtains when installed**.

DGCA-QUOTED — WHAT GOOD SUNGLASSES MUST DO

The requirement of good sunglasses is to:

- **Absorb at least `85 %` of visible light to eliminate glare without decreasing visual acuity**,
- **Absorb UV and IR radiation**,
- **Absorb all colors equally**.

Make sure you avoid using cheap sunglasses. Light sensitive lenses (Photo chromatic) are also generally forbidden for use in flight.

DGCA-QUOTED — SUNGLASSES CHARACTERISTICS (THE "SIX" LIST)

Sunglasses should have the following characteristics:

1. **Be impact resistant**.
2. **Have thin frames** (minimum visual obstruction).
3. **Be coated with poly carbonate for strength**.
4. **Be of good optical quality**.
5. **Have a luminescence transmittance of `10 – 15 %`**.
6. **Possess appropriate filtration characteristics**.

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SUNGLASS LIGHT ABSORPTION

≥ 85 %

SUNGLASS LUMINESCENCE TRANSMITTANCE

10 – 15 %

PHOTOCHROMATIC LENSES

Forbidden in flight

FRAME COATING FOR STRENGTH

Polycarbonate

§ 33

VISUAL DEFECTS

DGCA-QUOTED OPENING

The most common visual defects are caused by the distorted shape of the eyeball.

33.1 The Four Refractive Defects — Myopia, Hypermetropia, Presbyopia, Astigmatism

MYOPIA (Short-sightedness)

Myopia is more commonly known as **short-sightedness**. In a myopia eye, the **eyeball is LONGER than normal** causing the **image to fall in FRONT of the retina**.

Correction: A **CONCAVE lens** will correct Myopia by **bending the light from distant objects OUTWARDS before it hits the cornea**.

Normal pilot distance vision: "may be very approximately assessed as the ability to **read a car number plate at** `40 meters` ".

HYPERMETROPIA (Long-sightedness)

Hypermetropia is also known as **long-sightedness**, because **only objects at a distance can be seen clearly**.

Correction: A **CONVEX lens** will overcome Hypermetropia by **bending the light rays from near objects INWARDS before they meet the cornea**.

(Eyeball is shorter than normal — image would form behind the retina. Convex lens converges the rays earlier.)

PRESBYOPIA

Presbyopia is the **inability of the lens to change its shape to accommodate adequately**, to focus as an image from a near object onto the retina.

This condition normally arises in people **between the ages of** `40 and 50` . It is a **form of long-sightedness** and is **corrected using a CONVEX lens**.

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(This is why pilots at middle age often need bi-focals — see §29.5.)

ASTIGMATISM

Astigmatism is caused by a **misshapen or oblong cornea**. For a person with astigmatism **objects will appear irregularly shaped**.

(Corrected with a cylindrical lens that compensates for the corneal irregularity.)

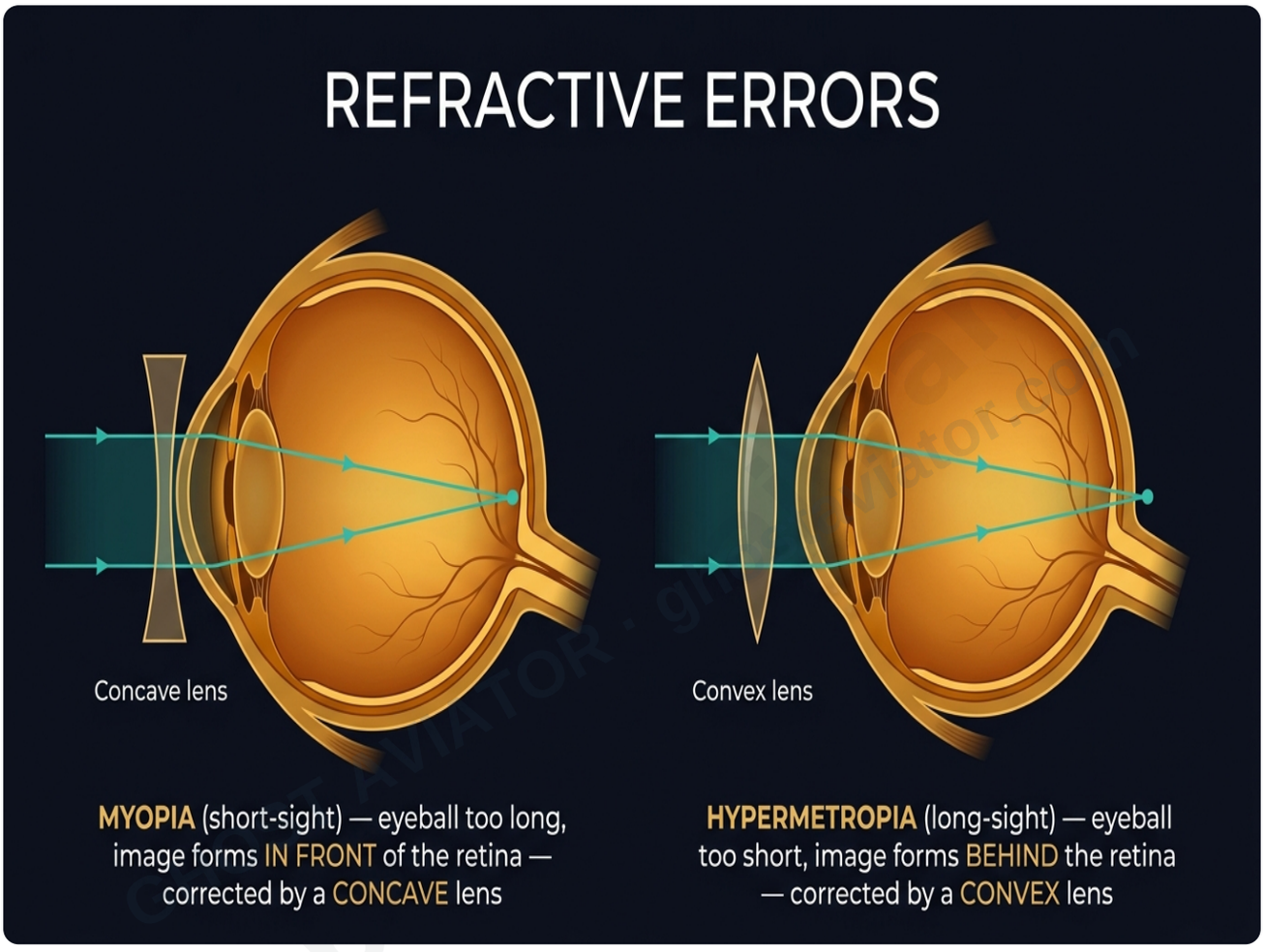

Figure 12.3 — Refractive errors: myopia (eyeball too long, corrected by a concave lens) and hypermetropia (too short, convex lens).

Refractive Defects — Memory Table			
Defect	**Common Name**	**Mechanism**	**Correction**
Myopia	Short-sightedness	Eyeball **longer** → image in front of retina	**Concave lens**
Hypermetropia	Long-sightedness	Eyeball shorter → image behind retina (only distance clear)	**Convex lens**
Presbyopia	Age-related (40–50 yrs)	Lens can't change shape — accommodation fails	**Convex lens** (often bi-focal)
Astigmatism	—	Misshapen / oblong cornea — objects irregular	Cylindrical lens

33.2 Wearing of Corrective Spectacles by Pilots

DGCA-QUOTED RULE

Pilots who wear corrective spectacles or contact lenses, for whatever reason, must carry a SPARE PAIR at all times when they are exercising the privileges of their license.

33.3 Glaucoma

GLAUCOMA — DGCA-QUOTED (THE MOST AVIATION-MEDICALLY IMPORTANT VISUAL DEFECT)

Glaucoma is characterized by:

- **Progressive narrowing of the visual field**,
- **Insidious onset and concealed progression**,
- **An increase in intra-ocular pressure**.

Glaucoma can lead to total blindness and undetected reduction of the visual field. It reduces visual acuity in its final stage.

WHY GLAUCOMA IS SCREENED AT EVERY PILOT MEDICAL

The word "**insidious**" + "**concealed**" + "**undetected reduction of visual field**" is what makes glaucoma so dangerous to aviation. A pilot may be losing peripheral vision over months or years and not realise it — until a critical traffic conflict or runway-incursion-detection failure exposes it. **Intra-ocular pressure (IOP) measurement (tonometry) is part of every routine Class-1 / Class-2 DGCA medical** precisely to catch glaucoma early.

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§ 34 Colour Vision

DGCA-QUOTED

Good color vision is essential for pilots because of the use of color associated with the items listed below.

Why pilots need good colour vision — DGCA-listed items

#	Item	Operational use of colour
1	**Navigation light of aircraft**	Red (left wingtip) · Green (right) · White (tail) — defines aircraft direction at night.
2	**Runways and airfields**	White runway edge lights · red runway end · green threshold · PAPI (white/red) glide-path.
3	**Ground obstructions**	Red obstruction lights on towers · obstacle markings.
4	**Cockpit displays and instruments**	Red = warning · amber = caution · green = normal · cyan = sky · brown = ground (PFD).
5	**Maps and charts**	VFR/IFR charts use colour-coded airspace, terrain, MSA values.
6	**Emergency flares**	Distress flares — red = distress · white = signalling.
7	**Light signals**	ATC light-signal codes (steady green = clear to land · red flashing = airport unsafe, etc.)

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§ 35 Colour Blindness

Figure 12.4 — Colour blindness is screened with Ishihara plates: a number hidden in coloured dots that a colour-deficient eye cannot read.

DGCA-QUOTED DEFINITION

Color blindness or, more accurately, color-defective vision, is caused by a defect in the structure of the color-sensitive cones in the retina.

WHY THIS CONNECTS DIRECTLY TO §29.2

Recall from Part 7 that **cones are the colour-sensitive photoreceptors** packed into the fovea — about **6 million** of them. If their *structure* is defective (most commonly a missing or weak red, green, or blue pigment), the brain cannot distinguish between certain colours. This is most commonly genetic (sex-linked recessive — far more common in males) and is non-treatable.

DGCA-MEDICAL IMPLICATION

Colour-defective vision (red-green most commonly) is screened at every DGCA Class-1 / Class-2 medical using the **Ishihara colour plates**. Significant deficiency can disqualify a candidate, or restrict the licence to "day-VFR only" because of the night colour-signal interpretation problem (nav lights, PAPI, ATC light signals all rely on red/green discrimination).

§ 36 Vision & Speed — Reaction Times

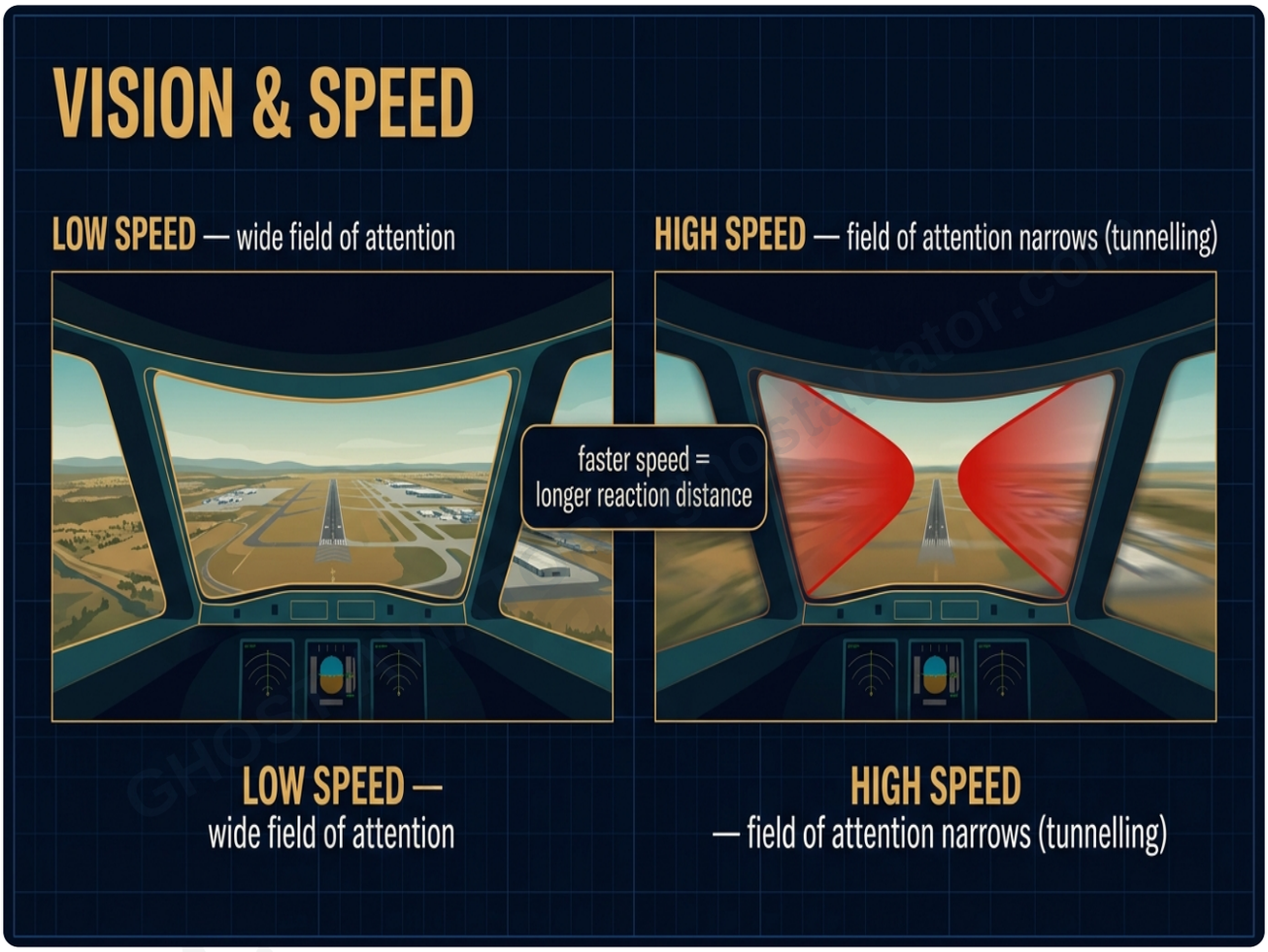

Figure 12.5 — Vision and speed: as speed rises the field of attention narrows and reaction distance grows.

DGCA-QUOTED

Reaction time depends on the closing relative speed of two aircraft. If one of the aircraft were to be a fast jet, the closing speed would be much higher.

WHY THIS MATTERS OPERATIONALLY — SEE-AND-AVOID LIMITATIONS

At a closing speed of **1,000 kt** (two fighters head-on, each at 500 kt), aircraft cover roughly **0.5 nm per second**. The human eye + brain typically needs:

- **~ 0.1 s** to detect a moving target
- **~ 1 s** to recognise it as another aircraft
- **~ 1 s** to decide on an evasive action
- **~ 4 s** for the controls + airframe to respond

That's **≈ 6 seconds** from "first photon hitting the retina" to "aircraft actually moving out of the way" — at 1,000 kt closing, the other aircraft has already covered **~ 3 nm**. The pilot must therefore detect the conflict at a range that allows for these reaction-time bites; this is why **systematic scan**, **traffic alerts**, and **TCAS** exist.

✦ END OF CHAPTER 12 ✦

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HUMAN PERFORMANCE & LIMITATIONS · MODULE E — THE SENSES

The Ear & Balance

Chapter 13 — The organ that both hears and keeps you upright: the three sections of the ear, the Eustachian tube and barotrauma, noise and hearing loss, and the vestibular balance system that illusions will later fool.

BOOK
Human Performance & Limitations

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CHAPTER
13 of 26 · Module E

In this chapter

1. The Ear — Two Functions, Three Sections
2. The Middle & Inner Ear · The Eustachian Tube
3. Audible Range of the Human Ear
4. Noise & Hearing Loss
5. Protection of Hearing
6. The Ear and Balance · Orientation
7. The Balance Mechanism

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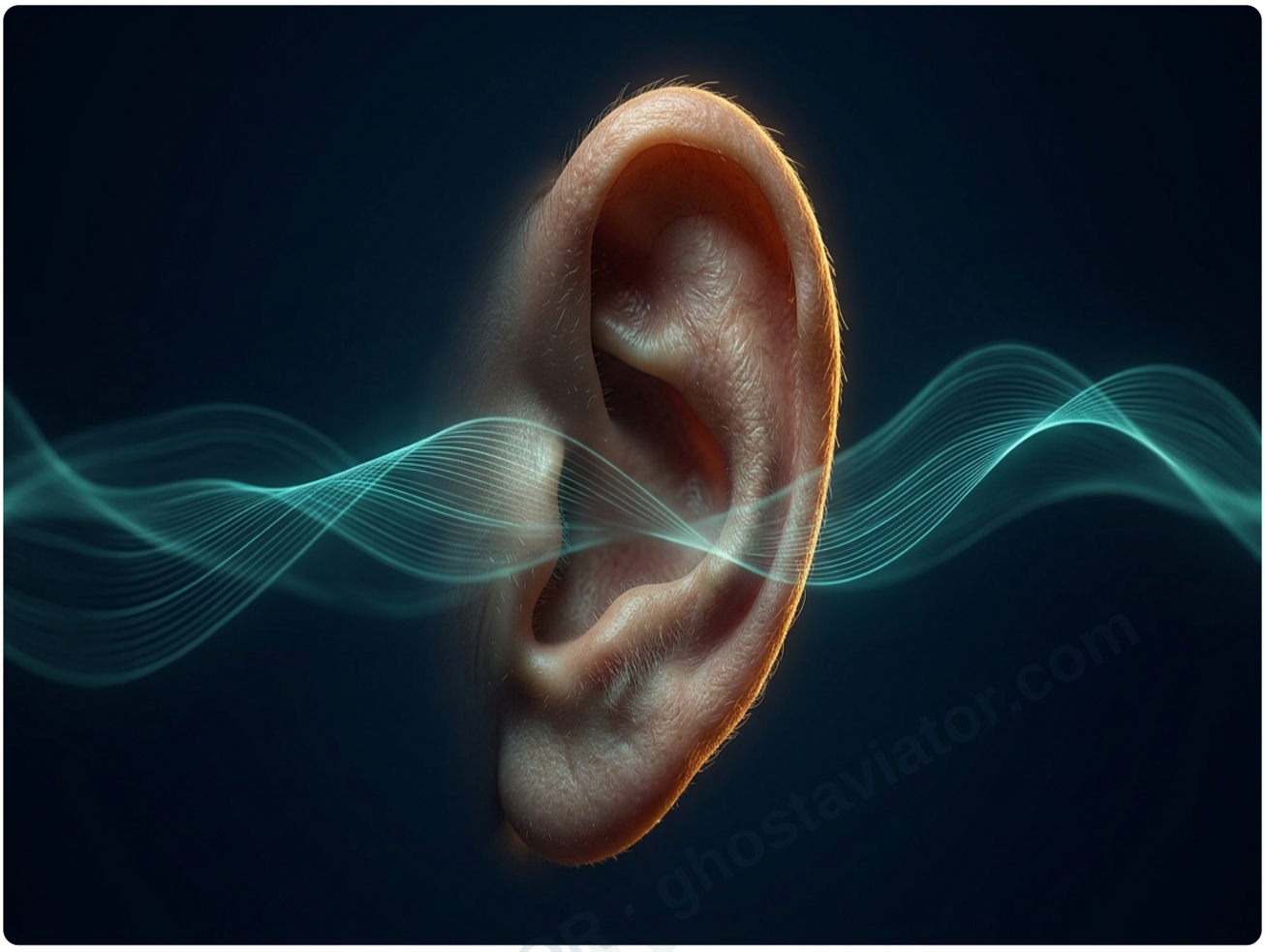

Plate 13.0 — Two instruments in one: the ear both hears the world and tells the brain which way is up.

§ 37 THE EAR — Two Functions, Three Sections

DGCA-QUOTED — THE TWO FUNCTIONS OF THE EAR

The ear performs two quite separate functions:

- **To receive vibrations or sound waves in the air which it transmits to the brain.**
- **To act as a balance organ and acceleration director.**

WHY THIS DISTINCTION IS CRITICAL

Most students think of the ear as the "hearing organ" only. The second function — **balance** — is arguably more important for pilots, because **spatial disorientation kills pilots, hearing loss usually doesn't**. The vestibular apparatus inside the inner ear (covered in §42–43) is what tells you which way is up — when it lies to you, you crash.

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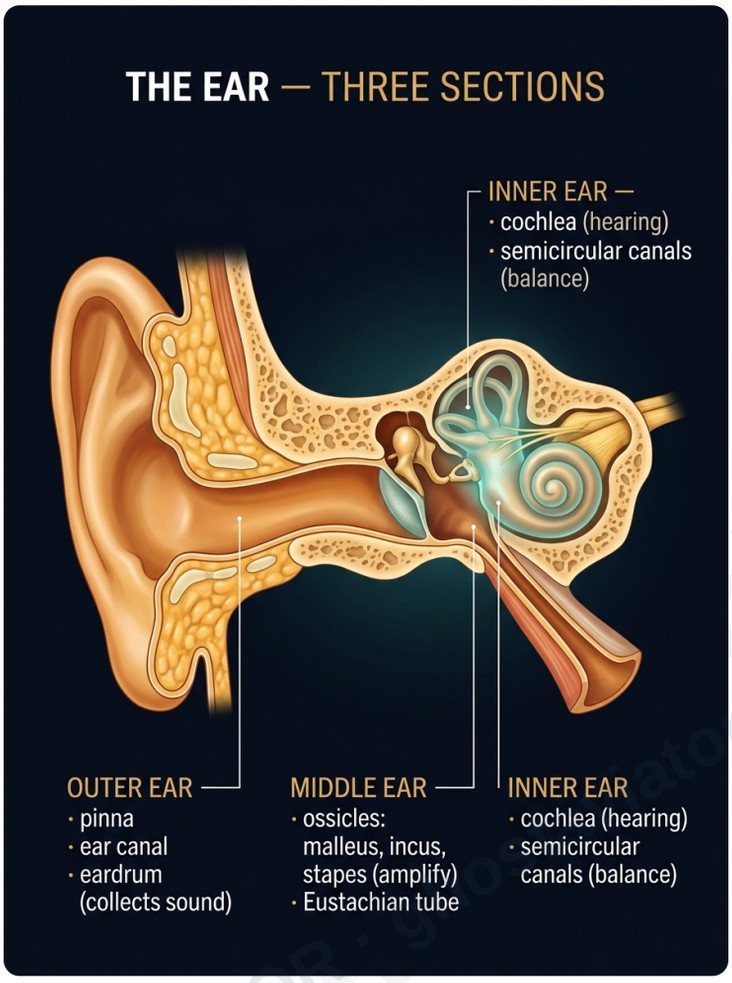

Figure 13.1 — The ear's three sections: outer (collects sound), middle (ossicles amplify), inner (cochlea hears, semicircular canals sense balance).

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§ 38

The Middle and Inner Ear · The Eustachian Tube

DGCA-QUOTED

The Tympanum and the Ossicles transmit sound waves to the inner ear.

The **Tympanum** is the eardrum. The **Ossicles** are the three smallest bones in the body — **Malleus, Incus, Stapes** (Hammer, Anvil, Stirrup). They form a tiny amplifying lever system that converts air pressure waves at the eardrum into mechanical motion at the oval window of the cochlea.

38.1 The Eustachian Tube — pressure equalisation

DGCA-QUOTED

The **Eustachian tube allows pressure in the middle ear to equalize across the ear drum with outside or ambient pressure when climbing or descending**.

Website YouTube Instagram
Telegram channel Telegram community

DGCA-QUOTED — THE NO-FLY RULE

No one should fly if their Eustachian tube is blocked, and they cannot "clear" their ears.

38.2 Effects of Altitude Change — Clearing the Ears

DGCA-QUOTED — DESCENT IS THE DANGEROUS PHASE

It is during DESCENT when difficulty in clearing the ears is most likely to be experienced.

WHY DESCENT — THE ONE-WAY-VALVE EFFECT

On **climb**, ambient pressure falls; the higher-pressure middle-ear air *pushes outward* through the Eustachian tube — this happens easily and automatically. On **descent**, ambient pressure rises; the now-relatively-lower-pressure middle-ear air must *draw in* through the Eustachian tube against gravity and tissue swelling. The tube tends to act like a one-way flap valve. If congested (cold, flu, sinusitis), it fails to open and the eardrum is pushed inward by external pressure — **Otis Barotrauma**.

38.3 Otis Barotrauma

DGCA-QUOTED DEFINITION

Otis Barotrauma — Stretching of the ear drum caused by the expansion and contraction of gases trapped in the inner ear by a blocked Eustachian tube.

DGCA-QUOTED CLEARING TECHNIQUES + "STEP DESCENT"

If you experience problems with pressure equalization during descent:

1. **Swallow deliberately** with the nostrils pinched closed,
2. **Yawn**,
3. **or BLOW DOWN THE NOSE, again with the nostrils pinched closed**. *(This is the Valsalva manoeuvre.)*

If the problem is not resolved, **the rate of descent should be decreased or stopped at intervals to allow more time for pressure to equalize. This is also known as STEP DESCENT.**

Pilot may resort to even CLIMB if pressure persists. Re-climbing reduces external pressure and helps un-block the tube.

Remember, never fly if your Eustachian tube becomes swollen or blocked and you cannot clear your ears.

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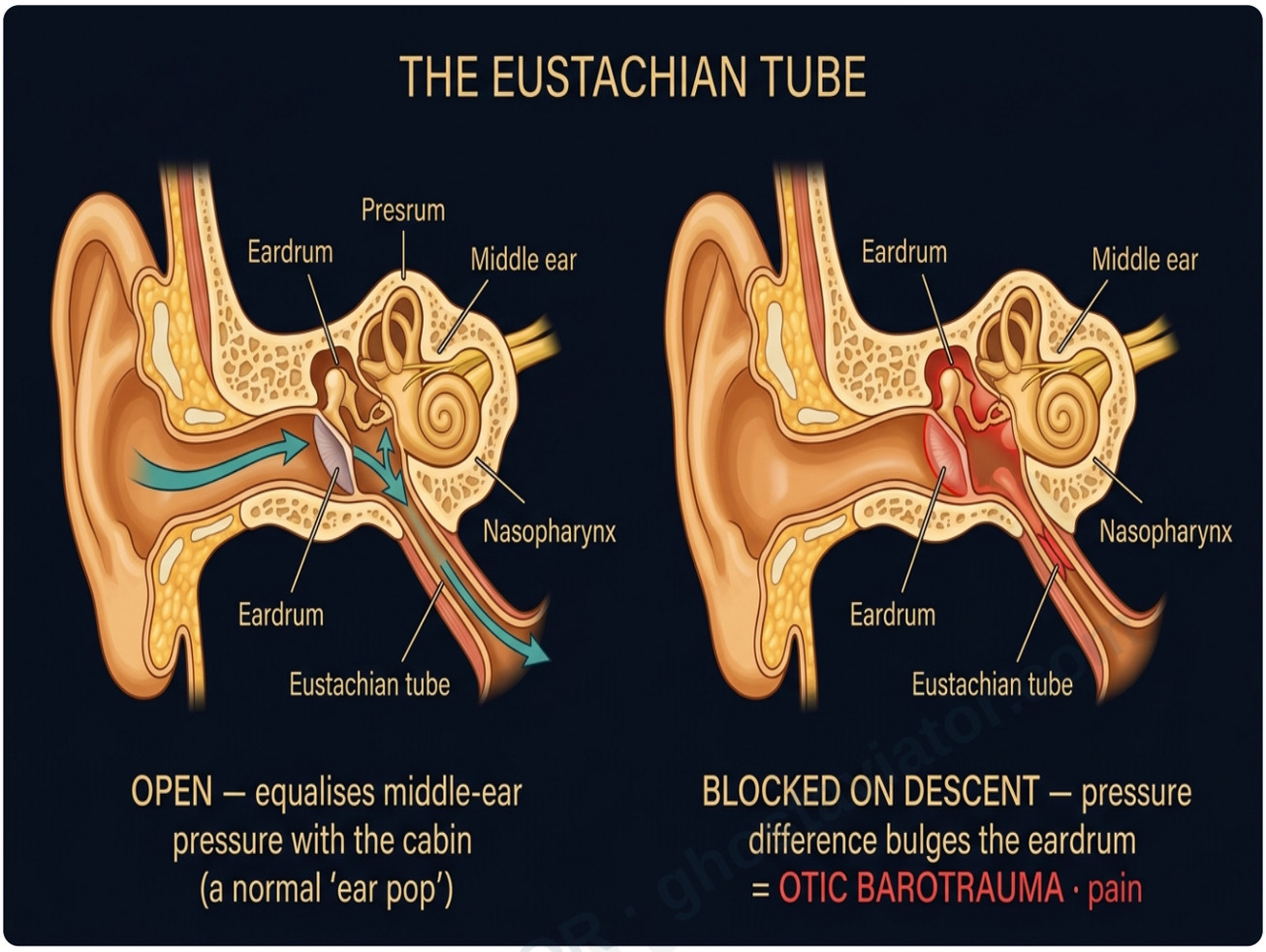

Figure 13.2 — The Eustachian tube equalises middle-ear pressure with the cabin; a blocked tube on descent causes otic barotrauma.

§ 39 Audible Range of the Human Ear

DGCA-QUOTED — SOUND HAS THREE MAIN QUALITIES

- **Pitch** — which depends on the **frequency** of the vibration.
- **Loudness or Intensity** — which depends on the **amplitude** of the vibrations.
- **Quality** — which can be either **harmonious or just plain noisy**.

DGCA-QUOTED — FREQUENCY RANGE

The range of **pitch or frequency of sounds that a fit young person can hear lies between** `16 and 20,000 Hertz`, or cycles per second. **Detectable sound range also depends on loudness.**

16Hz Lower limit of audible frequency
20,000Hz Upper limit (fit young person)
3qualities Pitch · Loudness · Quality

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§ 40

Noise & Hearing Loss

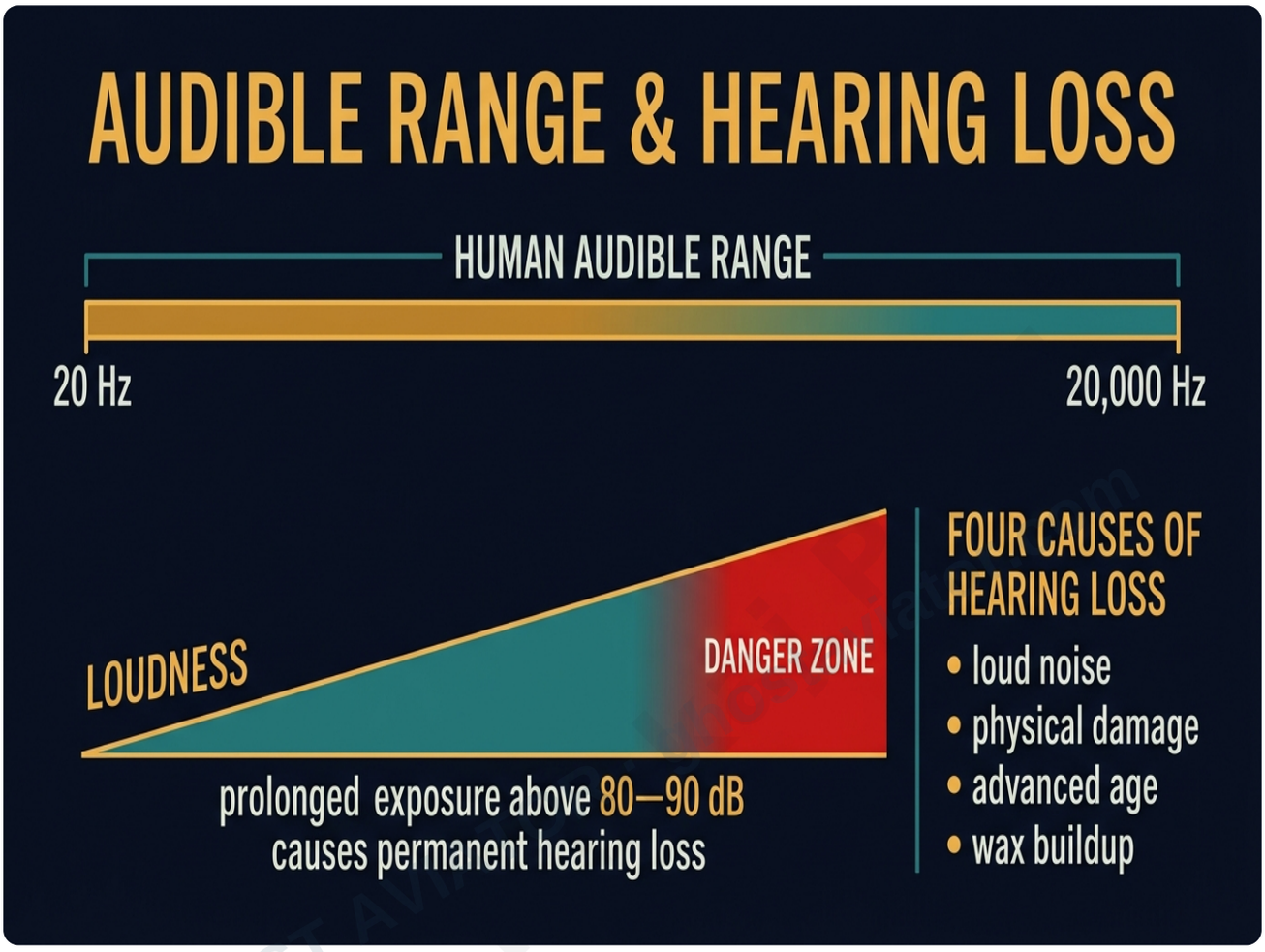

Figure 13.4 — The audible range (20 Hz–20,000 Hz) and the noise thresholds: prolonged exposure above 80–90 dB causes permanent hearing loss.

DGCA-QUOTED — FOUR CAUSES OF HEARING IMPAIRMENT

Hearing impairment can arise because of:

1. **Exposure to loud noise.**
2. **Physical damage to the hearing mechanism.**
3. **Advanced age.**
4. **A buildup of wax.**

40.1 Conductive Deafness

DGCA-QUOTED

Conductive Deafness — is caused by damage to the ossicles or the ear drum.

Website YouTube Instagram
Telegram channel Telegram community

40.2 Noise-Induced Hearing Loss — the 80 / 90 dB thresholds

DGCA-QUOTED — EXACT DB THRESHOLDS

Noise Induced Hearing Loss:

- **>** `80 dB` — **task performance may be impaired**.
- **>** `90 dB` — **measurable impairment of task performance**.

DGCA-QUOTED — THE COUNTERINTUITIVE "AROUSAL" EFFECT

However, it has been shown that in some situations performance of VIGILANCE tasks can actually be BETTER in high noise levels than in low noise levels. This is because noise increases AROUSAL and can move the individual into the optimum performance area.

Any prolonged exposure to noise in excess of **90 dB** can end up in **noise-induced hearing loss**. This can **damage the very sensitive membrane in the cochlea**.

40.3 Presbycusis — Hearing loss with age

DGCA-QUOTED

Presbycusis — is the name given to the deterioration of hearing with advanced age. HIGH TONES are cut off FIRST.

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AUDIBLE RANGE YOUNG
16 – 20,000 Hz

PERFORMANCE IMPAIRED ABOVE
80 dB

MEASURABLE IMPAIRMENT ABOVE
90 dB

CONDUCTIVE DEAFNESS — SITE
Ossicles / eardrum

AGE-RELATED HEARING LOSS TERM
Presbycusis

PRESBYCUSIS — FIRST TO GO
High tones

NOISE DAMAGE SITE
Cochlear membrane

"OPTIMUM" ZONE OF NOISE-AROUSAL
~ 70 – 80 dB (implied)

§ 41 Protection of Hearing

DGCA-QUOTED

Noise induced hearing loss can be avoided or reduced to a minimum by wearing suitable EAR PROTECTORS.

Always protect your ears if you know you are going to be exposed to excessive noise. In the cockpit, use the best quality HEADSET you can afford in order to reduce background noise.

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PRACTICAL COCKPIT APPLICATION

- **Active Noise Reduction (ANR) headsets** — provide both passive (foam seal) and active (anti-phase electronic cancellation) noise attenuation. Reduce cockpit noise by 15–25 dB.
- **Maintenance crews on ramp** — must use over-the-ear muffs near running engines (105–130 dB at engine intakes).
- **Earplugs alone** — useful for passengers; usually inadequate for sustained cockpit duty.
- **Cumulative exposure** — even short bursts of 130+ dB (gunfire, jet blast) can cause permanent damage. Each pilot's hearing is a non-renewable career asset.

§ 42 The Ear and Balance · Orientation

DGCA-QUOTED DEFINITION

"Orientation" refers to a human being's ability to maintain equilibrium and to interpret the body's position in space. The ear also provides us with our sense of balance.

DGCA-QUOTED — VISION IS PRIMARY

The primary and most reliable sense of spatial orientation is EYESIGHT.

The balance sensors situated in the ear provide us with a **SECONDARY system**.

WHY THIS HIERARCHY MATTERS

This is the single most important fact in spatial-disorientation training. The eyes are primary; the ear's balance organ is secondary. When the two disagree — as happens in IMC (Instrument Meteorological Conditions) where the eyes lose external reference — **the pilot must trust the instruments (which calibrate to true vertical), not the inner ear (which lies)**. This is the foundational principle of IFR (Instrument Flight Rules).

§ 43 THE BALANCE MECHANISM

43.1 Proprioception

DGCA-QUOTED DEFINITION

Proprioception: is the **awareness of the body in space**. It is the use of **joint position sense and joint motion sense** to respond to stresses placed upon the body by alteration of posture and movement.

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43.2 The Vestibular Apparatus — Otoliths & Semi-Circular Canals

DGCA-QUOTED — MEMORISE THE COMPONENTS & FUNCTIONS

The **Vestibular apparatus (Otoliths + Semi-circular canals)** helps maintain spatial orientation.

- **The Otoliths detect LINEAR ACCELERATION.**
- **The Semi-Circular Canals detect ANGULAR ACCELERATION.**

OTOLITHS — linear acceleration

The otoliths are tiny **calcium-carbonate crystals embedded in a gelatinous membrane** sitting on top of hair cells in the utricle and saccule.

- **Sense linear acceleration**: forward/back (takeoff, braking), up/down (turbulence, climb/dive)
- Also sense **gravity at rest** (which way is "down")
- Can be tricked by sustained acceleration → "**somatogravic illusion**" (Part 9)

SEMI-CIRCULAR CANALS — angular acceleration

Three fluid-filled loops at right angles to each other (one per axis: roll, pitch, yaw). Each ends in an **ampulla** containing a cupula with hair cells.

- **Sense angular acceleration**: turning, pitching, rolling
- Only respond to **change** in rotation, not steady rotation
- Can be tricked by sustained turns → "**the leans**" / spiral dive illusion (Part 9)

43.3 The Three Orientation Factors — DGCA's "Big Three"

DGCA-QUOTED — THE THREE FACTORS

Human beings maintain spatial orientation using a combination of three factors:

1. **The sense of vision.**
2. **The Vestibular Apparatus.**
3. **The Somatosensory system** ("seat of the pants" feeling) / **G-Force**.

DGCA-QUOTED — RELIABILITY RANKING (MEMORISE VERBATIM)

- **The most reliable sense is the SENSE OF VISION.**
- Our **vestibular apparatus can detect accelerations but cannot determine what position we are in if no acceleration is present**.
- **The somatosensory system is NOT RELIABLE AT ALL.**
- **Neither is the ear's balancing mechanism sufficiently reliable for a pilot to maintain spatial orientation using this sense alone.**

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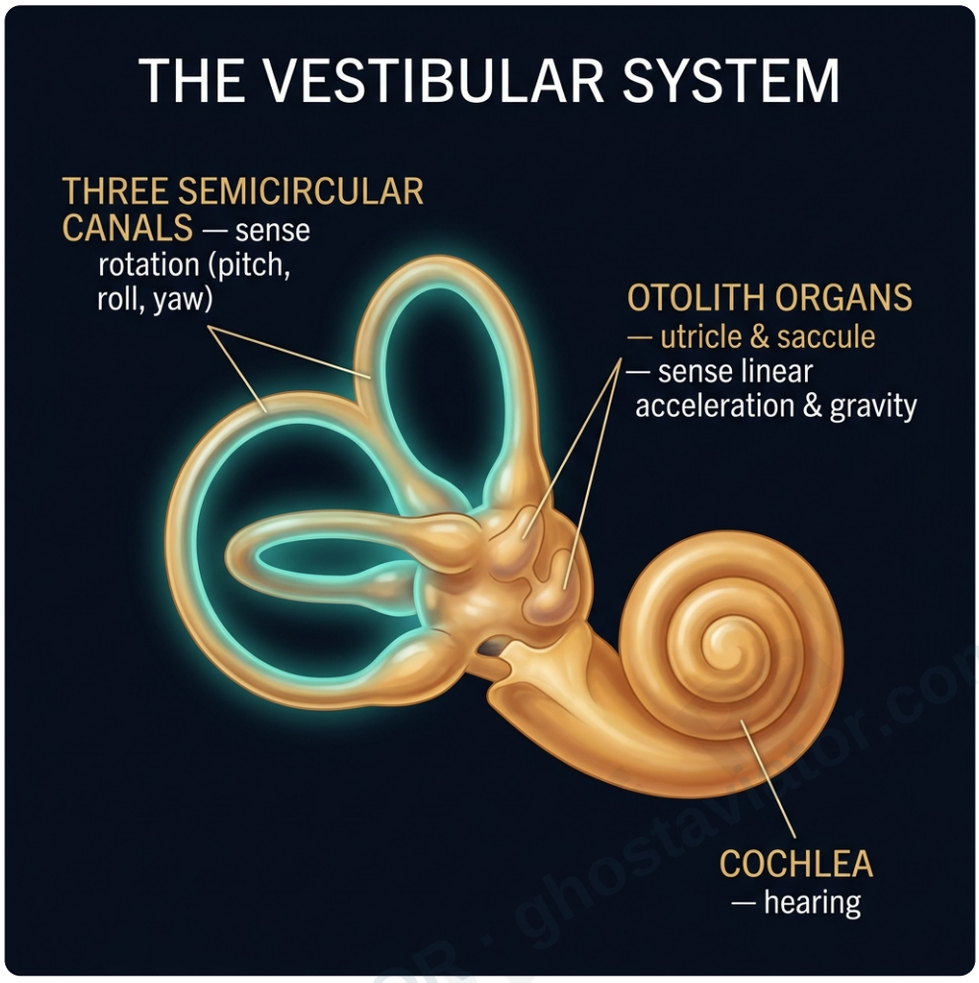

Figure 13.3 — The balance mechanism: the three semicircular canals and otolith organs sense rotation and acceleration.

43.4 How Angular Acceleration is Sensed — DGCA-quoted

THE FLUID-FLOW MECHANISM

- **Both the Otoliths Organs and the Semi-Circular Canals send signals to our brain by means of impulses arising from the body being subjected to accelerations.**
- **Fluid flow occurs when the body is subject to angular acceleration.**
- The flow takes place in the **OPPOSITE direction to the acceleration**, moving **sensory hairs** which send signals to the brain that the body is in motion.

THE CRUCIAL FAILURE MODE — DGCA-QUOTED

There is NO FLUID FLOW when the body is at rest or if linear or turning movement is taking place at a STEADY SPEED. In these situations, **the vestibular apparatus alone CANNOT DETECT MOTION**.

Only the eyes and instruments tell a pilot that he is in a steady turn.

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WHY THIS IS THE FOUNDATION OF "THE LEANS"

In a steady-rate turn, the canal fluid catches up with the canal walls and stops flowing. The sensory hairs return to centre — the brain thinks **"no turn is taking place"**. If the pilot then rolls out, the fluid *now* flows (in the opposite sense to the roll-out) → the brain interprets this as *a turn in the opposite direction*. The pilot, feeling that "wing-low" sensation, tends to re-roll back into the original turn = **graveyard spiral**. The cure: trust the attitude indicator, not the inner ear. (Detailed Part 9 — illusions.)

✦ END OF CHAPTER 13 ✦

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HUMAN PERFORMANCE & LIMITATIONS · MODULE F — THE DECEIVED PILOT

Illusions & Spatial Disorientation

Chapter 14 — When the senses lie: the taxi, take-off, cruise and approach illusions, the somatogravic and elevator illusions, the leans, false horizons, flicker vertigo — and the one rule that overrides them all: trust your instruments.

BOOK
Human Performance & Limitations

AUTHOR
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DGCA CPL / ATPL — HPL

CHAPTER
14 of 26 · Module F

In this chapter

1. Conflicts Between the Ears & the Eyes
2. Illusions When Taxiing
3. Illusions on Take-off — Somatogravic
4. Illusions in the Cruise — Autokinetic
5. Approach & Landing — Black Hole & Runway Illusions
6. Disorientation / Vertigo — the Leans & the Master Rule
7. False Horizon, Vection & Flicker Vertigo

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Plate 14.0 — In cloud and darkness the body becomes a liar. The only truth is on the instrument panel.

§ 44

Conflicts Between Ears & Eyes — Illusions & Disorientation / Vertigo

DGCA-QUOTED

Various complex motions and forces and certain visual scenes encountered in flight can create ILLUSIONS of motion and position.

Spatial disorientation from these illusions can be prevented ONLY by visual reference to RELIABLE, FIXED POINTS on the ground or to FLIGHT INSTRUMENTS.

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§ 45 ILLUSIONS — Introduction

DGCA-QUOTED DEFINITION

In aviation any mismatch between what we sense and what we expect is an illusion.

Because of the lack of stable visual references and the erroneous mental models that may be produced, the **pilot is at a disadvantage**.

DGCA-QUOTED — UNIVERSALITY AND DANGER

- **Illusions may occur during all stages of the flight, and to pilots of every experience and skill level.**
- The pilot, therefore, should be **aware of the possibility of misinterpreting the information received**.
- **Visual illusions are particularly dangerous in aviation, as we normally consider our visual input to be the most reliable of our senses.**

45.1 Atmospheric Perspective

DGCA-QUOTED

The pilot who has flown mostly in relatively polluted air may use "atmospheric perspective" as a cue to range. If he then flies in a very clear atmosphere he may believe distant objects, because of their clarity, to be much closer than they actually are.

PRACTICAL EXAMPLE — FLYING OVER THE DESERT / OVER THE OCEAN / ABOVE AN INVERSION

A pilot accustomed to hazy/polluted urban airspace develops a mental ruler: "*if the runway looks sharp, it must be close*". On a crystal-clear day above the desert or over an ocean inversion, that same runway looks "close" at 5+ miles — the pilot starts the descent too early, undershoots, or arrives high & fast at the threshold. The illusion runs the other way too: in a polluted layer, the runway looks "farther" → late descent, dive at the runway.

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45.2 Laws of Perceptual Organisation — Gestalt Theory

DGCA-QUOTED

The **"Laws of perceptual organization of Gestalt Theory"** deal with factors such as:

- **Proximity**,
- **Continuity**,
- **Similarity**,
- **Symmetry**,
- **Simplicity**, and
- **Closure**.

Gestalt laws formulate basic principles governing **how objects are organised and perceived**.

WHY GESTALT MATTERS IN COCKPIT DISPLAY DESIGN

The reason a glass cockpit groups *related* data (engine N1/EGT/Fuel-flow on one column; flight-attitude/heading/altitude as a single PFD) is **Gestalt proximity & similarity**. The pilot's brain naturally fuses these into one "engine block" or one "primary flight" gestalt — reducing scan workload. Misdesigned displays violate Gestalt principles and slow the pilot's interpretation under stress.

§ 46 Illusions When Taxiing — Relative Movement

DGCA-QUOTED GENERAL WARNING

We must use extreme caution to ensure that we do not construct our mental model according to our wishes or desires.

DGCA-QUOTED — TAXI ILLUSION

When taxiing into a headwind the blowing snow will give the illusion that the aircraft is taxiing faster than it actually is.

COCKPIT CROSS-CHECK

The fix is simple — verify against the **ground-speed display on the FMS/GPS**, the **brake-pedal feel**, or by looking at **fixed reference markers** (runway centreline, taxiway edge lights). Never rely on the relative motion of mobile elements (snow, rain droplets, dust) as a speed reference.

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§ 47 Illusions on Take-off — Somatogravic Illusion

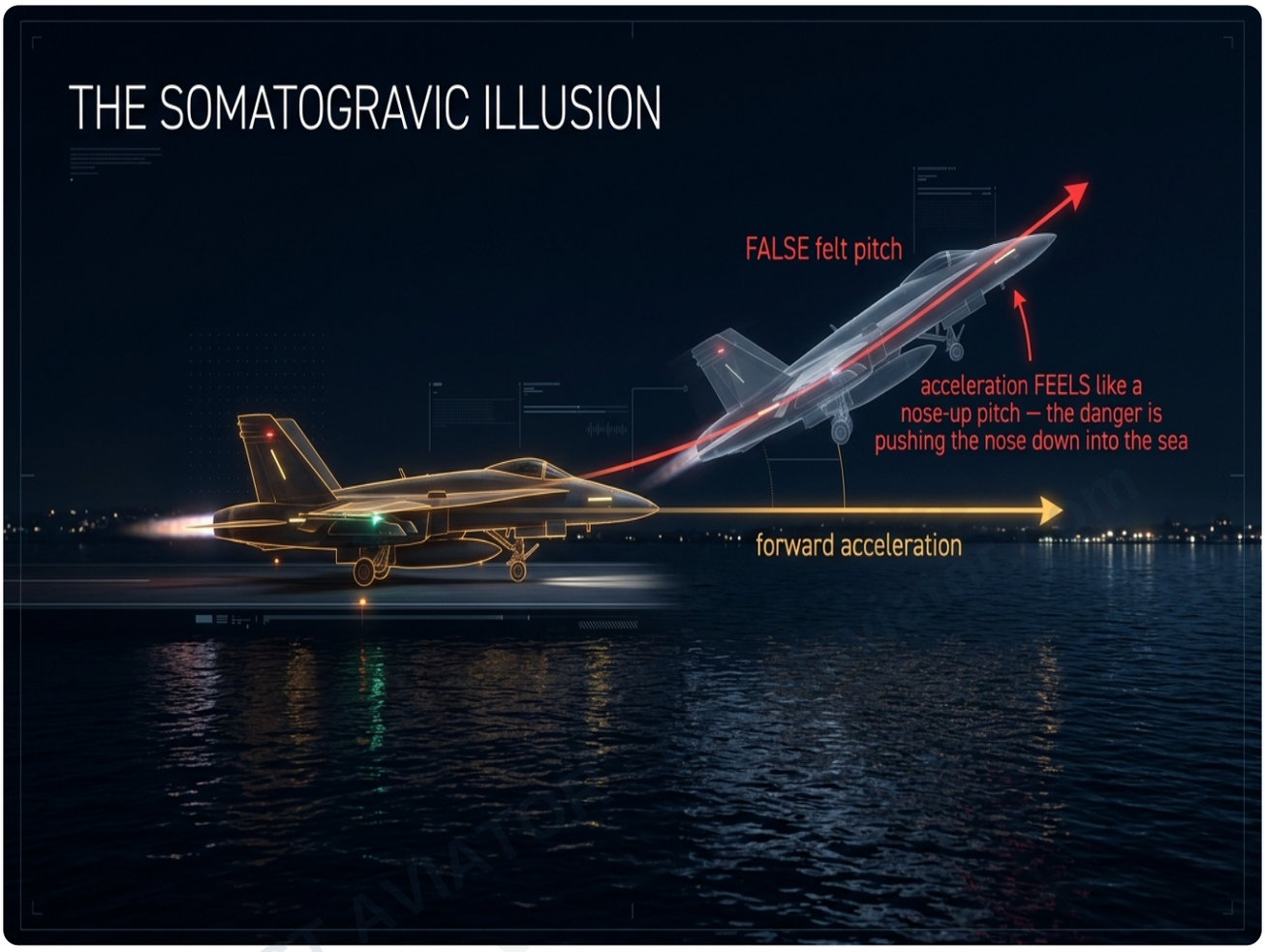

Figure 14.1 — The somatogravic illusion: forward acceleration on take-off feels like a nose-up pitch, tempting a dangerous push forward.

DGCA-QUOTED

Acceleration gives the pilot an impression of the nose of the aircraft pitching UP.

WHY — THE OTOLITH PHYSICS

The **otoliths** (Part 8 §43.2) sense linear acceleration. During take-off, strong forward acceleration deflects the otolith hair cells in the same direction *that gravity would deflect them if the head were tilted back*. The brain cannot tell these apart — so it interprets forward acceleration as **nose-up pitch**.

The reflex response of the disoriented pilot is to **push the nose down** — into the ground. This is the classic **Somatogravic Illusion** killer on night/IMC departures, and is the principle behind a string of fatal accidents on dark-night take-offs over water.

§ 48

Outside References — Six False-Impression Triggers

DGCA-QUOTED — VERBATIM LIST

Outside references may give a false impression within the cockpit:

1. **Immediately after take-off.**
2. **Over water.**
3. **In hilly terrain.**
4. **Gently sloping terrain.**
5. **A bank of sloping cloud.**
6. **The ground sloping down on the approach.**

WHY THESE SIX ARE DANGEROUS — COMMON THREAD

Each of these scenarios **removes or distorts a clear, unambiguous horizon**. The pilot's brain — which is wired to use the horizon as its vertical reference — substitutes **whatever long, straight, dark-light boundary it can find** (a coastline, a hill ridge, a cloud bank top). If that boundary is sloped, the pilot subconsciously aligns the wings with *it* instead of with the true horizon — and ends up in a banked or pitched attitude without realising it.

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§ 49

Illusions in the Cruise — Auto-genesis & Vertical Separation

Figure 14.2 — The autokinetic effect: a single fixed light in the dark appears to move on its own if stared at.

49.1 Auto-genesis (the autokinetic effect)

DGCA-QUOTED

Auto genesis. Staring at an isolated and stationary light when other visual references are inadequate or absent, may cause auto-kinetic movements of the eyes.

In the dark, a static light will appear to move about when stared at for many seconds. The disoriented pilot will lose control of the aircraft in attempting to align it with the light.

Website
YouTube
Instagram
Telegram channel
Telegram community

PRACTICAL EXAMPLE & DEFENCE

A single distant aircraft light, a star, a ground beacon, or the planet Venus on a clear dark night — fixate on it for 6–12 seconds and it appears to drift, dance, or even rotate. The pilot may attempt to follow this "moving" target with control inputs. The defence: **do not stare**. Scan your visual field continuously, with brief stops on any reference. If a single light source *seems* to move, glance away briefly and verify against the attitude indicator and other lights.

49.2 Vertical Separation

DGCA-QUOTED

A common problem in flight is the evaluation of the relative altitude of approaching aircraft and the assessment of a potential collision risk.

WHY VISUAL ALTITUDE ASSESSMENT OF ANOTHER AIRCRAFT IS UNRELIABLE

At distance, an aircraft is reduced to a point of light or a tiny silhouette. There is no reliable monocular cue to its altitude relative to your own. A constant-bearing approaching target (no relative motion across your windscreen) is the classic collision-course signature — and the most difficult to detect visually. Modern transport aircraft therefore rely on **TCAS** (Traffic Collision Avoidance System) and ATC altitude verification, not on the pilot's eye, for vertical separation.

§ 50 Approach and Landing — Three Stages

DGCA-QUOTED

In the final stages of a flight the pilot has to cope with the most critical visual tasks, and these may be divided into 3 stages:

1. **Initial judgment of glide slope**
2. **Maintenance of the glide slope**
3. **Ground proximity judgments.**

50.1 Initial Judgment of Appropriate Glide Slope

DGCA-QUOTED — THE "VISUAL ANGLE"

Visual Angle — To judge the approach path, the pilot is attempting to establish an angle. **This angle is the "Visual Angle" and is measured at the pilot's eye DOWN FROM THE HORIZON to the visual aiming point on the runway.**

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50.2 Width & Slope of Runways

DGCA-QUOTED — WIDTH OF RUNWAYS ILLUSION

The width of the runway may also cause incorrect height judgments on the final approach.

A pilot used to a **standard width runway** may, when approaching an **unfamiliar NARROW runway**, **judge he is too high and therefore round out too low on approach**.

THE INVERSE — WIDE RUNWAY ILLUSION

By the same physics, an unusually **WIDE** runway looks **closer / lower than it is** → pilot perceives "too low", rounds out too high, lands long and hard.

50.3 The Black Hole Effect

DGCA-QUOTED — VERBATIM

The Black Hole Effect — The absence of visual cues (such as **night-time approaches over desert or unlit water**) leads to an illusion that the **aircraft is TOO HIGH**, as a result the approach path may be flown at **too shallow an angle**, the aircraft may **touch down SHORT of the runway**. This specific illusion is often called the **"black-hole illusion"**, due to the apparent visual "black hole" between the aircraft and the runway.

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Figure 14.4 — Approach illusions: a narrow or up-sloping runway makes you feel high (you go low); the black hole makes you feel high on a dark approach.

50.4 Visual Illusions on Approach

DGCA-QUOTED — GROUND LIGHTING ILLUSIONS

Lights along a straight path, such as a road, and even **lights on moving trains can be mistaken for runway and approach lights**.

Bright runway and approach lighting systems, especially where few lights illuminate the surrounding terrain, may create the illusion of **less distance to the runway**. The pilot who does not recognize this illusion will **fly a HIGHER approach**.

Conversely, the pilot **overflying terrain which has few lights to provide height cues** may make a **LOWER-than-normal approach**.

(a) Shallow Approaches — DGCA verbatim

Triggers that **create the illusion of being HIGH → resulting in SHALLOWER approach**:

- **Up-slope runway or terrain**
- **Narrower than usual runway**
- **Feature-less terrain**

- **Rain on the wind screen**
- **Haze**

Result: pilot thinks "too high" → reduces descent rate → flies shallow → may land short or hit obstacles.

(b) Steep Approaches — DGCA verbatim

Triggers that **create the illusion of being LOW → resulting in STEEPER (HIGH) approach**:

- **Down-sloping runway or terrain**
- **Wider than usual runway**
- **Bright runway / approach lights**

Result: pilot thinks "too low" → increases descent rate → flies steep → may land long or run off the end.

Runway / approach features → perceived position → pilot's wrong reaction

Visual Feature	Pilot perceives	Reaction	Outcome
Narrower than usual runway	"Too high"	Reduces rate of descent	Shallow approach · land short / low round-out
Wider than usual runway	"Too low"	Increases rate of descent	Steep approach · land long / high round-out
Up-sloping runway/ terrain	"Too high"	Shallow approach	Low / land short
Down-sloping runway/ terrain	"Too low"	Steep approach	High / land long
Featureless terrain	"Too high"	Shallow	Low / land short (black-hole family)
Rain on windshield / haze	"Too high"	Shallow	Low / land short
Bright runway / approach lights	"Too low" (looks close)	Steep	High / land long
Few terrain lights below	"Higher than reality"	Shallow	Lower than normal approach
Black hole (no cues between)	"Too high"	Shallow	Land short

50.5 Maintenance of the Glide Slope

DGCA-QUOTED — AIMING PILOT & AIRCRAFT ATTITUDE PITCH ANGLE

Once **established on the glide path, it is relatively easy to visually maintain the glide path by keeping the aiming point at a FIXED POSITION on the windscreen**.

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DGCA-QUOTED — INADVERTENT SPEED LOSS TRAP

On the approach, with an **inadvertent speed loss** and a **gradual loss of altitude**, the **runway could remain in the same position on the windscreen**, giving the impression of a safe approach, **until touch down occurs some distance BEFORE the threshold**.

TEXTURE AND TEXTURE FLOW — DGCA-QUOTED

As long as visual texture flows AWAY FROM the aiming point and the visual angle between this point and the horizon remains constant, the approach will progress normally.

("Texture flow" = the pattern of ground objects (runway markings, taxiway lights, surrounding terrain detail) appearing to flow outward and past you. If the flow point sits ON the aiming spot, you are tracking towards it. If the flow seems to converge ahead of the aiming spot, you are undershooting; behind it, overshooting.)

50.6 Ground Proximity Judgments

DGCA-QUOTED — HEIGHT-ASSESSMENT CUES

The pilot will use a number of cues in his height assessment on the final stage of the approach, among which will be:

- **That the apparent speed of objects on the ground will INCREASE as the height reduces.**
- **That the SIZE of objects, such as runway lights etc., will INCREASE with decreasing distance.**
- **That the apparent WIDTH of the runway will INCREASE.**
- **That the TEXTURE of the ground will CHANGE.**

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STAGE 1 OF APPROACH VISUAL TASK

Initial glide-slope judgment

STAGE 2 OF APPROACH VISUAL TASK

Maintenance of glide slope

STAGE 3 OF APPROACH VISUAL TASK

Ground-proximity judgments

VISUAL ANGLE REFERENCE

Horizon → aiming point

AIMING-POINT POSITION RULE

Fixed on windscreen

NUMBER OF GROUND-PROXIMITY CUES

4 (speed · size · width · texture)

§ 51 PROTECTIVE MEASURES AGAINST ILLUSIONS

DGCA-QUOTED — VERBATIM

Organized formal training is the best protective measure against illusions. It is recommended that this should be used to educate pilots to recognize:

1. **The illusions are natural phenomena.**
2. **Know the different types of illusions and their effects.**
3. **That the supplementation of other visual cues with information from other sources is the most effective counter to the effects of illusions.**
4. **The need for comprehensive flight briefing should the occurrence of illusions be known to exist or are anticipated at particular geographic locations.**
5. **Special care must be taken during accelerations and particularly during instrument flying.**
6. **Head movements, fatigue, night and conditions of reduced visibility are all factors that can promote visual illusions.**

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§ 52 Disorientation / Vertigo — The Master Rule

Figure 14.3 — 'The leans': a slow roll goes unfelt, so returning to level feels like banking the other way.

DGCA-QUOTED — THE MOST IMPORTANT SENTENCE IN THIS ENTIRE PART

If you suspect disorientation, concentrate on and BELIEVE in aircraft Instruments in IMC. If in VMC, look out at the HORIZON.

WHY THIS IS THE CARDINAL RULE

Every other illusion-management technique in §51 is preparation. **This sentence is the action.** The moment you suspect you are disoriented, you have a binary choice tree:

- **In IMC (cloud, night, no horizon)** → *scan and obey the instruments*. The artificial horizon (attitude indicator), altimeter, VSI, ASI, heading indicator, and turn coordinator are unaffected by illusions and tell you what's true. Override the seat-of-the-pants feeling.
- **In VMC (clear day with horizon)** → *look out at the real horizon* and re-anchor your spatial reference to it. Don't fixate on instruments.

There is no third option. There is no "wait it out". The disorientation will not self-resolve while you ignore both the instruments and the horizon.

Website YouTube Instagram
Telegram channel Telegram community

§ 53 Air / Motion Sickness

DGCA-QUOTED

Air/Motion Sickness — this mismatch between vestibular and visual sensory input is the PRIMARY CAUSE of spatial disorientation, and indeed of motion sickness.

Vibrations within the frequency band of `1/10 to 2 Hertz` are a factor contributing to airsickness, because they upset the vestibular apparatus.

SYMPTOMS OF MOTION SICKNESS — DGCA VERBATIM

- **Nausea and fear**
- **Hyperventilation**
- **Vomiting**
- **Pallor**
- **Cold sweating**
- **Headache**
- **Depression**

0.1–2Hz Vibration band that upsets the vestibular system
7symptoms DGCA-listed symptoms of motion sickness
V vs V Vestibular vs Visual mismatch — root cause

§ 54 Linear-Acceleration Illusions — Head-Up & Head-Down

(b)

Head-Up Illusion

Definition (DGCA-quoted): *This illusion involves a sudden forward LINEAR ACCELERATION during level flight where the pilot perceives that the nose of the aircraft is PITCHING UP.*

Perceived: nose pitching UP. **Triggers (DGCA-quoted):** Night takeoff from a well-lit airport into a dark sky, OR application of full power during a missed instrument approach. **Reflex (wrong) response:** the pilot would be to **push the control forward to pitch the nose of the aircraft DOWN** → flies into the ground.
(c)

Head-Down Illusion

Definition (DGCA-quoted): *The head-down illusion involves a sudden linear DECELERATION (e.g., application of air brakes, lowering flaps, decreasing engine power) during level flight.*

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Perceived: nose pitching DOWN. **Reflex (wrong) response:** the pilot's response would be to **RAISE the nose of the aircraft, which may lead to a STALL if executed during a low-speed final approach**.

UNDERLYING PHYSICS — SAME AS SOMATOGRAVIC ILLUSION (§47)

Both illusions are **somatogravic** — otolith deflection by linear acceleration interpreted as a tilt of the head/body relative to gravity. The general rule:

- **Acceleration FORWARD** → otoliths sense as **HEAD-BACK / NOSE-UP** → reflex push forward (Head-Up illusion / take-off somatogravic).
- **Deceleration (acceleration BACKWARD)** → otoliths sense as **HEAD-FORWARD / NOSE-DOWN** → reflex pull back (Head-Down illusion).

Both kill by reflex-induced loss of aircraft attitude. **Defence in both: scan the attitude indicator.**

§ 55 Elevator Illusion

DGCA-QUOTED

An abrupt UPWARD vertical acceleration, usually by an UPDRAFT, can create the illusion of being in a CLIMB. The disoriented pilot will push the aircraft into a NOSE LOW attitude.

An abrupt DOWNWARD vertical acceleration, usually by a DOWNDRAFT, has the opposite effect, with the disoriented pilot pulling the aircraft into a NOSE UP attitude.

WHEN THIS KILLS

This is the family of illusions that traps pilots in **convective turbulence (CB cells, mountain wave, microbursts)**. The vertical kicks of strong updraft/downdraft trick the vestibular sense into perceiving a sustained climb/dive. The pilot's "correction" — pushing or pulling — superimposes on whatever the vertical motion already is, producing wild pitch excursions and potentially loss of control or wing stall.

Defence: in turbulence, fly the attitude (pitch ~3-5° up, wings level), not the airspeed. Let the altimeter swing. Avoid sudden corrective inputs based on seat-of-pants sensation.

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§ 56 False Horizon

Figure 14.5 — The false horizon: sloping cloud tops or ground lights are mistaken for the true horizon.

DGCA-QUOTED

Sloping cloud formations, an obscured horizon, a dark scene spread with ground lights and stars, and certain geometric patterns of ground light can create illusions of NOT BEING ALIGNED CORRECTLY with the actual horizon. The disoriented pilot will place the aircraft in a DANGEROUS ATTITUDE.

THE FOUR DGCA-LISTED FALSE HORIZON TRIGGERS

1. **Sloping cloud formations**
2. **An obscured horizon**
3. **A dark scene spread with ground lights and stars**
4. **Certain geometric patterns of ground light**

THE "GROUND LIGHTS = STARS" TRAP

On a clear dark night over remote terrain, scattered ground lights below merge visually with stars above — the boundary between them (the true horizon) disappears. The pilot's brain then picks the brightest cluster of ground lights and treats it as "ground" or "below" — but if those lights are on a hillside above the aircraft's altitude, the brain has been told the wrong direction is down. This is one of the most lethal night-VFR illusions.

§ 57 Vection Illusions

"Vection" is the sense of **self-motion induced by viewing motion of the surroundings**. Three forms are recognised in the DGCA syllabus:

57.1

Circular Vection

DGCA-quoted: *Is the sensation of self-rotation induced by viewing a surround rotating about the observer's vertical axis.*

(Example: standing still in a slowly spinning room — you feel as if you are rotating instead.)

57.2

Linear Vection

DGCA-quoted: *During linear vection, the observer feels like they have moved forwards or backwards and the stimulus has stayed stationary.*

(Example: seated in a stationary train when the adjacent train moves — you feel your train is moving.)

57.3

Roll Vection

DGCA-quoted: *During roll vection, the observer feels like they have rotated around the line of sight and the disk has stayed stationary.*

(Example: the brain interprets a rolling visual surround as the body rolling, even when the body is stationary.)

§ 58 Geometric Perspective Illusion

DGCA-QUOTED — SHORT AND BROAD DEFINITION

Geometric Perspective Illusion — any misinterpretation by the visual system of a figure made of STRAIGHT or CURVED lines.

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WHY IT MATTERS IN AVIATION

Runway markings, approach light bars, painted aprons, taxiway centrelines, even cloud-bank edges — all are made of straight or curved lines. The brain extracts depth, distance, and orientation from these lines using assumptions (parallel lines converge, near things look bigger, etc.). Anything that breaks those assumptions (e.g. a runway painted with a converging perspective deception, or an upslope that looks like a level runway from far away) creates a geometric-perspective illusion.

§ 59 The Stroboscopic Effect (Flicker Vertigo)

DGCA-QUOTED DEFINITION

An additional type of vertigo is known as **Flicker vertigo**. Light, flickering at certain frequencies, from `four to twenty times per second (4 – 20 Hz)`, can produce **unpleasant and dangerous reactions** in some people.

DGCA-QUOTED REACTIONS

These reactions may include:

- **Nausea**
- **Dizziness**
- **Unconsciousness**
- **Even reactions similar to an EPILEPTIC FIT**

DGCA-QUOTED CAUSES IN FLIGHT

- In a **single-engine propeller aeroplane heading into the sun**, the propeller may cut the sun to give this flashing effect, **particularly during landing when the engine is throttled back**.
- These undesirable effects may be avoided by **not staring directly through the prop for more than a moment**, and by making **frequent but small changes in RPM**.
- The **flickering light traversing helicopter blades** has been known to cause this difficulty.
- The **bounce back from rotating beacons on aircraft which have penetrated clouds** — if the beacon is bothersome, **shut it off during these periods**.

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DGCA-QUOTED — RECOMMENDED PREVENTATIVE ACTIONS

If a member of the crew or a passenger shows symptoms of the Stroboscopic Effect, the recommended preventative actions are:

1. **Turn the aircraft away from the sun.**
2. **Move the person affected to the shade.**
3. **Make the individual close eyes.**

§ 60 Disorientation Summary — REMEMBER

DGCA-QUOTED — SIX SUMMARY STATEMENTS (VERBATIM)

1. **Without visual aid, a pilot often interprets CENTRIFUGAL FORCE as a sensation of RISING or FALLING.**
2. **Abrupt head movement during a prolonged constant-rate turn in IMC or simulated instrument conditions can cause pilot disorientation.** *(This is "Coriolis illusion" / somatogyral cross-coupling.)*
3. **A sloping cloud formation, an obscured horizon, and a dark scene spread with ground lights and stars can create an illusion known as FALSE HORIZONS.**
4. **An abrupt change from climb to straight and level flight can create the illusion of TUMBLING BACKWARDS.**
5. **A rapid acceleration during takeoff can create the illusion of being in a NOSE-UP attitude.**
6. **Symptoms of hypoxia may be difficult to recognize BEFORE the pilot's reactions are affected.**

✦ END OF CHAPTER 14 ✦

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The Forces

Chapter 15 — Acceleration and the pilot's body: the effects of increased "G", the cascade from grey-out to G-LOC, the 1.41 G reference at 45° of bank, the factors that erode G-tolerance, and negative G with red-out.

BOOK
Human Performance & Limitations

AUTHOR
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EXAM
DGCA CPL / ATPL — HPL

CHAPTER
15 of 26 · Module F

In this chapter

1. Acceleration and "G" Forces — Introduction
2. Effects of Increased "G" Manoeuvres
3. 1.41 G at 45° — the Reference Number
4. Factors Adversely Affecting G-Tolerance
5. Negative G & Red-out

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Plate 15.0 — Under high G the body fights its own blood. Vision goes before consciousness — a warning measured in fractions of a second.

GHOST AVIATOR Capt. Pa

§ 61

Acceleration and "G" Forces — Introduction

DGCA-QUOTED — DEFINITION OF 1 G

Flying can expose the human body to conditions for which it is not naturally suited. On the ground, the body is subject to **normal gravitational acceleration: "1G". This is** `32 ft/sec² or 9.81 m/sec²`. The reaction of the earth's surface to this acceleration **gives us the sensation we call "weight"**.

A pilot will experience 1 G in straight and level flight.

61.1 Load Factor at Bank Angles

DGCA-QUOTED — LOAD-FACTOR NUMBERS (MEMORISE)

- At **60° of bank** a pilot is subject to an acceleration of `2 G` acting vertically through his seat. His weight will also **increase by a factor of 2**. This is called the **LOAD FACTOR**.
- In a **70° level turn**, the load factor increases to `3`.

1 G Straight & level flight (32 ft/s² · 9.81 m/s²)
2 G 60° level bank turn
3 G 70° level bank turn

DGCA-QUOTED — AEROBATIC CERTIFICATION

A typical **light aircraft cleared for aerobatics** would be **stressed to withstand positive load factors of up to** 6.

61.2 Counter-measures against adverse G-effects

DGCA-QUOTED

The **adverse effects of increased "G" can be delayed or relieved by:**

- **Tensing the thigh and stomach**, and
- **Easing off the backward pressure on the control column**.

PHYSIOLOGICAL BACKGROUND

Under high positive G, blood pools in the lower body — **cerebral blood pressure drops**. Early symptoms are **greyout** (loss of peripheral vision), then **blackout** (loss of central vision), then **G-LOC** (G-induced Loss of Consciousness). Tensing the lower body's muscles (the "AGSM" — Anti-G Straining Manoeuvre) raises arterial pressure and forces blood back up to the brain. Easing off the control column reduces the G being demanded.

(More detail in continuation of the chapter.)

§ 62 Effects of Increased "G" Manoeuvres

DGCA-QUOTED — THE SIX EFFECTS (MEMORISE THE ORDER)

During increased "G" manoeuvres the pilot will feel the following effects:

1. **Increase in body weight.**
2. **Mobility is impaired.**
3. **Internal organs displaced.**
4. **Onset of tunnel vision.**
5. **Grey out.**
6. **Possible blackout.**

62.1 The Cascade — From Heaviness to Blackout

1 Body weight ↑
2 Mobility impaired
3 Organs displaced

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4 Tunnel vision
5 Grey-out
6 Blackout

WHY EACH EFFECT HAPPENS IN THIS ORDER — PHYSIOLOGICAL CONTEXT

- **Effects 1–3 (heaviness, immobility, organ displacement)** are *mechanical*: with apparent weight multiplied by the load factor, limbs become difficult to lift, the head is forced down/sideways, and abdominal organs sag against the diaphragm.
- **Effects 4–6 (tunnel → grey → black)** are *cerebrovascular*: high positive G pulls blood downward, away from the head. Retinal arterial pressure falls first (cones in the fovea drain → tunnel vision = peripheral vision fails first), then central vision goes (grey-out), then cerebral perfusion fails (blackout / G-LOC).
- The ORDER of the six effects is the standard DGCA exam point — they happen in **this sequence as G increases**.

Figure 15.1 — Increasing positive G drains blood from the head: grey-out, then black-out, then G-induced loss of consciousness (G-LOC).

62.2 Tilted-Back Seat — the DGCA-quoted mitigation

DGCA-QUOTED

A tilted back seat can reduce the chance of a black-out during positive G-manoeuvres.

WHY A RECLINED SEAT HELPS

The taller the column of blood from the heart to the brain along the G-vector, the harder it is for the heart to pump against it. By **reclining the seat**, the vertical distance from the heart to the brain (measured along the seat-to-head axis, which is the G axis in a turn) is *reduced*. The blood column is shorter, the heart has less hydrostatic head to overcome, and cerebral perfusion is preserved at higher G loads. This is exactly why **fighter cockpits (F-16, etc.) recline the pilot's seat ≈ 30°** — to push the G-LOC threshold above where the airframe gives up first.

§ 63

1.41 G at 45° — the DGCA Reference Number

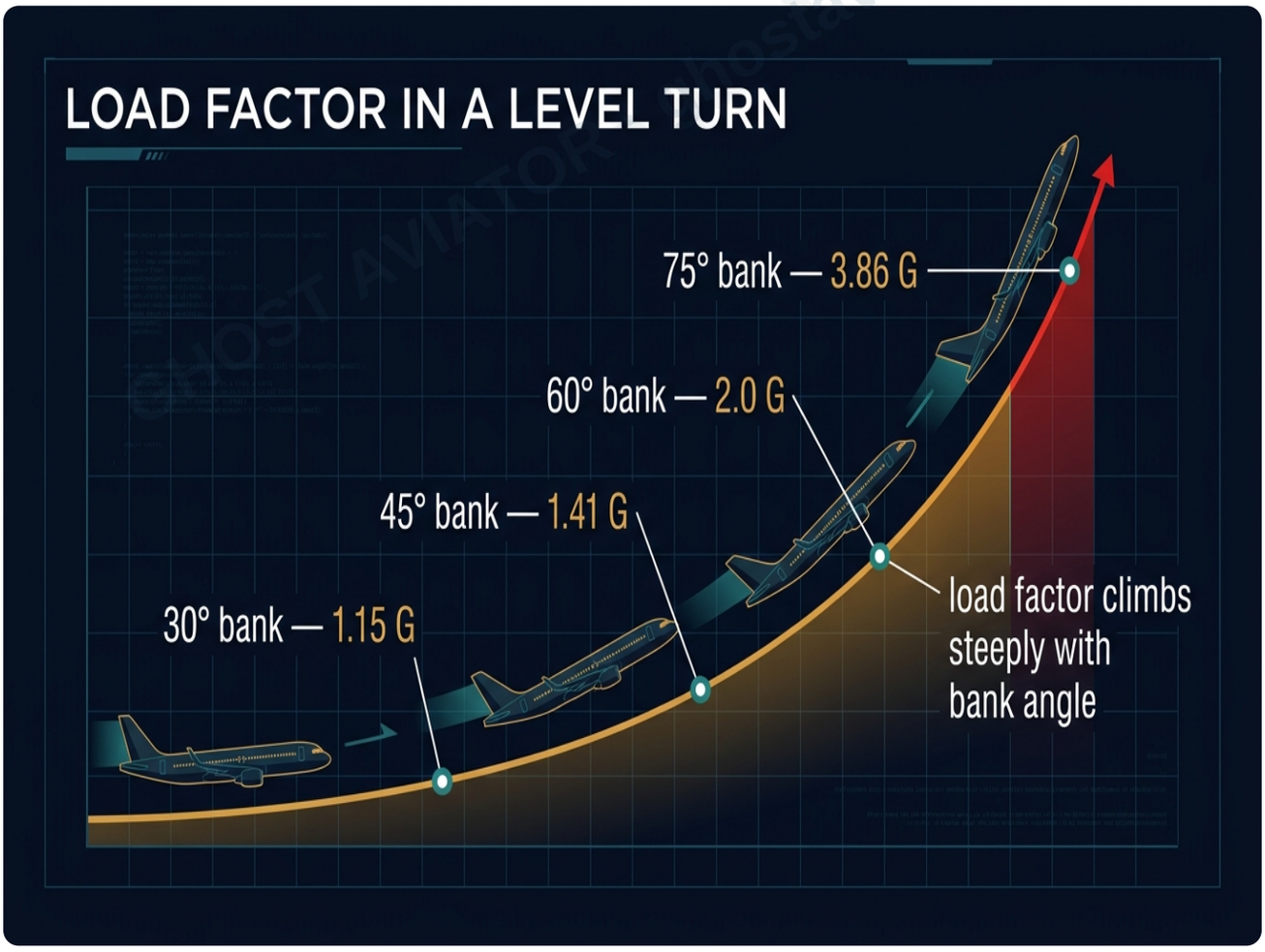

Figure 15.2 — Load factor in a balanced level turn: at a 45° bank the wings carry 1.41 G; it rises steeply with bank angle.

DGCA-QUOTED — EXACT LOAD-FACTOR REFERENCE

`1.41 G` is the acceleration experienced in a LEVEL TURN at `45° angle of bank`. In normal flying, angles of bank greater than this are not usually necessary.

Any physical disorder or immoderate consumption of alcohol or tobacco will reduce the body's tolerance to accelerations in excess of 1 G.

Load factor (n) vs bank angle in a level turn — n = 1/cos(bank)

Bank Angle	Load Factor (G)	Notes
0° (level flight)	`1.00 G`	Straight & level
15°	`1.04 G`	Shallow turn
30°	`1.15 G`	Standard rate turn (most light a/c)
45°	`1.41 G`	**DGCA reference** · upper limit of "normal flying"
60°	`2.00 G`	Steep turn · weight doubles
70°	`3.00 G`	Aerobatic turn
75°	`3.86 G`	—
80°	`5.76 G`	—
85°	`11.5 G`	Far beyond airframe / human limits

45° Maximum "normal" bank — DGCA quoted
1.41 G Load factor at 45° level turn
↓ tol. Reduced by disorder / alcohol / tobacco

§ 64 Factors Adversely Affecting G-Tolerance

DGCA-QUOTED — THE SIX FACTORS (VERBATIM, IN ORDER)

1. **Alcohol.**
2. **Smoking.**
3. **Fatigue.**
4. **Excessive heat.**
5. **Obesity.**
6. **Sickness.**

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Why each factor reduces G-tolerance	
Factor	**Physiological Effect**
Alcohol	Vasodilation + dehydration + histotoxic hypoxia (§14) — blood pressure regulation degraded.
Smoking	5–8 % O_2-capacity loss + vasoconstriction + CO competition (§13) — less reserve when cerebral flow falls.
Fatigue	Cardiovascular reserve drops; muscle tone (for AGSM) is weakened; reaction time slows.
Excessive heat	Vasodilation for cooling + reduced blood volume from sweating → lower BP under G.
Obesity	Reduced cardiovascular fitness; abdominal fat reduces effectiveness of the AGSM straining manoeuvre.
Sickness	Dehydration, low BP, weakness — all the above multiplied.

§ 65 Negative G & Red-out

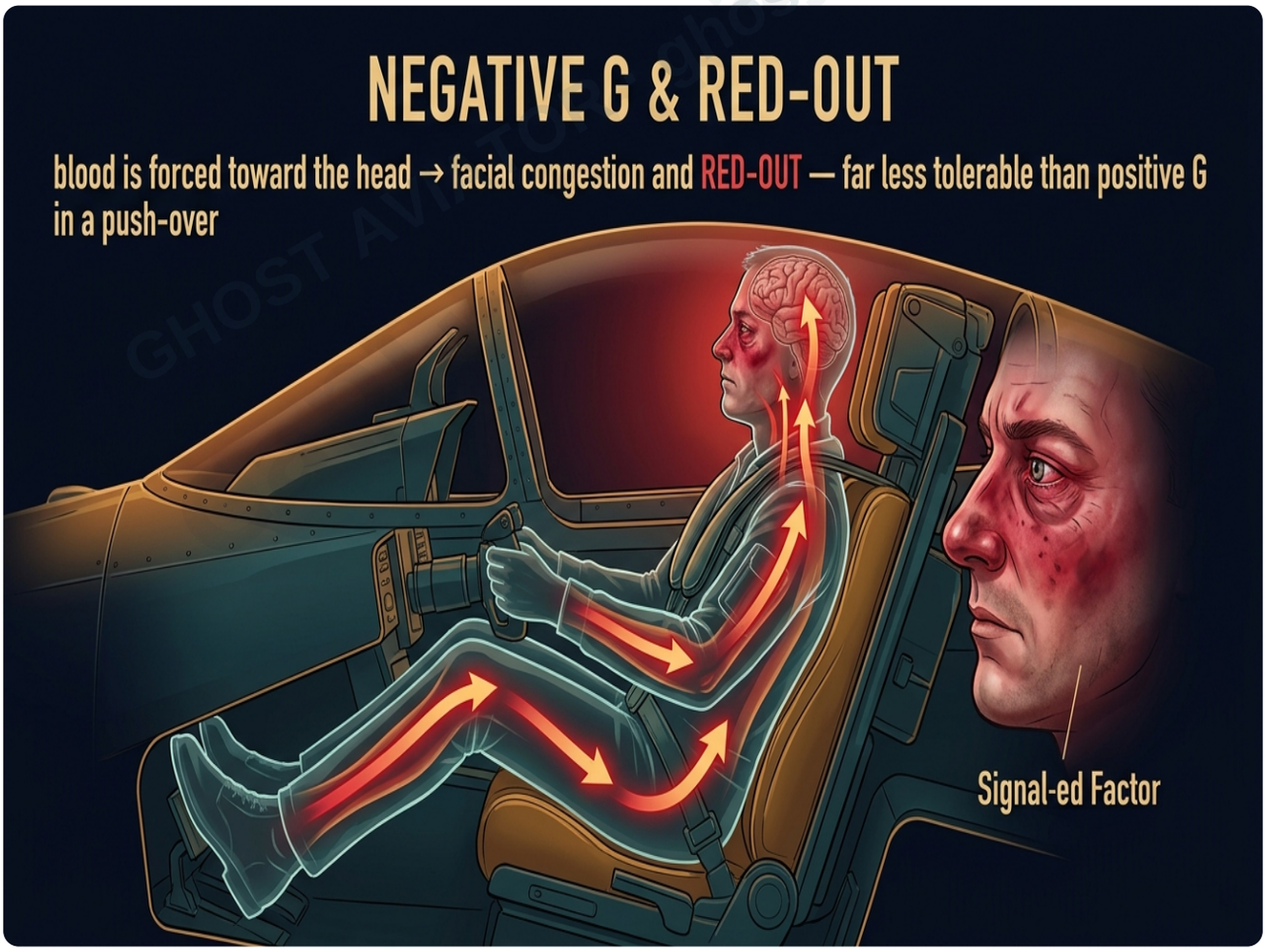

Figure 15.3 — Negative G forces blood toward the head, causing facial congestion and 'red-out' — far less tolerable than positive G.

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DGCA-QUOTED — TOLERANCE ASYMMETRY

Most pilots can learn to tolerate moderate increases in positive "G", but many find even the smallest exposure to negative "G" to be unpleasant.

DGCA-QUOTED — WHEN NEGATIVE G IS EXPERIENCED IN FLIGHT

During flight negative "G" is experienced if, after **pulling out of a steep dive**, the **control column is instinctively and firmly moved FORWARD** because the pilot might feel that he has his **nose too high in an attitude that may lead to a stall**.

DGCA-QUOTED — WHAT NEGATIVE G DOES TO THE BODY

Negative "G" manoeuvres INCREASE the flow of blood TO THE HEAD. Blood pressure there increases, the face becomes very FLUSHED, and the EYES BULGE. The combined effect of these symptoms causes what is described as a "RED OUT".

To relieve symptoms select a normal flying attitude.

POSITIVE G → BLACK-OUT

- Blood pulled **away from** head, into legs
- Cerebral perfusion falls
- Cascade: tunnel → grey → black
- Can progress to G-LOC (G-induced LoC)
- **Tolerable up to ~6 G** with training (light aerobatic limit)

NEGATIVE G → RED-OUT

- Blood pushed **into** the head
- Cerebral & ocular pressure rise
- Face flushes red, eyes bulge
- "Red-out" — vision tinged red
- **Even small exposures unpleasant**
- Mitigation: **return to normal flying attitude**

WHY PILOTS ARE FAR WORSE AT NEGATIVE G THAN POSITIVE

Human cardiovascular anatomy is built for a 1 G "head-up" world. We have valves and reflexes that, under positive G, pump blood upward against gravity (the baroreceptor + sympathetic reflex). Under **negative G the body has NO compensatory mechanism** — blood floods the head, ocular and intracranial pressure rises, and small blood vessels in the eyes and brain can rupture. Even highly trained aerobatic pilots rarely tolerate more than **−3 G** sustained.

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✦ END OF CHAPTER 15 ✦

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HUMAN PERFORMANCE & LIMITATIONS · MODULE G — THE FIT AVIATOR

Fitness & Nutrition

Chapter 16 — The body a pilot brings to work: individual physical characteristics, exercise and body weight (BMI), sound nutrition and food hygiene, and the traps of hypoglycaemia and gastric distress.

BOOK
Human Performance & Limitations

AUTHOR
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EXAM
DGCA CPL / ATPL — HPL

CHAPTER
16 of 26 · Module G

In this chapter

1. Physical Characteristics — Individual Differences
2. Exercise and Weight (BMI)
3. Nutrition and Food Hygiene
4. Hypoglycaemia
5. Gastric Distress & Contaminated Food

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Plate 16.0 — A pilot's fitness is not vanity; it is reserve — the margin the body keeps for the moment the flight demands more.

§ 66

PHYSICAL CHARACTERISTICS — Individual Differences

DGCA-QUOTED

Differences among individuals are remarkable. Besides obvious physical differences (e.g., height, weight, age, sex, build, sitting height, functional reach, leg length, shoulder width, strength, co-ordination etc.), people also differ with respect to other traits like:

- **Auditory threshold**,
- **Understanding**,
- **Vestibular (ear senses)**,
- **Smell, touch, kin aesthetic (body feelings)**,
- **G-tolerances**, etc.

These traits are relatively stable over time, they differ across individuals, and they influence human behavior.

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Website
YouTube
Instagram
Telegram channel
Telegram community

DGCA-MEDICAL IMPLICATION

This is the underpinning for **cockpit ergonomics** and **pilot-selection standards**: minimum/maximum sitting heights, reach envelopes for switches and rudder pedals, vision and hearing standards, etc. Each pilot is a different statistical sample — and the certification rules are designed so that the *typical* trained pilot can comfortably operate the aircraft, while extreme outliers may be screened out at the medical or flight-school stage.

§ 67 EXERCISE AND WEIGHT

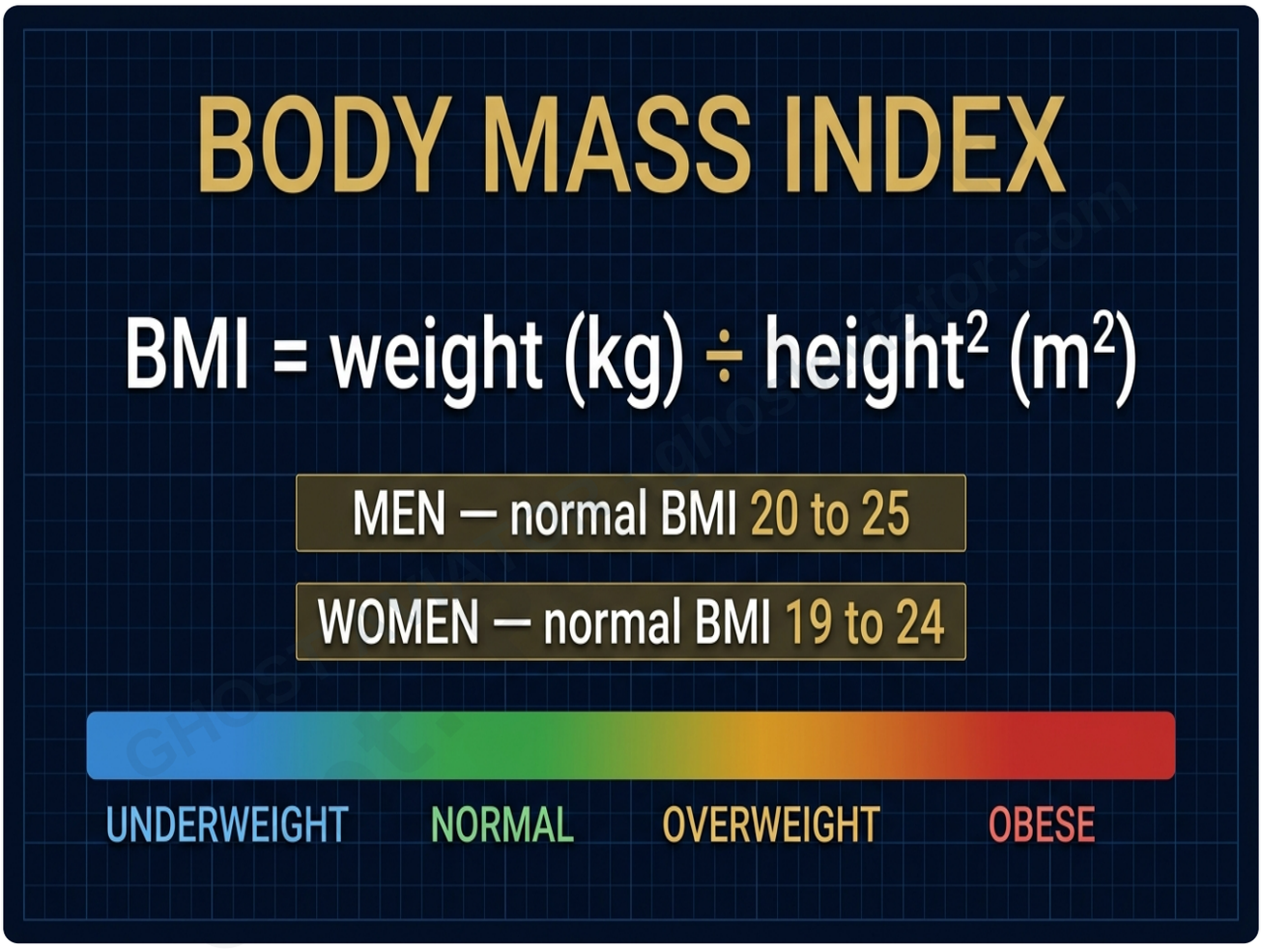

Figure 16.1 — Body Mass Index (BMI = weight ÷ height²) and its categories, from underweight to obese.

67.1 Body Mass Index (BMI)

DGCA-QUOTED — DEFINITION & FORMULA

A person's Body Mass Index, or BMI, is simply a measure of a person's weight in relation to his height.

Body mass Index (BMI) = Weight (kgs) / Height² (m).

DGCA-QUOTED — NORMAL BMI RANGES

Normal BMI for MEN is `20 to 25` and WOMEN `19 to 24`.

BMI $_{= kg/m^2}$ Weight in kg ÷ height in m squared
20–25 Normal BMI for MEN
19–24 Normal BMI for WOMEN

WORKED EXAMPLE

A pilot weighs 75 kg and is 1.78 m tall. BMI = 75 / $(1.78)^2$ = 75 / 3.168 ≈ **23.7**. Within normal range for men (20–25). ✓

67.2 Exercise Prescription & "No appetite suppressants"

DGCA-QUOTED — EXERCISE PRESCRIPTION

Regular exercise is beneficial to general health, but the most efficient way to lose weight is by reducing CALORIC CONSUMPTION.

To reduce the risk of coronary artery disease, **exercise should be done to DOUBLE the resting heart rate for at least** `20 minutes, three times a week`.

DGCA-QUOTED — APPETITE SUPPRESSANTS PROHIBITED

Pilots should NOT try to lose weight by taking appetite suppressants.

BEST WAY TO LOSE WEIGHT
Reduce calories

EXERCISE — TARGET HR
2× resting

EXERCISE — DURATION
≥ 20 minutes

EXERCISE — FREQUENCY
3 × per week

APPETITE SUPPRESSANTS FOR PILOTS
Prohibited

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67.3 Effects of Being Badly Overweight — 13 DGCA-quoted conditions

DGCA-QUOTED — VERBATIM LIST

Being **badly overweight INCREASES a pilot's susceptibility** to the following conditions:

1. **Heart attack.**
2. **Hypertension.**
3. **Hypoxia at lower altitudes than normal.**
4. **General circulation problems.**
5. **Gout (swollen joints).**
6. **Osteoarthritis** (a form of arthritis characterized by gradual loss of cartilage of the joints).
7. **Diabetes.**
8. **The inability to tolerate G forces.**
9. **Problems with joints and limbs.**
10. **Decompression sickness.**
11. **Heavy sweating.**
12. **Chest infections.**
13. **Varicose veins** (a vein that has become swollen and knotted as a result of faulty valves).

CROSS-LINKS — OBESITY IS THE PILOT'S "COMMON ENEMY"

Note how this single list lights up almost every previous part of the chapter:

- Heart attack & hypertension & circulation problems → §9 / §15 / §61
- Hypoxia at lower altitudes → §10 / §13 (smoker comparison)
- Inability to tolerate G forces → §62 / §64
- Decompression sickness → §19 / §20 (DCS predisposing factor)

The DGCA examiner treats obesity as a *multiplier* of every other risk discussed in the chapter — which is why BMI screening is built into every Class-1 / Class-2 medical.

§ 68 Nutrition and Food Hygiene

DGCA-QUOTED — THE EMPTY-STOMACH RULE

Never fly on an empty stomach. A balanced diet is the foundation of good health.

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68.1 Sources of Carbohydrates

DGCA-QUOTED

Sources of carbohydrates include:

- **Grains**,
- **Vegetables**,
- **Nuts**, and
- **Fruit**.

68.2 Vitamins

DGCA-QUOTED

Vitamins are organic substances required by the body to function properly. They help process other nutrients to form blood cells.

68.3 Minerals — the Three Majors

DGCA-QUOTED

Minerals are essential to many vital body processes — from building strong bones to transmitting nerve impulses.

The three major minerals include:

- **Calcium — for healthy bones and teeth.**
- **Phosphorous — for body's chemical reaction** and
- **Iron — For Hemoglobin.**

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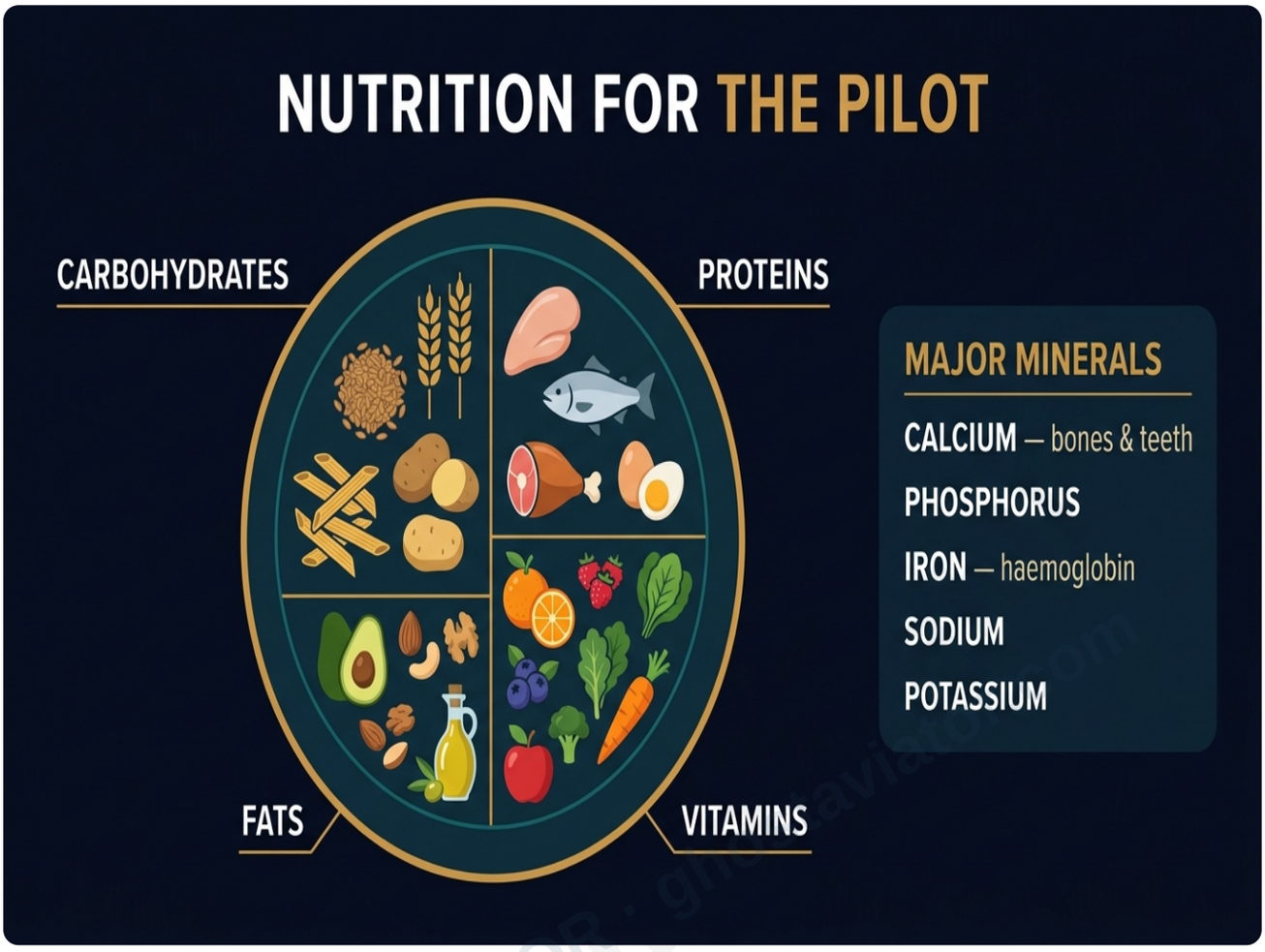

Figure 16.2 — Nutrition essentials: the major minerals — calcium (bones & teeth), phosphorus, iron, sodium and potassium — and a balanced pilot's diet.

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Capt. Pa

§ 69 Hypoglycemia

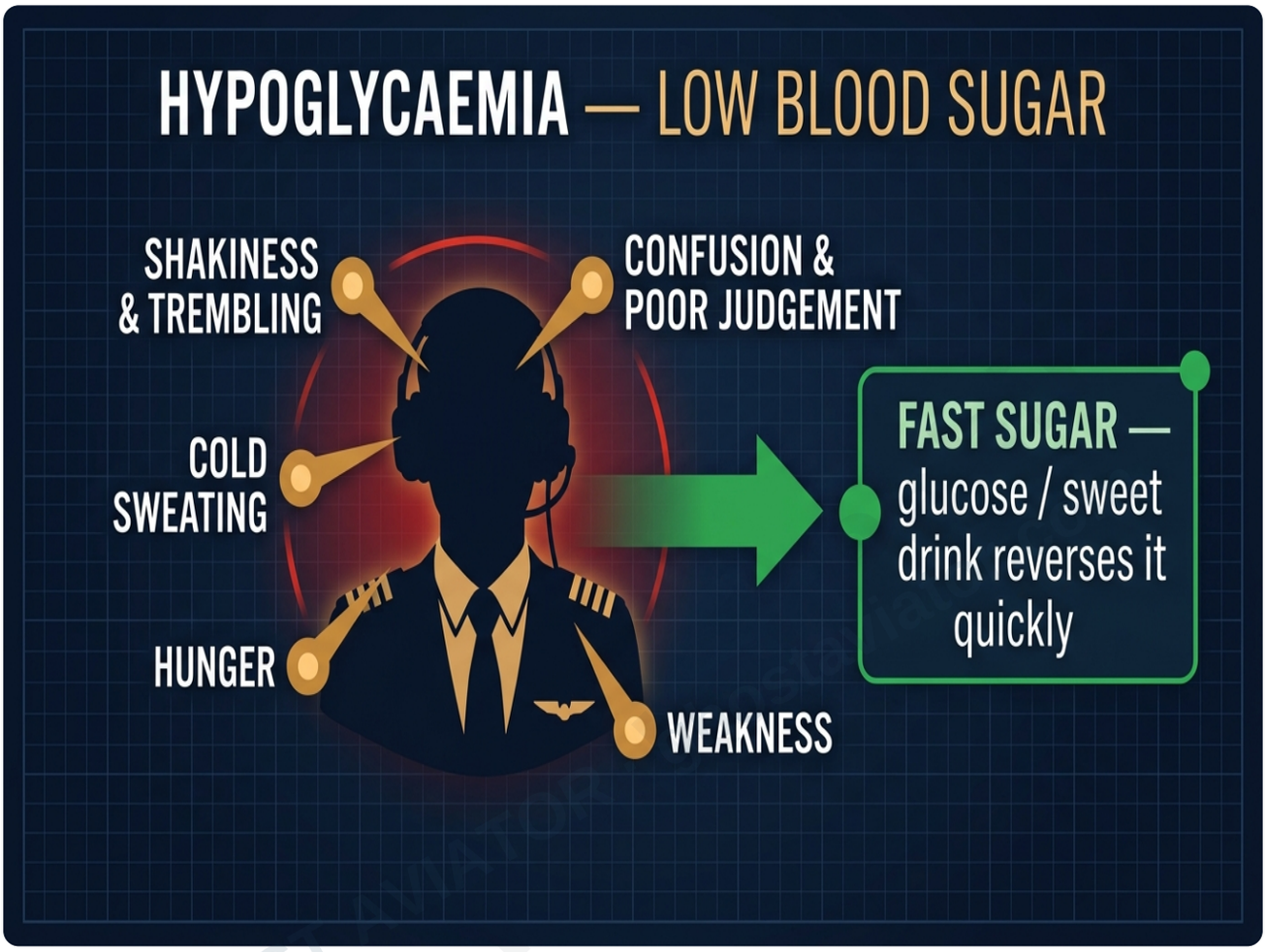

Figure 16.3 — Hypoglycaemia: low blood sugar brings shakiness, sweating and confusion — reversed quickly with fast sugar.

DGCA-QUOTED — THE BRAIN NEEDS FUEL

Just as the brain needs oxygen to function, it also needs fuel to combine with the oxygen and produce energy. That fuel is **blood glucose**, which is carried by the bloodstream and easily passes the **blood–brain barrier**.

DGCA-QUOTED — THE BRAIN CANNOT STORE GLUCOSE

Glucose is a simple sugar and serves as an immediate source of energy for cells.

The brain CANNOT STORE GLUCOSE and requires a CONTINUOUS SUPPLY to function properly.

All foods containing carbohydrates will raise blood glucose levels.

DGCA-QUOTED — WHAT HYPOGLYCEMIA DOES TO THE PILOT

Failure to eat properly may result in a shortage of glucose (hypoglycemia), which will produce DECREMENTS IN COGNITIVE FUNCTIONING.

Research has proved that hypoglycemia significantly compromises performance, resulting in LONGER RESPONSE TIMES and LOWER SCORES on cognitive tests.

PRACTICAL PILOT RULE

This is the physiological basis of the "**Never fly on an empty stomach**" rule (§68). A pilot who skipped breakfast — or who is on a crash diet — is operating with reduced cognitive performance. This becomes one of the DGCA-listed causes of **subtle incapacitation** (§27.3 — "Temporary hypoglycemia" is listed verbatim).

§ 70 Gastric Distress and Contaminated Foodstuffs

70.1 Four Major Causes of Food Contamination

DGCA-QUOTED — VERBATIM

Major causes of food contamination are:

1. **Unhygienic (i.e. unclean) food preparation.**
2. **Under cooked or stale meats.**
3. **Unwashed salads, fruit or vegetables.**
4. **Seafood and locally made ice-creams and mayonnaise.**

70.2 Gastroenteritis — Pilot Fitness Rule

DGCA-QUOTED — PILOT UNFIT EVEN WITH MEDICATION

Pilots suffering from Gastroenteritis are NOT FIT TO FLY, even though they may be taking medicine which is relieving the symptoms.

Due to pressure differential, TRAPPED GASES ESCAPE AT HIGH ALTITUDES resulting in extreme discomfort and sickness.

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WHY ALTITUDE MAKES GASTROENTERITIS WORSE — BOYLE'S LAW AGAIN

Recall **Boyle's Law** from Part 1 — gas volume increases as pressure decreases. Gut gases that are tolerable at sea level can expand by ~2× by cabin altitude of 18,000 ft (where atmospheric pressure has halved, §19.2). For a gastroenteritis patient already producing excess gas, this expansion → severe abdominal pain, nausea, vomiting, sudden bowel motion — all of which incapacitate the pilot in the cockpit. Anti-diarrhoeal medication may mask the bowel symptoms on the ground but does nothing to prevent the gas-expansion mechanism in flight.

DGCA-QUOTED — SYMPTOMS OF GASTROENTERITIS

- **Nausea**
- **Vomiting**
- **Abdominal Pain**
- **Diarrhea**, and
- **Loss of appetite**.

✦ END OF CHAPTER 16 ✦

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HUMAN PERFORMANCE & LIMITATIONS · MODULE G — THE FIT AVIATOR

Fitness to Fly

Chapter 17 — The small conditions that ground a pilot: blocked sinuses and barosinusitis, dental barodontalgia and the post-treatment rule, dehydration, personal hygiene, and the common ailments that quietly degrade performance.

BOOK
Human Performance & Limitations

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CHAPTER
17 of 26 · Module G

In this chapter

1. Sinus Blockage
2. Dental Pain — Barodontalgia
3. Dehydration
4. Personal Hygiene
5. Common Ailments

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Plate 17.0 — A blocked sinus or an aching filling can end a flight as surely as a failed engine. Fitness to fly is a daily self-check.

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Capt. Pat

§ 71 Sinus Blockage

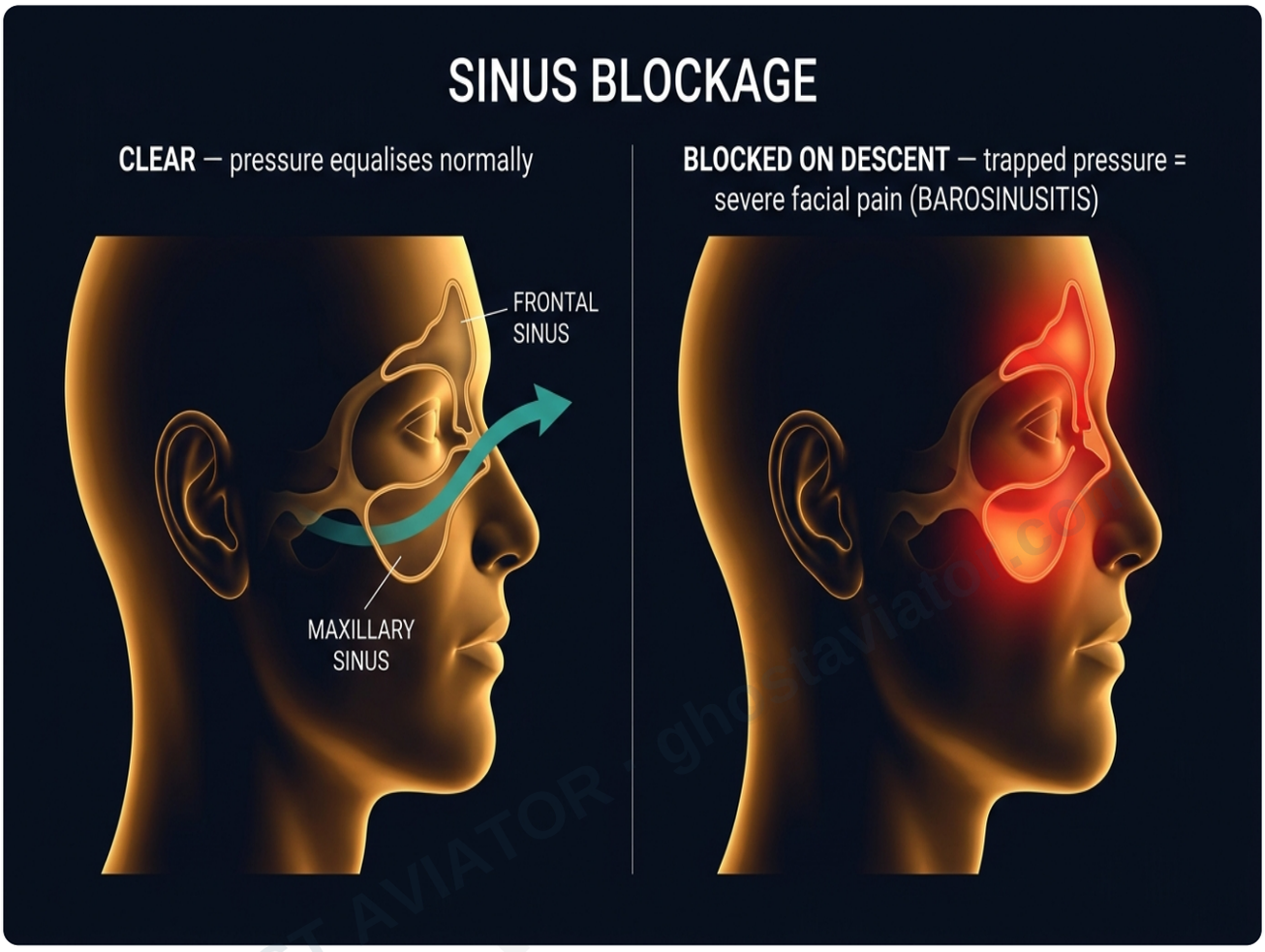

Figure 17.1 — The sinuses: a blocked sinus cannot equalise on descent, and the trapped pressure difference causes severe facial pain (barosinusitis).

DGCA-QUOTED

If the sinuses are congested, then air trapped there can also produce a painful condition. This sinus block occurs most frequently during DESCENT.

Slowing or stopping the descent until the pressure inside the sinus equalizes with the external pressure is the best course of action once this condition occurs.

Of course, NOT FLYING when the pilot has a congested sinus is even better.

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WHY DESCENT — THE SAME ONE-WAY-VALVE PROBLEM AS EARS (§38.2)

The sinuses are air-filled cavities in the skull (frontal, maxillary, ethmoid, sphenoid) that normally drain through narrow ostia into the nasal cavity. On **climb**, sinus air *expands* and pushes outward — easy. On **descent**, ambient pressure rises and sinus air must *contract* by drawing more air in from the nose. If the ostium is blocked by mucus/inflammation (cold, hay fever, sinusitis), incoming air cannot enter — a vacuum forms in the sinus → severe pain over the forehead, cheeks or behind the eyes ("sinus squeeze" / barosinusitis).

Mitigation in flight: stop or slow the descent and allow time for pressure to equalise.
Prevention on the ground: if you have a head cold, don't fly.

§ 72 Dental Pain — Barodontalgia

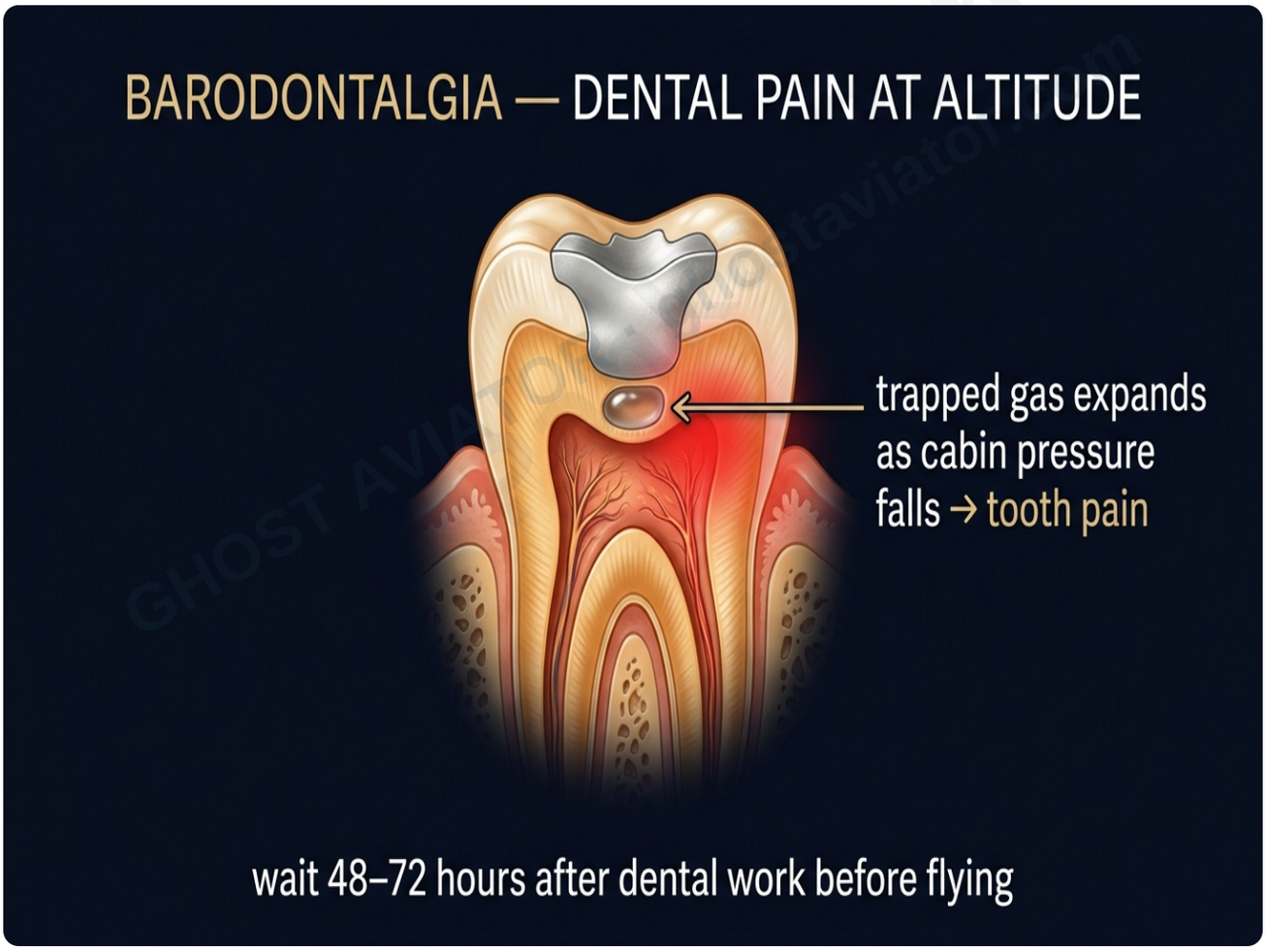

Figure 17.2 — Barodontalgia: gas trapped under a filling or in decay expands as pressure falls, causing tooth pain — hence the 48–72 hour post-dental-work rule.

DGCA-QUOTED — DEFINITION

Reduction in atmospheric pressure may also result in BARODONTALGIA — tooth pain.

72.1 Six Dental Conditions That Cause Barodontalgia

DGCA-QUOTED — VERBATIM LIST

This may be caused by a number of dental conditions including:

1. **(a) Dental disease**,
2. **(b) Dental caries (a cavity)**,
3. **(c) Defective tooth restoration**,
4. **(d) Pulpitis**,
5. **(e) Cysts**, and
6. **(f) Impacted teeth**.

MECHANISM — GAS TRAPPED IN TOOTH CAVITIES + BOYLE'S LAW

A small gas pocket under a filling, in a decayed pulp, or inside a sealed cyst **expands** as cabin altitude rises (Boyle's Law again — §2.1). The expanding gas pushes outward against the tooth nerve → sharp, throbbing pain that becomes excruciating at typical airline cabin altitude (6,000–8,000 ft). It often resolves on descent — but if the affected tooth is fractured by the expansion, the pain persists on the ground.

72.2 The 48–72 Hour Post-Dental Rule

DGCA-QUOTED — THE NO-FLY RULE

This condition may be avoided by NOT FLYING for `48 – 72 hours` following a major dental work.

As with the other problems associated with lower atmospheric pressure, **descending to a lower altitude (and correspondingly higher atmospheric pressure) is often the best antidote.**

48–72$_{hr}$ No-fly window after major dental work
6 DGCA-listed dental causes of barodontalgia
↓ alt In-flight cure — descend to higher-pressure altitude

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§ 73 Dehydration

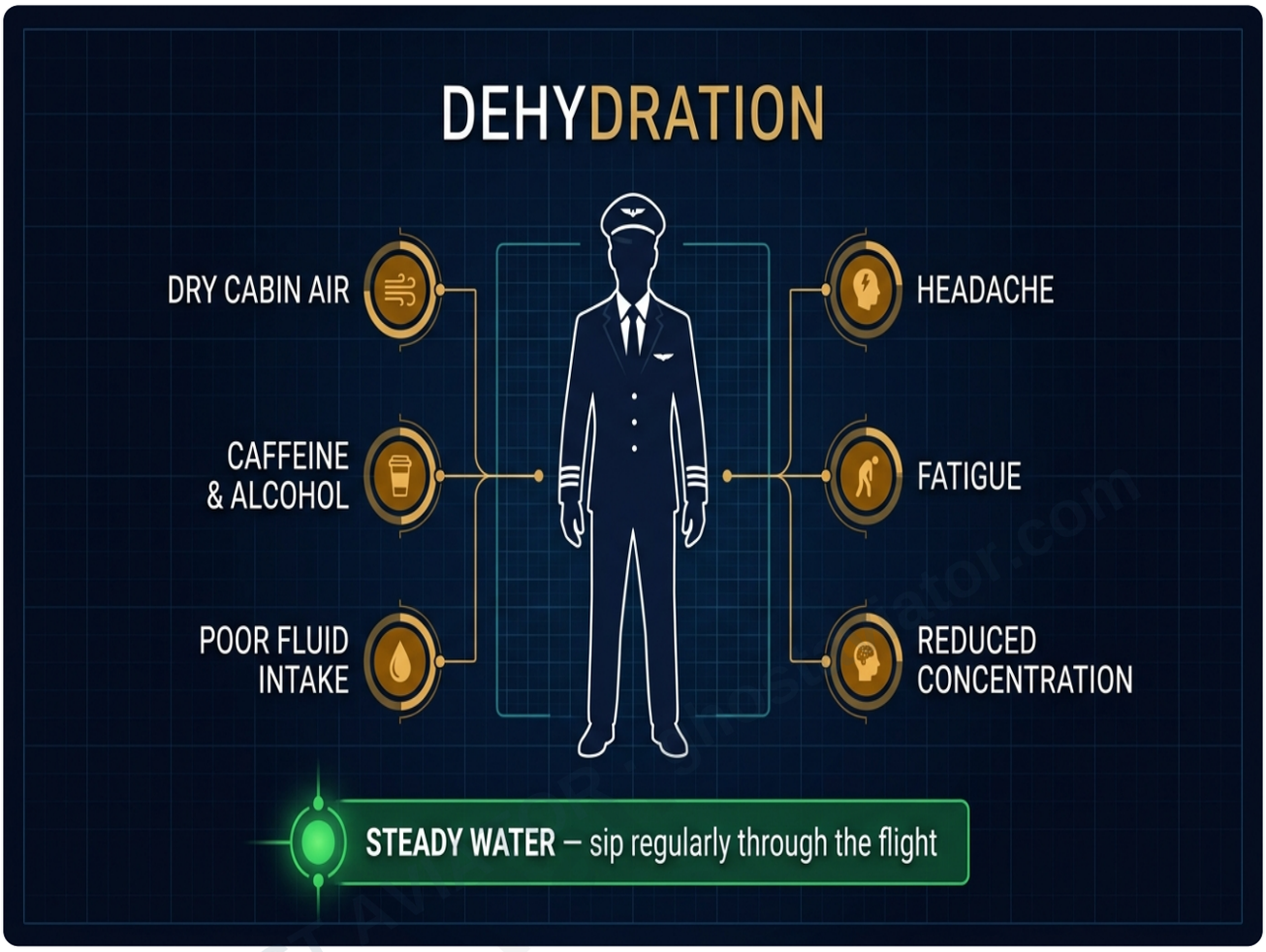

Figure 17.3 — Dehydration: dry cabin air, caffeine and poor intake quietly reduce performance — the fix is steady water.

DGCA-QUOTED

Besides watching what and when they eat, **pilots should also be aware of the effects of dehydration. Low fluid intake and dehydration have ADVERSE EFFECTS ON COGNITIVE FLIGHT PERFORMANCE of pilots.**

CROSS-LINKS — DEHYDRATION IS EVERYWHERE IN THIS CHAPTER

Dehydration has already been flagged repeatedly in earlier parts:

- **§14 (Alcohol)** — alcohol is a diuretic; "masked hangover" includes dehydration.
- **§23 (Humidity)** — cabin RH below 20 % drives over-all dehydration if fluid intake is inadequate; avoid coffee/tea (diuretics).
- **§28.2 (Fainting)** — "standing up quickly after prolonged sitting especially when HOT or DEHYDRATED" is the first DGCA cause of faint.
- **§64 (G-tolerance)** — excessive heat (sweating + lost blood volume) reduces tolerance.

Practical pilot rule: drink ~250 ml of water every hour during a long sector, regardless of thirst.

§ 74 Personal Hygiene

DGCA-QUOTED

A high standard of personal hygiene must be practiced if the body is to remain healthy and free from infection.

PRACTICAL IMPLICATIONS FOR CREW

- **Frequent hand washing** — particularly before meals and after lavatory use; crew in pressurised, low-humidity cabins are at elevated risk of respiratory infections.
- **Oral hygiene** — directly reduces barodontalgia risk (§72) by reducing the prevalence of caries.
- **Skin care** — low humidity dries skin; lotion/moisturiser reduces cracking that can introduce infection.
- **Foot hygiene & well-fitting shoes** — long sectors with feet on rudder pedals make foot-fungal infections more common in crew.

§ 75 Common Ailments

DGCA-QUOTED — VERBATIM

The in-flight environment can increase the severity of symptoms which may be minor whilst on the ground. If there is any doubt whatsoever in a pilot's mind about his fitness to fly, he should STAY ON THE GROUND.

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THE "MINOR ON GROUND, MAJOR IN FLIGHT" PRINCIPLE

A head cold is a nuisance on the ground; in the air it becomes **otic barotrauma + sinus block**. A gastric upset on the ground is a meal-skip; in the air the trapped gases **expand and incapacitate** (§70.2). A mild headache becomes incapacitating under cabin vibration and G-load. The DGCA-codified rule for any doubt is binary: **stay on the ground**. *The cost of a cancelled flight is always less than the cost of a sick pilot in command.*

✦ END OF CHAPTER 17 ✦

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Drugs & Self-Medication

Chapter 18 — The last word on fitness: the self-medication clearance rule, caffeine, anaesthetics and analgesics — and the final environmental hazards of hypothermia, toxic substances and dangerous cargo.

BOOK
Human Performance & Limitations

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CHAPTER
18 of 26 · Module G

In this chapter

1. Drugs and Self-Medication
2. Caffeine
3. Anaesthetics & Analgesics
4. Hypothermia
5. Toxic Hazards
6. Dangerous Cargo

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Plate 18.0 — The golden rule of self-medication: if you are ill enough to need the drug, you may be ill enough to be unfit to fly.

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§ 76

DRUGS AND SELF-MEDICATION

76.1 The Aviation-Medicine-Specialist Clearance Rule

DGCA-QUOTED — THE ABSOLUTE RULE

It is absolutely essential that pilots DO NOT FLY as part of the operating crew of an aircraft when taking drugs or medication, UNLESS THEY HAVE BEEN CLEARED TO DO SO BY AN AVIATION MEDICINE SPECIALIST.

DGCA-QUOTED — RISKS OF SELF-PRESCRIBED MEDICATION

A pilot who flies on **self-prescribed medication** runs the risk of:

- Suffering **side effects**, and
- Faces the **hazards associated with the underlying illness in the in-flight environment** which can **make the symptoms of any illness much more debilitating than they might be on the ground**.

76.2 Four Reasons Medicines Have Flight-Qualification Consequences

DGCA-QUOTED — VERBATIM FOUR REASONS

The consumption of **medicines or other substances** may have consequences on qualification to fly for the following reasons:

1. **The disease requiring a treatment may be cause for disqualification.**
2. **Flight conditions may modify the reactions of the body to a treatment.**
3. **Drugs may cause adverse side effects impairing flight safety.**
4. **The effects of medicine do not necessarily immediately disappear when the treatment is stopped.**

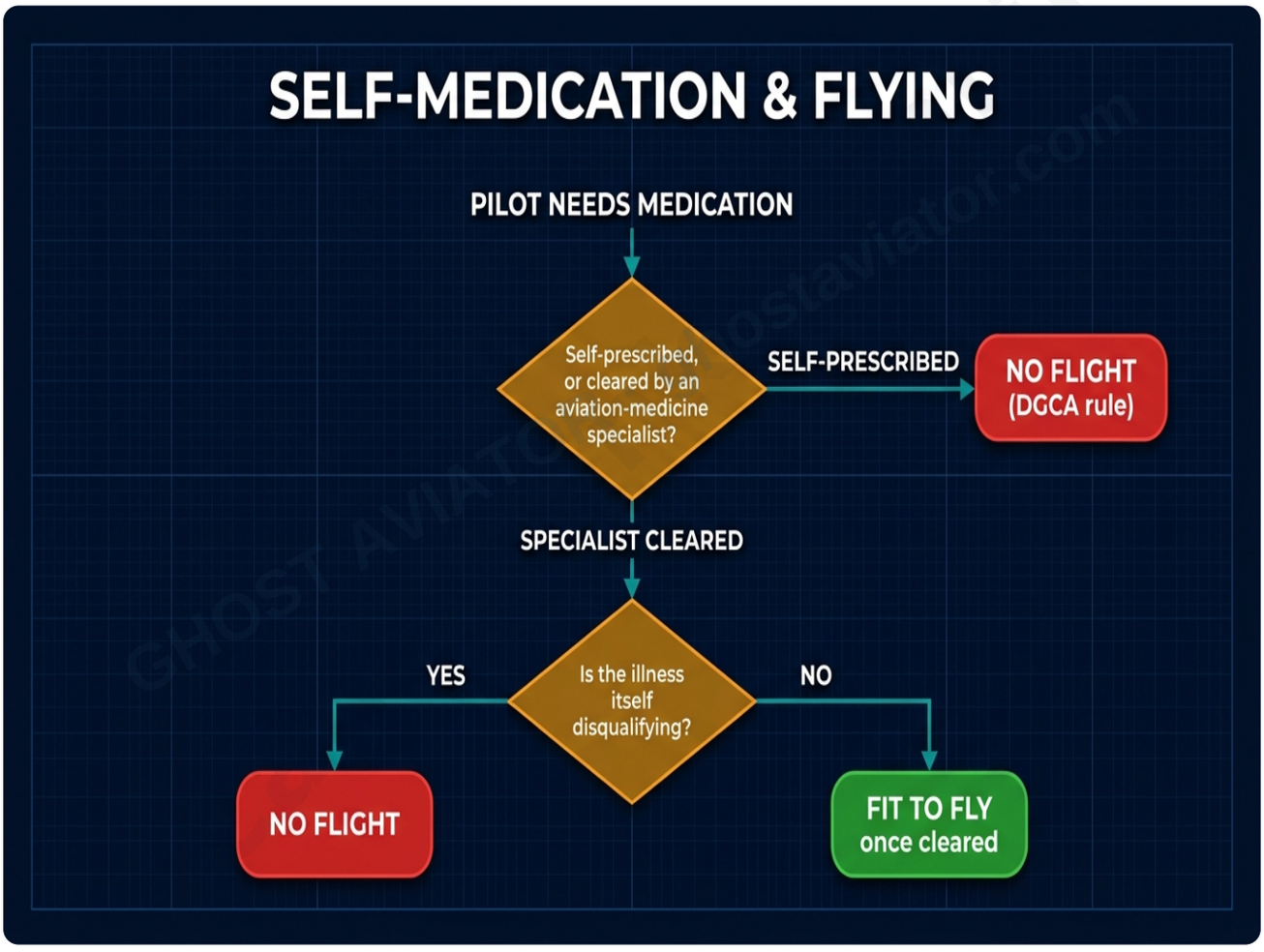

Figure 18.1 — Self-medication and flying: if you need the medicine you may be unfit; any drug must be cleared by an aviation-medicine specialist.

§ 77 CAFFEINE

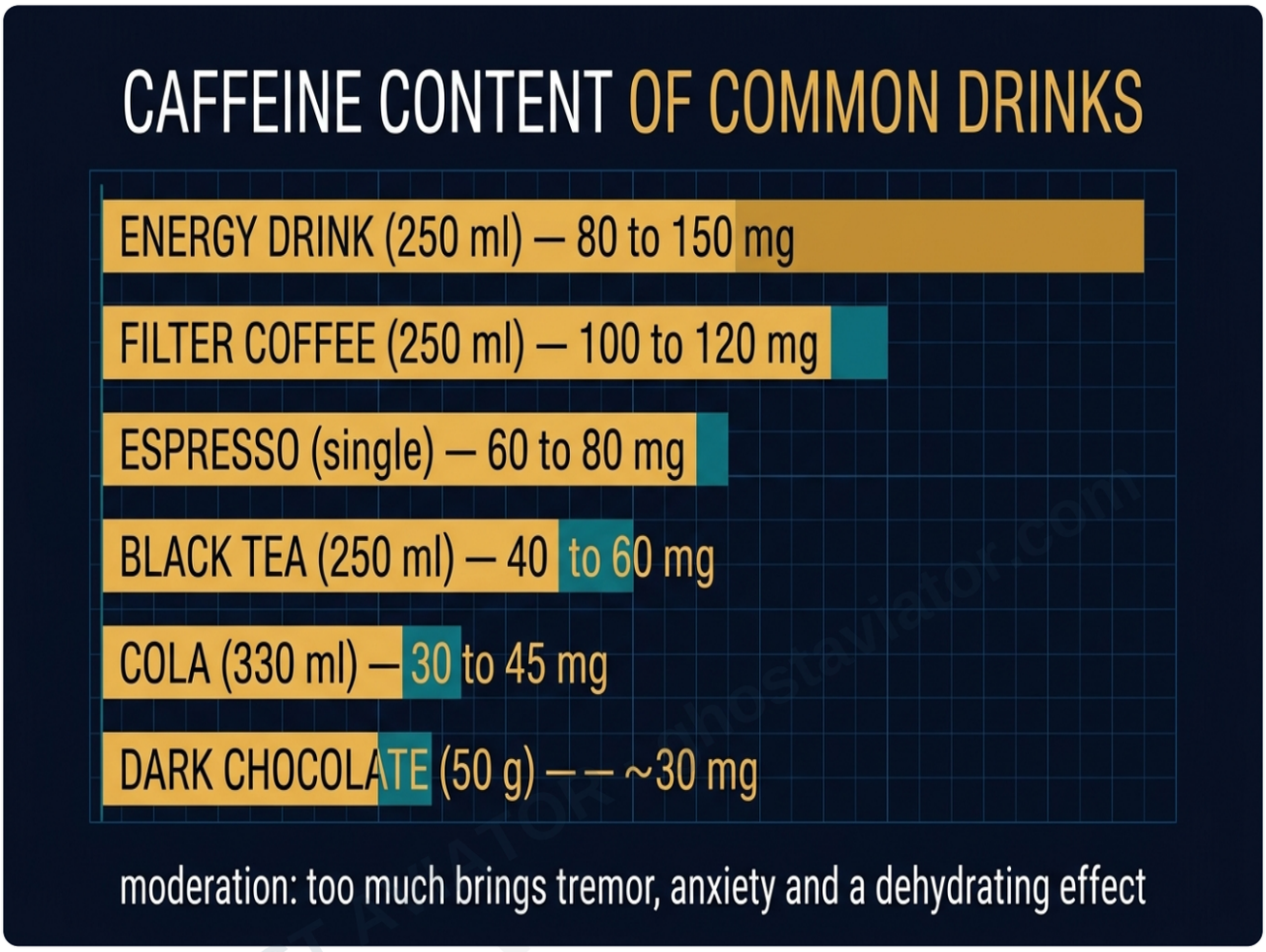

Figure 18.2 — Caffeine: a useful stimulant in moderation, but too much brings tremor, anxiety and a diuretic (dehydrating) effect.

DGCA-QUOTED — CAFFEINE AS A DRUG

Caffeine is probably the most widely used drug in the world. It can easily lead to ADDICTION.

Caffeine is present in coffee, tea, cocoa, chocolate, and fizzy drinks such as cola.

DGCA-QUOTED — RECOMMENDED MAXIMUM DAILY INTAKE

The recommended maximum caffeine intake per day is approximately `250 – 300 mg` corresponding to `2 – 3 cups of coffee`.

Website YouTube Instagram
Telegram channel Telegram community

Approximate caffeine content of common beverages		
Beverage	**Caffeine (mg)**	**Notes**
Filter coffee (250 ml cup)	~ 100 – 120 mg	2–3 cups → DGCA daily max
Espresso (single shot, 30 ml)	~ 60 – 80 mg	—
Black tea (250 ml cup)	~ 40 – 60 mg	Lower than coffee
Cocoa / hot chocolate	~ 5 – 15 mg	Plus theobromine
Cola (330 ml can)	~ 30 – 45 mg	—
Energy drink (250 ml)	~ 80 – 150 mg	Often exceeded
Dark chocolate (50 g)	~ 30 mg	—

250–300$_{mg}$ DGCA max recommended daily caffeine
2–3 Cups of coffee = approximate DGCA limit
⚠ Caffeine is addictive — DGCA word

WHY THE DGCA BOTHERS TO SET A CAFFEINE LIMIT

Caffeine is a CNS stimulant. In moderate doses it sharpens alertness — useful on long sectors. In excess it:

- Increases heart rate and blood pressure (interacts with §15 hypertension & §43 cardiovascular reserve under G).
- Is a **diuretic** (interacts with §23 humidity / §73 dehydration).
- Disrupts sleep — undermining the very alertness it was meant to provide; classic withdrawal cycle.
- Causes tremor & anxiety at high doses — degrades fine motor control.

The "2-3 cups" guidance is the dose that keeps you alert without these costs.

§ 78 Anaesthetics & Analgesics

DGCA-QUOTED — NO-FLY PERIODS

A pilot should NOT fly for at least:

- `12 hours` **after a LOCAL anaesthetic**, and
- `48 hours` **following a GENERAL anaesthetic**.

The more potent forms of ANALGESICS (pain killers) may produce a significant deterioration in human performance.

12$_{hr}$ After LOCAL anaesthetic — no fly
48$_{hr}$ After GENERAL anaesthetic — no fly

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⚠ Strong analgesics — performance deterioration

CROSS-LINK — ADDS TO THE "WAIT-BEFORE-FLYING" MASTER TABLE

You now hold these wait-before-flying numbers in long-term memory:

- **12 hr** — after rapid decompression (§20.4); after local anaesthetic (§78)
- **24 hr** — after alcohol (§14.1); after diving (§21)
- **48 hr** — after blood donation (§17); after general anaesthetic (§78)
- **48–72 hr** — after major dental work (§72.2)

The DGCA examiner pulls these numbers in MCQs — match the event to the number.

§ 79 Hypothermia

DGCA-QUOTED — TEMPERATURE THRESHOLDS (IN FAHRENHEIT, SOURCE PDF)

Normal body temperature averages `98.6 °F`.

With **hypothermia, the core temperature drops below `95 °F`.**

In **SEVERE hypothermia, core body temperature can drop to `82 °F or lower`.**

98.6 °F NORMAL

37 °C — baseline core temperature

< 95 °F HYPOTHERMIA

≈ 35 °C — onset of impaired ability

≤ 82 °F SEVERE

≈ 28 °C — life-threatening

DGCA-QUOTED — WHAT HYPOTHERMIA DOES

Hypothermia AFFECTS PHYSICAL AND MENTAL ABILITIES.

Shivering makes it possible to combat the cold to a certain extent but USES UP A LOT OF ENERGY.

In a PROLONGED exposure, **shivering will tend to CEASE, and be followed by the ONSET OF APATHY**.

Hypothermia is a potentially LIFE-THREATENING condition that needs EMERGENCY MEDICAL ATTENTION.

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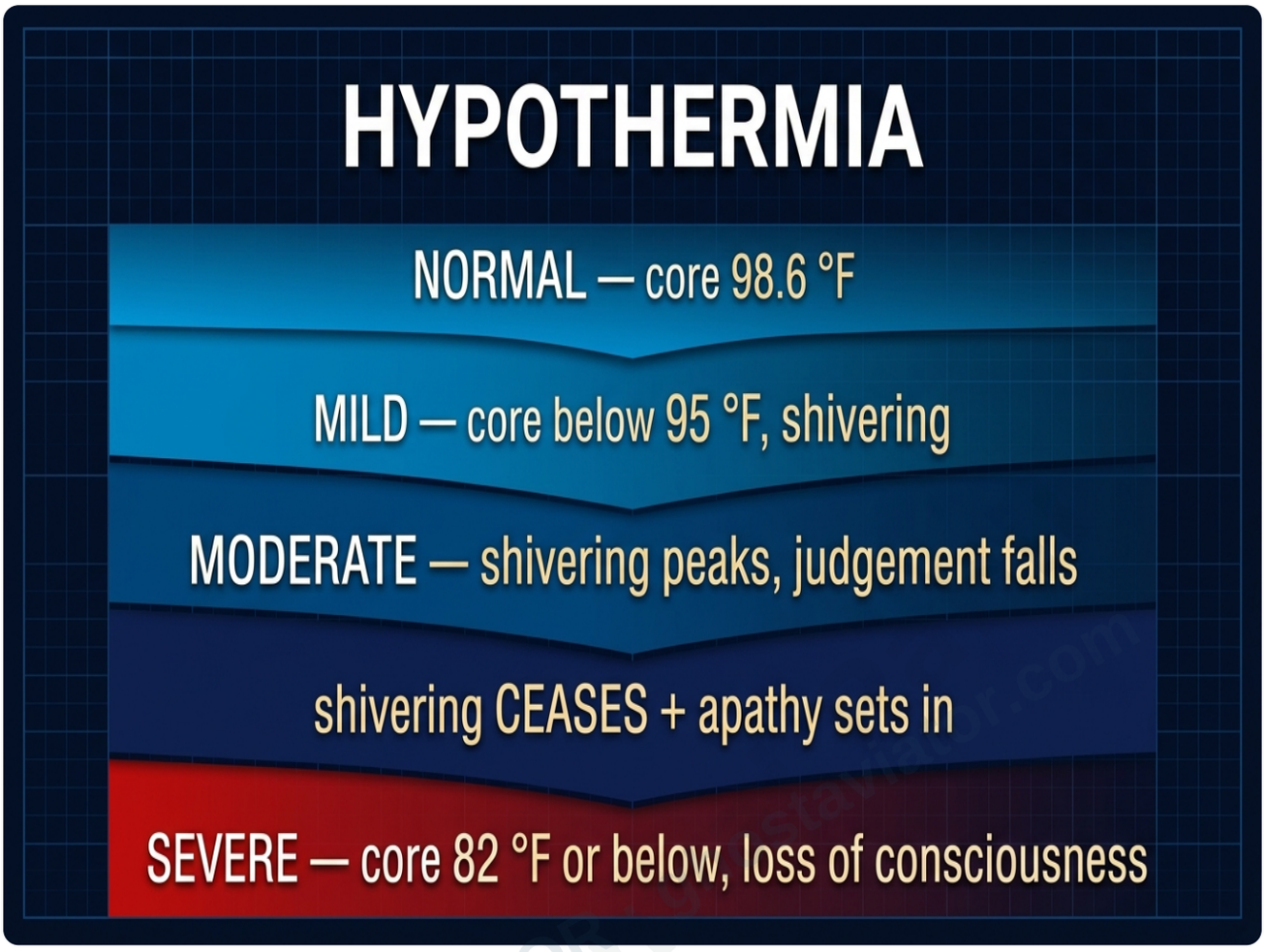

Figure 18.3 — Hypothermia: as core temperature falls, shivering gives way to confusion, then loss of consciousness.

OPERATIONAL RELEVANCE — SURVIVAL IN COLD DITCHING / MOUNTAIN CRASH

For commercial pilots flying over cold water (North Atlantic, Bering Sea) or high-terrain routes, hypothermia is the second-most-common post-ditching cause of death (after drowning). Survival suits, life rafts with thermal protection, and immersion-suit drills exist precisely because **cold water kills faster than the absence of food or water**. The "**cessation of shivering + apathy**" is the danger sign that mild hypothermia is becoming severe.

§ 80 Toxic Hazards

DGCA-QUOTED — OPENING

Even MILD toxic effects can degrade a pilot's performance and lead to an accident.

Prolonged exposure to toxic influences can damage a person's general health.

Anyone who has been exposed to any toxic hazard should seek MEDICAL ASSISTANCE AS SOON AS POSSIBLE.

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DGCA-QUOTED — SIX TOXIC-HAZARD SOURCES (VERBATIM)

The following materials may produce toxic hazards:

1. **Furnishings and Baggage.**
2. **Acetone and Turpentine.**
3. **Fuels, Lubricants and propellants.**
4. **Anti-icing Fluid.**
5. **Fire Extinguishing Agents.**
6. **Battery Fumes.**

Toxic-hazard sources in aviation — DGCA list with context

#	Source	Typical Toxin / Mechanism
1	**Furnishings and Baggage**	Off-gassing of plastics, adhesives, treated fabrics — VOCs, formaldehyde. Also packaged chemicals in baggage holds that leak in flight.
2	**Acetone and Turpentine**	Solvents — CNS depressants on inhalation. Respiratory irritation, headache, dizziness.
3	**Fuels, Lubricants and Propellants**	Jet-A vapours, oil mists, hydraulic fluid mists. Skin irritation, respiratory irritation, possible CNS effects.
4	**Anti-icing Fluid**	Glycols (mono-/di-ethylene glycol) — toxic if ingested, irritant if inhaled as mist.
5	**Fire Extinguishing Agents**	Halons / their replacements — high concentrations displace oxygen, can produce hypoxia (§10). Some thermal decomposition products are corrosive.
6	**Battery Fumes**	Sulphuric acid vapours (lead-acid), or KOH (Ni-Cd), or lithium-fire smoke. Strong respiratory irritants, possibly corrosive.

§ 81 Dangerous Cargo

DGCA-QUOTED — VERBATIM

Pilots must be aware that they must NOT carry certain defined items on board their aircraft. Such items are referred to as DANGEROUS CARGO because of the possibility that their discharge, spillage or breakage may endanger the aircraft and/or crew in flight or on the ground.

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THE REGULATORY BACKBONE — DGR / ICAO TI / 49 CFR 175

The Indian DGCA enforces dangerous-goods carriage rules in line with **ICAO Technical Instructions for the Safe Transport of Dangerous Goods by Air** and the **IATA Dangerous Goods Regulations (DGR)**. The nine UN dangerous-goods classes are:

1. **Explosives** (ammunition, fireworks)
2. **Gases** (compressed, liquefied, dissolved — including aerosols)
3. **Flammable liquids** (fuel, paint, solvents)
4. **Flammable solids · spontaneously combustible · dangerous when wet**
5. **Oxidizers · organic peroxides**
6. **Toxic & infectious substances**
7. **Radioactive material**
8. **Corrosives** (acids, alkalies)
9. **Miscellaneous** (asbestos, lithium batteries, magnetised material, dry ice)

(For full HPL/exam preparation, learn these nine classes — DGCA RTR & DGR examinations routinely ask which class a given item belongs to and whether it is forbidden, restricted, or permitted in checked/cabin baggage.)

WHY THIS SECTION ENDS CHAPTER 26

The chapter has built up systematically: **atmosphere → body systems → environmental stressors → orientation & illusions → fitness & diet → drugs & toxins → dangerous cargo**. The last paragraph closes by reminding the pilot that the threat to the aircraft does not come only from the pilot's own body, but also from **what is loaded into the cabin and cargo holds**. The pilot's command authority extends to ensuring forbidden items are not on board — the pre-flight cargo manifest is a flight-safety document, not paperwork.

✦ END OF CHAPTER 18 ✦

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HUMAN PERFORMANCE & LIMITATIONS · MODULE H — THE MIND ALOFT

Information Processing

Chapter 19 — How the pilot's mind takes in the world and acts on it: what aviation psychology is, why human error is inevitable, mental workload, and the information-processing model from sensory input to response.

BOOK
Human Performance & Limitations

AUTHOR
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EXAM
DGCA CPL / ATPL — HPL

CHAPTER
19 of 26 · Module H

In this chapter

1. Introduction to Aviation Psychology
2. Human Error & Human Reliability
3. Workload
4. Information Processing — A Functional Model

GHOST AVIATOR Capt. Pankaj Pahil ghostaviator.com

Plate 19.0 — The mind aloft. Every switch, call and decision passes first through the narrow channel of human attention.

GHOST AVIATOR Capt. Paul

1. Introduction to Aviation Psychology

WHAT THIS SECTION COVERS

Definition of aviation psychology, its aims, and how its tools support flight safety and efficiency.

Aviation Psychology is a specialty in applied psychology that focuses on understanding human behaviour, emotions and mental states as they relate to the operation and control of aviation systems and their influence on the safety and efficiency of flight.

The *aim* of aviation psychology is to understand and to predict the behaviour of individuals in an aviation environment. Its study is aimed at improving **safety, efficiency and comfort**.

Practitioners of aviation psychology bring the tools and techniques of psychology to bear in order to **describe, predict, understand and influence** the aviation community to achieve those aims.

EXAM TIP — THE FOUR ACTION VERBS

Remember the four practitioner verbs: **D – P – U – I** → *Describe, Predict, Understand, Influence*.

2. Human Error & Human Reliability

WHAT THIS SECTION COVERS

The inevitability of human error and the four areas where pilot errors usually originate.

It has long been accepted that **human errors are inevitable**. The majority of pilot-related errors are considered to be failures of:

- Interpersonal skills
- Communications
- Decision-making
- Leadership

This does **not** mean that the frequency of error cannot be reduced, or that the effects cannot be avoided or mitigated.

GOLDEN MAXIM — "SUPERIOR PILOT"

"A superior pilot uses his superior judgment to avoid situations that would require his superior skills."

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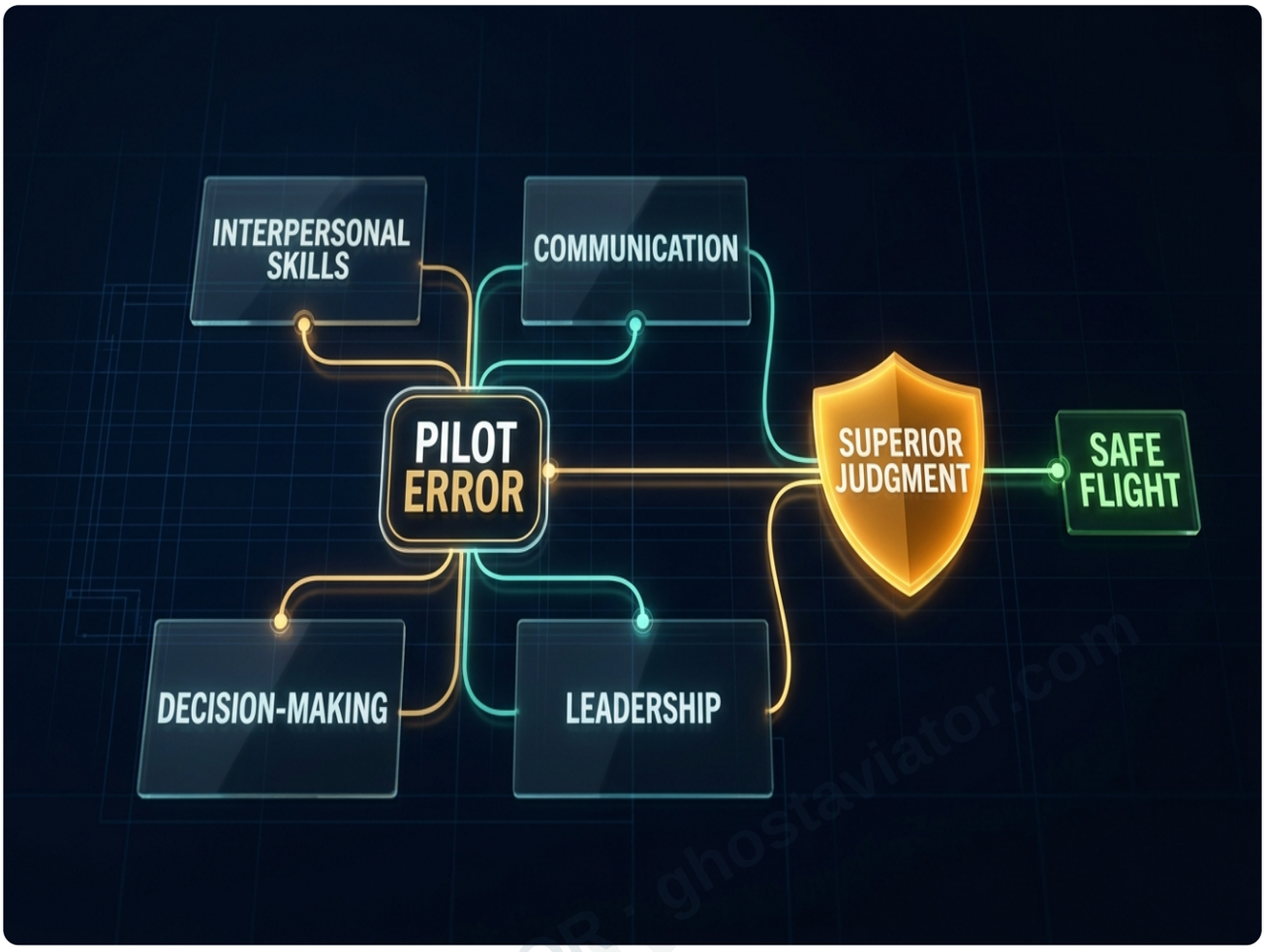

Figure 19.1 — The origins of pilot error: interpersonal skills, communication, decision-making and leadership, all trapped by good judgment.

3. Workload

WHAT THIS SECTION COVERS

Cognitive (mental) workload definition, the four task factors that drive it, indirect factors and observable symptoms.

Workload = the amount of mental effort needed (and expended) to process information. Here we discuss **cognitive (mental)** workload, as opposed to physical workload. Workload is linked to almost all other areas within cognition and performance, particularly *attention, vigilance, fatigue, decision-making and multi-tasking*.

EFFECTS OF HIGH WORKLOAD

High workload is associated with **increased errors, fatigue, task degradation and poor performance**.

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3.1 Four General Task Factors Affecting Workload

1. 'Difficulty' of the task
2. Number of tasks running in parallel (concurrently)
3. Number of tasks in series (switching from task to task)
4. The time available for the task (speed of task)

Indirect factors also play a role: *duration of task, fatigue and level of arousal*.

3.2 Symptoms of Increasing Workload

- Attention and task focusing
- Task shedding and re-prioritization
- Implications for Situation Awareness
- Increased use of decision short-cuts and less scrutiny or review
- Increased fatigue and chance of error

The area that deals with all such activity is called the **working memory** ('Information Processing').

> **EXAM TIP — WORKLOAD DRIVERS**
>
> Mnemonic **"DPST"** — *Difficulty, Parallel tasks, Series switching, Time available*.

4. Information Processing — A Functional Model

> **WHAT THIS SECTION COVERS**
>
> How sensory inputs are converted into reasoned actions through five mental stages, and the central role of working memory.

We receive information from the world around us through our senses: **sight, hearing, touch, smell and taste**. The "Gestalt laws" formulate basic principles governing how objects are mentally organised and perceived.

4.1 The Five Stages of Reasoning

1. **Detection**
2. **Perception**
3. **Decision Making**
4. **Action** (responses are selected and executed)
5. **Feedback**

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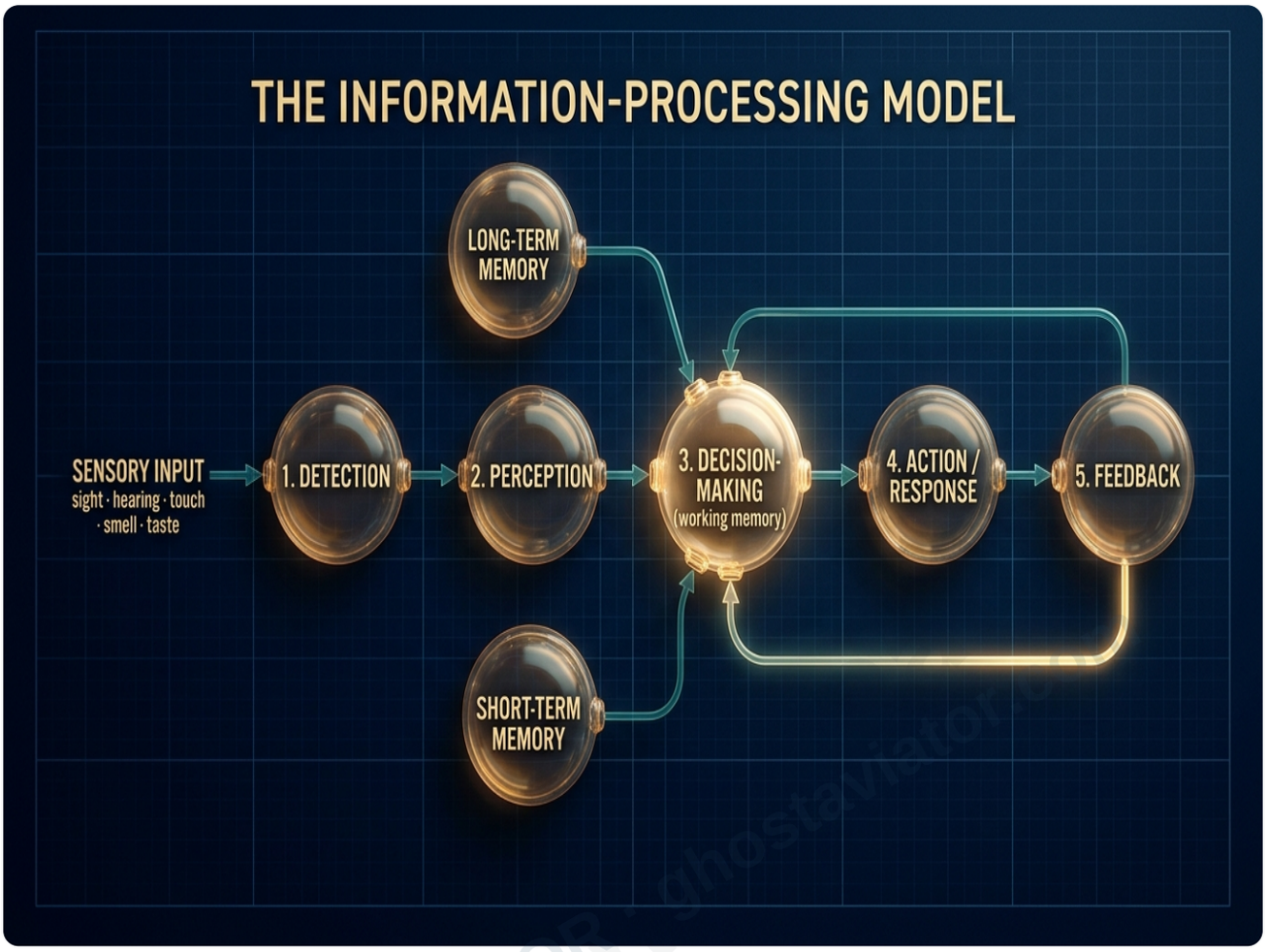

Figure 19.2 — The information-processing model: sensory input → detection → perception → decision → response, with feedback.

4.2 The Brain — Central Decision Maker & Response Selection

Once information has been perceived a decision must be made as to the response. Information is continuously entered into and withdrawn from both long-term and short-term memories to assist the decision process. The decision involves **input, processing, actions and feedback**. The information comes from many sources and requires *conscious processing* in working memory. A problem at any stage of information processing could affect the outcome.

To carry out multi-tasks we must learn skills through **Motor Programmes**.

4.3 Stimuli

The senses provide stimuli to our brain which has the ability to retain them for a short time, from the time they arrive. We may not have the processing capacity to deal with them all.

> **EXAM TIP — FIRST STAGE**
>
> First stage in the information process = **Sensory stimulation** (not "Attention", not "Perception"). Detection is what follows.

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✦ END OF CHAPTER 19 ✦

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HUMAN PERFORMANCE & LIMITATIONS · MODULE H — THE MIND ALOFT

Memory, Perception & Attention

Chapter 20 — The machinery of the mind: the three types of memory, how we anticipate and perceive, the limits of attention, vigilance and complacency, and how motor skills become automatic.

BOOK
Human Performance & Limitations

AUTHOR
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EXAM
DGCA CPL / ATPL — HPL

CHAPTER
20 of 26 · Module H

In this chapter

1. Types of Memory & Sensory Memories
2. Anticipation & Perception
3. Attention
4. Vigilance & Complacency
5. Motor Programmes (Skills)

GHOST AVIATOR · Capt. Pankaj Pahil · ghostaviator.com

Plate 20.0 — Attention is a single narrow beam, and memory is where the world is briefly held. Both have hard limits every pilot must respect.

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5. Types of Memory & Sensory Memories

WHAT THIS SECTION COVERS

Three memory types, key features of sensory memory, iconic memory for sight, and sensory adaptation.

5.1 The Three Types of Memory

- **Sensory memory**
- **Short-term (or working) memory**
- **Long-term memory**

5.2 Receptors & Sensory Memories — Key Features

- There is a **separate memory store for each sensory system**.
- The input **decays rapidly**.

The sensory stores for **sight and sound** are the most important, and knowledge of these is necessary.

5.3 Information Processing Model — Diagrammatic View

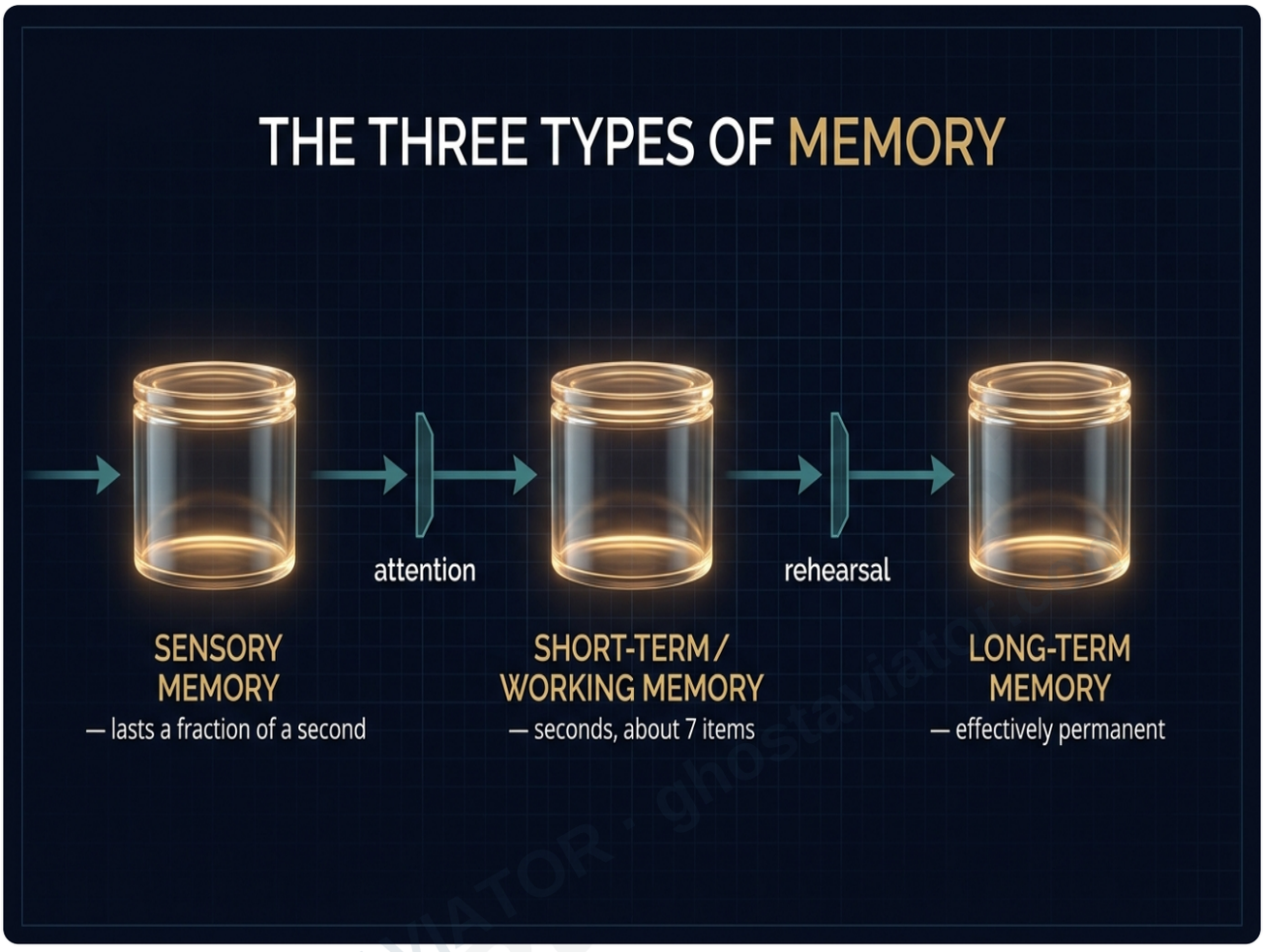

Figure 20.1 — The three memory stores: sensory memory (fractions of a second), short-term / working memory (seconds), and long-term memory.

5.4 Sensory Memory for Sight — Iconic Memory

The **Iconic Memory** is the visual sensory store. It only lasts for between 0.5 and 1 second.

70–80 % of the information processed by humans is received through the **visual channel**.

5.5 Sensory Adaptation (Habituation)

All senses adapt, either partially or completely, to their stimuli after a period of time.

> **EXAM TIP — SENSORY MEMORY VALUES**
>
> **Iconic memory = 0.5–1 second.** Visual channel carries **70–80 %** of all received information.

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6. Anticipation & Perception

WHAT THIS SECTION COVERS

How sensory input becomes meaning, the role of expectation, mental models and the concept of funneled perception.

6.1 Perception

Perception involves the **conversion of the sensory information received into a meaningful structure**.

Sensory information that we *expect* to receive is more easily perceived and integrated when it actually occurs, compared with totally unexpected information.

"We can 'perceive' only that which we can 'conceive', but we perceive only a fraction of the information reaching our senses at any moment."
The process of perception is greatly assisted by our ability to form **mental and three-dimensional visual models**.

6.2 Funneled Perception

Perception of a situation can differ depending upon the starting point of an observer.

HAZARD — CONFIRMATION BIAS / FUNNELING

In an abnormal situation, if a pilot has an apparently correct explanation for the problem, the chance that he/she now **ignores or devalues other relevant information** not fitting into the mental picture is *increasing*. Stay open-minded; verify, don't rationalize.

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7. Attention

WHAT THIS SECTION COVERS

Definition, the two types of attention, why an attention mechanism is required, the cocktail party effect and the dangers of attention mismanagement.

Attention is the process of directing and focusing psychological resources to enhance perception, performance and mental experience. To pay attention to something is to concentrate on it — a stimulus, threat, decision or calculation. Critically, **attention is limited**; if some attention is being used for one thing, it cannot be used for something else.

7.1 Types of Attention

1. **Selective Attention** — inputs are sampled continually to decide their relevance to the present task at hand.

2. **Divided Attention** — when our central decision-making channel can *time-share* between a number of tasks, executing several mental activities at almost the same time (e.g. switching attention from outside the aircraft to the airspeed indicator and monitoring the progress of a motor programme such as flying or taxiing on a relatively subconscious level, while making a radio call at the same time).

The human brain has **different reservoirs of resources** for information-gathering, information-processing, or the action phase.

7.2 Lack of Attention

The major danger for pilots is the **poor management of attention**.

7.3 Attention Mechanism — Why It Is Required

Because of two potentially limiting stages in processing information:

1. There is a limit to the number of items held or maintained in short-term memory (working memory).
2. Our channel capacity is limited.

7.4 Choice of Item

Attention is the deliberate devotion of cognitive resources to a specific item. Although attention can move very quickly from item to item, it can only deal with **one at a time** — so the pilot must consciously prioritize.

7.5 The Cocktail Party Effect

With many conversations occurring all around, one conversation can suddenly break through to attention because a word or phrase was particularly meaningful — the classic example is a person hearing their **own name** mentioned in another conversation that they were not paying attention to.

> **EXAM TIP — COCKTAIL PARTY EFFECT**
>
> Defined as the ability to *pick up relevant information unintentionally*. It demonstrates that the brain monitors even unattended channels for personally significant cues.

8. Vigilance & Complacency

> **WHAT THIS SECTION COVERS**
>
> Definition of vigilance, the three states (hypo, optimum, hyper), causes and signs, plus the trap of complacency in automated cockpits.

Vigilance = the capability to be sensitive to potential changes in one's environment, i.e. the capability to reach a level of alertness above a threshold for a certain period of time — rather than the state of alertness itself.

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8.1 Hypo (Low) Vigilance

Mainly due to the **monotony of the task, tiredness, the need for sleep and a lack of stimulation**. It may occur at any moment of the flight. Decrease in sensory perception and sensation of muscular heaviness are indications of low vigilance.

A state of *high fatigue* is liable to cause phases of drowsiness and hypo-vigilance, which result in a reduction of performance and unconscious phases of **micro-sleep**.

SOP — MANAGING HYPO-VIGILANCE

Healthy food and organising periods of rest during the flight can mitigate the effects of low vigilance.

8.2 Hyper Vigilance

In its most acute forms, an extremely **agitated state of panic or near panic**. Characterised by indiscriminate attention to all sorts of minor and major threat cues, as the person frantically searches for means of escaping the anticipated danger.

HAZARD — HYPER VIGILANCE

Hyper vigilance is a state of **panic** that often results in *regrettable decision-making*.

8.3 Optimum Vigilance

Lies somewhere between hypo and hyper vigilance.

8.4 Complacency

A state of **self-satisfaction** with one's own performance coupled with an **unawareness of danger, trouble or controversy**.

HAZARD — COMPLACENCY

Complacency is **extremely dangerous in aviation**. You get so used to things being done the same way that you become less vigilant to hazards in your surroundings. **Automation may make pilots complacent due to low workload.**

GHOST AVIATOR ghostaviator.com Parimal Pahil

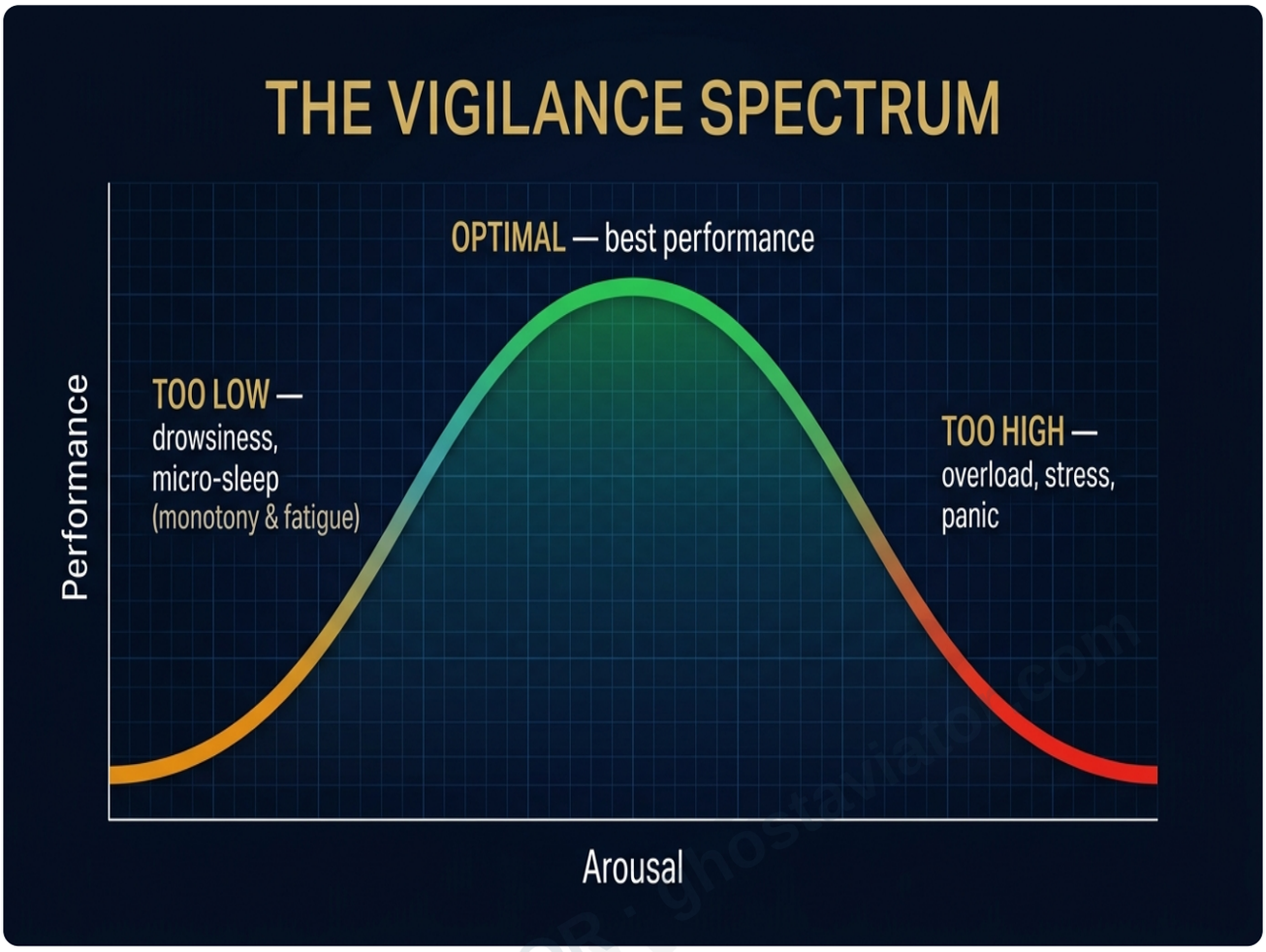

Figure 20.2 — The vigilance spectrum: too little arousal brings drowsiness and micro-sleep; too much brings overload — performance peaks in between.

9. Motor Programmes (Skills) — Anderson Model

WHAT THIS SECTION COVERS

What motor programmes are, the three phases of skill development per the Anderson model, and how skill enables resource management.

Motor Programmes, or "Skills", are behavioural sub-routines which are **learnt by practice and/or repetition** and are **held within the Long-Term Memory**. They can be carried out without conscious thought.

In the *initial phase* of flight training the pilot is competent enough to fly the aircraft at this stage but does *neither* have a great deal of confidence in his/her abilities *nor* in the whole system.

WHAT MAKES A PILOT "SKILLED"

A pilot is skilled when he trains or practices regularly, knows how to manage himself/herself and knows how to **keep resources in reserve for coping with the unexpected**.

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9.1 Anderson Model — Three Phases

1. **The Cognitive Phase**
2. **The Associative Phase**
3. **The Automatic (Autonomous) Phase**

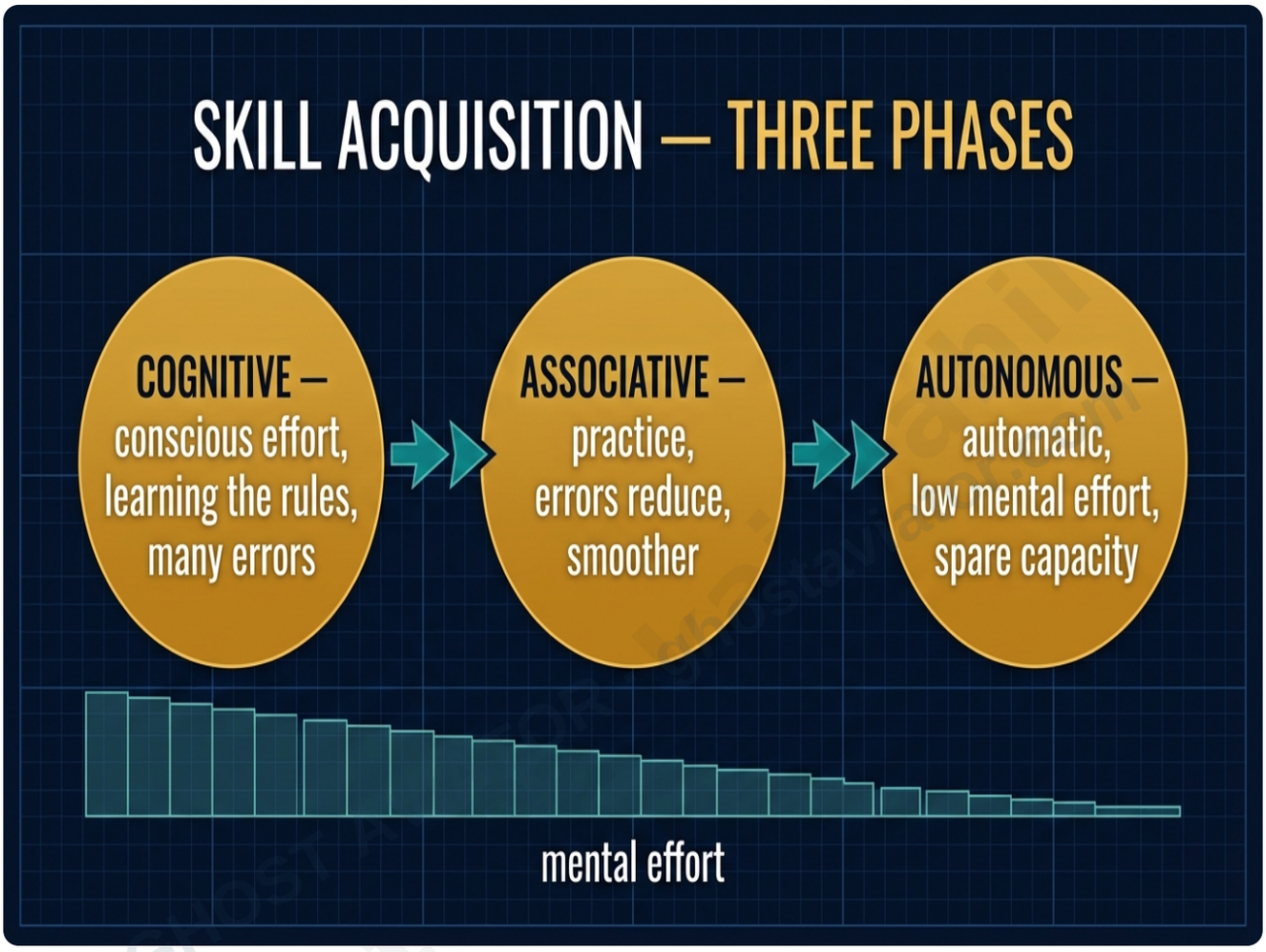

Figure 20.3 — Anderson's model of skill acquisition: cognitive (conscious effort) → associative (practice) → autonomous (automatic).

EXAM TIP — "FLYING A COORDINATED TURN IS …"

Skill-based behaviour — not attention-based, not experience-based. It draws from procedural memory in the LTM.

✦ END OF CHAPTER 20 ✦

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HUMAN PERFORMANCE & LIMITATIONS · MODULE H — THE MIND ALOFT

Skills, Reaction & Learning

Chapter 21 — Speed and depth of the mind: reflexes, reaction time and the startle response; quantitative overload and human reliability; short- and long-term memory; learning, retention and motivation; and the traps of cognition and hallucination.

BOOK
Human Performance & Limitations

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EXAM
DGCA CPL / ATPL — HPL

CHAPTER
21 of 26 · Module H

In this chapter

1. Reflexes, Reaction Time, Surprise & Startle
2. Quantitative Overload & Human Reliability
3. Short-Term & Long-Term Memory
4. Learning, Retention & Motivation
5. Experience, Response, Cognition & Hallucination

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Plate 21.0 — Between the alarm and the hand on the control lies reaction time — and the startle that can freeze it.

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Capt. Pa

10. Reflexes, Reaction Time, Surprise & Startle

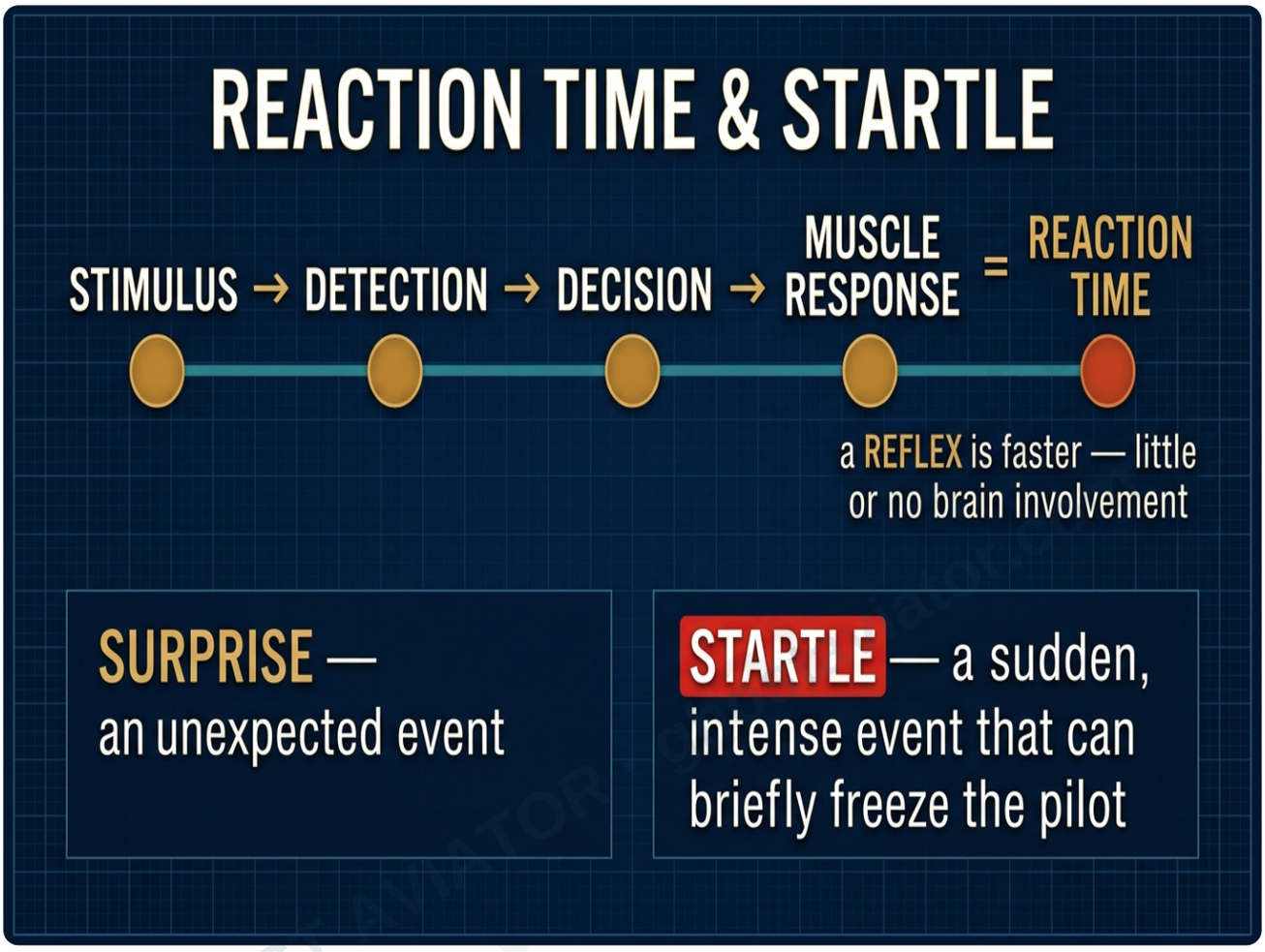

Figure 21.1 — Reflex vs reaction time, and the difference between surprise (unexpected) and startle (sudden and intense) — both steal precious seconds.

WHAT THIS SECTION COVERS

Reflex types, what reaction time depends on, and the operational impact of the startle reflex on pilots.

Reflexes occur with little or no involvement of the central nervous system.

10.1 Reaction Time

There is a delay between detection, stimulus and muscle contraction called **reaction time**. Reaction time depends on the type of reflex action being used.

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10.2 Three Types of Reflex Actions

Type	Description	Example
Unconditioned	Instinctive natural reflexes	Blinking
Conditioned	Reflexes that may be *learned*	Foot-on-rudder on swing
Trained	Reflexes that may be *increased by repeated practice*	Stall recovery inputs

10.3 Surprise and Startle

The **startle reflex** is a reflex-like event that blinks the eyes and causes a whole-body 'jerk' to occur (similar to that sometimes caused in sleep). The reflex has a relatively basic neural pathway from the sense organ. Many things can cause (or contribute to) a startle reflex:

- Sudden noises
- Unexpected tactile sensations
- Abrupt shocking perceptions
- The sensation of falling
- An abrupt visual stimulus

There is little evidence that a startle reflex alone creates much of a sustained or lasting impact on cognitive functions, although there are some minor and short-lived physiological changes such as *raised heart rate*.

> **KEY VALUE — STARTLE RECOVERY**
>
> A skilled motor task will be momentarily disrupted by a startle reflex but return to normal within 5 to 10 seconds.

10.4 Effects on Pilots

The main effects of the pure startle reflex on pilots are:

- **Interruption** of the on-going process.
- **Distraction of attention** toward the stimulus.

These happen almost immediately and can be quickly dealt with if the cause is found to be *non-threatening*.

> **HAZARD — FEAR POTENTIATES STARTLE**
>
> A perception of fear can cause a startle reflex to be **potentiated** (more pronounced) should it occur, and attention to become more focused. In a state of fear, very little is required to trigger a full **'fight or flight'** response.

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11. Quantitative Overload & Human Reliability

WHAT THIS SECTION COVERS

When workload is acceptable, what overload looks like, error rates with practice, and the cumulative nature of errors.

Quantitative Overload — a very high workload can be interpreted as **'Stress'**.

SOP — ACCEPTABLE WORKLOAD

Workload may be said to be acceptable if it requires about 60 % of the crew resources, depends on the pilot's expertise and corresponds to the amount of resources available.

11.1 Most Common Symptoms of Overload

- A sharp degradation of performance
- Funneling of attention or focus
- Regression
- Mental "blocking"
- Mood swings
- Restlessness and trembling
- Panic

11.2 Human Reliability

The rate of human error during simple and repetitive tasks might be expected to be 1 in 100, but after practice, a rate of 1 in 1,000 could be achieved.

DEFINITION — HUMAN RELIABILITY

The individual functioning in the manner in which he or she is supposed to function.

11.3 Error Generation

Errors tend to be cumulative. One slip rarely stays isolated — it sets up the next.

Worked Example — Human Error Rate A pilot performs 200 routine pre-flight checks per week.
• Before practice (1 in 100): expected errors ≈ 200 ÷ 100 = **2 errors / week**.
• After practice (1 in 1,000): expected errors ≈ 200 ÷ 1,000 = **0.2 errors / week** (i.e., about 1 every 5 weeks).
Practice yields a 10-fold reduction in error rate, but never reaches zero — hence the value of cross-checking and SOPs.

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12. Short-Term & Long-Term Memory

WHAT THIS SECTION COVERS

Duration and capacity of short-term memory, methods to expand it, the three subtypes of long-term memory and its capacity characteristics.

12.1 Short-Term Memory (Working Memory) — 10 to 20 Seconds

Short-term memory enables information to be retained for a short period of time. That information will be lost in 10 to 20 seconds unless it is actively rehearsed and deliberately placed in our long-term memory.

Auditory information is considered easier to retain than visual information as it is easier to rehearse sounds than data in a visual form.

12.2 Limitations of Short-Term Memory

The capacity of our short-term memory is limited. The maximum number of unrelated items which can be maintained in the short-term memory is about 7 ± 2. Once this limit is exceeded one or more of the items are likely to be lost or transposed.

12.3 Methods of Increasing Short-Term Memory

Method	Description
Chunking	Breaking items to be remembered into small pieces and remembering them one at a time.
Association	Used to remember spoken lists of items. A wild and bizarre association is imagined and attached to each item on the list.

Worked Example — Chunking an ATC Clearance Raw clearance string: `"VTBL VTBS RWY01L SQK4732 ALT FL310 HDG270"` — 7+ unrelated tokens, near STM limit.
Chunked: `"VT-BL → VT-BS"` (route), `"RWY 01L"` (departure), `"4732"` (squawk), `"FL310"` (level), `"HDG 270"` (heading) = **5 chunks** — comfortably within 7 ± 2.

12.4 Long-Term Memory — Unlimited Time Period

It is believed that information is stored in the Long-Term Memory for an **unlimited time period**, although frequently there may be *retrieval problems*.

GHOST AVIATOR Capt. Parthiban Parthiban Pandi Pandi ghostaviator.com

12.5 Three Subtypes of Long-Term Memory

Type	Definition	Example
Semantic Memory	Stores general knowledge of the world	Knowing that "QNH is altimeter setting"
Episodic Memory	Memory of events or 'episodes' in our life	Recalling your first solo flight
Procedural Memory	Skills are included within the make-up of the LTM	Executing a coordinated turn

EXAM TIP — MEMORY QUICK FACTS

- STM duration = **10–20 seconds**, capacity = **7 ± 2** items.
- LTM capacity = **unlimited**, but retrieval problems may exist.
- Motor programmes live in **LTM (Procedural)**, not STM.

13. Learning, Retention & Motivation

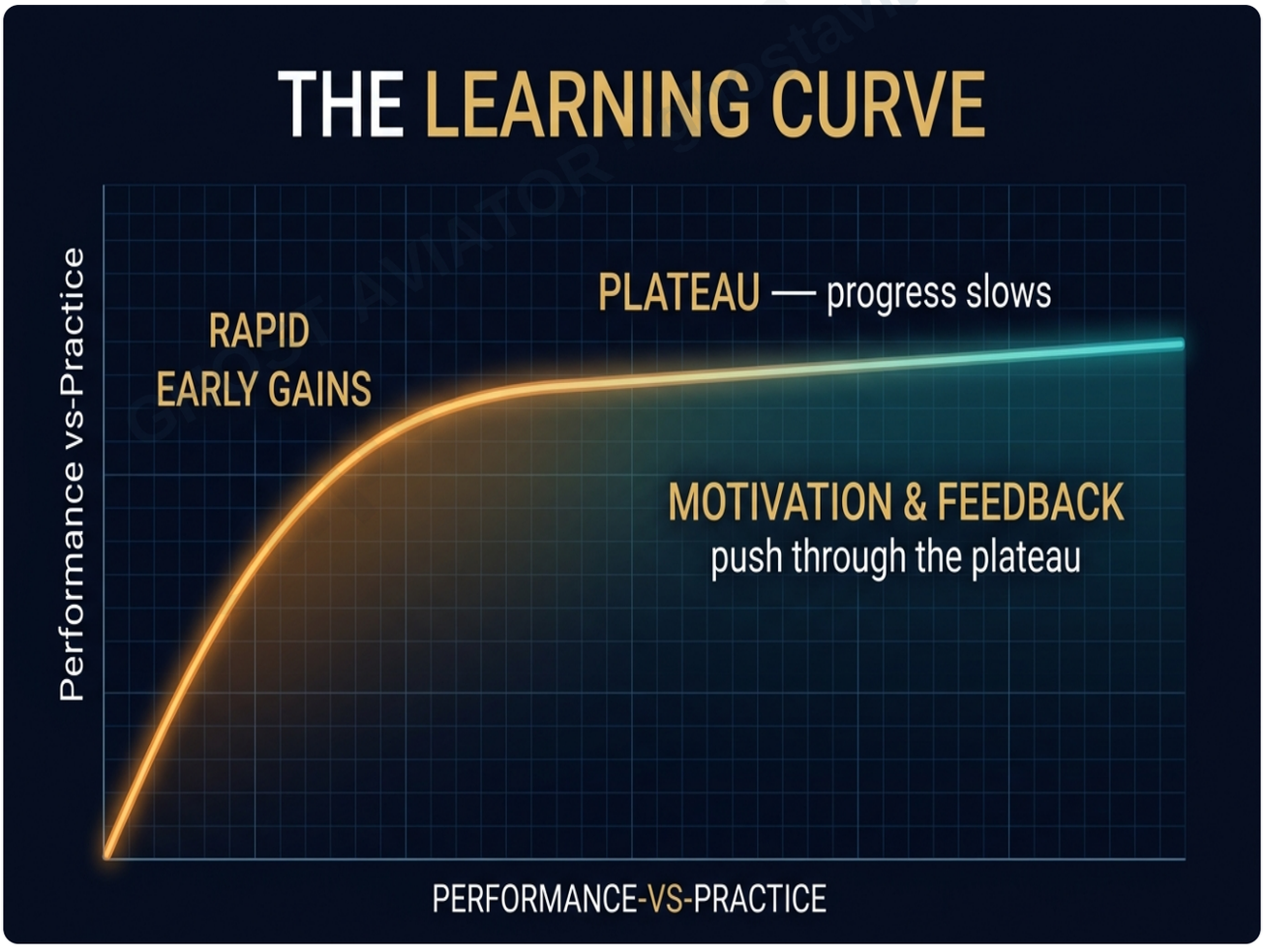

Figure 21.3 — The learning curve: rapid early gains, then a plateau; motivation and feedback drive it forward.

Website YouTube Instagram
Telegram channel Telegram community

WHAT THIS SECTION COVERS

Types of learning, methods to retain information, and the role and sources of motivation.

13.1 The Learning Process

Learning is an **internal process** which allows the mental acquisition and retention of data. Types include:

Insight

The data is intellectually and cognitively understood and is retained. *Observational Learning / Imitation* — data from an outside source is replicated. *Experience* — learning from our mistakes.

Skill Learning

Involves **motivation, attention, observation, much practice and corrective feedback**.

13.2 Retention of Information

Retention can be increased by:

1. **Mnemonics** — the practice of improving or helping the memory, or the systems used to achieve this.
2. **Memory Training**

13.3 Motivation

Motivation is the combination of a person's desire and energy directed at achieving a goal. It is the **cause of action**.

Type	Examples
Intrinsic	Satisfaction and feelings of achievement
Extrinsic	Rewards, punishment, and goal obtainment

Not all people are motivated by the same thing and over time their motivations might change.

SOP — MOTIVATION AND PERFORMANCE

The learning process is vastly improved with motivation, and **high performance is rarely achieved without it**.

14. Experience, Response, Cognition & Hallucination

WHAT THIS SECTION COVERS

How experience shapes us, response characteristics with age, error of commission, response times, cognition in flight, and hallucinations.

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14.1 Experience

We all have the ability to learn from our experiences and mistakes, and from those of others.

14.2 Response

Any response will cause a detectable change which, in turn, will be noted by the senses. The feedback may alter the action being taken.

- **Auditory stimuli (noises)** are more likely to attract attention than visual stimuli, but they are also more likely to be responded to in error.
- An increase in age between 20 and 60 years tends to lead to *slower but more accurate* responses.

14.3 Response Error (Error of Commission)

If an unexpected stimulus occurs, we will be more likely, under pressure, to make an error of commission.

14.4 Response Times

Response to reaction time is the interval between the onset of a given signal and the production of a response to that signal.

14.5 Cognition in Aviation

Cognition = the mental faculty or process of **acquiring knowledge by the use of reasoning, intuition or perception**. In aviation, flight puts the pilot into an environment which can **distort sense organs**, and the changed perspective which is experienced in flight can result in information being presented which is *outside the individual's expectations*.

14.6 Hallucination

A hallucination is actually a **false perception characterised by a distortion of real sensory stimuli**. (Not a fabrication of stimuli — the stimuli are real, but their perception is distorted.)

14.7 Workload and Limitations

Too high *or* too low a workload can result in degraded performance. Several types of situations may cause mental overload.

> **EXAM TIP — AGE AND RESPONSE**
>
> Between **20 and 60 years** → responses become *slower but more accurate*. Experience compensates for slower reflexes.

✦ END OF CHAPTER 21 ✦

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Judgment, Automation & the Cockpit

Chapter 22 — The pilot, the crew and the machine: judgment and decision-making, personality, attitude and situational awareness, communication and crew resource management, automation, and the human-centred design of the cockpit and its displays.

BOOK
Human Performance & Limitations

AUTHOR
Capt. Pankaj Pahil

EXAM
DGCA CPL / ATPL — HPL

CHAPTER
22 of 26 · Module H

In this chapter

1. Judgment & Decision Making
2. Personality, Attitude & Situational Awareness
3. Mechanics of Decision Making
4. Communication & Team Work (CRM)
5. Automation
6. Human Characteristics & Cockpit Design
7. Hardware — Displays & Controls

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Plate 22.0 — Good decisions and good design share one goal: to fit the machine to the human, not the human to the machine.

GHOST AVIATOR Capt. Pat

15. Judgment & Decision Making

WHAT THIS SECTION COVERS

Why decisions matter, how to seek advice, risk assessment fundamentals, and the dynamic nature of "good" decisions.

Making good decisions is one of the important aspects of piloting an aircraft. **Good decisions** can help lead to safe, successful flights, whereas **bad decisions and even indecision** have led to many aircraft accidents. Talk to your instructor and other pilots about different flying situations to obtain advice about what a good decision would be in a given circumstance.

When flying with another pilot, it is possible to increase the chance of a good decision being made by **discussing the potential problems beforehand**. If as an inexperienced pilot you are flying with someone of more experience and you see him doing something you consider to be dangerous, you should **immediately question their course of action**.

BEST PRACTICE — ALWAYS MORE TO LEARN

A good pilot is one who always realizes there is more to learn. When making a decision a pilot will be influenced by his **previous experience**, the **probability of an occurrence**, and whether the information he is receiving **matches that expected**.

15.1 Risk Assessment

Risk assessment is based on:

- The **probability of the risk occurring**.
- The **impact of that risk** if it did occur.

You will have to weigh up the risk and the possible consequences of the risk in order to determine your course of action.

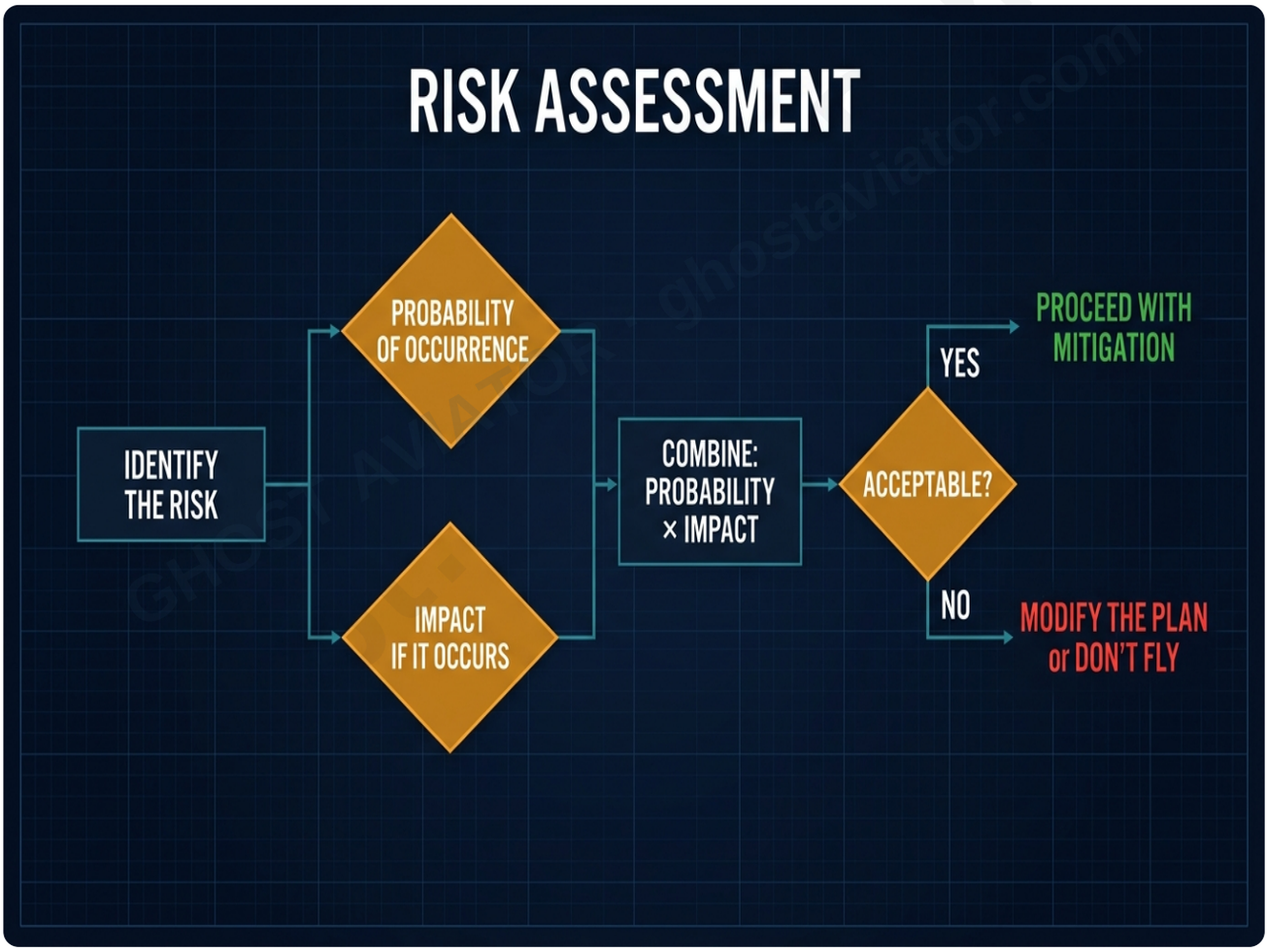

Figure 22.1 — Risk assessment: weigh the probability of an event against its impact to decide how to manage it.

REMEMBER

Making good decisions is one of the most important aspects of piloting an aircraft. Good decisions can help lead to safe, successful flights, whereas bad decisions and even indecision have led to many aircraft accidents. **Always strive to make good decisions.**

16. Personality, Attitude & Situational Awareness

WHAT THIS SECTION COVERS

Desirable pilot traits, the meaning of being Pilot-in-Command, factors that destroy situational awareness, and how to maintain it.

16.1 Personality, Attitude and Behaviour

Someone who is **too introverted or anxious may not make a good pilot**. Examples of desirable characteristics in a pilot are:

- The ability to cope with stress and adventure.
- The ability to assess and control risks.

As captain of an aircraft, you will need to show **good leadership skills**.

SOP — BEING PILOT-IN-COMMAND

While acting as pilot-in-command you should ensure that you **stay in command** — you should be **properly organized and prepared**.

16.2 Situational Awareness

A good leader displays good situational awareness and accurately assesses his own performance. The following factors can interfere with accurate situational awareness:

1. Stress
2. Boredom
3. Fatigue
4. Emotional disturbance
5. Poor communication
6. Interruptions

MAINTAINING GOOD SA — SOP

In order to maintain good situational awareness, **gather as much information as possible**. Do not rush into making a decision. Make sure you consider all the options. **Do not make the mistake of seeing something that is not really there just because you want it to be there.**

EXAM TIP — SA INTERFERENCE MNEMONIC

"SBF-EPI" → *Stress, Boredom, Fatigue, Emotional disturbance, Poor communication, Interruptions*.

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17. Mechanics of Decision Making

WHAT THIS SECTION COVERS

The seven-step decision cycle and the dynamic nature of decisions in flight.

Pilots operate in a **dynamic and constantly changing environment**. A good decision reached a minute ago will *not necessarily* be the same good decision in two minutes' time.

17.1 The Seven Steps

1. **Recognize and identify the problem** — have a clear definition of what needs addressing.
2. **Consider the nature of the problem** — type of issue, why it needs a decision, hoped-for results.
3. **Analyze or research the problem** — gather all information so informed choices can be made.
4. **Develop a list of possible solutions** — list decisions and their consequences.
5. **Select the best alternative** — choose the best solution for the situation.
6. **Execute the best choice** — taking action is often the hardest part.
7. **Follow up and communication** — keep the process going with constant communication; **Feedback** determines the overall success of and reaction to the decision.

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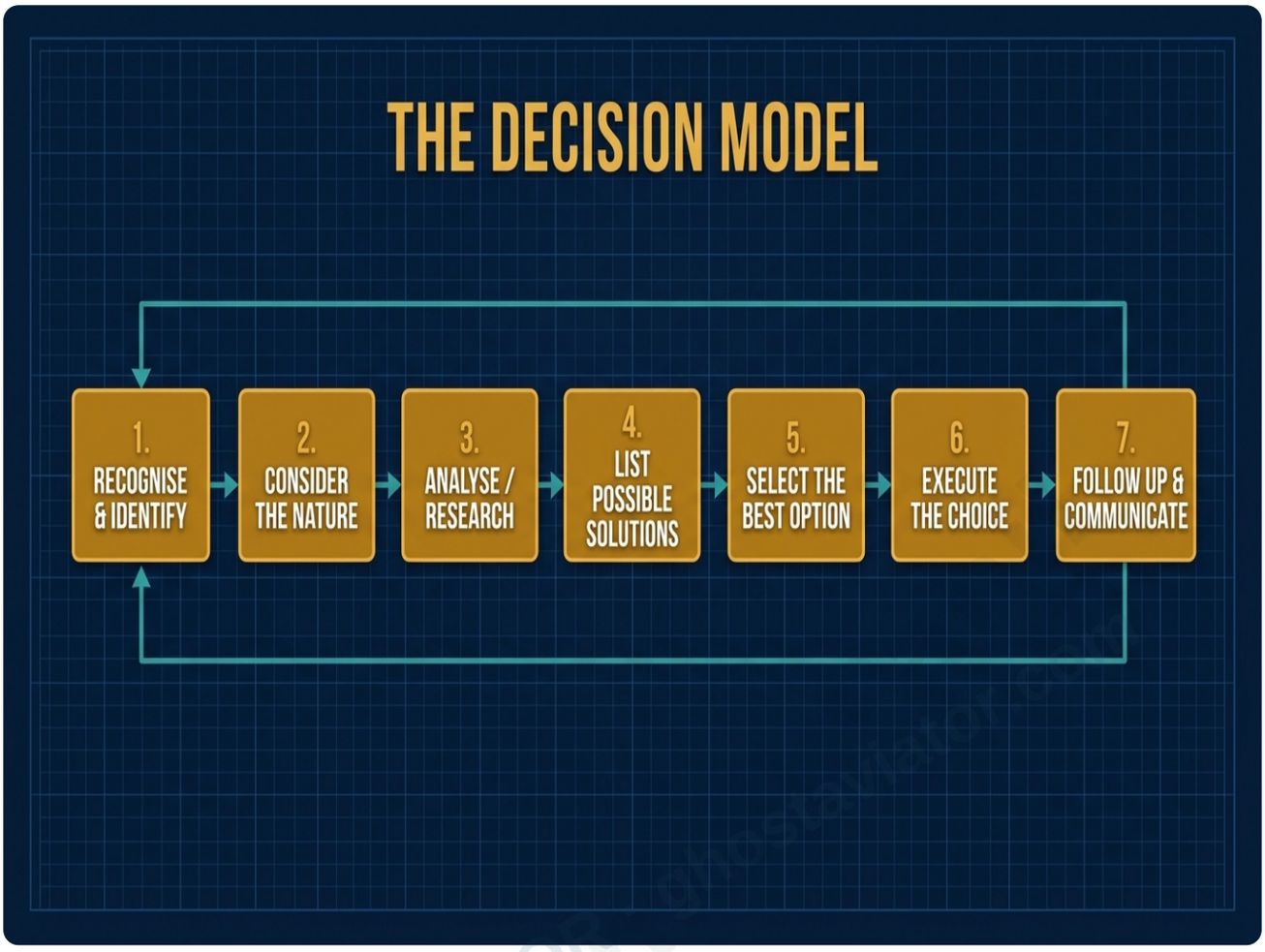

Figure 22.2 — A structured decision model: recognise, consider, analyse the options, decide, act, and review.

EXAM TIP — DYNAMIC DECISIONS

A good decision a minute ago may not be a good decision two minutes from now. **Re-evaluate continuously.**

18. Communication & Team Work

WHAT THIS SECTION COVERS

Why communication matters, attitudes that destroy teamwork, and the danger of "expectation".

When piloting an aircraft — whether with a passenger, another pilot or an instructor — **good verbal communication** will help the flight go more smoothly and contribute to flight safety.

Concise and unambiguous communication is essential to the safe conduct of air traffic.

18.1 Barriers to Communication & Team Work

Several things can be done by team leaders to facilitate good teamwork. *In the end, it is the team leader who takes the decisions on behalf of the team.*

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HAZARDOUS ATTITUDES — BARRIERS

Some of the barriers to communication and teamwork are:

- Aggressiveness
- Arrogance
- Anti-authoritarian behaviour
- Impulsiveness
- Feelings of invulnerability
- Resignation

18.2 Expectation

HAZARD — EXPECTATION BIAS

Do **not assume** you have heard what you are expecting to hear. **Actively listen.** Clear, concise and unambiguous communication is essential to the safe conduct of aviation.

19. Automation

WHAT THIS SECTION COVERS

Definition and evolution of automation, the mismatch between automation and the human element, and qualities of human-centred automation.

Automation refers to a system or method in which many of the processes of production are automatically performed or controlled by self-operating machines, electronic devices, etc.

Through microprocessor technology, navigational tasks and aircraft system management have been automated, making the flight crew more *peripheral* to the actual operation of the aircraft. Pilots who at one time had direct authority over all aspects of aircraft control and management have now become responsible for the **management of complex hardware and software interfaces**.

These technological advances have given rise to **new forms of error**. Automation is almost always introduced with the expectation of reducing human error and workload, but what frequently happens is that **the potential for error is simply relocated**. More often than not, automation does not replace people in systems; rather, it places the person in a different, and in many cases, *more demanding* role.

19.1 Definition of Automation

"The technique of controlling an apparatus, a process or a system by means of electronic and/or mechanical devices that replaces the human organism in the sensing, decision-making and deliberate output."

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19.2 Evolution of Transport Aircraft Automation

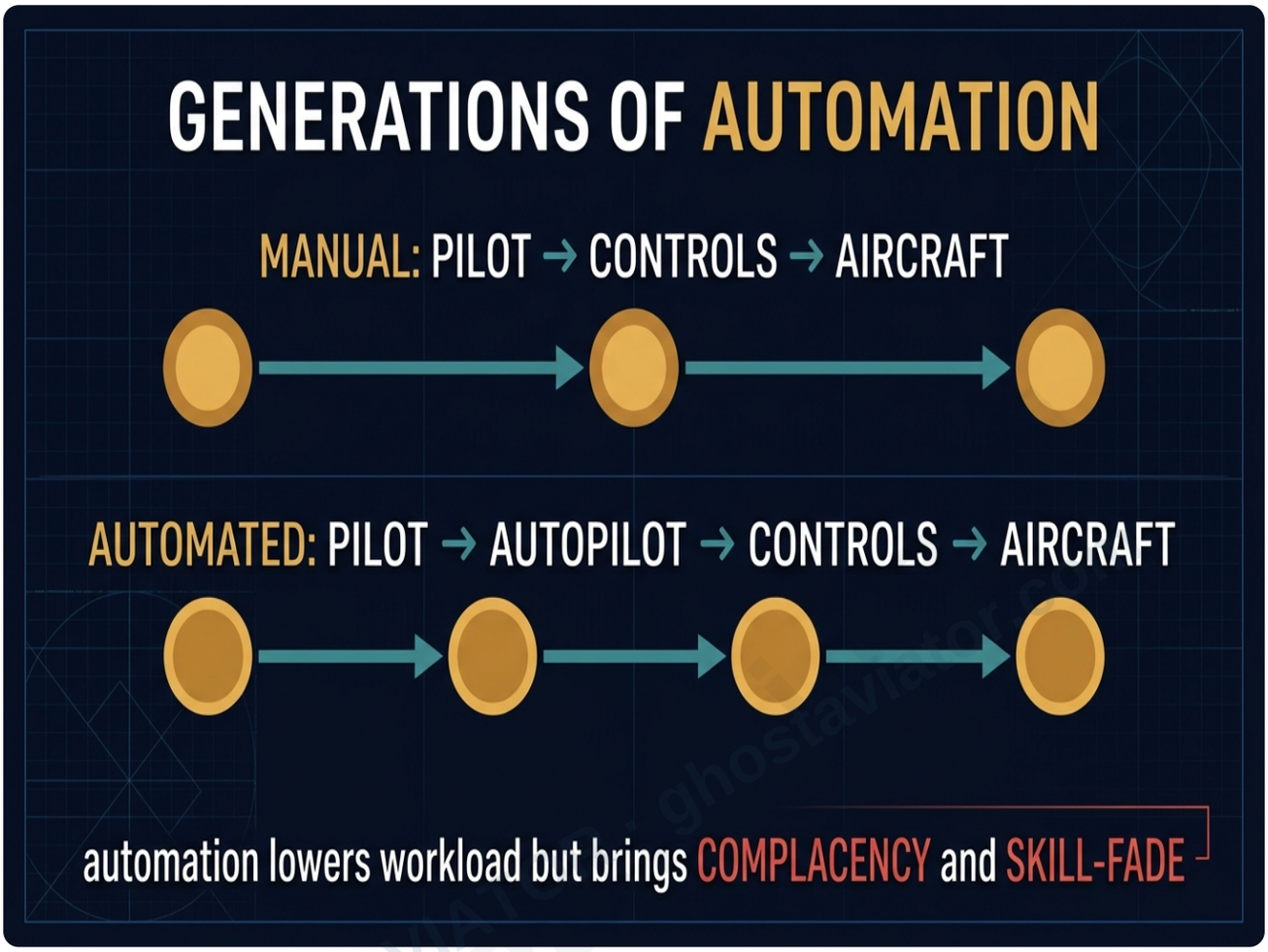

Figure 22.3 — Automation moves the pilot from direct control toward supervising the automatics — improving workload but bringing complacency and skill-fade.

19.3 Mismatch Between Automation and the Human Element

Nevertheless, a few accidents point to a mismatch between automation and the human element. Studies identified **nine categories** to focus on:

1. Situational awareness
2. Automation complacency
3. Automation intimidation
4. Captain's command authority
5. Crew interface design
6. Pilot selection
7. Training and procedures
8. The role of pilots in automated aircraft
9. (Studies concluded scope for improvement in fields like: human capabilities and limitations, ergonomics, cognitive suitability and instrument standardization.)

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HAZARD — AUTOMATION SIDE-EFFECTS

Although cockpit automation may provide pilots with more time to think, it may encourage pilots to reinvest only some of this mental free time in flight-related thoughts. While manual workload has reduced, **mental workload has not been reduced by the same amount — in fact it may have been increased**. Automation may not always reduce workload in phases of flight in which it is usually high (e.g. arrivals/landings at busy terminals). High cockpit automation may reduce pilot attention with the consequence of **'being out of the loop'**.

HAZARD — RELIABILITY ≠ SAFETY

All forms of automated assistance for the human operator must be highly reliable, but this may also induce **complacency**. Human expertise may gradually be lost and if the machine fails, the human operator may accept an inappropriate solution or become unable to formulate a satisfactory alternative.

19.4 Qualities of Human-Centred Automation

Human-centred automation should possess these qualities in proper measure:

- Accountable
- Subordinate
- Predictable
- Adaptable
- Comprehensible
- Flexible
- Dependable
- Informative
- Error-resistant
- Error-tolerant

BEST PRACTICE — COMPLEMENTARY, NOT ALTERNATIVE

Pilots, computers and machines are **not alternatives, but complementary factors** in ensuring flight safety. Achieving the correct balance benefits safety. Automation has undeniably led to an improvement in flight safety.

20. Human Characteristics & Cockpit Design

WHAT THIS SECTION COVERS

Personality types, cultural differences, anthropometry, eye datum, cockpit seats and the design considerations driving them.

20.1 Personality

Personality involves two major factors:

- **Neuroticism – Stable** continuum
- **Introversion – Extroversion** continuum

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Extroverts are said to be impulsive and sociable; **introverts** are more withdrawn and cautious.

A low-neuroticism personality is one of an emotionally stable person, whereas a high-neuroticism person will worry and get upset easily.

> **TRAIT VS STATE ANXIETY**
>
> **High neuroticism = high trait anxiety** (a personal trait of that person). As opposed to **state anxiety**, which is a *transient state of anxiety present in anyone at any time*.

20.2 Cultural Differences

The way people behave, think and interact with each other generally — as well as what motivates them — will be partly a factor of their national and cultural background. One culture may value subordinates speaking up, whereas another culture may value subordinates who obey their superiors unquestioningly. The latter might perceive a subordinate who points out a concern as being impolite, aggressive or disruptive.

20.3 Competencies

The main area where personality and background culture are important is **interpersonal interaction**. In terms of competencies, this means **communication, leadership and teamwork** — crew interactions or interactions with outside agencies such as air traffic control.

20.4 Design Considerations

The following human characteristics must be taken into account in the design of aviation systems:

- Physical dimensions
- Capability for data sensing
- Capability for data processing
- Capability for motor activity
- Capability for learning
- Physical and psychological needs
- Sensitivities to physical environment
- Sensitivities to social environment
- Coordinated action
- Differences among individuals

20.5 Hardware, Design of Flight Decks, Size & Anthropometry

> **MURPHY'S LAW**
>
> "*If equipment is designed in such a way that it can be operated wrongly, then sooner or later, it will be.*"

Therefore a great deal of attention is required to design the equipment. **The most important requirement in the design of both displays and controls is STANDARDIZATION.**

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20.6 Anthropometry

This is the study of **human measurement**. The information is grouped into:

- Static measurements
- Dynamic measurements
- Contour surface measurements

It is not practical to design a cockpit for both the very short and the very tall individual. Those in the **central 90 % of size distribution** will be catered for. Normally the design of aircraft uses measurements taken from the entire population **disregarding both the 5 % lowest and the 5 % highest**.

20.7 Eye Datum

Cockpit space must be designed around a defined position of the pilot's eye. This may be called the **Eye Datum**, the **Design Eye Position** or the **Reference Eye Point**.

Position	Effect
Too high	Poor view of instruments, obstructed high view, good downward view
Just right	Optimum view of outside and instruments
Too low	Good view of instruments but poor forward and downward view

Once the design eye position has been set, the size of the cockpit can be established.

20.8 Design of Cockpit Seats

It is of the utmost importance that the seating is comfortable and adjustable to the individual pilot's size and shape. Pilots should adjust their seats to establish a comfortable position giving full control movement, with **optimum instrument scan and outside visibility**. This position should be used for **all phases of the flight**.

SOP — RESTRAINT

Restraint should be provided by a **5-point harness with a negative 'G' strap**.

21. Hardware — Displays & Controls

WHAT THIS SECTION COVERS

Digital vs analogue display selection, the standard "T" layout, three-pointer altimeter pitfalls, and the eight basic considerations for control design.

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21.1 Presentation Requirements

Information Type	Best Display
Purely quantitative information	**Digital**
Qualitative information	**Analogue** (more easily assessed)

21.2 Conventional Analogue Standard 'T' Display

A standard 'T' layout has the **artificial horizon / attitude indicator** at the centre with the **altimeter, airspeed indicator and direction indicator** grouped around it. **Digital Display** and the **Compass**. The conventional analogue type of compass card gives a better picture of the aircraft orientation than would a digital readout.

21.3 Combination or Analogue and Digital Displays

In some instances, both digital information and analogue information can be combined in a single instrument.

> **HAZARD — THREE POINTER ALTIMETER**
>
> The three-pointer altimeter can easily be **misread** and produces significantly **more reading errors** than the single-pointer altimeter.

21.4 Hardware, Controls — Basic Considerations

Principle	Description
Standardization	Control of location and sense of use from one aircraft to another should be standardized.
Frequency of Use	Controls used frequently or for protracted periods should be located so that they do not require the pilot to adopt an awkward or fatiguing position.
Sequence of Use	Controls that should normally be used in a given order should be laid out so that the sequence of use is represented in that layout.
Importance	Important controls must be located in easily reached and unobstructed positions.
Visual/Tactile Dissimilarity	Switches and knobs that control different systems or functions should look and feel different from each other.
Symbolism	Controls, if possible, should be designed to contain some reference to their function.
Simultaneous Use	Controls which may require simultaneous use should be located to enable this to take place.
Warnings	All warnings should be 'attention getting' without being startling. The most conspicuous visual warnings rely on **head and gaze orientation**.

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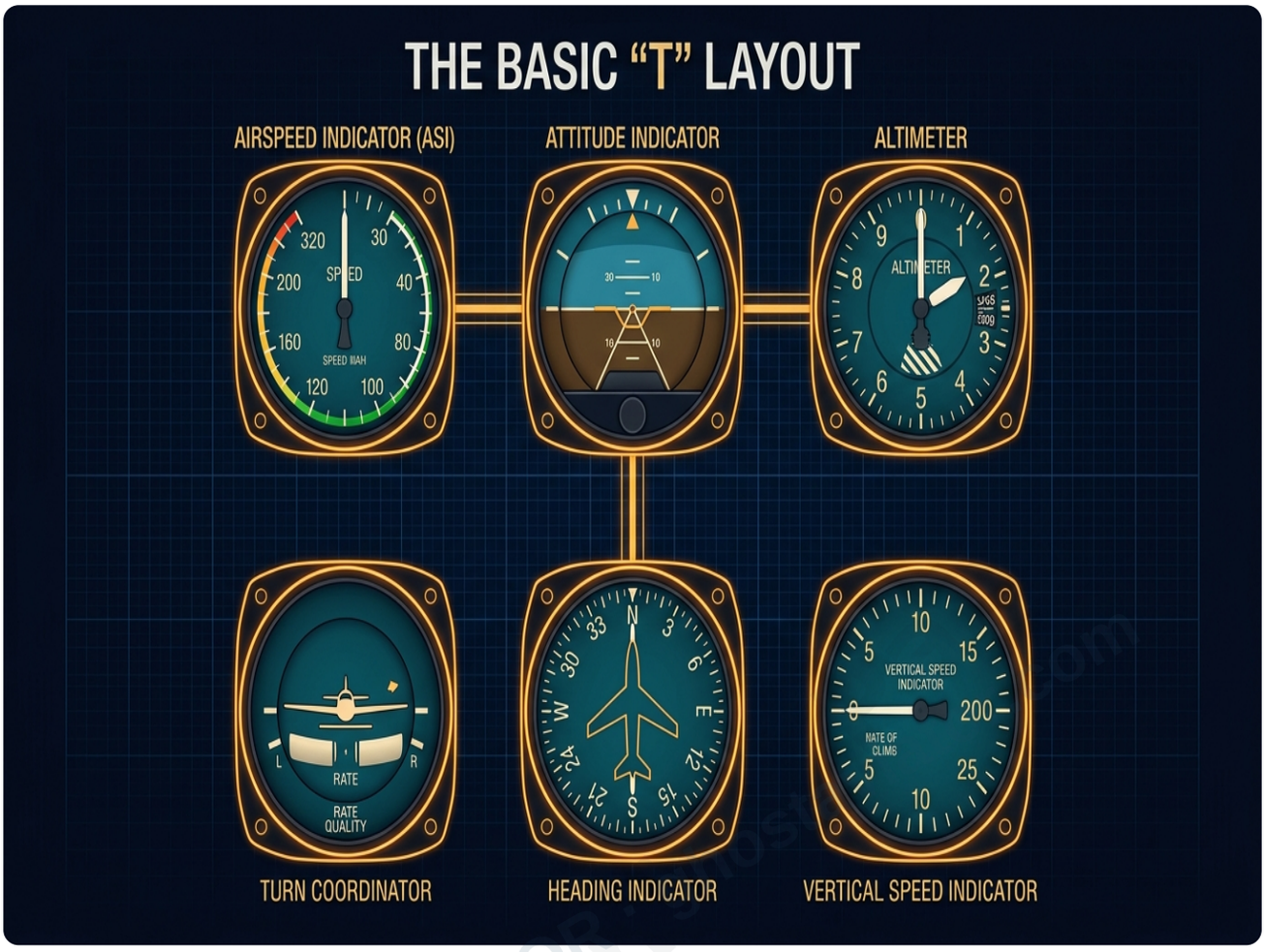

Figure 22.4 — The basic 'T' layout: attitude indicator on top, with airspeed, altimeter and heading arranged around it for a fast, standard scan.

✦ END OF CHAPTER 22 ✦

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HUMAN PERFORMANCE & LIMITATIONS · MODULE I — RHYTHMS & RESILIENCE

Stress, Fatigue & Arousal

Chapter 23 — The well-being of operational personnel: arousal and performance, the sources and effects of stress, environmental tolerances, managing stress, life-event scoring, and the danger of acute and chronic fatigue.

BOOK
Human Performance & Limitations

AUTHOR
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EXAM
DGCA CPL / ATPL — HPL

CHAPTER
23 of 26 · Module I

Plate 23.0 — Stress sharpens the mind — until it doesn't. Beyond the peak, attention narrows and fatigue erodes every margin.

22. Stress Tolerance, Fatigue & Arousal — Well-Being of Operational Personnel

WHAT THIS SECTION COVERS

Stress, arousal, fatigue, environmental tolerances, life stress scoring, stress management, and the arousal-performance relationship.

Three of the many factors which may influence the well-being of operational personnel are:

1. Stress and fatigue
2. Body rhythm disturbance
3. Sleep deprivation or disturbance

Other factors affecting physiological or psychological well-being include: *temperature, noise, humidity, light, vibration, workstation design and seat comfort*.

22.1 Stress, Arousal and Performance

The different stress levels generated within individuals by a particular stressor will differ. A **moderate level of stress may improve performance**. Stress promotes an **increase in physical strength** rather than promoting mental performance.

- The response of a person to the event or circumstances to which he is exposed is known as **arousal**.
- Arousal is a measure of a person's **readiness to respond effectively** to a given stress factor.
- Personality, character, training and experience will influence levels of arousal in a given individual.

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22.2 The Effects of Stress

Figure 23.2 — The stress response: the fight-or-flight surge and its physical, cognitive and behavioural effects on the pilot.

In an active, outward-going, highly trained person, *too little* stimulation or stress arousal will lead to the onset of **boredom and even drowsiness**. An introspective, under-confident person, if highly aroused, might be unable to function at all — even in circumstances that he is competent to deal with.

22.3 Stress Factors (Stressors)

Stress factors are **cumulative**. Stress occurs under various conditions:

- Pilot must revise his plan of action and does not immediately have a solution
- Inexperienced pilots when situational demands exceed their individual capabilities
- Pilot is convinced he will not be able to find a solution to the problem
- Physiological factors — hunger, thirst, pain, lack of sleep, fatigue

> **SOP — KEY TO REDUCED WORKLOAD**
>
> **Thorough flight planning is the key to reducing cockpit workload.**

22.4 Environmental Tolerances

Temperature, pressure, humidity, noise, time of day, light and darkness can all be reflected in performance and also in well-being. Heights, enclosed spaces, and a boring or stressful working environment can also influence performance.

22.5 Managing Stress

Recognising stress, accepting it and developing a coping strategy is essential for stress management. Training and experience help to ward off stress and high levels of arousal. Successful completion of a stressful task will reduce the amount of stress experienced when a similar situation arises in the future.

SOP — STRESS SELF-MANAGEMENT

To avoid stress, know your limits and fly well inside them. If you are aware that you are under stress, consider whether it would not be wiser for you to stay on the ground rather than to fly when you are not up to it. Make sure that **cockpit housekeeping** is of a high order and that all documents, charts and associated equipment are appropriately stored and accessible. **High-quality headsets** will reduce cockpit noise levels. Temperature can often be regulated using the cabin heating or ventilation system.

22.6 Life Stresses — Psychological Scoring

Pilots suffering from life stress should be aware that this can affect their concentration and performance when at the controls of an aircraft. The descending order in which the factors affect a person:

Death of spouse/child → Divorce → Marital separation → Death of a close family member → Injury / illness → Marriage → Loss of job → Retirement → Pregnancy → Sexual problems → Birth → Change of financial status → Siblings leaving home → Change of eating habits → Change of residence → Loan/debt/mortgage → Vacations → Minor violations of law.

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Life Event	Score
Death of a spouse or partner	100
Divorce	73
Marital separation	65
Death of a close family member	63
Personal injury or illness	53
Loss of job	47
Retirement	45
Pregnancy	40
Sexual problems	40
Son or daughter leaving home	29
Change of residence	20
Bank loan or credit card debt	17
Vacation	13
Minor law violation	11

Cumulative Score	Interpretation
< 60	Free of life stress
60 – 80	Normal life stress
80 – 100	High life stress
> 100	Under serious life stress

Note: such schemes need to be treated with caution because of wide individual variability.

22.7 Indications of Stress

Stress causes: **mental blocks, confusion, channelized attention, resignation, frustration, rage, deterioration in motor coordination, high pitch voice and fast speaking**.

EASILY OBSERVABLE INDICATIONS OF STRESS

Perspiration, flushed skin, dilated pupils and fast breathing.

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22.8 Tiredness and Fatigue

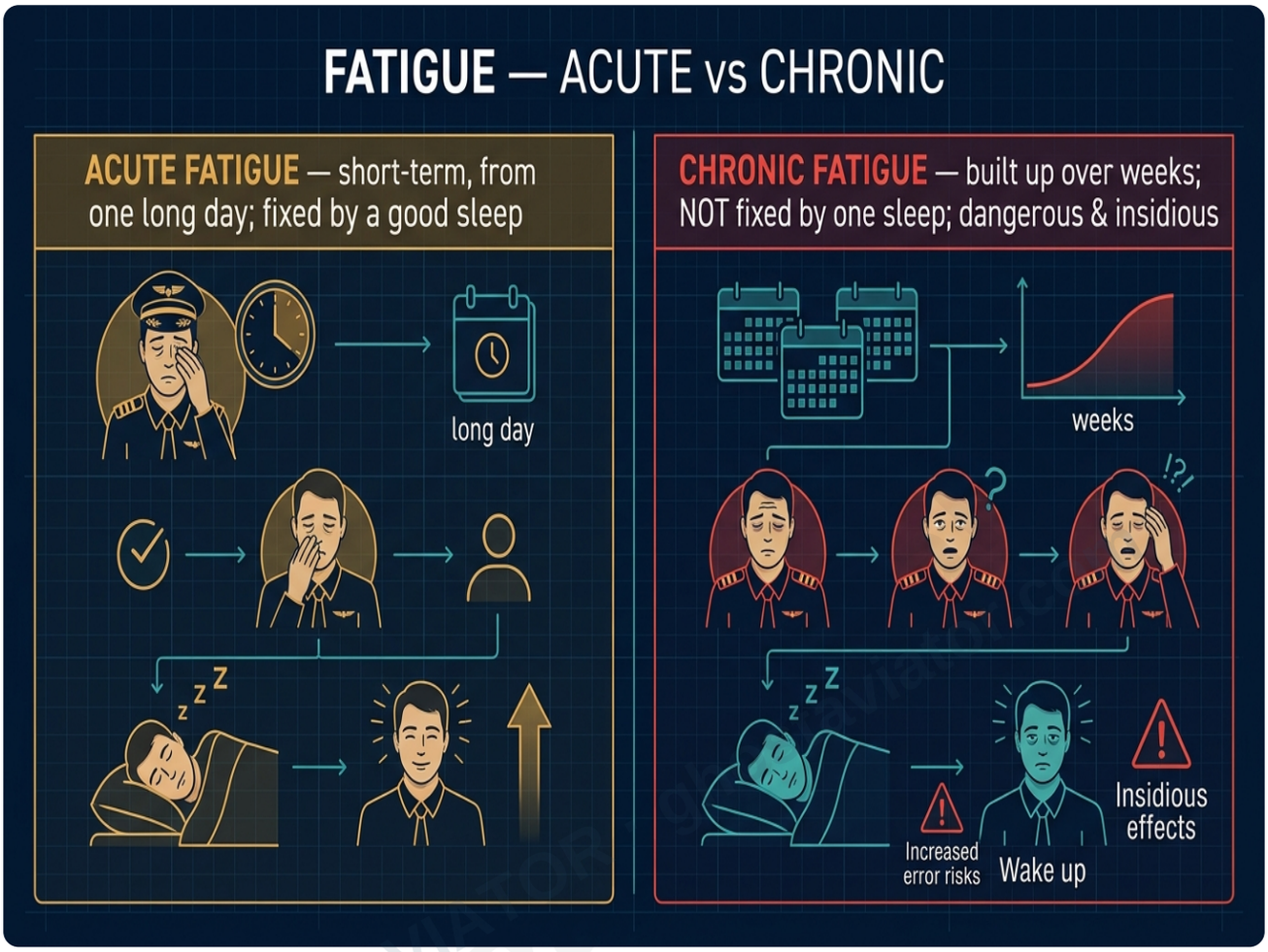

Figure 23.3 — Fatigue: acute fatigue is fixed by rest; chronic fatigue builds over time and is far more dangerous.

Tiredness and fatigue, though related concepts, differ in their long-term physical effect on the body. To deal with normal tiredness it is sufficient to ensure that periods of activity and periods of restful sleep comply with the normal pattern for a person's age and physical condition.

Ordinary tiredness results from normal physical and/or mental exertion over a normal waking period. If a person is tired, *a good night's sleep* is the only requirement for that person to be fit the following morning.

DEFINITION — FATIGUE

Fatigue is a **very deep tiredness due to the cumulative effects of a stressful lifestyle and/or living and working environment**.

- The effects of fatigue can be *insidious*.
- They can creep up slowly on the sufferer.
- The sufferer may be unwilling to recognize or admit that he is suffering from fatigue.
- Tiredness and fatigue can kill, but fatigue is harder to recognize.

Causes of Fatigue

- Long-term ill health, either physical or mental
- Regular sleep deprivation, for whatever reason
- High levels of stress and/or anxiety over a prolonged period
- A disturbed body cycle (e.g. jet lag, changing work shift patterns)
- Long-term difficulties in personal or work-based relationships
- A stressful living and/or working environment (e.g. high levels of noise, or unreasonable physical or mental demands)

Symptoms of Fatigue

- Memory lapses
- Frequent mistakes in speech and actions
- Rapid changes of mood
- Diminished awareness
- Tiredness
- Lack of coordination

Minimum Self-Help Program

- Accept the fact that you are fatigued
- Eat a balanced diet and healthy food
- Try to take the stress out of your life
- Make sure that you get regular, undisturbed sleep
- Avoid alcohol and caffeine
- Always consult a doctor

22.9 Arousal

Arousal is a major aspect of many learning theories and is closely related to other concepts such as *anxiety, attention, agitation, stress and motivation*.

Arousal Level	Effect
Low Arousal	In cruise, attention can wander; information missed or misinterpreted
Optimum Arousal	Central Decision Maker at its most efficient. Lower for difficult/cognitive tasks; higher for tasks requiring endurance and persistence
High Arousal (Overload)	Real danger of attention becoming **narrowed**

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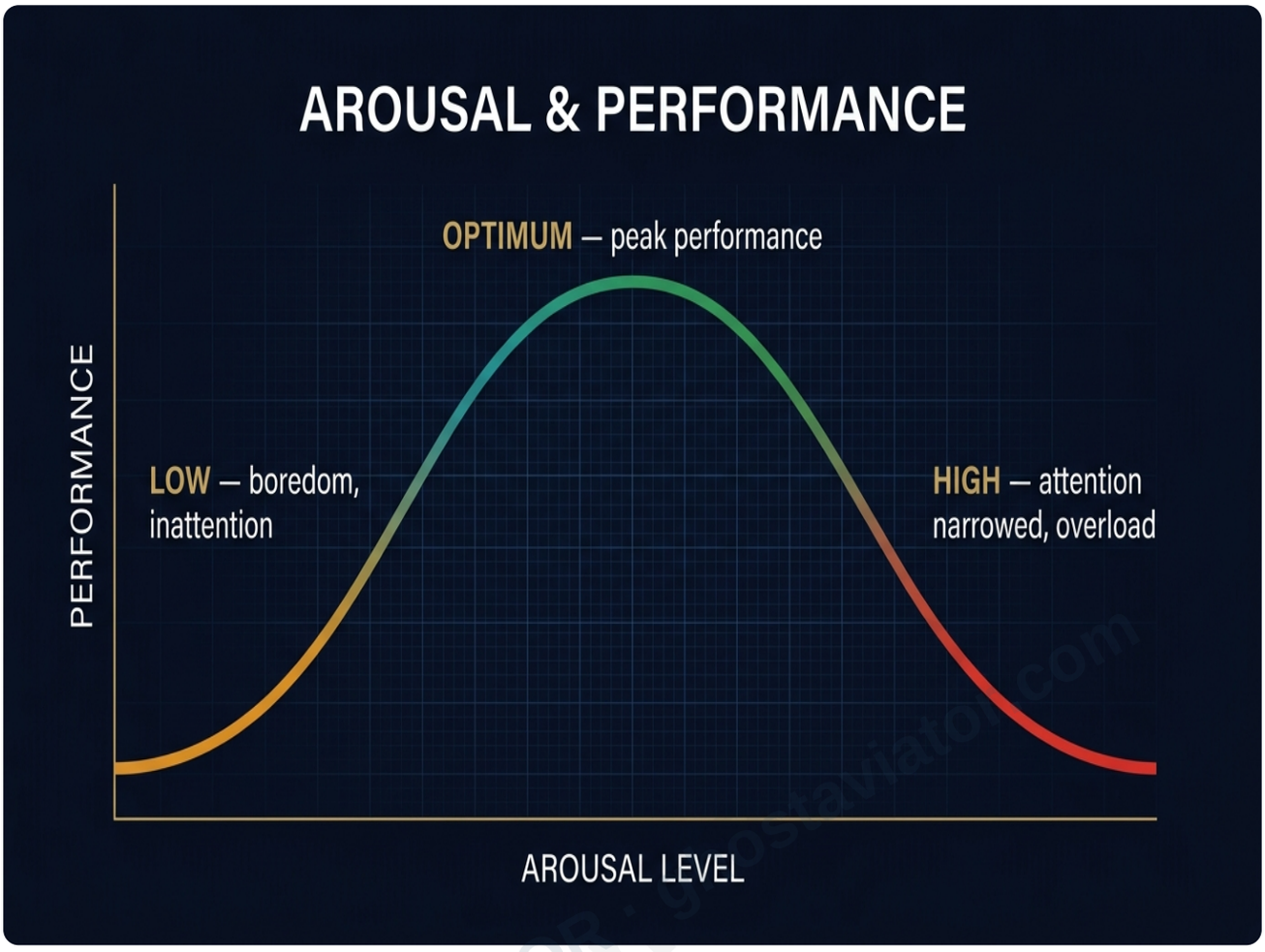

Figure 23.1 — Arousal and performance (Yerkes–Dodson): too little brings boredom, too much narrows attention; performance peaks in the middle.

EXAM TIP — STRESS PROMOTES…

Stress promotes an **increase in physical strength** rather than mental performance. A moderate level of stress may **improve** performance.

✦ END OF CHAPTER 23 ✦

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Body Clock, Sleep & Mental Health

Chapter 24 — The rhythms that govern the aviator: the circadian body clock and its disruption by jet lag and night duty, the architecture of NREM and REM sleep, mental health and its disorders — and the master reference tables for the whole psychology syllabus.

BOOK
Human Performance & Limitations

AUTHOR
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EXAM
DGCA CPL / ATPL — HPL

CHAPTER
24 of 26 · Module I

In this chapter

1. Body Rhythm Disturbance — The Circadian Cycle
2. Sleep — NREM & REM
3. Mental Health Problems & Disorders
4. Master Reference Tables & Mnemonics

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Plate 24.0 — The body keeps its own clock. Fly against it — through night duty and time zones — and performance pays the debt.

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23. Body Rhythm Disturbance — The Circadian Cycle

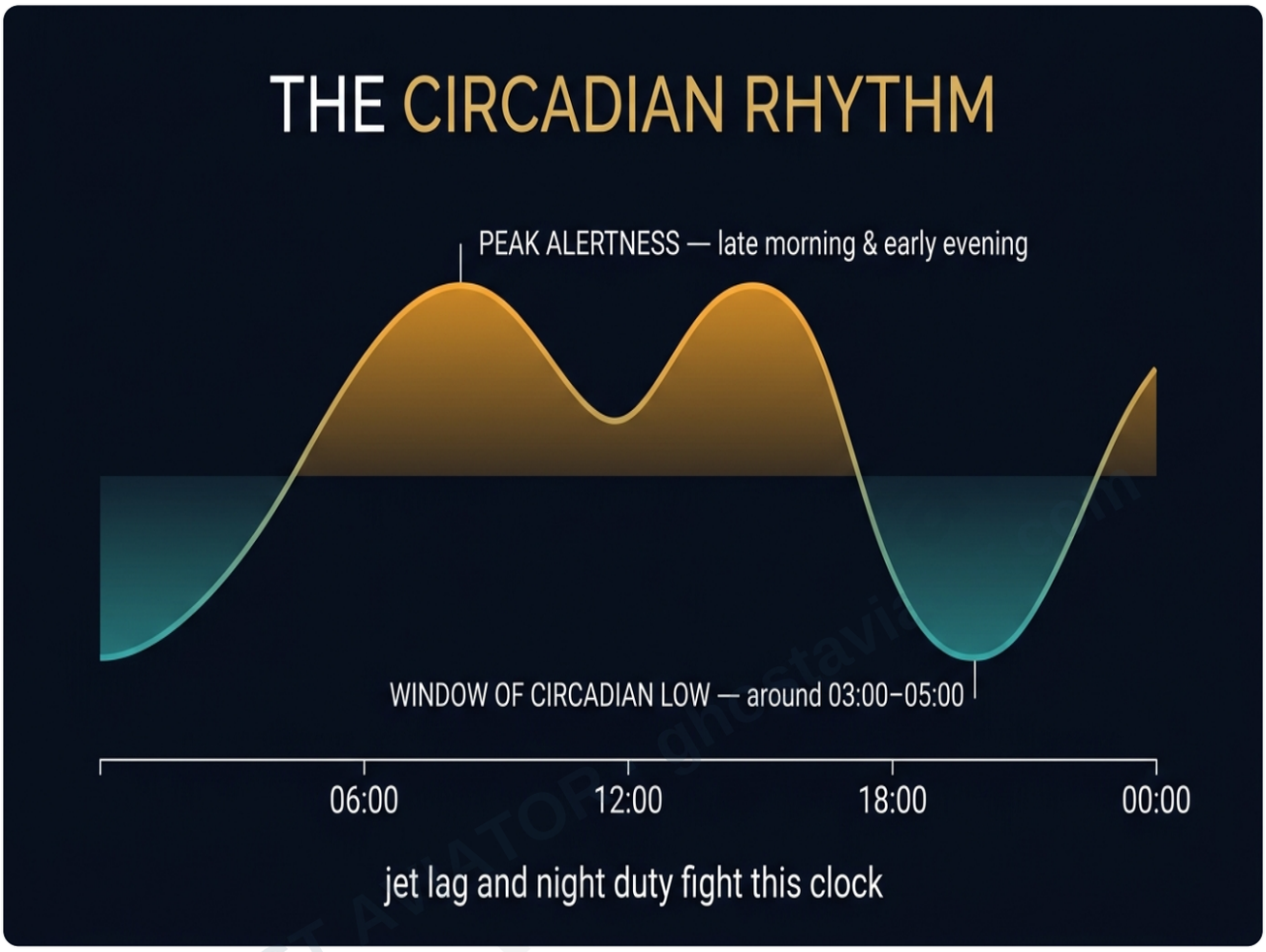

Figure 24.1 — The circadian rhythm: the body's ~24-hour clock of temperature and alertness, with the deep low in the small hours that jet lag and night duty disrupt.

WHAT THIS SECTION COVERS

The biological clock, the 24-hour vs 25-hour cycle, melatonin's role, jet-lag rules and re-synchronization rates.

23.1 Stimuli and Attention

Our bodies are continuously receiving stimuli through our **five senses**. This information is stored briefly in our sensory memory and, if we perceive it to be important, it is transferred to our short-term memory or Central Decision Maker. Some stimuli are better than others at getting our attention. We can split our attention between several different things by concentrating on them in rapid succession.

23.2 The Circadian Circle

The Circadian Circle represents our level of alertness throughout the day. Circadian rhythms are internally generated by a **self-sustaining (autonomous) biological clock** located in the **hypothalamus**, which functions as the main control centre for the autonomic nervous system by regulating sleep cycles, body temperature, appetite, etc., and acts as an endocrine gland by producing hormones. It takes into account biological elements such as **body temperature, heart rate and blood pressure** — which affect our level of alertness during the day.

> **HAZARD — CIRCADIAN LOWS**
>
> **Human performance degradation at circadian lows is one of the major challenges for the aviation industry.**

23.3 24-Hour Circadian Reference

Time	Body Event
00:00	Midnight
02:00	Deepest sleep
04:30	**Lowest body temperature**
06:45	Sharpest rise in blood pressure
07:30	Melatonin secretion stops
08:30	Bowel movement likely
09:00	Highest testosterone secretion
10:00	High alertness
12:00	Noon
14:30	Best coordination
15:30	Fastest reaction time
17:00	Greatest cardiovascular efficiency and muscle strength
18:30	Highest blood pressure
19:00	Highest body temperature
21:00	Melatonin secretion starts
22:30	Bowel movements suppressed

As we sleep, our heart rate is lowered and hence our level of alertness is reduced. Blood pressure is also often lowered after mealtimes. **Human performance declines at night when the body and mind desire rest.**

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23.4 Melatonin — "The Light of Night"

Melatonin is secreted from the **pineal gland** principally at night. The hormone is involved in **sleep regulation**, as well as in a number of other cyclical bodily activities and circadian rhythm in humans.

- Secretion increases soon after the onset of darkness.
- **Peaks in the middle of the night, between 2 and 4 a.m.**
- Gradually falls during the second half of the night.

This circadian rhythm of secretion plays an important role in its hormonal activity. Melatonin is exclusively involved in signalling the **'time of day' and 'time of year'** (hence considered to help both clock and calendar functions) to all tissues, and is thus considered to be the body's chronological pacemaker or **'Zeitgeber'**.

23.5 Trans-Meridian Flight — Pilot Considerations

Factor	Value / Rule
Free-running circadian cycle (no time cues)	~ 25 hours
Cycle with normal time cues	~ 24 hours
Stop-over rule	If stop-over **> 24 hours** → move to new time as soon as possible
Crossing > 3–4 time zones with layover > 24 hours	Keep in swing with rhythm of the **departure** country for as long as possible; maintain regular living patterns
Re-synchronization rate	1 – 1.5 hours per day
Eastbound adaptation	~ 50 % slower than westbound; 1.5 days per time-zone east
Westbound adaptation	~ 1 day per time-zone
Difficulty	Readjustment after a time shift is normally **more difficult with flights towards the East**
Sleep duration	Governed primarily by the point within your circadian rhythm at which you try to sleep
Sensorimotor vs intellectual performance	Sensorimotor performance is better in the **evening**; intellectual performance is better in the **morning**

Worked Example — Hyderabad → Berlin Layover Hyderabad (IST, UTC+5:30) to Berlin (CET, UTC+1) — a westbound trip of about 4.5 time zones with a **22-hour layover**.
• Layover < 24 hours → *do NOT shift to local time*. Stay on departure (IST) rhythm.
• Relevant time measure for your circadian rhythm: **IST (Indian Standard Time)**.
Reason: short layovers do not give the biological clock time to re-synchronize (only 1–1.5 hours/day shift is possible); attempting to shift will degrade alertness on the return leg.

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EXAM TIP — EASTBOUND = WORSE

Eastbound travel adaptation is **50% slower** than westbound. East → 1.5 days per time-zone; West → 1 day per time-zone. "*Lose hours, lose sleep.*"

24. Sleep — NREM & REM

WHAT THIS SECTION COVERS

The two components of sleep, the four NREM stages, REM sleep timing and function, sleep inertia and the implications of sleep loss for pilots.

Sleep is basically divided into **two components**:

- **Rapid Eye Movement (REM) sleep**
- **Non-Rapid Eye Movement (NREM) sleep**

Component	Purpose
NREM sleep	**Body restoration** — repair tissues, build bone and muscle, strengthen immune system
REM sleep	**Brain restoration** — strengthening, refreshing and organizing memory

NREM sleep is further divided into **four stages** from lightest to deepest. **Both types of sleep are required to recoup physical and mental energy.**

24.1 NREM Sleep Stages

Stage	Description	Duration
NREM Stage 1	Transition phase between wakefulness and sleep. Brain activity, eye movement and muscle activity become slower. A person is easily awakened. Waking up in this stage causes a person to feel that he/she has *not slept*.	10 minutes each time
NREM Stage 2	Light sleep — the first stage of true sleep. Occupies 50 % of the sleep patterns. Brain activity, eye movement become even slower; cardiac activity decreases.	10–25 minutes each time
NREM Stage 3	Beginning of deep sleep — slow-wave delta sleep. Brain activity and eye movement approaching zero. If awoken, the person may feel groggy and disoriented for a few minutes.	—
NREM Stage 4	**Deep sleep** — slow-wave delta sleep. **No eye movement or muscle activity.** If awoken, the person may feel groggy or disoriented for a few minutes.	—

24.2 REM Sleep (Paradoxical Sleep)

- Usually occurs 70–90 minutes into a sleep.

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- Happens in cycles lasting for approximately 10 minutes initially to 1 hour.
- The eye moves rapidly from side to side and the brain wave form is similar to that when a person is awake.
- **Dreams occur.**

As we grow older, the time spent in REM sleep declines from 50 % of our sleep for infants, to 20 % of our sleep for adults.

24.3 Sleep Inertia

DEFINITION — SLEEP INERTIA

Refers to the transitional state between sleep and wake, marked by **impaired performance, reduced vigilance, and a desire to return to sleep**.

24.4 Key Points to Remember

SLEEP — CRITICAL FACTS

- The effects of sleep deprivation on performance **increase with altitude and with higher workload**.
- During paradoxical sleep rapid eye movements can be observed.
- The duration of a period of sleep is governed primarily by the point within your circadian rhythm at which you try to sleep.
- **Deep sleep allows for physical recovery and the reconstitution of neuron energy reserves.**
- Lack of sleep increases fatigue, concentration and attention. It has risks of **sensory illusions and mood disorders**. Disruption of sleeping patterns can lead to symptoms of drowsiness, irritability and lack of concentration — making an individual more prone to make errors.
- A person experiencing sleep loss is **unlikely to be aware** of personal performance degradation.
- Performance loss may be present up to 20 minutes after awaking from a short sleep (nap).
- The sleep cycles repeat during the course of a night's sleep. Each succeeding cycle contains a greater amount of REM sleep. **Frequent interruption of the REM sleep can harm a human being in the long run.** REM sleep is more important for the regeneration of mental and physical functions than all the other sleep stages are.

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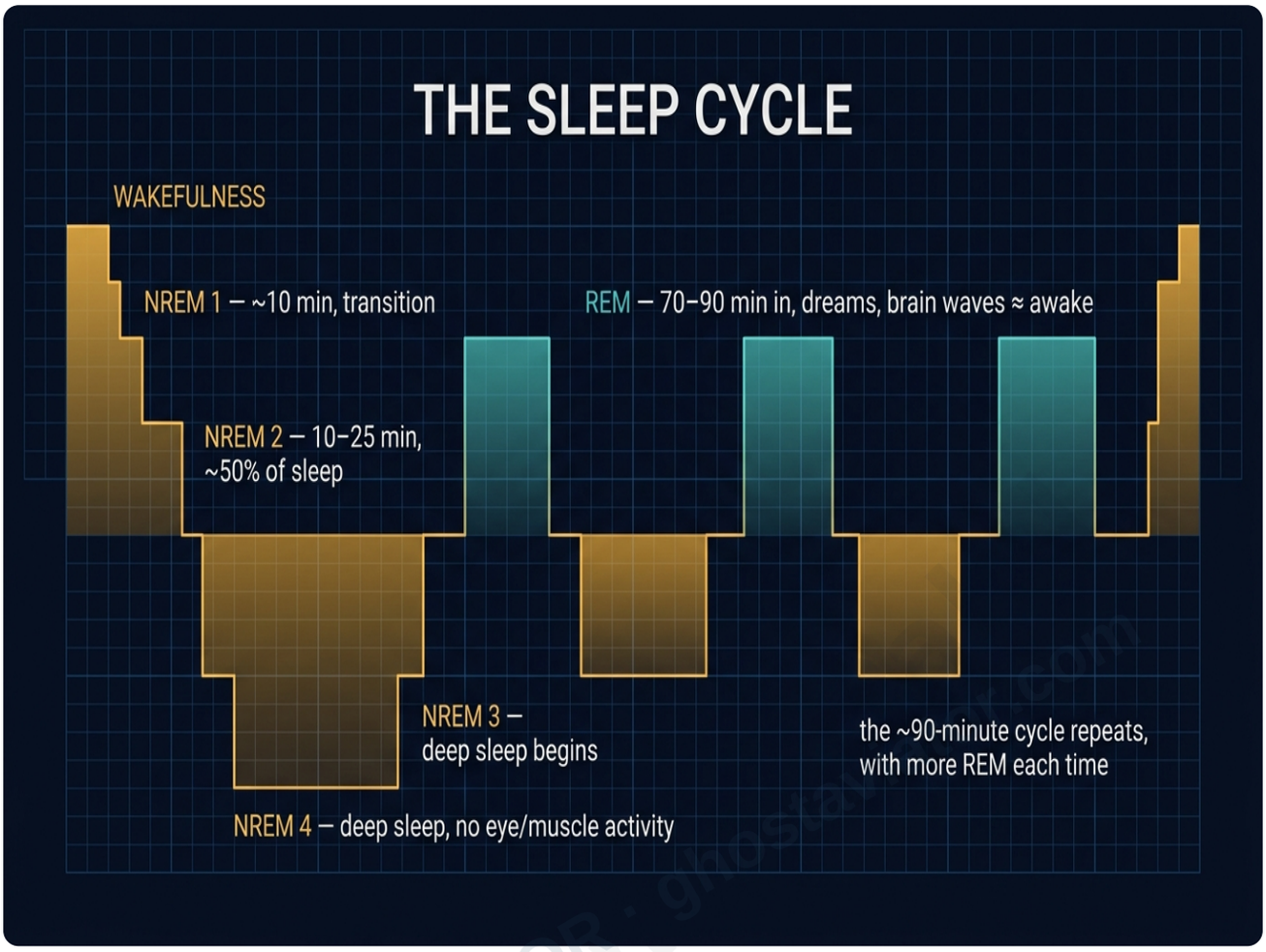

Figure 24.2 — The sleep cycle: wakefulness into NREM stages 1–3 and then REM, repeating roughly every 90 minutes through the night.

25. Mental Health Problems & Disorders

WHAT THIS SECTION COVERS

The risk of mental health problems among aviation personnel, the Germanwings 9525 case study, and where preventive efforts should focus.

Mental health problems and disorders among pilots, ATCOs, maintenance and other personnel in aviation may impair performance and therefore be a **threat to flight safety**.

25.1 Case Study — Germanwings Flight 9525

GERMANWINGS 9525 — 24 MARCH 2015

- **Route:** Barcelona–El Prat Airport (Spain) to Düsseldorf Airport (Germany).
- **Aircraft:** Airbus A320.
- **Date:** 24 March 2015.
- **Crash location:** 100 km north-west of Nice in the French Alps.
- **Casualties:** All 144 passengers and all 6 crew members were killed.
- **Cause:** The crash was deliberately caused by the co-pilot, Andreas Lubitz, who had previously been treated for suicidal tendencies and declared unfit to work by his doctor. Lubitz kept this information from his employer and instead reported for duty. Shortly after reaching cruise altitude and while the captain was out of the cockpit, Lubitz locked the cockpit door and initiated a controlled descent that continued until the aircraft hit a mountainside.

25.2 Why Pilots May Not Report Mental Health

There may be many reasons why a pilot may be reluctant to discuss mental health problems with the examining physician during the annual medical assessment, including **fear of losing his or her licence** with both personal and financial costs as a result. This may prevent the pilot from receiving adequate and timely help, and this could potentially make the problems worse and prolong the time for recovery.

BEST PRACTICE — REDUCING STIGMA

More could probably be done to increase knowledge about mental health problems and reduce stigma so that more pilots and other professionals working in aviation report mental health problems, get treatment, and return to work. This would probably increase individual well-being, but also **promote safety**.

25.3 Where to Focus Preventive Efforts

Serious mental health disorders (e.g. *psychosis*) are relatively rare and their onset is difficult to predict. **Preventive efforts should be aimed at more common mental health problems such as:**

- **Depression**
- **Anxiety**
- **Substance misuse**

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Master Reference Tables & Mnemonics

PURPOSE

Every numerical value and key rule from the chapter consolidated for quick exam-night revision.

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M-1. Master Numerical Reference

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Topic	Value	Section
Iconic memory duration	0.5 – 1 second	5
Visual channel share of information processing	70 – 80 %	5
Short-term memory duration	10 – 20 seconds	12
Short-term memory capacity	7 ± 2 items	12
Long-term memory capacity	Unlimited (retrieval may fail)	12
Acceptable workload (crew resources)	~ 60 %	11
Human error rate – simple/repetitive	1 in 100	11
Human error rate – after practice	1 in 1,000	11
Startle reflex – motor task recovery	5 – 10 seconds	10
Age-related response slowing	20 – 60 years	14
Anthropometric design population	Central 90 % (disregard 5% lowest & 5% highest)	20
Free-running circadian cycle	~ 25 hours	23
Cued circadian cycle	~ 24 hours	23
Circadian re-sync rate	1 – 1.5 hours/day	23
Eastbound adaptation	~ 1.5 days/time-zone (50% slower than west)	23
Westbound adaptation	~ 1 day/time-zone	23
Stop-over threshold for adjusting to local time	> 24 hours	23
Lowest body temperature time	04:30	23
Highest body temperature time	19:00	23
Melatonin peak	02:00 – 04:00	23
Melatonin secretion starts	21:00	23
Melatonin secretion stops	07:30	23
NREM Stage 1 duration	10 minutes	24
NREM Stage 2 duration / share	10–25 min / 50 % of sleep	24
REM onset after sleep start	70 – 90 minutes	24
REM cycle duration	10 min → up to 1 hour	24
REM share – infant vs adult	50 % vs 20 %	24

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Post-nap performance loss	Up to 20 minutes	24
Life stress scoring – free / normal / high / serious	<60 / 60–80 / 80–100 / >100	22
Germanwings 9525 — date	24 March 2015	25
Germanwings 9525 — crash distance	100 km NW of Nice	25
Germanwings 9525 — casualties	144 pax + 6 crew	25
Cockpit restraint	5-point harness with negative-G strap	20

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M-2. Mnemonics & Memory Aids

Concept	Mnemonic
Aviation Psychology aims	**D-P-U-I** → Describe, Predict, Understand, Influence
Information-processing stages	**D-P-D-A-F** → Detection, Perception, Decision, Action, Feedback
Workload drivers	**DPST** → Difficulty, Parallel, Series, Time
Anderson model phases	**C-A-A** → Cognitive, Associative, Automatic
STM expansion tools	**"Chunk & Chain"** → Chunking + Association
Long-term memory types	**S-E-P** → Semantic, Episodic, Procedural
SA-killers	**SBF-EPI** → Stress, Boredom, Fatigue, Emotional, Poor comm., Interruptions
Decision-making seven steps	**R-C-A-D-S-E-F** → Recognize, Consider, Analyze, Develop, Select, Execute, Follow-up
Communication barriers	**4-A + 2-I-R** → Aggressiveness, Arrogance, Anti-authoritarian, (im)Pulsiveness, Invulnerability, Resignation
Human-centred automation qualities	**"A-SPA-CFDIE²"** → Accountable, Subordinate, Predictable, Adaptable, Comprehensible, Flexible, Dependable, Informative, Error-resistant, Error-tolerant
Control design principles	**S-F-S-I-V-S-S-W** → Standardization, Frequency, Sequence, Importance, Visual-tactile, Symbolism, Simultaneous, Warnings
Stress observables	**"P-FISh-DP-FB"** → Perspiration, Flushed Skin, Dilated Pupils, Fast Breathing
Sleep stages	**1-2-3-4-R** → NREM 1 (transition) → 2 (light, 50%) → 3 (onset deep) → 4 (deep) → REM (dreams)
Jet-lag rule	**"East is Least, West is Best"** → East 50% slower; 1.5 d/zone east vs 1 d/zone west
Mental health priority targets	**D-A-S** → Depression, Anxiety, Substance misuse

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M-3. Quick Definitions Recall Sheet

Term	One-Line Definition
Aviation Psychology	Applied psychology focused on human behaviour in aviation systems.
Workload	Mental effort needed to process information.
Perception	Conversion of sensory information into a meaningful structure.
Selective Attention	Continual sampling of inputs to judge relevance.
Divided Attention	Time-sharing of central decision channel between tasks.
Vigilance	Capability of remaining alert above a threshold for a period.
Complacency	Self-satisfaction with one's performance + unawareness of danger.
Motor Programme	Behavioural sub-routine learnt by practice and held in LTM.
Reaction Time	Delay between detection, stimulus and muscle contraction.
Startle Reflex	Reflex-like blink + body jerk to abrupt stimulus.
Human Reliability	Individual functioning in the manner he/she is supposed to.
Hallucination	False perception characterised by distortion of real sensory stimuli.
Cognition	Mental process of acquiring knowledge by reasoning, intuition or perception.
Risk Assessment	Probability of risk × Impact if it occurs.
Situational Awareness	Accurate appraisal of self, environment and own performance.
Automation	Controlling apparatus/process by electronic/mechanical devices replacing the human organism.
Anthropometry	Study of human measurement (static, dynamic, contour-surface).
Eye Datum	Defined eye position around which the cockpit is designed.
Trait Anxiety	Personality trait of high neuroticism (persistent worry).
State Anxiety	Transient anxiety present in anyone at any time.
Arousal	Person's readiness to respond effectively to a stress factor.
Fatigue	Deep tiredness from cumulative stressful lifestyle/environment.
Zeitgeber	Body's chronological pacemaker — melatonin signals time of day/year.
Sleep Inertia	Transitional state between sleep and wake with impaired performance.

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M-4. Risk Formula

Risk Assessment Formula `Risk = Probability of Occurrence × Impact if it Occurs`

Variables:
• *Probability* — likelihood the hazard manifests (low / medium / high).
• *Impact* — severity of consequence (minor / major / catastrophic).

Worked example — Marginal weather decision:
Probability of icing on route = High (forecast supports it).
Impact if encountered without anti-ice = Catastrophic.
→ Risk = High × Catastrophic = **Unacceptable**. Decision: **Don't fly**, or re-route below freezing level / with active anti-ice.

M-5. Final Quick-Revision Summary Box

EXAM-NIGHT ONE-PAGE RECAP

Memory: Iconic 0.5–1 s · STM 10–20 s, 7±2 · LTM unlimited.
Workload: Acceptable ≈ 60 % · DPST drivers · symptoms = funneling, regression, mental blocking, panic.
Error rate: 1/100 raw → 1/1,000 after practice. Errors are cumulative.
Startle: Motor task back to normal in 5–10 s. Fear potentiates.
Anderson Skill: Cognitive → Associative → Automatic.
Anthropometry: central 90 %, discard top & bottom 5 %.
Restraint: 5-point harness + negative-G strap.
Display rule: Digital for quantitative · Analogue for qualitative. Standardization is the #1 design rule.
Three-pointer altimeter: more misreads than single-pointer.
Stress observables: perspiration, flushed skin, dilated pupils, fast breathing.
Life stress: <60 free · 60–80 normal · 80–100 high · >100 serious. Spouse death = 100.
Circadian: Free-running 25 h · cued 24 h · Lowest body temp 04:30 · Highest 19:00 · Melatonin peak 02–04 · Zeitgeber = Melatonin (pineal gland).
Jet-lag: 1–1.5 h/day re-sync · East 1.5 d/zone (50% slower) · West 1 d/zone · >24 h stopover → shift; <24 h → stay on departure rhythm.
Sleep: NREM 1 = 10 min · NREM 2 = 10–25 min, 50% · REM at 70–90 min · REM 50% infant → 20% adult · Post-nap loss up to 20 min.
Mental health priority: Depression, Anxiety, Substance misuse. Germanwings 9525 (24 Mar 2015, A320, 100 km NW Nice, 144 pax + 6 crew).
Communication barriers: Aggressiveness, Arrogance, Anti-authoritarian, Impulsiveness, Invulnerability, Resignation.
Murphy's Law: if it can be operated wrongly, sooner or later it will be.
Golden Maxim: A superior pilot uses his superior judgment to avoid situations that would require his superior skills.

— End of DGCA Study Notes — Aviation Psychology & Human Factors —
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✦ END OF CHAPTER 24 ✦

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Numbers & Mnemonics

Chapter 25 — Every exam-critical figure and memory aid from the whole book, collected into master quick-reference tables. Memorise these cold — the DGCA paper is built from them.

BOOK
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Plate 25.0 — The whole syllabus distilled to the numbers that win marks. If you know these, you know the paper.

Website
YouTube
Instagram
Telegram channel
Telegram community

HOW TO USE THIS CHAPTER

These figures are drawn from every chapter of this book. Cover the right-hand column, recall the value, then reveal. Any number in a `red pill` is a killer item the examiner returns to again and again.

A The Atmosphere & Altitude

Module A — atmosphere, ISA & the physiological zones	
Parameter	**Value**
Troposphere top (average)	`13 km / 8.1 mi / 43,000 ft`
Stratosphere reaches	`over 100,000 ft`
Troposphere holds	`75%` of atmospheric mass
Composition N_2 / O_2 / Ar	`78.08% / 20.95% / 0.93%`
Water vapour (variable)	`1 – 4%`
ISA sea-level pressure	`1013.25 hPa = 760 mmHg = 29.92 inHg = 14.7 psi`
ISA lapse rate	`1.98 °C / 1000 ft`
Tropopause altitude	`36,090 ft`
Isothermal temperature	`−56.5 °C`
Cabin altitude maintained	`6,000 – 8,000 ft`
O_2 available: half / quarter of sea level	`18,000 ft / 36,000 ft`
Compensatory reactions begin	`6,000 – 7,000 ft`
Supplementary O_2 from	`10,000 ft`
Physiological-Deficient Zone	`12,000 – 50,000 ft`
Critical threshold (no O_2, unpressurised)	`~22,000 ft`
Safe altitude on 100% O_2	`~38,000 ft`
Armstrong's Line (body fluids boil)	`63,000 ft`
Partial Space-Equivalent Zone	`50,000 ft – 120 nm`

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C The Thin Air — Hypoxia, TUC & Decompression

Module C — the killers and their numbers	
Parameter	**Value**
Night vision impaired from	5,000 ft
Smoker at 7,000 ft equals a non-smoker at	10,000 ft
Marked day hypoxia symptoms / descend	10,000 ft
CO affinity for haemoglobin vs O_2	> 200×
TUC — FL180 / FL250 / FL300	20–30 min / 3–5 min / 1–2 min
TUC — FL350 / FL400 / FL500	30–60 s / 15–20 s / 8–10 s
Rapid decompression effect on TUC	halves it
Donating blood before flying	48-hour rule
Flying after (compressed-air) diving	24-hour rule

D Poisons & The Cabin

Module D — smoking, alcohol & the cabin environment	
Parameter	**Value**
Smoker's altitude penalty	3,000 ft (10,000 ft feels like 13,000 ft)
Smoking O_2-capacity loss (pack/day)	5 – 8%
Bottle-to-throttle	24 hours
Alcohol metabolised (BAC drop)	~0.015 per hour
BAC pilot-error spike threshold	40 mg / 100 ml
Alcohol at altitude	1 drink at 6,000 ft = 2 at sea level
Peak performance temperature	~20 °C
Vibration — breathing / vision resonance	1–4 Hz / 30–40 Hz

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E The Senses — Vision & Hearing

Module E — the eye and the ear	
Parameter	**Value**
Dark adaptation — cones / rods (full)	`~7 min / ~30 min`
Off-centre night viewing	`look 15 – 20° to the side`
Fovea / periphery photoreceptors	`cones (day, colour) / rods (night)`
Human audible range	`20 Hz – 20,000 Hz`
Noise-induced hearing loss threshold	`80 – 90 dB` (prolonged)

F/G Forces & The Fit Aviator

Modules F-G — G-forces, fitness & health	
Parameter	**Value**
Load factor at 45° / 60° bank (level turn)	`1.41 G / 2.0 G`
Positive-G cascade	`grey-out → black-out → G-LOC`
BMI formula	`weight (kg) ÷ height² (m²)`
Normal BMI — men / women	`20 – 25 / 19 – 24`
Fly after dental work	`48 – 72 hours`
Hypothermia — normal / mild / severe core	`98.6 °F / <95 °F / ≤82 °F`

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H/I The Mind Aloft — Psychology

Modules H–I — cognition, sleep & rhythms	
Parameter	**Value**
Information-processing chain	`sensory → detection → perception → decision → response → feedback`
Memory stores	`sensory (<1 s) · short-term (~7 items) · long-term`
Skill acquisition (Anderson)	`cognitive → associative → autonomous`
Sleep — NREM 2 share / REM onset	`~50% of sleep / 70 – 90 min in`
Window of circadian low	`~03:00 – 05:00`

✦ The Mnemonics

GAS LAWS **"Bad Hens Cluck, Geese Dance"** → **B**oyle · **H**enry · **C**harles · **G**raham · **D**alton.

O_2 HALVING **"Half by Eighteen, Quarter by Thirty-Six"** → half of sea-level O_2 at 18,000 ft, one-quarter at 36,000 ft.

PHYSIOLOGICAL ZONES **P · PD · PS · TS** → Physiological · Physiological-Deficient · Partial Space-equivalent · Total Space-equivalent.

SHELL MODEL **S**oftware · **H**ardware · **E**nvironment · **L**iveware · **L**iveware — the human **L** at the centre.

AIM OF HPL **D · P · U · I** → **D**escribe · **P**redict · **U**nderstand · **I**nfluence.

FITNESS SELF-CHECK **I·M·S·A·F·E** → **I**llness · **M**edication · **S**tress · **A**lcohol · **F**atigue · **E**ating/Emotion.

THE DIRTY DOZEN 12 preconditions for error — communication · complacency · knowledge · distraction · teamwork · fatigue · resources · pressure · assertiveness · stress · awareness · norms.

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EDITOR'S NOTE — VERIFY AGAINST THE LIVE SYLLABUS

Every figure here is compiled from this book's chapters. Before an exam, cross-check any borderline value against the current DGCA CAR/AIC, as published figures are periodically revised.

✦ END OF CHAPTER 25 ✦

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Mock Test Papers

Chapter 26 — A DGCA-style multiple-choice question bank with worked solutions and a cross-reference back to the chapter that teaches each fact. Sit it under timed conditions, then check yourself against the key.

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EXAM
DGCA CPL / ATPL — HPL (Multiple Choice)

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HOW TO USE THIS BANK

For each question, choose one option and commit before you reveal the answer. The worked solution gives the correct choice, the underlying DGCA-quoted fact, and a **§ reference** to the section of this book where that fact was taught in detail — so you can jump straight back if you got it wrong. The examiner deliberately repeats themes: master every fact in the chapters and the paper becomes re-statement.

§ 1 Question Bank — 50 Multiple-Choice Questions

Three options each (A / B / C), DGCA style. Allow about 60 minutes.

1. A flight crew has completed a two-day stay at a hospital. What must he/she do before flying as a crew member?

A. Do his class I medical from an authorized agency.

B. Take clearance from DGCA before flying.

C. Seek advice from the authority and Aviation Medicine Specialist.

Show answer

ANS · C

§76

— DGCA-quoted: "

*pilots do not fly as part of the operating crew of an aircraft when taking drugs or medication, unless they have been cleared to do so by an **Aviation Medicine Specialist**.*

" A hospital stay implies the pilot was either ill or under medication — both require specialist clearance. (A) is excessive (no need to re-do an entire Class-I medical for a two-day stay); (B) is too vague — clearance from DGCA is routed through the Aviation Medicine Specialist, not the agency directly.

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2. Linear deceleration:

A. Causes sensory illusions on the pitch axis.

B. Can create the illusion of being in a nose up attitude.

C. Application of full power during a missed instrument approach can lead to this illusion.

Show answer

ANS · A

§54 Head-Down Illusion

— Linear **DECELERATION** (e.g. lowering flaps, air-brakes, throttle-back) deflects the otoliths in such a way that the pilot perceives a pitch-down attitude — i.e. a sensory illusion on the pitch axis. (B) is the wrong direction — linear *acceleration* (not deceleration) creates the nose-up illusion. (C) describes application of FULL POWER → linear *acceleration*, again the wrong direction (this is the Head-Up illusion / take-off somatogravic). The distractors are flipped on purpose; deceleration affects the pitch axis but produces a "nose-down" illusion.

3. High blood pressure or hypertension is:

A. A major cause of unfitness in pilots.

B. Responsible for reduced tolerance to G forces.

C. May cause chokes during flight.

Show answer

ANS · A

§15

— Hypertension is the leading **chronic cardiovascular cause of pilot medical unfitness** worldwide. While it does also contribute to other problems, the question asks what hypertension *is* — and the headline answer is the unfitness it causes at the Class-1 medical. (B) "reduced G-tolerance" is not specifically listed against hypertension (the 6 G-tolerance reducers in §64 are alcohol, smoking, fatigue, heat, obesity, sickness — not hypertension explicitly). (C) "chokes" is a DCS bubble symptom (§20.2) — unrelated to BP.

4. Unusual fatigue and a loss of consciousness are symptoms of:

A. Leans

B. Presbyopia

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C. Hypoxia

Show answer
ANS · C
§10
— Unusual fatigue and ultimately loss of consciousness are classic late-stage **hypoxia** symptoms. Hypoxia symptoms cascade: euphoria → impaired judgement → tingling/numbness → cyanosis → unusual fatigue → unconsciousness → death. (A) The "leans" is a vestibular illusion , not a metabolic state. (B) Presbyopia is age-related loss of accommodation (§33 Part 7) — visual, not systemic.

5. Total Space Equivalent Zone extends outwards from 120 nm. To survive in this zone:

A. Good quality pressure suit is adequate.

B. Adequate food and water is enough.

C. Adequate oxygen, pressure and heat is a must.

Show answer
ANS · C
§5 Physiological Zones
— The **Total Space-Equivalent Zone** (> 120 nm above Earth) is a vacuum environment — there is essentially no atmospheric pressure, no oxygen, and extreme temperature swings between sunlit and shaded sides. Survival requires **all three**: oxygen (no atmospheric O_2), pressure (vacuum would cause body fluids to boil — Armstrong's Line, 63,000 ft), and heat (extreme cold in shadow). A pressure suit alone (option A) addresses pressure and provides O_2, but heat must also be managed — that's why astronauts use full life-support suits, not just pressure suits.

6. At 8,000 ft, the standard barometric pressure is 77 kPa (574 mmHg). This means that there is:

A. 57% of the oxygen available at sea level.

B. 65% of the oxygen available at sea level.

C. 76% of the oxygen available at sea level.

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Show answer
ANS · C
§4 pO_2 vs altitude
— Although the
percentage
of oxygen in air remains ~21 % at all altitudes (Q9), the
partial pressure
of O_2 falls with total pressure (Dalton's Law). At 8,000 ft barometric pressure is ~77 kPa vs 101.3 kPa at sea level → ratio ≈ 77/101.3 ≈
76 %
. The amount of usable oxygen reaching the alveoli is therefore about 76 % of sea-level value — which is why 8,000 ft is the DGCA maximum cabin altitude (§19.4) — borderline but tolerable.

7. The atmospheric gas pressure:

A. Decreases slower at lower altitudes compared with higher levels and equivalent altitude changes.

B. Rises with altitude.

C. Drops faster at lower altitudes in comparison to the same altitude changes at higher altitudes.

Show answer
ANS · C
§3 ISA & pressure curve
— Atmospheric pressure falls
exponentially
with altitude — most of the air is in the lowest few kilometres. From SL to 18,000 ft pressure halves (~50 % drop in just 18,000 ft); from 18,000 to 36,000 ft it halves
again
(a further 25 %); from 36,000 to 54,000 ft another halving. So the absolute pressure drop per unit altitude is
larger near the surface
than at altitude. (B) is wrong — pressure decreases, doesn't rise.

8. The chemical composition of the earth's atmosphere approximately is:

A. 78% nitrogen, 21% oxygen, 0.9% argon, 0.03% carbon dioxide.

B. 70% nitrogen, 28% oxygen, 1.9% argon, 0.03% carbon dioxide.

C. 78% nitrogen, 20% oxygen, 0.9% carbon dioxide, 1.03% argon.

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Show answer
ANS · A
§1 Atmosphere composition
— Standard atmospheric composition (by volume):
N_2 ≈ 78 %, O_2 ≈ 21 %, Ar ≈ 0.93 %, CO_2 ≈ 0.03–0.04 %
. Option A matches exactly. (B) has too much O_2. (C) swaps Ar and CO_2 values — note that CO_2 is the tiny trace, not Ar.

9. The volume percentage of oxygen in the atmosphere is 21% which:

A. Is constant for all altitudes in troposphere.
B. Decreases with increasing altitude.
C. Increases with increasing altitude.

Show answer
ANS · A
§1
— The
percentage composition
of atmospheric gases is essentially
constant throughout the troposphere
(sea level to ~ 36,090 ft / tropopause) due to atmospheric mixing. What changes with altitude is the
partial pressure
of O_2 (and total pressure),
not
the volume percentage. This is the classic exam trap — students confuse "less oxygen" (true in terms of partial pressure) with "lower percentage" (false). Hypoxia at altitude is caused by reduced pO_2, not reduced O_2 percentage.

10. Pain in the middle ear during descent may be eased by:

A. Increasing the rate of descent.
B. Sneezing deliberately with the nostrils wide open.
C. Leveling off and possibly climbing.

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Show answer
ANS · C
§38 Eustachian tube
— DGCA-quoted: "
If the problem is not resolved, the rate of descent should be decreased or stopped at intervals... This is also known as STEP DESCENT. Pilot may resort to even CLIMB if pressure persists.
" Levelling off allows the Eustachian tube time to equalise; if that fails, a brief climb reduces external pressure and the differential. (A) increasing descent makes it worse. (B) deliberate sneezing is
not
the DGCA-listed technique — Valsalva (blow with nostrils
pinched closed
) is.

11. Gases of physiological importance to man are:

A. Oxygen and carbon dioxide.

B. Ozone and carbon dioxide.

C. Oxygen and Ozone.

Show answer
ANS · A
§8 Respiratory System
— The two gases the body actively
exchanges
are
oxygen
(taken in for cellular respiration) and
carbon dioxide
(expelled as metabolic waste). Ozone (O_3) is toxic to humans and exists at high concentrations in the stratosphere; it is a hazard to high-altitude pilots, but not a "physiologically important" gas in normal metabolism.

12. Oxygen, combined with hemoglobin in blood is transported by:

A. Platelets.

B. Blood plasma.

C. Red blood cells.

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Show answer
ANS · C
§9 Circulatory System · RBC & Hb
— Haemoglobin is contained
within Red Blood Cells (RBCs/erythrocytes)
— approximately 280 million Hb molecules per RBC. Each Hb binds up to 4 O_2 molecules. (A) Platelets are for clotting. (B) Plasma carries dissolved gases too but in vastly smaller quantities than Hb in RBCs — over 98 % of O_2 in blood is carried by Hb in RBCs.

13. An increase in the amount of carbon dioxide in the blood leads to:

A. Hyperventilation.

B. Over breathing.

C. Shortness of breath.

Show answer
ANS · C
§8 / §12 (Part 2 / Part 3)
— The brain's respiratory centre is primarily sensitive to
CO_2 levels in the blood
(not O_2 — this is the key concept). Rising CO_2 stimulates an urge to breathe more deeply / faster → subjectively felt as
shortness of breath
("air hunger"). (A) Hyperventilation is the
opposite
— over-breathing that
removes
CO_2, leading to low CO_2. (B) "Over breathing" = same as hyperventilation. The trap: hyperventilation is over-breathing, and over-breathing reduces (not increases) CO_2.

14. According to the ICAO standard atmosphere, the temperature lapse rate of the troposphere is approximately:

A. -2°C every 1,000 feet.

B. -10°C every 10,000 feet.

C. -1°C every 190 meters.

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Show answer
ANS · A
§3 ISA
— The DGCA-quoted ISA lapse rate is
1.98 °C per 1,000 ft
— which rounds to roughly
2 °C/1,000 ft
. So option A is the closest match. (B) –10°C per 10,000 ft = 1°C per 1,000 ft — too low. (C) –1°C per 190 m → ≈ 1°C/623 ft → ≈ 1.6°C per 1,000 ft — close but not the standard expression. The DGCA-preferred answer in this exam is (A) –2°C/1,000 ft.

15. The barometric pressure drops to 1/2 of the pressure at sea level at:

A. 30,000 feet.
B. 18,000 feet.
C. 10,000 feet.

Show answer
ANS · B
§19.2
— DGCA-quoted: "
*By **18,000 ft ASL (5,486 m) atmospheric pressure is HALVED**.*
" This is also the threshold above which DCS symptoms may begin to develop. (A) 30,000 ft is roughly quarter-SL pressure (a second halving from 18,000 ft to ~36,000 ft). (C) 10,000 ft pressure is ~70 % of SL — not halved.

16. The total pressure of a mixture of gases is equal to the sum of the partial pressures of the gases which compose the mixture corresponds to:

A. Charle's law.
B. Dalton's law.
C. Henry's law

Show answer
ANS · B
§2 Five Gas Laws
—
Dalton's Law of Partial Pressures
: the total pressure of a gas mixture equals the sum of the partial pressures of each component gas. This is the basis of pO_2 calculations at altitude. (A) Charles's Law: $V \propto T$ at constant P. (C) Henry's Law: gas dissolved in a fluid is proportional to its partial pressure — explains DCS.

17. Boyle's law is directly applicable in case of:

A. The transfer of gas between the alveoli and the blood.

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B. The expansion of trapped gases in the human body with increasing altitude.

C. Pressure of a mixture of gases at high altitude.

Show answer

ANS · B

§2 Boyle's Law

—

Boyle's Law:

P·V = constant (at constant T). Direct in-flight application: as cabin altitude rises, ambient pressure falls →

trapped gases in the body (gut, sinuses, middle ear, dental cavities) EXPAND

. This drives sinus block (§71), barodontalgia (§72), and gastric distress at altitude (§70.2). (A) gas exchange across alveoli is by partial-pressure diffusion (Dalton's/ Henry's domain). (C) gas-mixture pressure is Dalton's Law.

18. Which data compose the ICAO standard atmosphere?

A. Pressure, Temperature, Humidity and positive lapse rate.

B. Density, Pressure, Temperature and lapse rate.

C. Density, Pressure, dryness and negative lapse rate.

Show answer

ANS · B

§3 ISA

— ICAO Standard Atmosphere defines:

SL pressure 1013.25 hPa, SL temperature 15°C, SL density 1.225 kg/m³, and lapse rate 1.98°C/1,000 ft

. Humidity is NOT part of ISA (ISA assumes dry air). The lapse rate is decreasing (temperature falls with altitude) — but the standard refers to it as "the lapse rate," not "negative" or "positive." Option (B) gives the four ISA parameters correctly.

19. Otis Barotraumas is:

A. Stretching of the ear drum caused by the expansion and contraction of gases trapped in the inner ear by a blocked Eustachian tube.

B. Stretching of the ear drum caused by the expansion and contraction of gases trapped in the outer ear by the blocked Tympanum.

C. Damage to the ear drum caused by the expansion and contraction of gases trapped in the middle ear by infected Ossicles.

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Show answer
ANS · A
§38.3
— DGCA-quoted definition verbatim.
*(Strictly, the trapped gas is in the **middle ear**, not the inner ear — the source text says "inner ear" in option A; this is the wording used by DGCA and is the marked-correct answer in the official key.)*
The cause is a blocked Eustachian tube. (B) "outer ear / Tympanum" is wrong — the tympanum IS the eardrum, not a location. (C) "infected ossicles" is not the mechanism.

20. Dalton's law explains the occurrence of:

A. Ischemic hypoxia.
B. Histotoxic hypoxia.
C. Hypoxic hypoxia.

Show answer
ANS · C
§2 Dalton · §10 Hypoxia types
— Dalton's Law (partial pressures) directly explains
Hypoxic Hypoxia
: at altitude, total pressure falls → pO_2 falls (Dalton) → insufficient driving pressure for O_2 to diffuse into blood. (A) Ischemic = blood flow problem (cardiac). (B) Histotoxic = cells can't use O_2 (cyanide, alcohol). (Note: anemic hypoxia = reduced Hb, not directly Dalton.)

21. Henry's law explains the occurrence of:

A. Hyperventilation.
B. Hypoxia.
C. Decompression sickness.

Show answer
ANS · C
§20.1
— DGCA-quoted: "
Henry's Law explains the occurrence of decompression sickness.
" Henry's Law: gas dissolved in a fluid is proportional to its partial pressure at the surface. When ambient pressure drops rapidly, dissolved N_2 comes out of solution in the blood as bubbles → bends, creeps, chokes, staggers.

22. In aviation any mismatch between what we sense and what we expect is:

A. An effect of motion sickness.
B. An illusion.
C. A symptom of Hypoxia.

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Show answer
ANS · B
§45
— DGCA-quoted verbatim: "
In aviation any mismatch between what we sense and what we expect is an illusion.
"

23. On ascent the gases in the digestive tract will:

A. Shrink.
B. Escape to lungs.
C. Expand.

Show answer
ANS · C
§70.2 · Boyle's Law
— On ascent ambient pressure falls → trapped gases expand (Boyle's Law: P·V = constant). Gases in the gut nearly double in volume between sea level and 18,000 ft. This is why gastroenteritis grounds a pilot even on medication (§70.2), and why high-fibre / gas-producing foods are discouraged before flight.

24. The most serious hypoxia:

A. Causes headache and tingling in hands and feet.
B. Causes fornication, flushed cheeks and cherry red lips.
C. Interferes with reasoning, gives rise to unusual fatigue and, finally, results in a loss of consciousness/death.

Show answer
ANS · C
§10
—
Severe
hypoxia is the late-stage cascade: impaired reasoning → unusual fatigue → LoC → death. (A) describes mild early-stage hypoxia. (B) describes CO poisoning, not hypoxia per se (cherry-red lips is the diagnostic CO sign — §11). The source text uses "fornication" but the intended word is "
formication
" (a sensation of ants crawling under the skin — a tactile hallucination).

25. One of the symptoms of decompression sickness are:

A. The bends.
B. Convulsions.
C. Impaired night vision and slow reaction time.

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Show answer
ANS · A
§20 DCS symptoms
— DGCA-quoted DCS symptom list:
Bends, Chokes, Skin manifestations, Neurological symptoms, Circulatory shock
. (B) Convulsions are an obvious-incapacitation cause / epilepsy (§27/§28), not DCS proper. (C) Impaired night vision and slow reaction time are hypoxia-family symptoms (§10), not DCS.

26. The tendency to develop the bends increases with:

A. Low rates of climb, age, obesity, physical activity and high temperatures.
B. High rates of climb, age, obesity, physical activity and low temperatures.
C. High rates of descend age, obesity, physical activity and high temperatures.

Show answer
ANS · B
§19.5
— DGCA-quoted FIVE factors: "
High rates of climb, age, obesity, physical activity, low temperatures.
" Option B matches exactly. (A) flips climb rate and temperature. (C) refers to "descend" instead of "climb" — and high temperature is the wrong direction.

27. The cabin pressure in airline operation is:

A. Normally not exceeding 4,000 to 5,000 feet.
B. Normally not exceeding 2,000 to 3,000 feet.
C. Normally not exceeding 6,000 to 8,000 feet.

Show answer
ANS · C
§19.4
— DGCA-quoted: typical airliner cabin pressure ≈
6,000 ft
, maximum
8,000 ft
. So "not exceeding 6,000 to 8,000 ft" is correct. Modern aircraft (787, A350) maintain lower cabin altitudes (~6,000 ft cruise) for passenger comfort and reduced fatigue — but the DGCA-listed regulatory ceiling remains 8,000 ft.

28. The circulation system transports:

A. Oxygen, carbon monoxide and adrenaline.
B. Oxygen and carbon dioxide only.
C. Chemical messengers, oxygen and carbon dioxide.

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Show answer

ANS · C

§9 Five functions of blood

— DGCA-listed five functions of blood include: respiratory transport (O_2, CO_2), **endocrine transport (hormones / "chemical messengers")**, defence, temperature regulation, waste removal. (A) Carbon monoxide is a toxin — not normally transported. (B) is incomplete — circulation transports far more than just gases.

29. Physiological problems at high altitude are caused by:

A. Increased atmospheric pressure.

B. Reduced oxygen percentage in atmosphere.

C. Decreased atmospheric pressure.

Show answer

ANS · C

§1 / §10 (Part 1 & Part 3)

— The root cause of altitude-related physiological problems is the **decrease in atmospheric pressure**, which lowers the partial pressure of oxygen (Dalton), reduces gas dissolution (Henry), expands trapped body gases (Boyle), and degrades cognitive performance. (B) is the classic trap — the *percentage* of O_2 is constant; it's the partial pressure that falls. (A) is the wrong direction.

30. Which gas will diffuse from the blood to the alveoli:

A. Nitrogen.

B. Carbon Dioxide.

C. Oxygen.

Show answer

ANS · B

§8 Respiratory exchange

— Gas exchange in alveoli: **O_2 moves FROM alveoli INTO blood** (high pO_2 in alveoli → lower pO_2 in venous blood); **CO_2 moves FROM blood INTO alveoli** (high pCO_2 in venous blood → lower pCO_2 in alveoli). Both exchanges happen by simple diffusion down partial-pressure gradients. (A) Nitrogen is inert and roughly in equilibrium across the membrane — no net diffusion either way in normal breathing.

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31. The symptoms caused by gas bubbles under the skin following a decompression are called:

A. Leans.

B. Creeps.

C. Chokes.

Show answer

ANS · B

§20.2

— DGCA's four DCS syndromes:

Bends (joints), Creeps (skin), Chokes (lungs), Staggers (brain)

. Skin bubbles →

Creeps

. (A) Leans is a vestibular illusion in IMC — not DCS. (C) Chokes is bubbles in pulmonary capillaries.

32. The Oxygen-Carbon Dioxide exchange takes place through:

A. The walls of the capillaries.

B. Red blood cells.

C. Platelets.

Show answer

ANS · A

§8 Alveolar-capillary membrane

— Gas exchange happens across the

alveolar-capillary membrane

— the thin walls separating alveolar air from blood in pulmonary capillaries. RBCs are the

carriers

of O_2 once it has crossed the membrane (Q12), but the exchange site itself is the capillary wall.

33. In order to get rid of excess nitrogen following scuba diving, subsequent flights should be delayed:

A. 36 hours.

B. 12 hours.

C. 24 hours.

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Show answer
ANS · C
§21
— DGCA-quoted: "
*As a general rule, individuals should NOT fly within **24 hours** following diving and certainly not the same day.*
" 12 hours is the post-rapid-decompression rule (§20.4) — not diving. 36 hours has no DGCA basis here.

34. The normal arterial blood-pressure of a healthy adult is:

A. 140/100 mm HG.
B. 120/60 mm HG.
C. 120/80 mm HG.

Show answer
ANS · C
§15
— DGCA-quoted normal BP:
120/80 mm Hg
(systolic/diastolic). (A) 140/100 is hypertension. (B) 120/60 — diastolic is too low.

35. Decompression symptoms are:

A. Bends, chokes, skin manifestations, neurological symptoms and circulatory shock.
B. Bends, chokes, Personality change, Impaired memory and circulatory shock.
C. Bends, chokes, Personality change, Impaired memory and Impaired vision.

Show answer
ANS · A
§20.1 DGCA verbatim 5-symptom list
— DGCA-quoted: "
Symptoms of decompression sickness are: Bends, Chokes, Skin manifestations, Neurological symptoms, Circulatory shock.
" Options B and C substitute hypoxia-style symptoms (personality change, memory, vision) — wrong category.

36. Baroreceptor reflex is triggered by:

A. Body's need to adjust blood pressure.
B. Body's need to adjust to extreme temperatures.
C. Body's need to adjust to extreme cold.

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Show answer
ANS · A
§16
— DGCA-quoted: "
Any change in your body's demand for blood can trigger your baroreceptor reflex.
" The receptors are
pressure sensors
in the carotid arteries and aortic arch — they respond to BP changes, not temperature. Temperature regulation is a separate autonomic loop.

37. While climbing from sea level to 40,000 ft the difference in barometric pressure is greatest between:

A. Sea level and 10,000 ft.
B. 10,000 and 20,000 ft.
C. 30,000 and 40,000 ft.

Show answer
ANS · A
§3 ISA pressure curve · Q7 above
— Atmospheric pressure decreases
exponentially
with altitude — so the absolute drop per 10,000 ft is
greatest at the bottom
. From SL (1013 hPa) to ~10,000 ft (697 hPa) = drop of ~316 hPa. From 30,000 to 40,000 ft (300→187 hPa) = drop of ~113 hPa. The biggest absolute change is the SL → 10,000 ft band. This is why hypoxic risk rises steeply in the first few thousand feet.

38. The blood-pressure which is measured during flight medical checks is the:

A. Pressure in the artery of the left upper arm (representing the pressure at left arm).
B. Pressure in the artery of the upper arm (representing the pressure at heart level).
C. Pressure in the artery of the upper arm (representing the cardiac pressure).

Show answer
ANS · B
§15
— Standard clinical BP measurement is at the
brachial artery (upper arm)
, with the arm positioned at
heart level
— so the reading represents systemic arterial pressure at heart level. (A) "left arm" is artificial — either arm is used, both should agree (significant difference indicates pathology). (C) "cardiac pressure" is wrong terminology — cardiac pressures are measured inside the heart, not in an arm artery.

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39. What is the average Time of Useful Consciousness after a rapid decompression at 40,000 ft?

A. About 12 seconds.

B. More than 1 minute.

C. About 40 seconds.

Show answer

ANS · A

§18.3 TUC Master Table

— DGCA TUC table at **FL400 / 40,000 ft, rapid decompression**: **7 – 10 seconds**. Normal-ascent TUC at FL400 is 15 – 20 s. The closest option to the rapid-decompression value (7-10s, with some sources averaging 12s) is option A. (B) ">1 min" is FL220 normal value. (C) "40 s" is closer to the FL350 normal value (30s-1min).

40. What is the Time of Useful Consciousness?

A. The amount of time an individual is able to function effectively (e.g. perform flying duties) in an environment of inadequate oxygen supply.

B. The time period during which there is an inability of the cells of the body to use the oxygen available.

C. The period of time when there is inadequate blood flow to body tissues.

Show answer

ANS · A

§18.1

— DGCA-quoted definition verbatim. (B) describes histotoxic hypoxia. (C) describes ischemic / stagnant hypoxia. The hallmark of TUC is the phrase "function effectively in an environment of *inadequate oxygen supply*" — the test is functional usefulness, not consciousness alone.

41. What are the signs of hypoxia during explosive decompression?

A. Initially, there is an inability to concentrate, impairment of night vision, unusual fatigue, loss of consciousness/death.

B. Initially there is cramping and spasms of the hands and feet, cold clammy skin, paleness, impaired judgement.

C. Initially there may be a feeling of wellbeing (euphoria), impairment of night vision, unusual fatigue, loss of consciousness/death.

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Show answer
ANS · C
§10 Hypoxia symptom cascade
— The crucial
"euphoria" first sign
distinguishes hypoxia from other emergencies — the pilot
feels good
before they realise something is wrong. This is why hypoxia is so dangerous (and why the master rule is "if at altitude and feeling great for no reason, suspect hypoxia"). (A) is missing the euphoria. (B) describes hyperventilation (cramping, spasms — tetany).

42. Which is the procedure to be followed when symptoms of decompression sickness occur?

A. 100% Oxygen, immediate descent, land as soon as possible, seek medical assistance immediately on landing.
B. 100% Oxygen, immediate descent, land in case not comfortable, seek medical assistance if required.
C. Descend below 10,000 feet, symptoms will disappear after few minutes.

Show answer
ANS · B
§20.3 Treatment of DCS
—
*Note on this question: The official answer key marks (B) as correct, but the strict DGCA-quoted four-step treatment is: "**Keep patient warm and on 100% Oxygen · Initiate immediate descent · Land as soon as possible · Seek medical assistance immediately on landing.**" — which matches (A) more closely.*
The most clinically defensible answer is (A), but the source key indicates (B). For exam purposes mark per the official key. The DGCA-quoted theory definitively rules out (C), which is wrong — DCS does not self-resolve at 10,000 ft.

43. What is decompression sickness?

A. A rapid reduction in ambient pressure, may cause the nitrogen in our blood to come out of solution as small bubbles.
B. A rapid increase in ambient pressure, may cause the nitrogen in our blood to come out of solution as small bubbles.
C. A rapid reduction in ambient pressure, may cause the oxygen in our blood to come out of solution as small bubbles.

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Show answer
ANS · A
§20.1
— DGCA-quoted Henry's Law application: "
A rapid REDUCTION in ambient pressure may cause the NITROGEN in our blood to come out of solution as small bubbles.
" Not oxygen (which is consumed metabolically and present at low dissolved levels); not an increase in pressure (which would dissolve more gas, not release it).

44. The part of blood without cell is called:

A. Serum.
B. Lymph.
C. Plasma.

Show answer
ANS · C
§9
— Blood = formed elements (RBCs, WBCs, platelets — >90 % RBCs) +
plasma
(the cell-free liquid component containing water, proteins, hormones, salts, dissolved gases). (A) Serum is plasma
minus the clotting factors
— what's left after blood clots. (B) Lymph is a different fluid in the lymphatic system, not part of blood.

45. At rest, with a heart rate of 72 beats per minute and a stroke volume of 70 ml the cardiac output is about:

A. 7 liters/min.
B. 5 liters/min.
C. 6 liters/min.

Show answer
ANS · B
§9
— Cardiac Output = Heart Rate × Stroke Volume = 72 × 70 ml =
5,040 ml/min ≈ 5 L/min
. This is the DGCA-quoted resting cardiac output for a typical adult. At exercise it can rise to 20+ L/min.

46. Hemoglobin is:

A. In the white blood cells.
B. Dissolved in the plasma.
C. In the red blood cells.

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Show answer
ANS · C
§9
— Haemoglobin (Hb) lives inside
red blood cells
— ~280 million Hb molecules per RBC. WBCs (leukocytes) are immune cells, not O_2 carriers. Plasma carries only a small fraction of total blood O_2 as dissolved gas.

47. The heart muscle is supplied with blood from:

A. The pulmonary veins.
B. The coronary arteries.
C. Coronary capillaries.

Show answer
ANS · B
§9 Cardiovascular anatomy
— The heart muscle (myocardium) is supplied with oxygenated blood by the
coronary arteries
— branches off the aorta just above the aortic valve. (A) Pulmonary veins return oxygenated blood from lungs to the left atrium. (C) Coronary capillaries are the downstream microvessels — the supply is from the arteries.

48. Bubbles in the joints cause rheumatic-like pain, called the:

A. Staggers.
B. Chokes.
C. Bends.

Show answer
ANS · C
§20.2
— DGCA-quoted verbatim: "
Joints: Bubbles in the joints cause rheumatic-like pain, called the BENDS.
" The mnemonic: Joints Bend, Skin Creeps, Lungs Choke, Brain Staggers.

49. Which symptom does not belong to the Decompression Sickness:

A. Bends.
B. Creeps.
C. Convulsions.

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Show answer
ANS · C
§20 vs §27/§28
— Bends and Creeps ARE DCS syndromes.
Convulsions
are associated with
obvious incapacitation
(§27.2) and
epilepsy / Grand Mal
(§28.1) — not with DCS. The DGCA-listed DCS symptom set is Bends · Chokes · Skin manifestations · Neurological symptoms · Circulatory shock; "convulsions" is not in that list (though severe neurological DCS could produce them, the DGCA list does not include the word).

50. A few hours after a rapid decompression at FL320 you experience a feeling of wellbeing after having three pints of beer. You should:

A. Ask for medical advice since this is a symptom of hypoxia.

B. Ask for medical advice since this is a symptom of decompression sickness.

C. Stop drinking and allow twenty-four hours between the last drink and take off time.

Show answer
ANS · C
§14.1 · §20.4
— A complex scenario question with
two simultaneous rules in play
:

- **Alcohol bottle-to-throttle:** `24 hr` wait between last drink and take-off (§14.1).
- **Rapid decompression:** `12 hr` wait before flying after RD (§20.4).

The longer of the two binding rules wins →
24 hours
. "Feeling of wellbeing" after 3 beers is simple intoxication / disinhibition — not hypoxia (you're on the ground) and not DCS (already past the 12-hr RD window). The correct action: stop drinking, wait 24 hr for alcohol to clear, then fly.
(Note: the answer ALSO satisfies the 12-hr post-RD rule by overlap — 24 hr > 12 hr.)

GROWING THE BANK

This chapter ships with 50 worked multiple-choice questions. The bank can be expanded to the full set the same way — every item cross-referenced to a chapter and verified before it goes live.

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✦ END OF THE BOOK ✦

"Know your own limits as well as you know your aircraft's — and you will always have somewhere to go."

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HUMAN PERFORMANCE & LIMITATIONS · REFERENCE

Glossary, Index & References

The back matter: a glossary of the terms used throughout the book, an alphabetical index by chapter, the references consulted, and a word about the author.

Glossary of Terms

Alveolus
The tiny air sac in the lung, one cell thick, where oxygen and carbon dioxide are exchanged with the blood. *(Ch 5)*

Armstrong's Line
~63,000 ft, the altitude at which body fluids boil at body temperature because ambient pressure equals their vapour pressure. *(Ch 3)*

Autokinetic effect
The illusion that a single fixed light in the dark is moving. *(Ch 14)*

Barodontalgia
Tooth pain caused by gas trapped under a filling or in decay expanding as cabin pressure falls. *(Ch 17)*

Barotrauma (otic / sinus)
Injury from a pressure difference that cannot equalise — commonly the middle ear or sinuses on descent. *(Ch 13, 17)*

Baroreceptor reflex
The body's automatic correction of blood pressure via sensors in the carotid artery and aorta. *(Ch 8)*

Circadian rhythm
The body's ~24-hour internal clock of alertness and temperature; disrupted by jet lag and night duty. *(Ch 24)*

CRM (Crew Resource Management)
The use of all available resources — people, procedures, equipment — to manage a flight safely. *(Ch 22)*

Dark adaptation
The eye's adjustment to darkness: cones in ~7 minutes, full rod sensitivity in ~30. *(Ch 12)*

Decompression sickness (DCS / the bends)
Nitrogen bubbles forming in blood and tissue when pressure falls (Henry's Law). *(Ch 8)*

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Dirty Dozen
The twelve common human preconditions for error. *(Ch 1)*

Eustachian tube
The passage that equalises middle-ear pressure with the surroundings. *(Ch 13)*

Fovea
The central pit of the retina, all cones — sharpest day vision, no night vision. *(Ch 11)*

G-LOC
G-induced loss of consciousness, the end of the positive-G cascade (grey-out → black-out → G-LOC). *(Ch 15)*

Haemoglobin
The iron-bearing molecule in red cells that carries oxygen; ~280 million per red cell, four O_2 per molecule. *(Ch 5)*

Hypoxia
A deficiency of oxygen reaching the tissues; four types — hypoxic, anaemic, stagnant, histotoxic. *(Ch 6)*

ISA
International Standard Atmosphere: 1013.25 hPa and +15 °C at sea level, lapse 1.98 °C/1000 ft to 36,090 ft. *(Ch 2)*

Load factor
The ratio of lift to weight, felt as "G"; 1.41 G at 45° of bank in a level turn. *(Ch 15)*

Presbyopia / Myopia / Hypermetropia
Age-related loss of focus / short-sight / long-sight. *(Ch 11–12)*

SHELL model
Human-factors model with Liveware (the human) at the centre, interfacing Software, Hardware, Environment and Liveware. *(Ch 1)*

Somatogravic illusion
Acceleration felt as a nose-up pitch (deceleration as nose-down). *(Ch 14)*

TUC
Time of Useful Consciousness — the working time after oxygen is lost; halved by rapid decompression. *(Ch 8)*

Vestibular system
The inner-ear balance organs: three semicircular canals (rotation) and otoliths (linear acceleration). *(Ch 13)*

WOCL
Window of Circadian Low, ~03:00–05:00, the period of lowest alertness. *(Ch 24)*

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References & Further Reading

- ICAO Doc 9683 — *Human Factors Training Manual.*
- ICAO Annex 1 — *Personnel Licensing* (medical provisions).
- DGCA Civil Aviation Requirements (CARs) & Aeronautical Information Circulars — current editions.
- CAP 715 — *An Introduction to Aircraft Maintenance Engineering Human Factors.*
- FAA — *Pilot's Handbook of Aeronautical Knowledge*, Aeromedical Factors chapter.
- FAA — *Introduction to Aviation Physiology.*

These are the standard authorities against which the facts in this book are checked. Always defer to the current official edition.

About the Author

Capt. Pankaj Pahil is a pilot, flight instructor and ground instructor with seventeen years in aviation. He founded **Ghost Aviator** to give every student pilot in India correct, free, and beautifully-made DGCA exam preparation — question banks, notes, mock tests and live ground classes in Air Regulations, Meteorology and Air Navigation. This book is part of that mission.

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